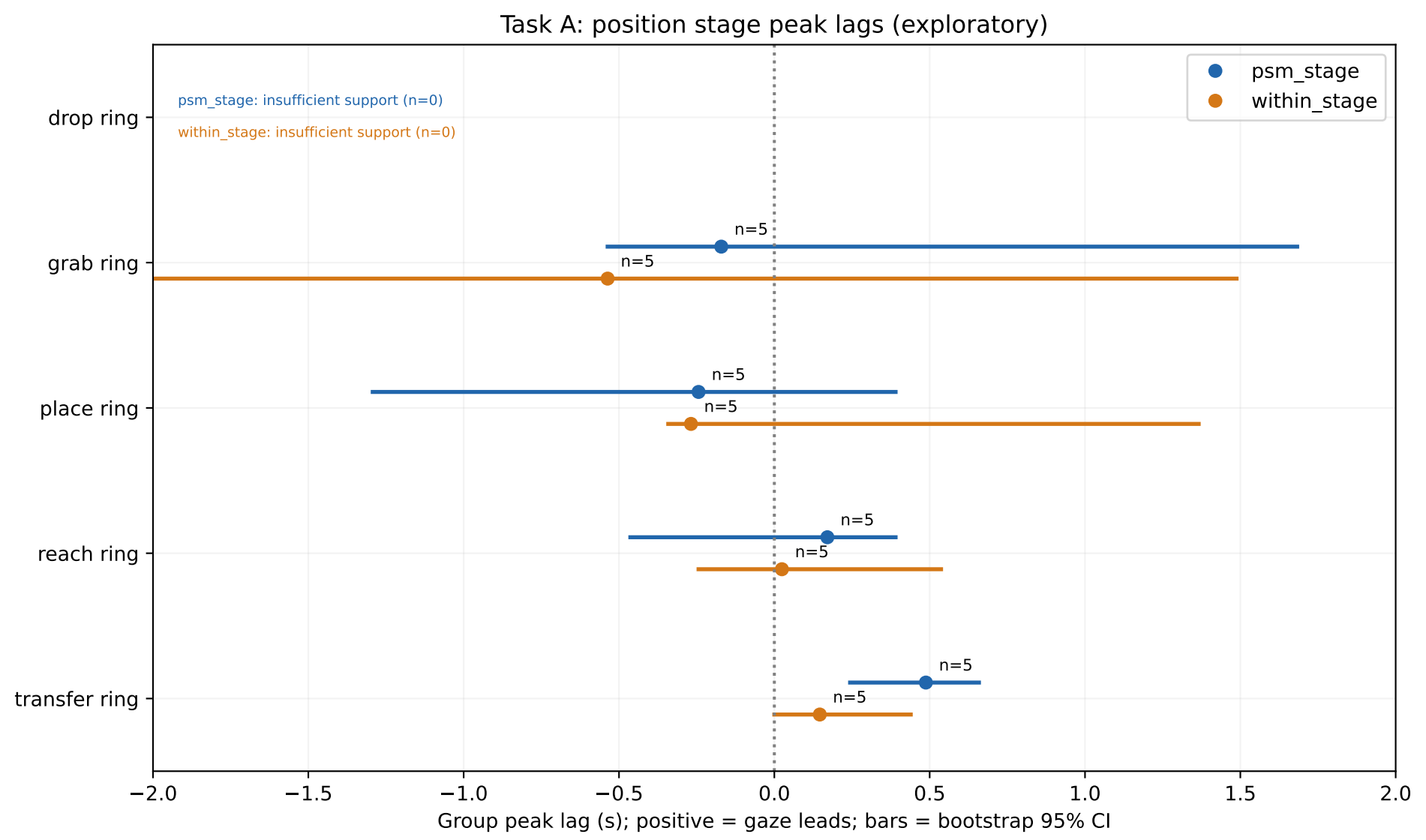
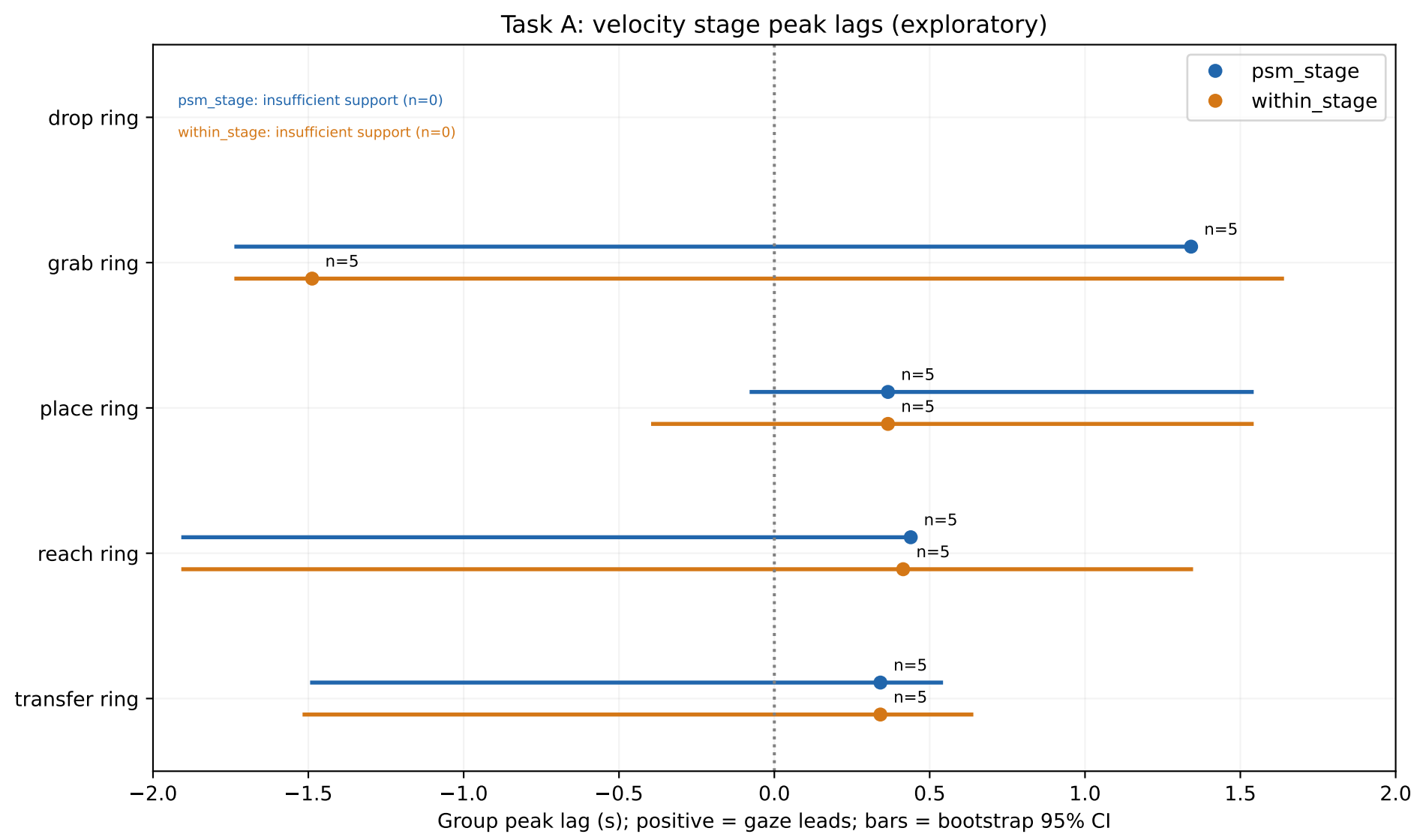
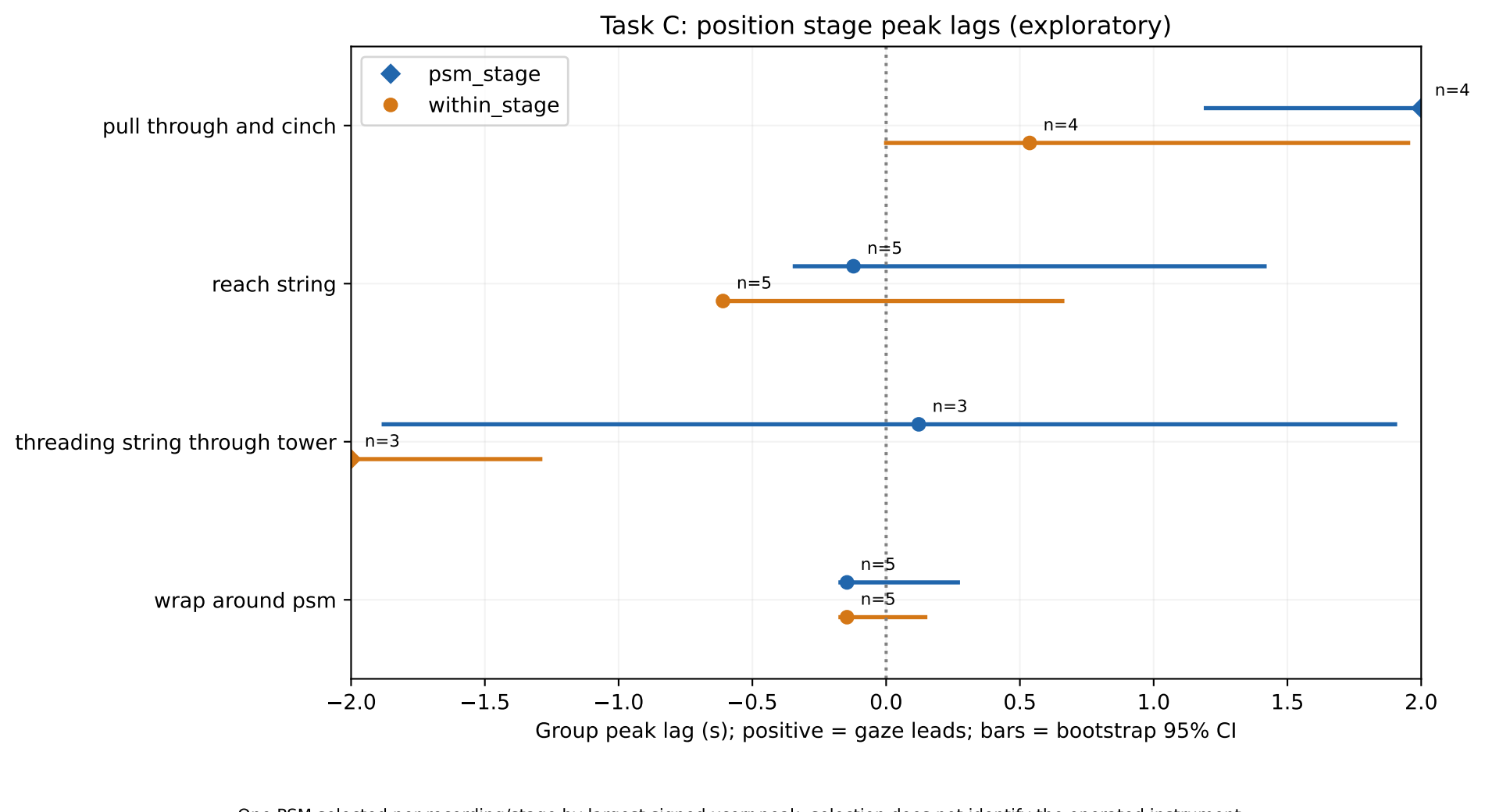


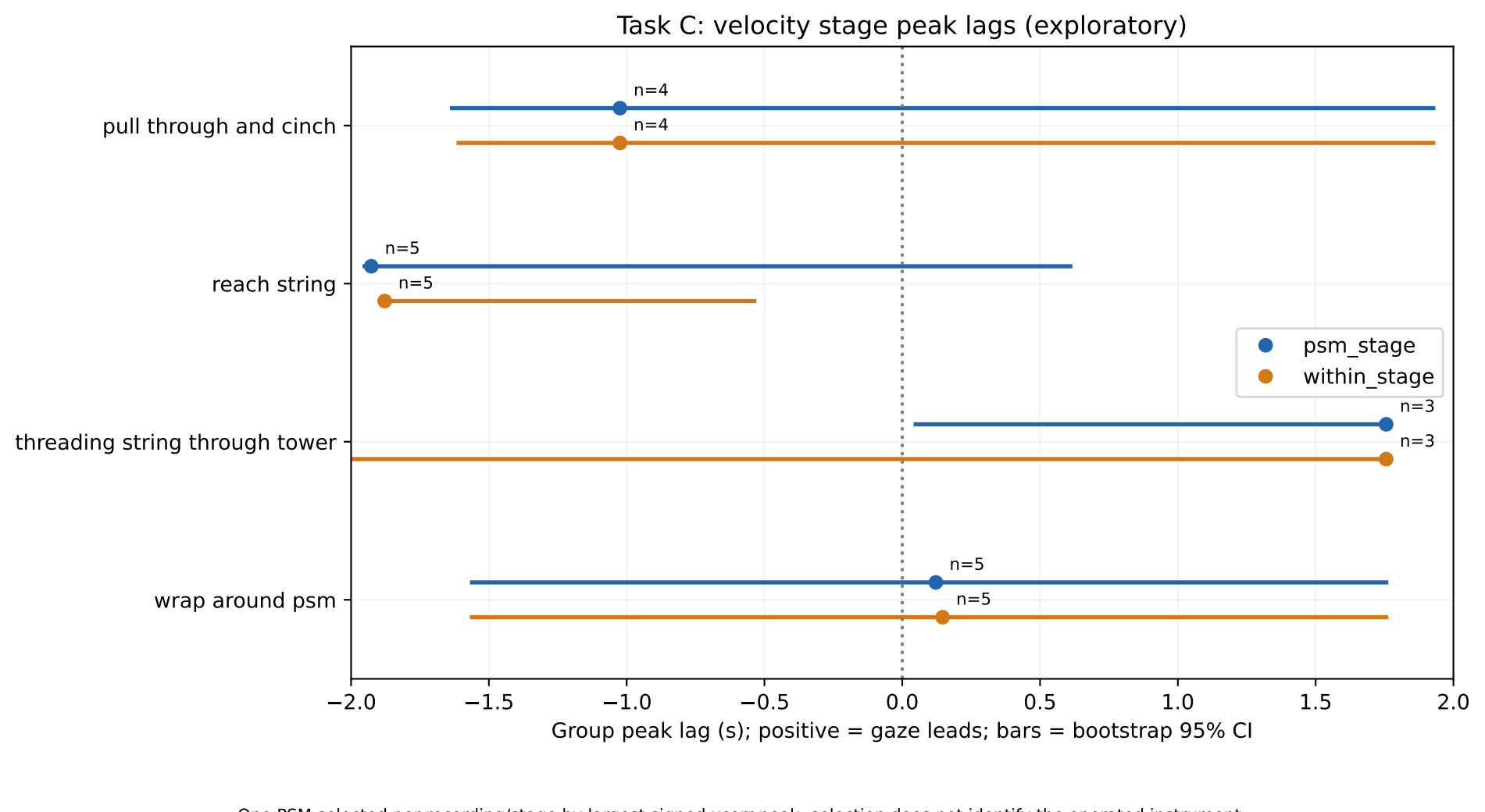
One PSM selected per recording/stage by largest signed xcorr peak; selection does not identify the operated instrument.



One PSM selected per recording/stage by largest signed xcorr peak; selection does not identify the operated instrument.



One PSM selected per recording/stage by largest signed xcorr peak; selection does not identify the operated instrument.



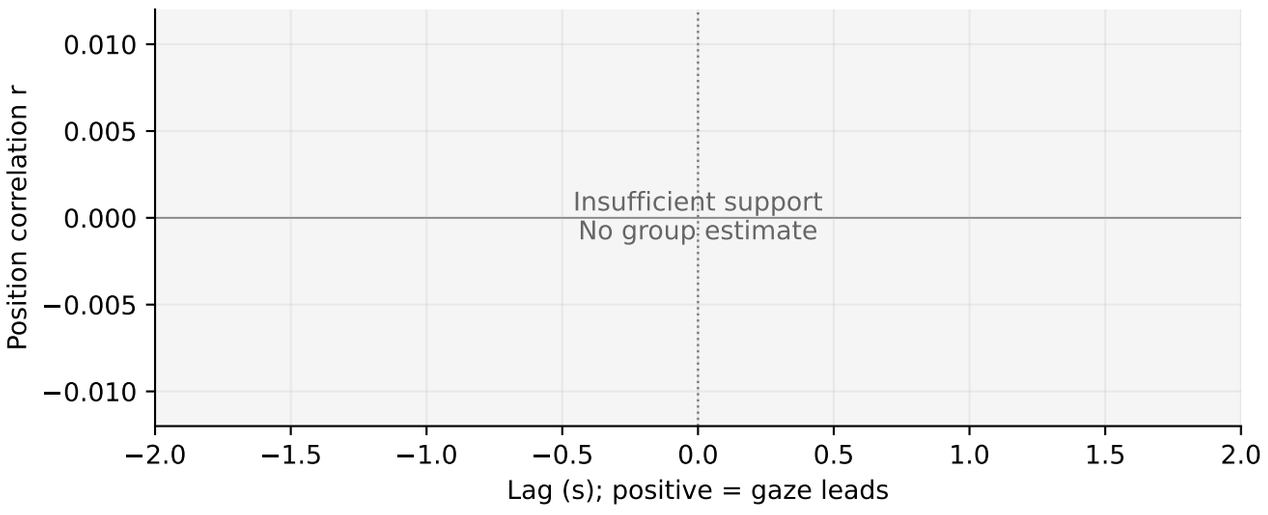
One PSM selected per recording/stage by largest signed xcorr peak; selection does not identify the operated instrument.

Task A: position, maximum-xcorr PSM per stage

Unpaired reference (95%) · Participant bootstrap (95%) · Group mean

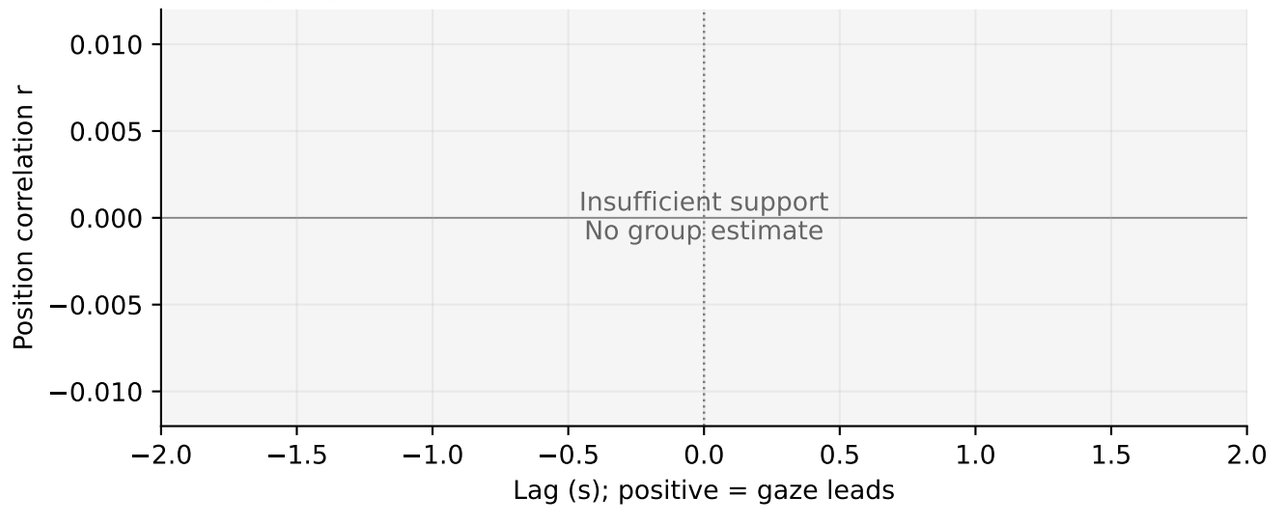
drop ring / psm_stage

n = 0; no_valid_curve



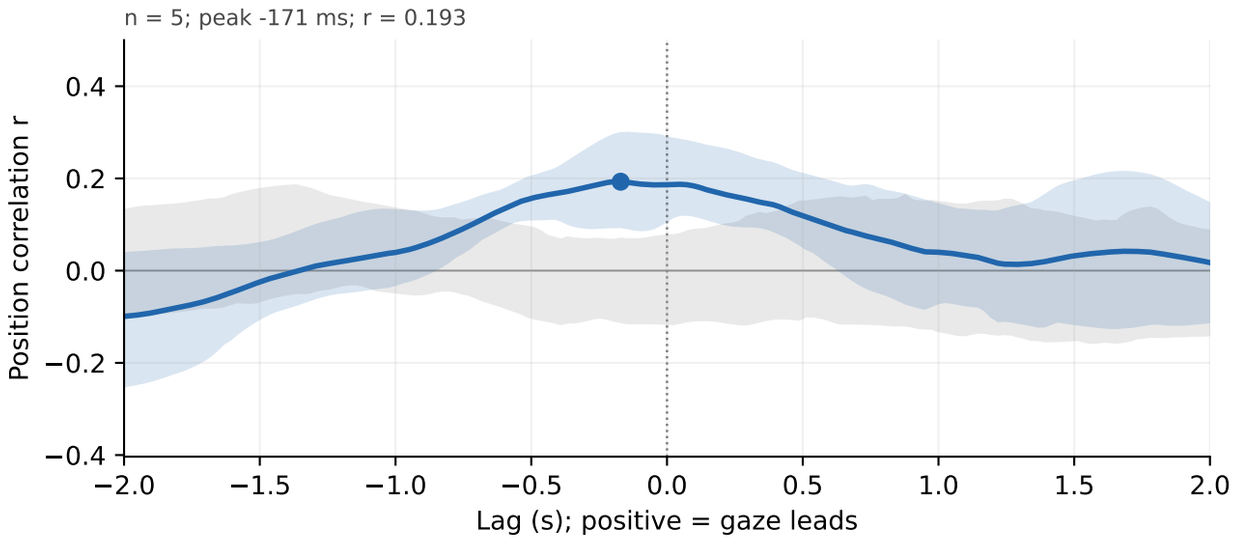
drop ring / within_stage

n = 0; no_valid_curve



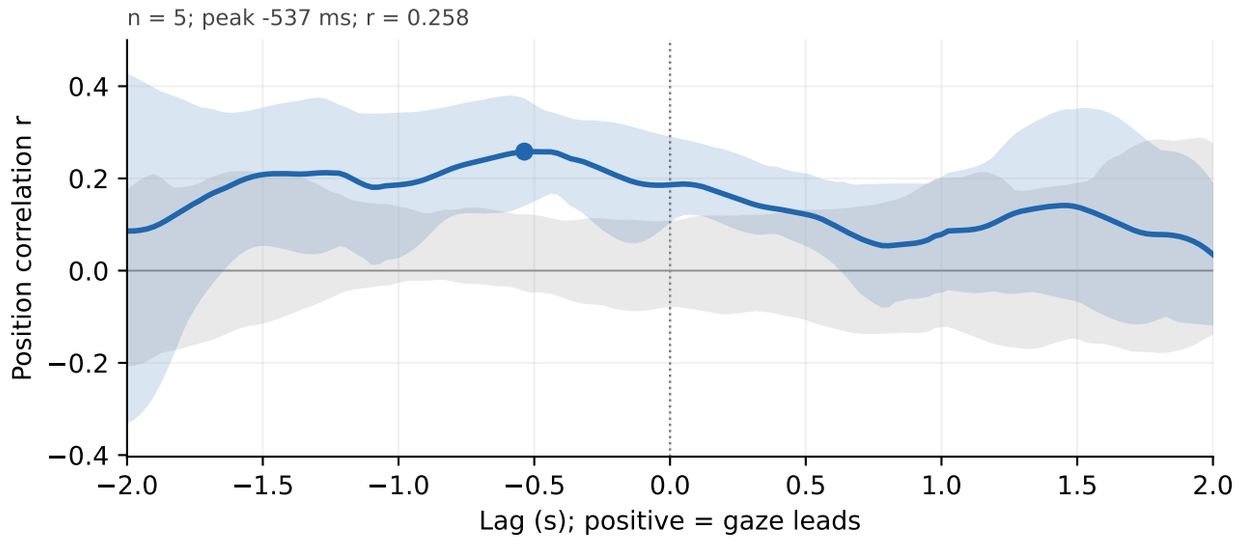
grab ring / psm_stage

n = 5; peak -171 ms; r = 0.193



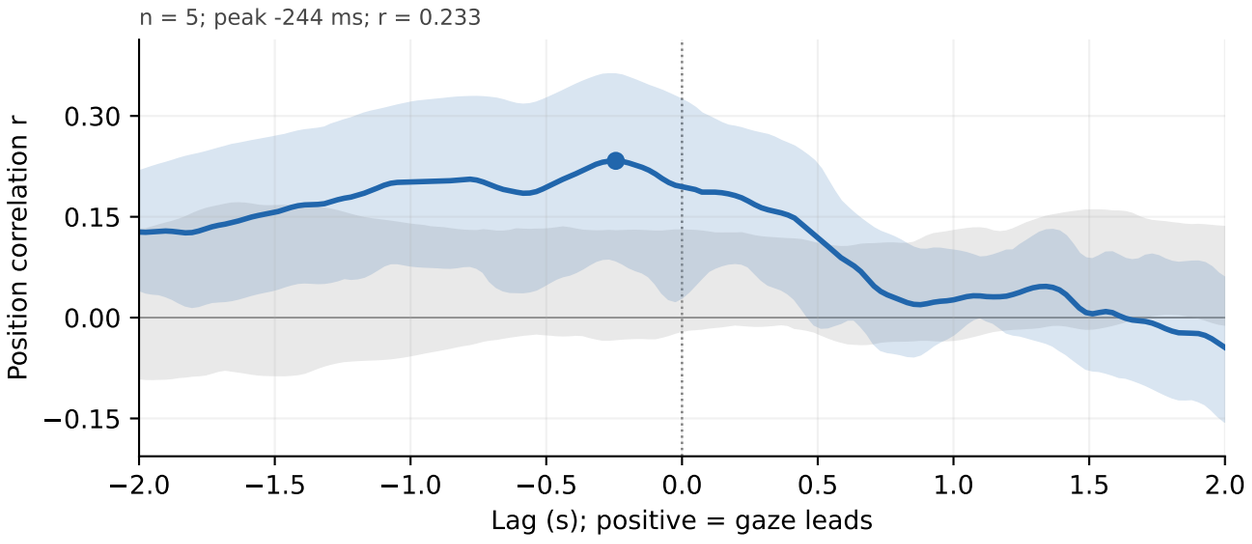
grab ring / within_stage

n = 5; peak -537 ms; r = 0.258



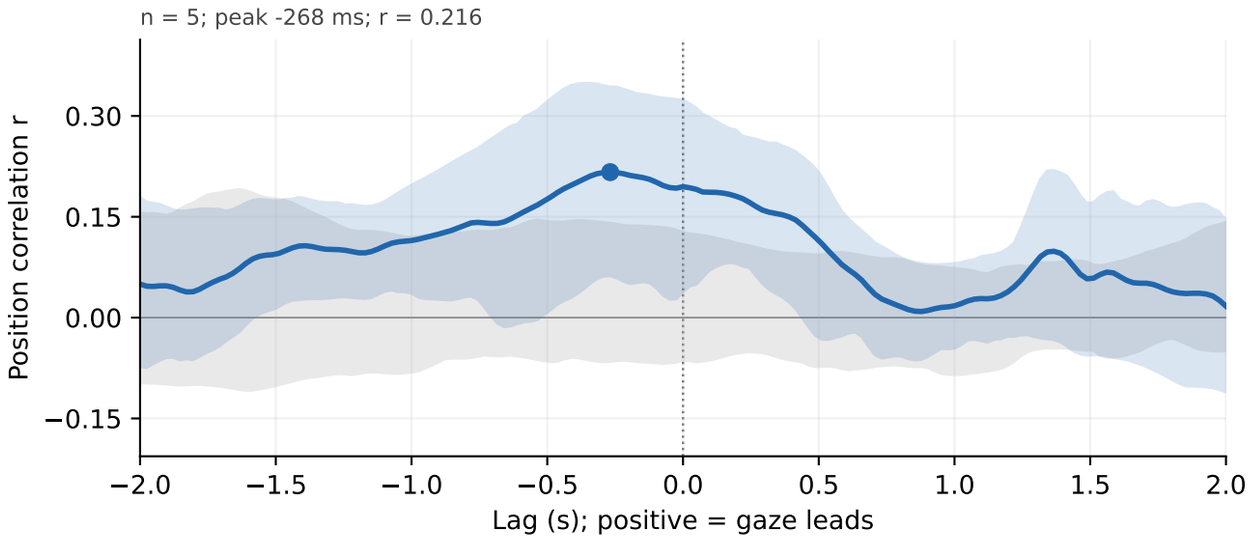
place ring / psm_stage

n = 5; peak -244 ms; r = 0.233



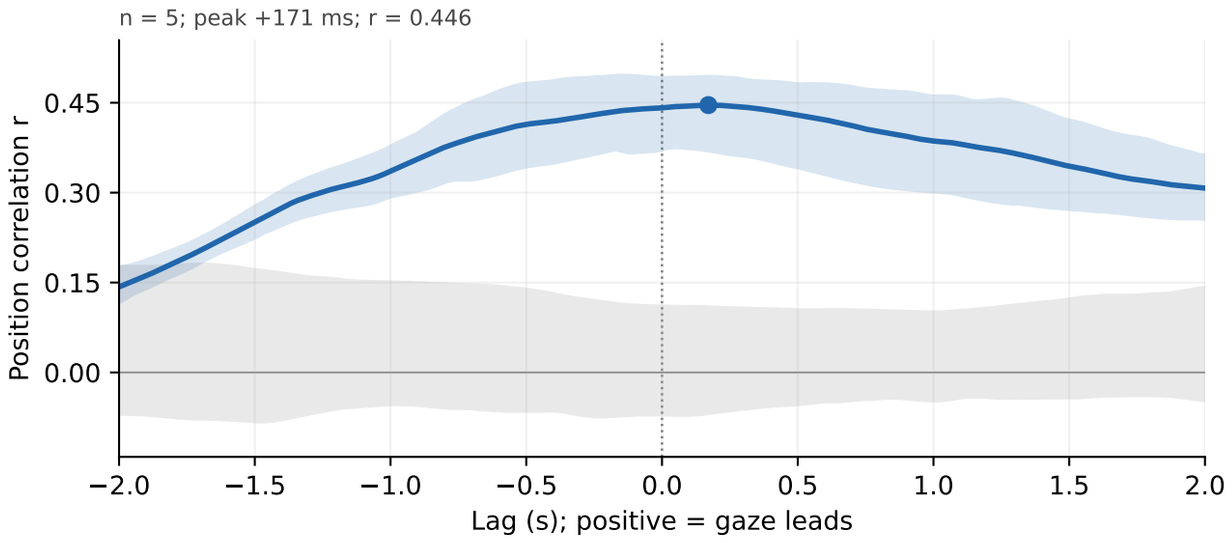
place ring / within_stage

n = 5; peak -268 ms; r = 0.216



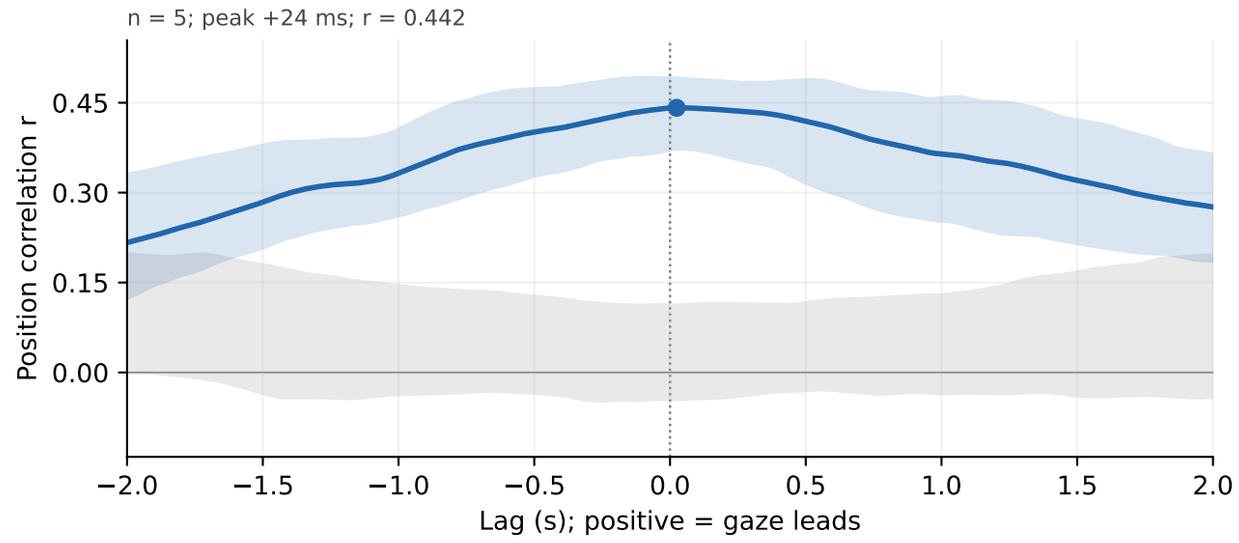
reach ring / psm_stage

n = 5; peak +171 ms; r = 0.446



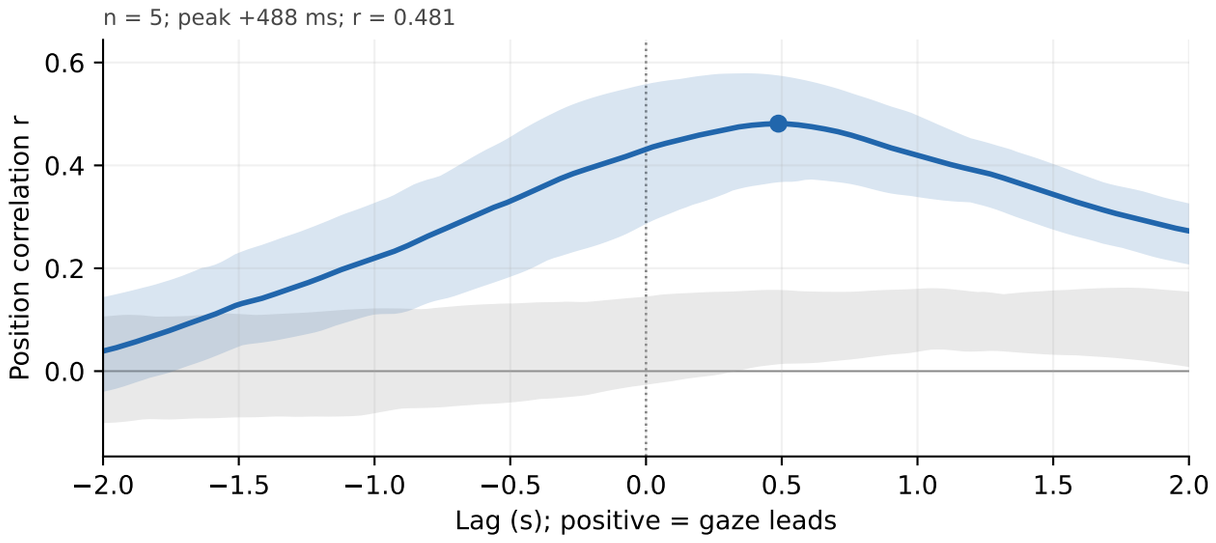
reach ring / within_stage

n = 5; peak +24 ms; r = 0.442



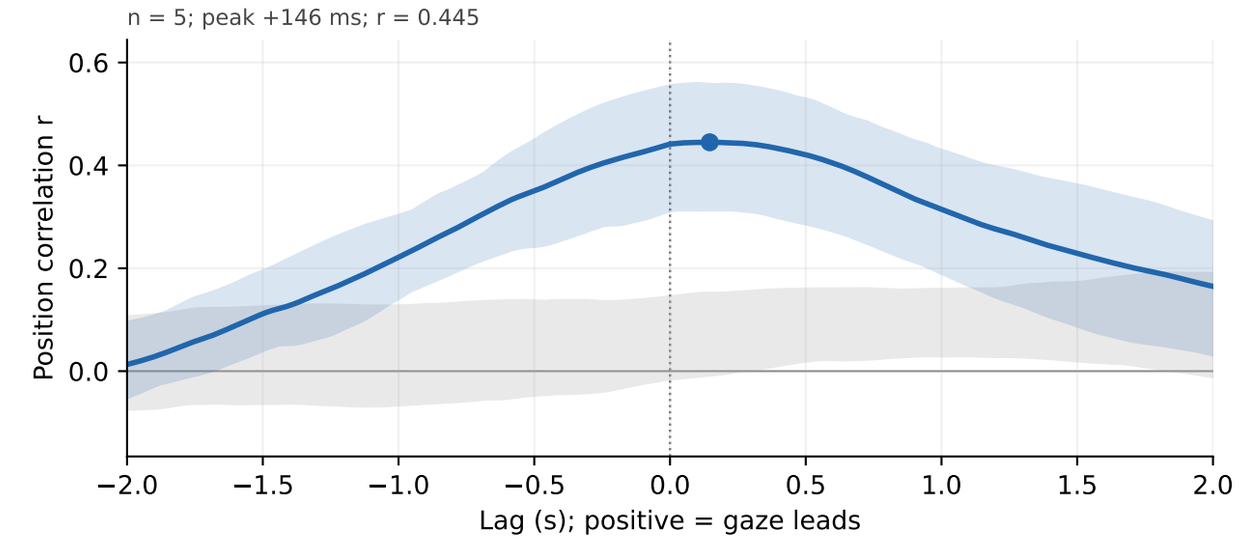
transfer ring / psm_stage

n = 5; peak +488 ms; r = 0.481



transfer ring / within_stage

n = 5; peak +146 ms; r = 0.445



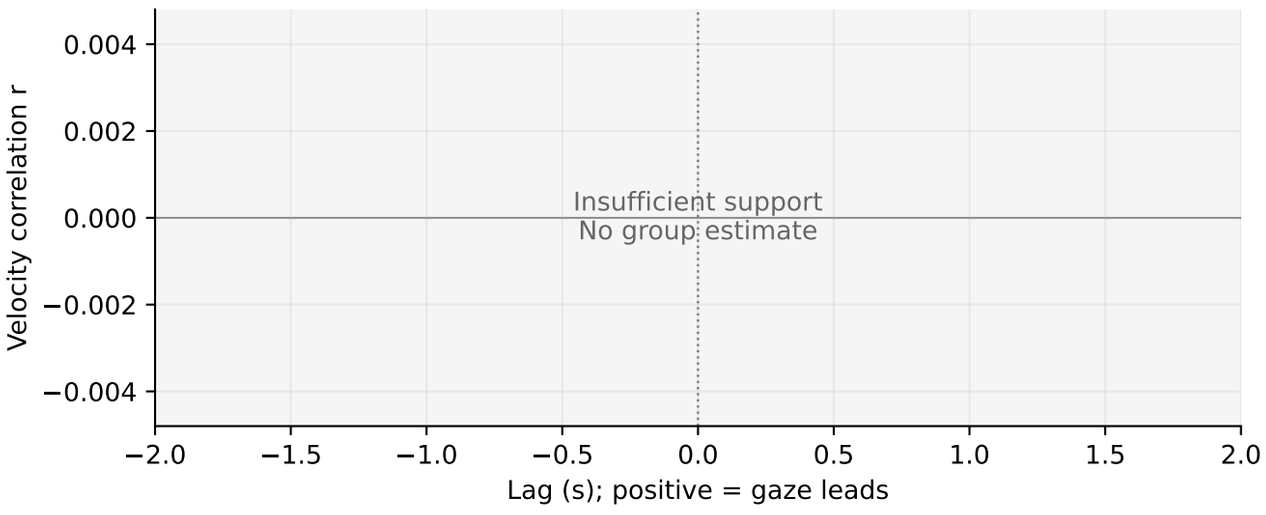
One PSM selected per recording/stage by largest signed xcorr peak; selection does not identify the operated instrument.

Task A: velocity, maximum-xcorr PSM per stage

Unpaired reference (95%) | Participant bootstrap (95%) | Group mean

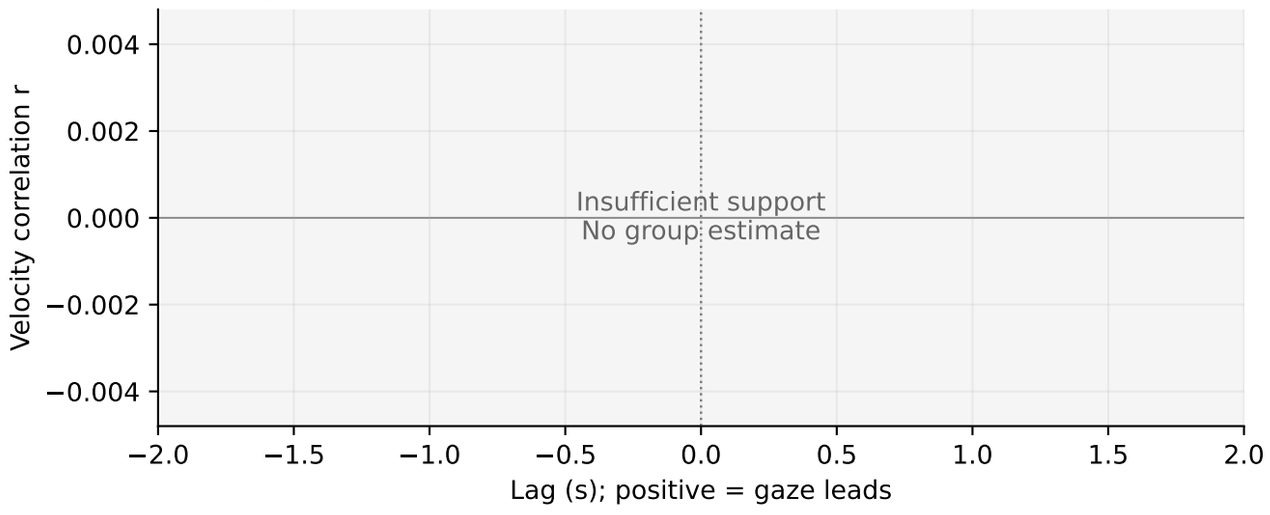
drop ring / psm_stage

n = 0; no_valid_curve



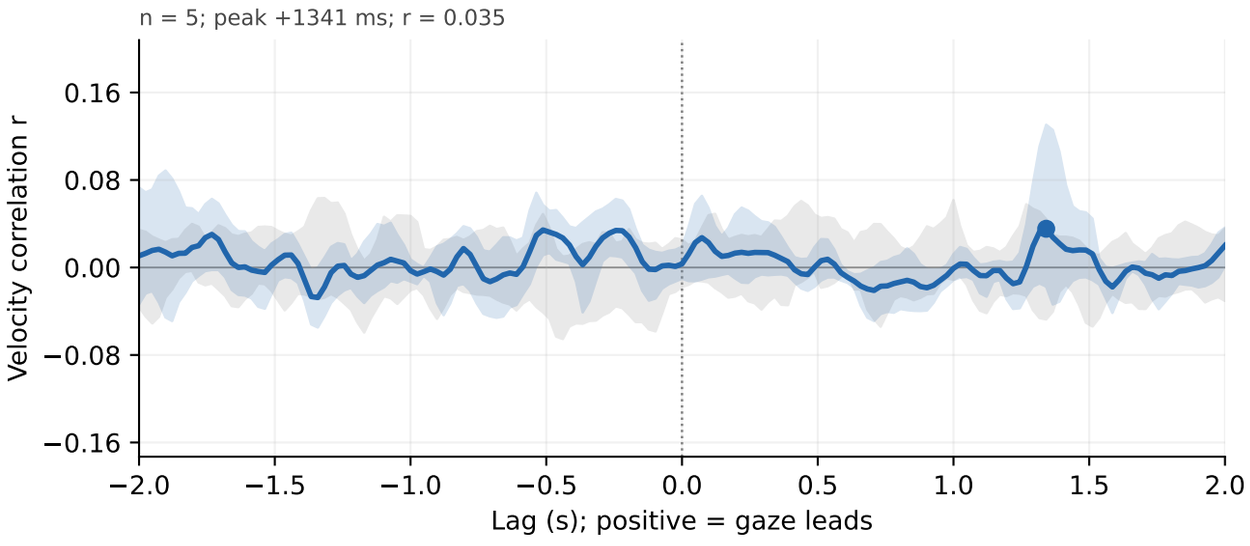
drop ring / within_stage

n = 0; no_valid_curve



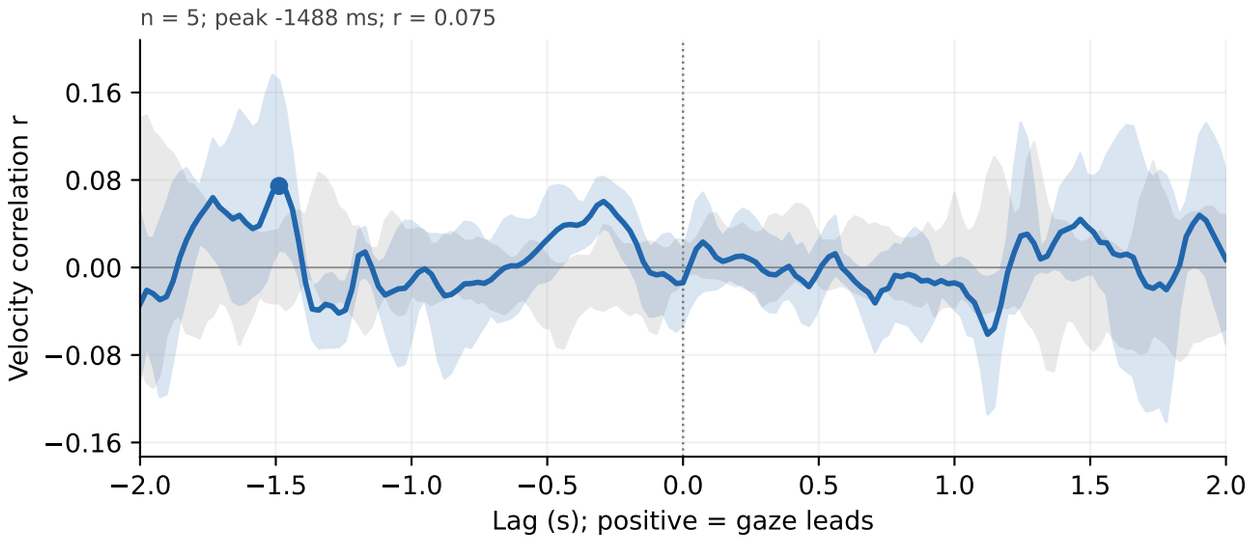
grab ring / psm_stage

n = 5; peak +1341 ms; r = 0.035



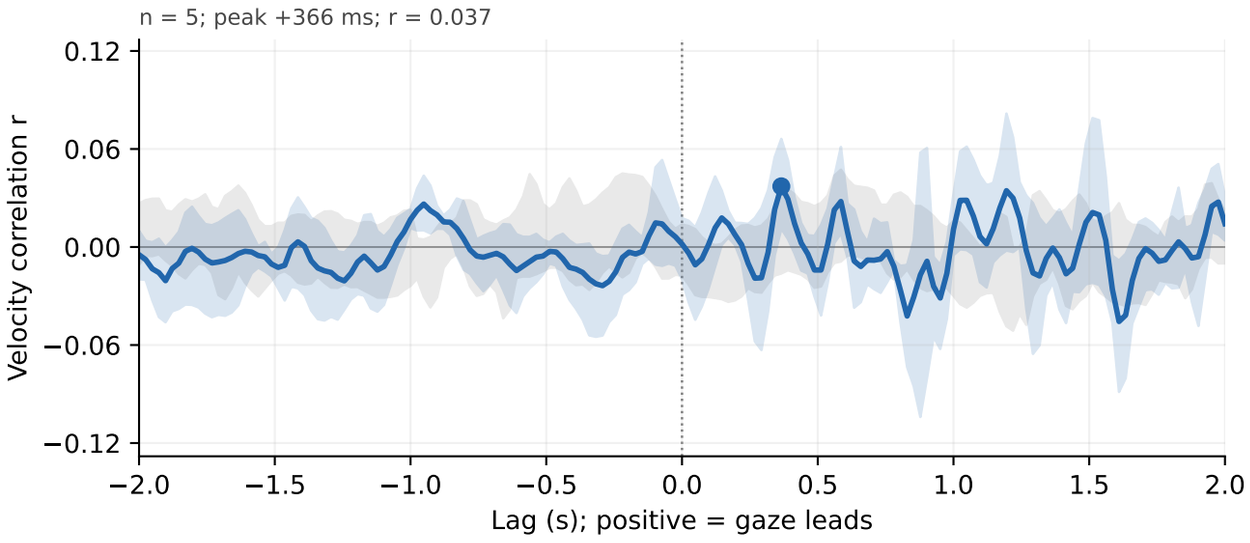
grab ring / within_stage

n = 5; peak -1488 ms; r = 0.075



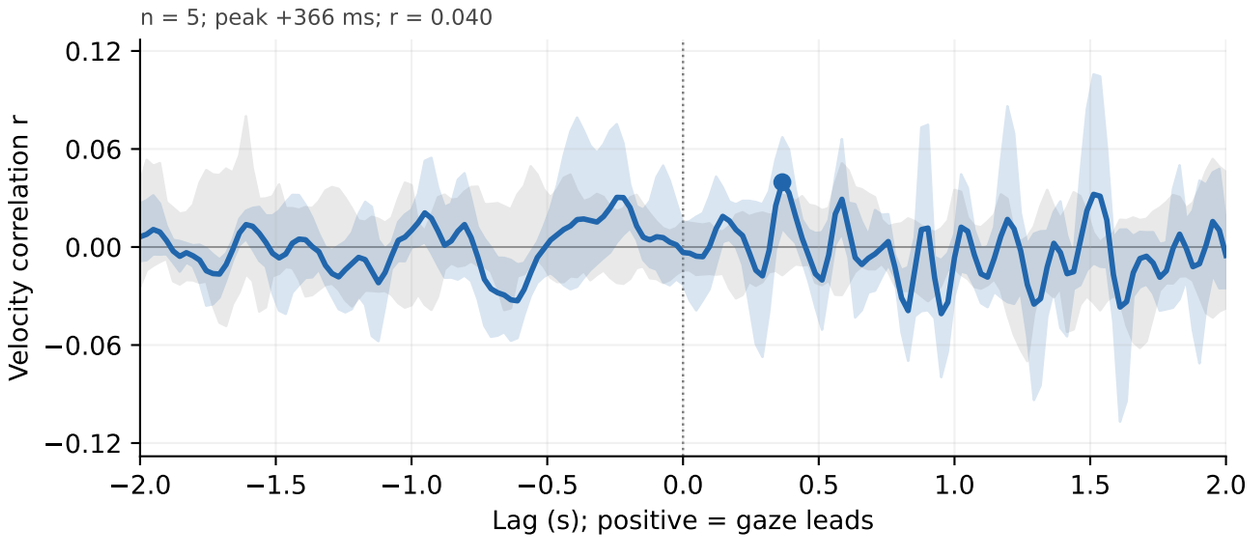
place ring / psm_stage

n = 5; peak +366 ms; r = 0.037



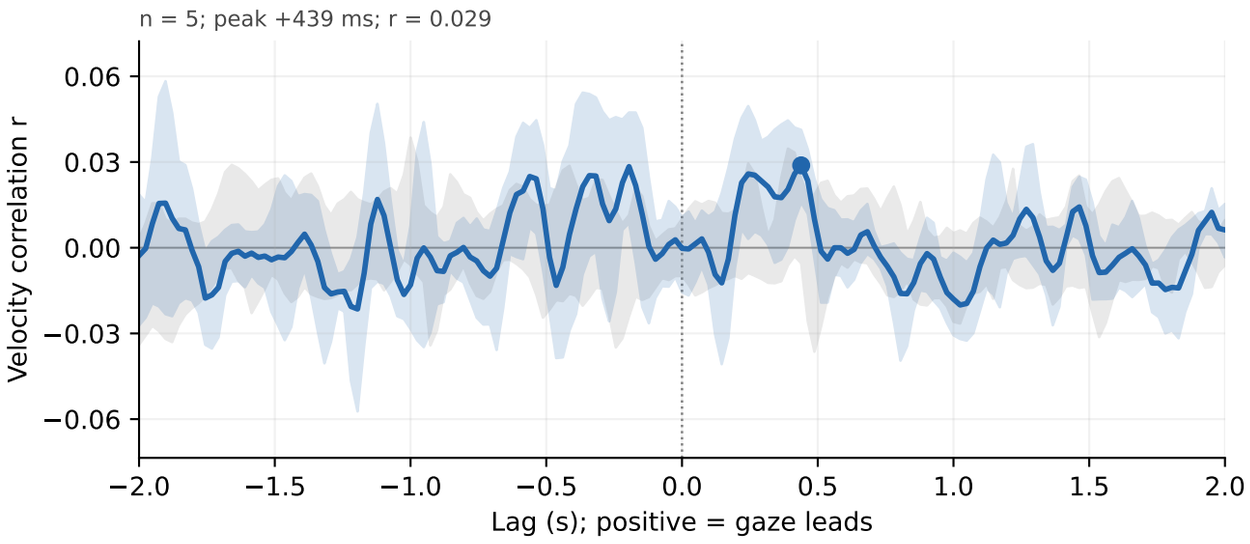
place ring / within_stage

n = 5; peak +366 ms; r = 0.040



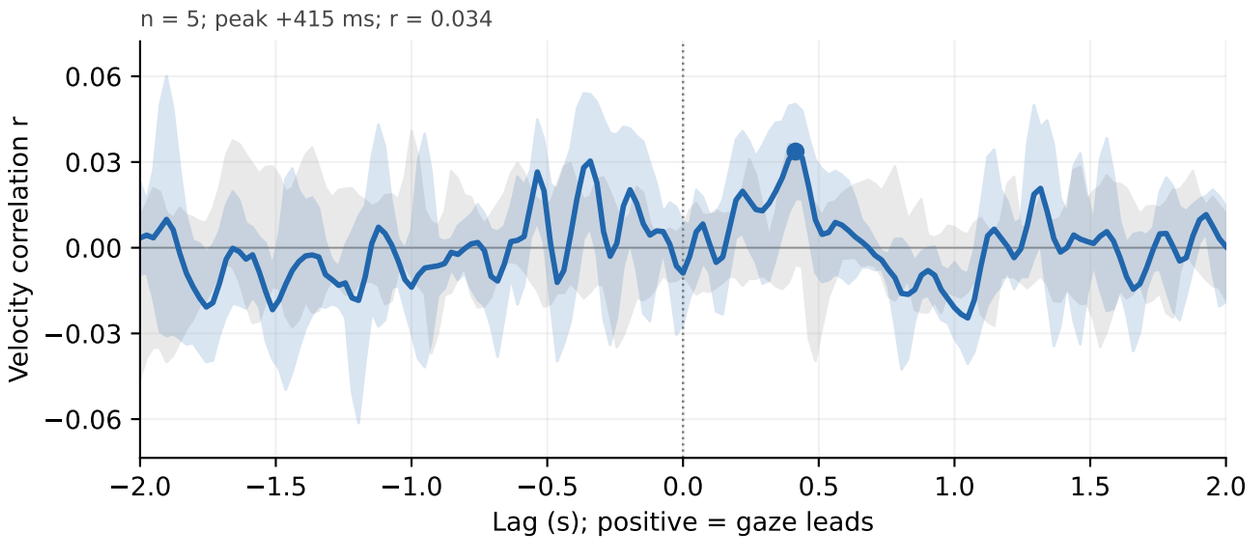
reach ring / psm_stage

n = 5; peak +439 ms; r = 0.029



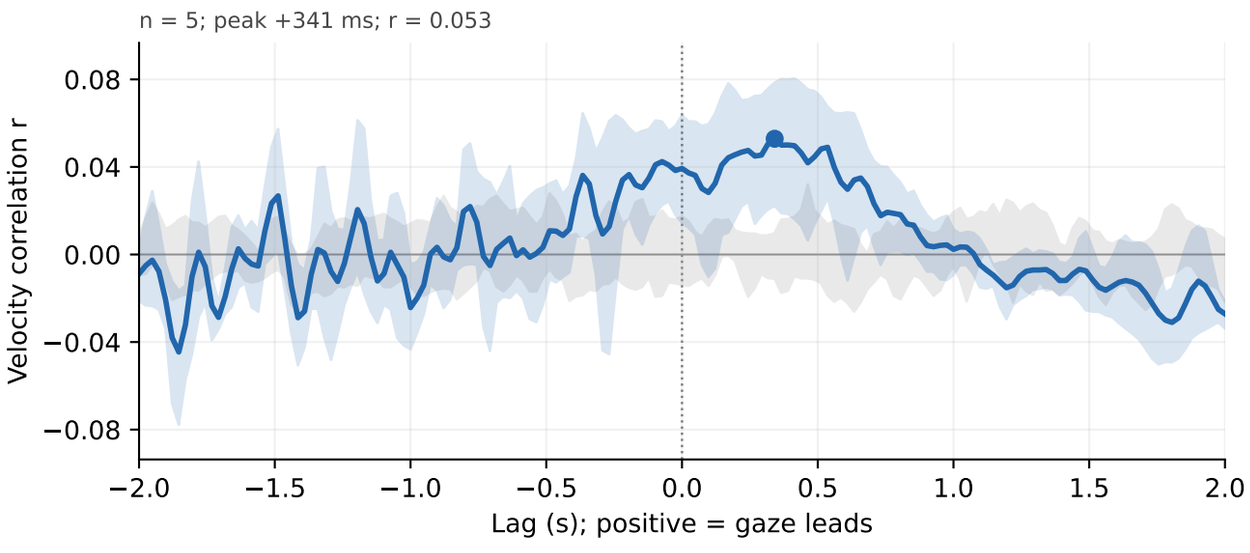
reach ring / within_stage

n = 5; peak +415 ms; r = 0.034



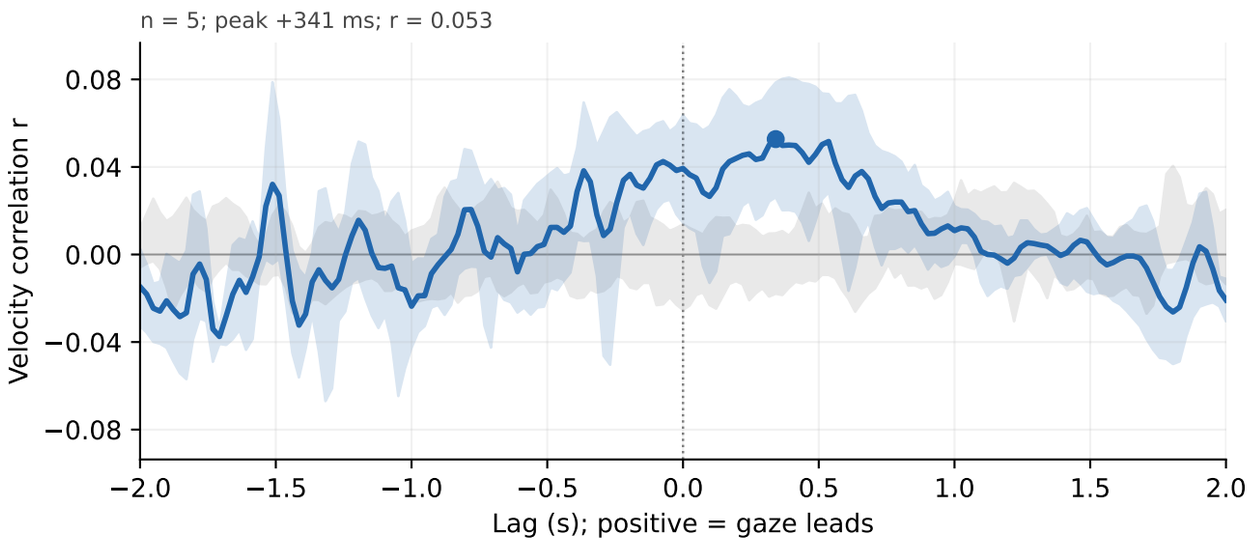
transfer ring / psm_stage

n = 5; peak +341 ms; r = 0.053



transfer ring / within_stage

n = 5; peak +341 ms; r = 0.053

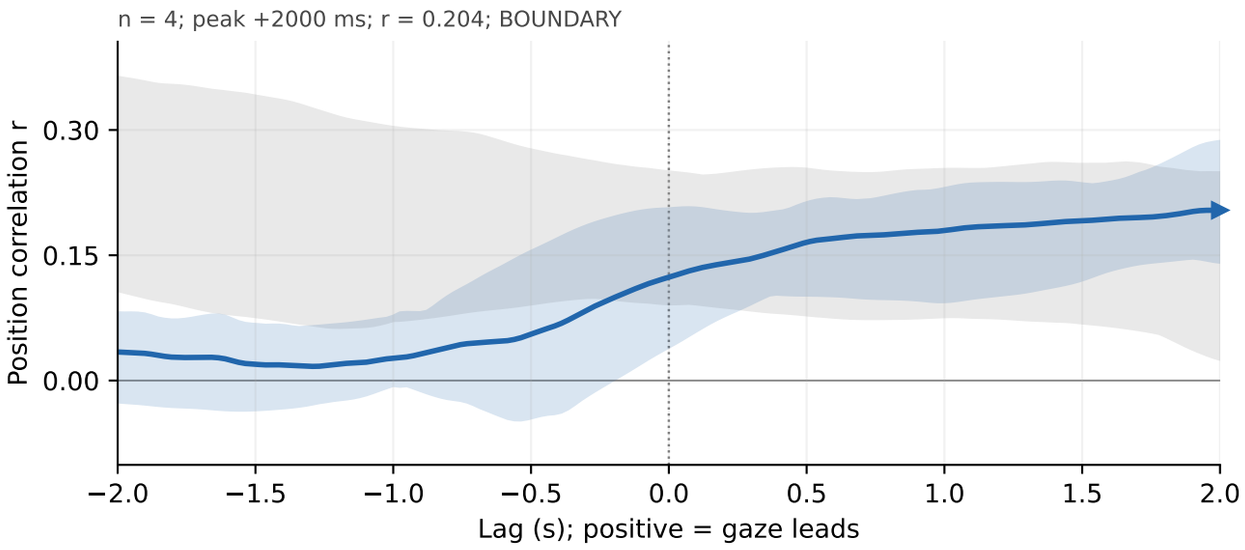


One PSM selected per recording/stage by largest signed xcorr peak; selection does not identify the operated instrument.

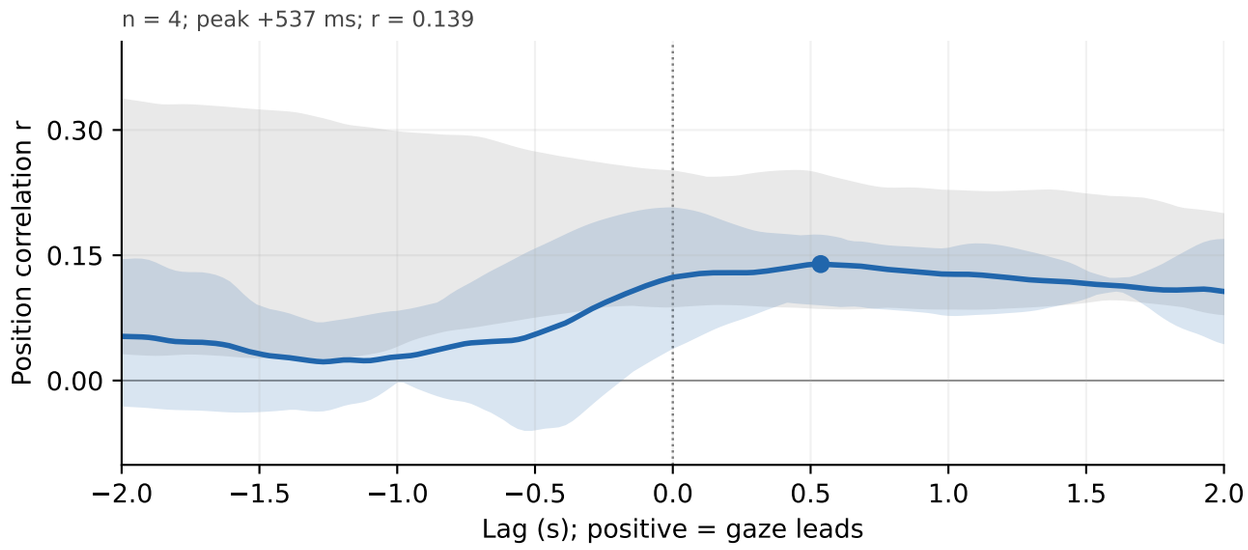
Task C: position, maximum-xcorr PSM per stage

Unpaired reference (95%) | Participant bootstrap (95%) | Group mean

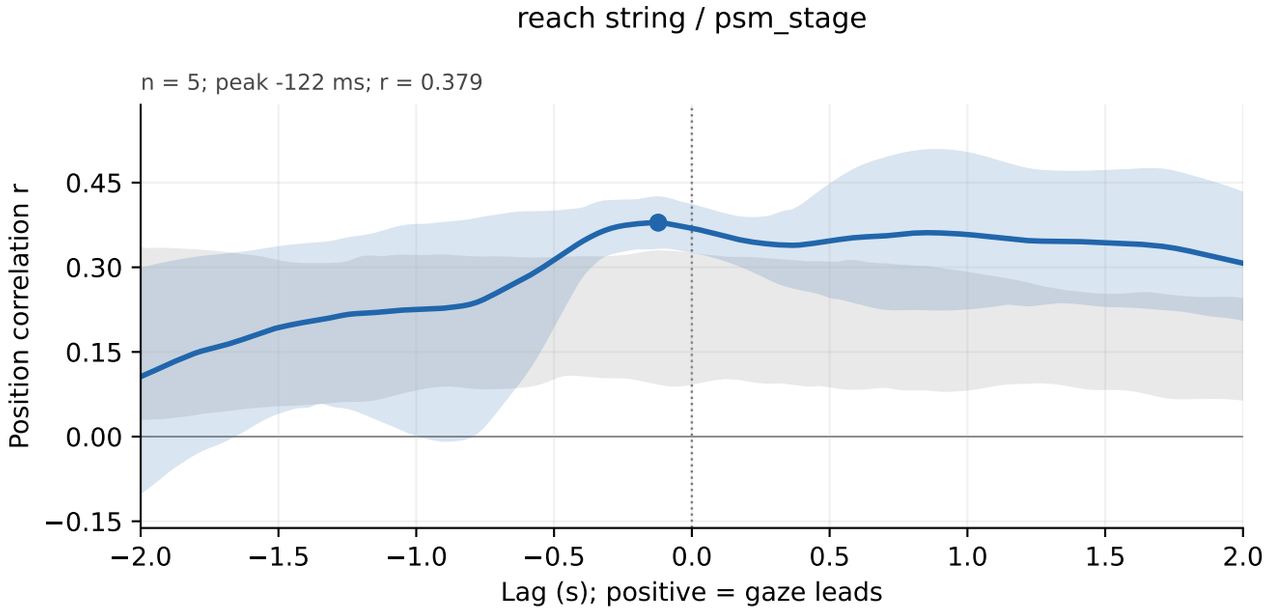
pull through and cinch / psm_stage



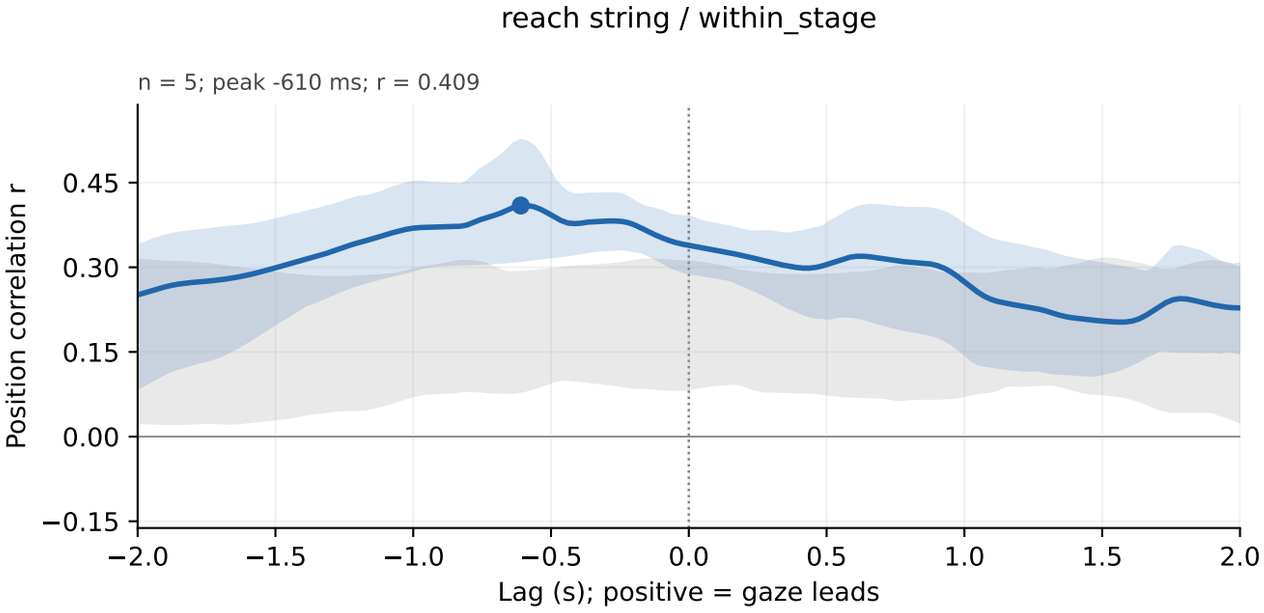
pull through and cinch / within_stage



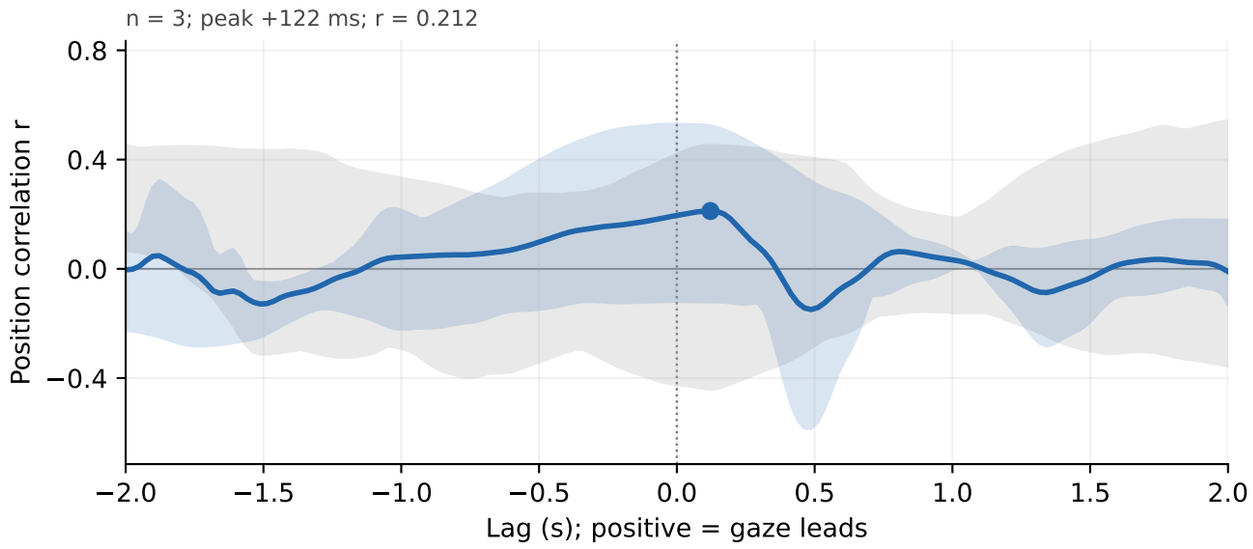
reach string / psm_stage



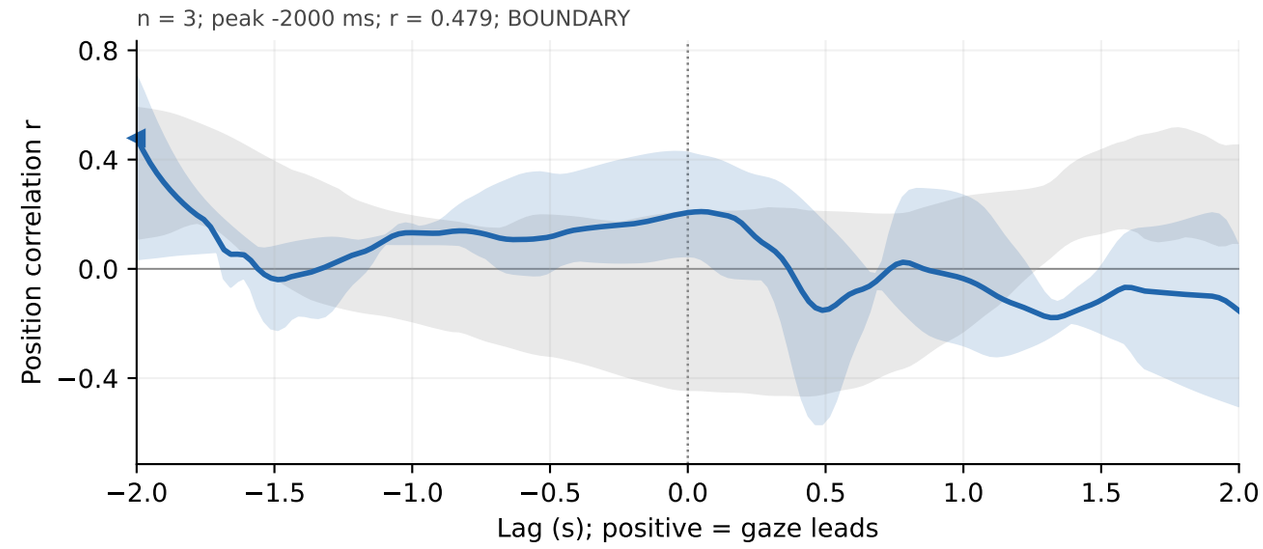
reach string / within_stage



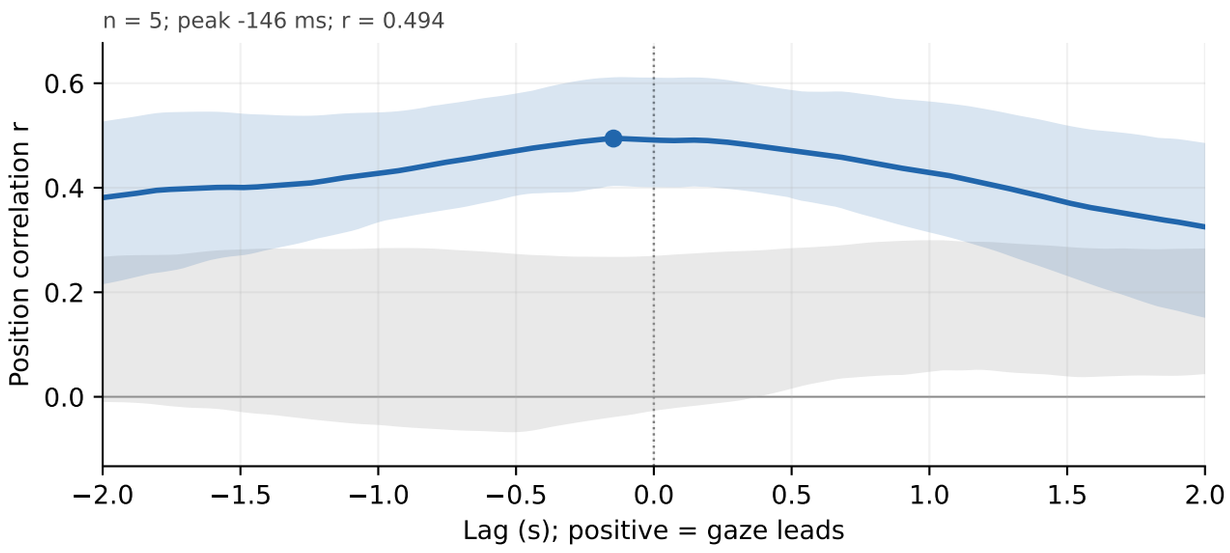
threading string through tower / psm_stage



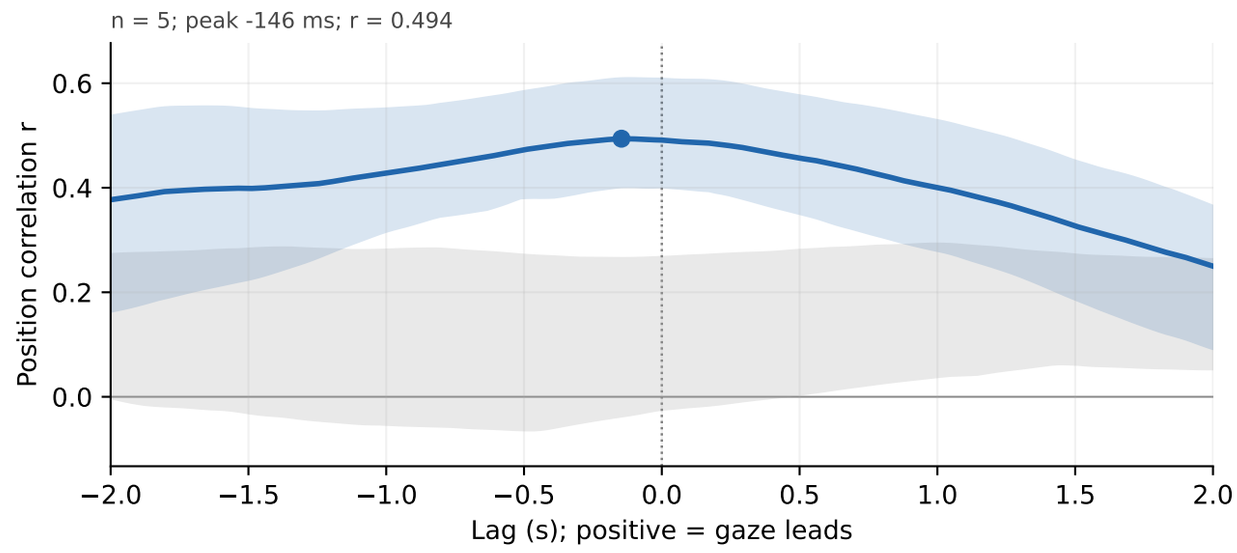
threading string through tower / within_stage



wrap around psm / psm_stage



wrap around psm / within_stage

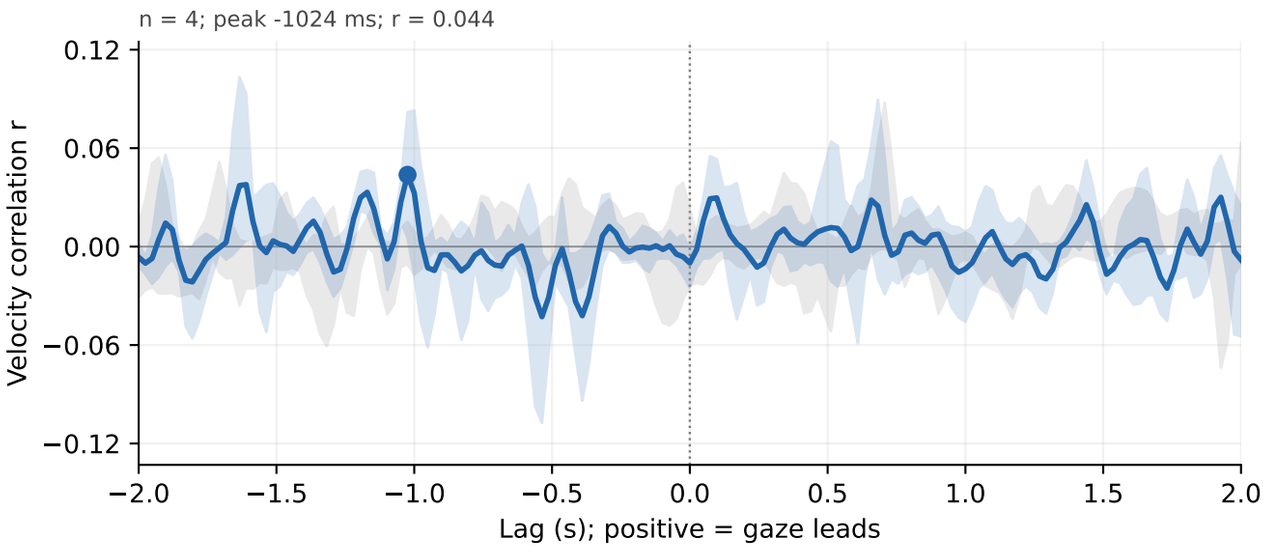


One PSM selected per recording/stage by largest signed xcorr peak; selection does not identify the operated instrument.

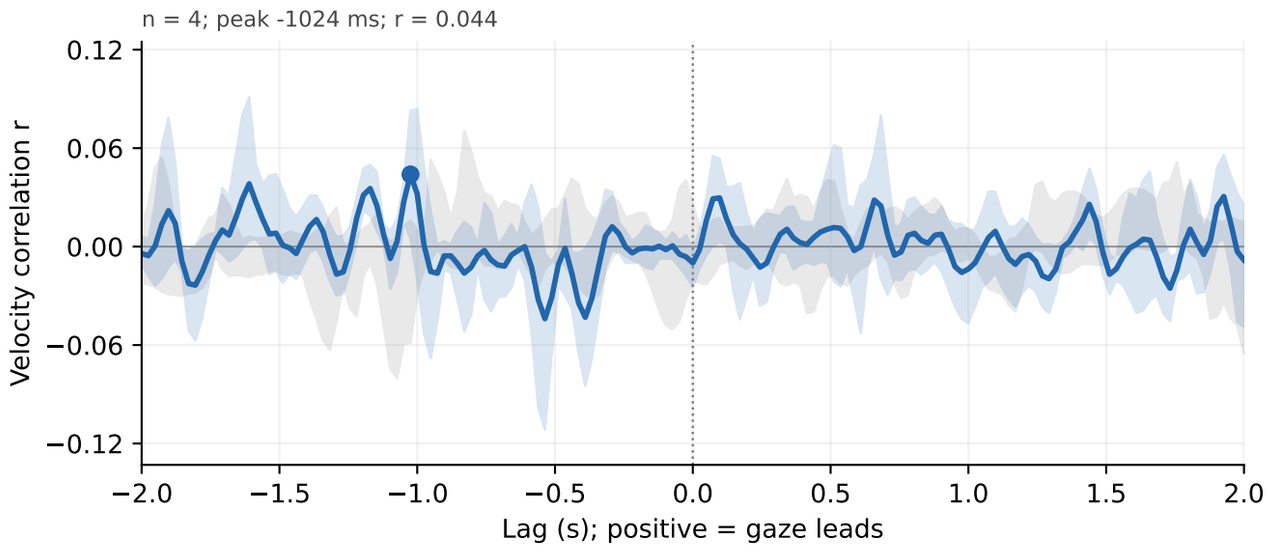
Task C: velocity, maximum-xcorr PSM per stage

Unpaired reference (95%) | Participant bootstrap (95%) | Group mean

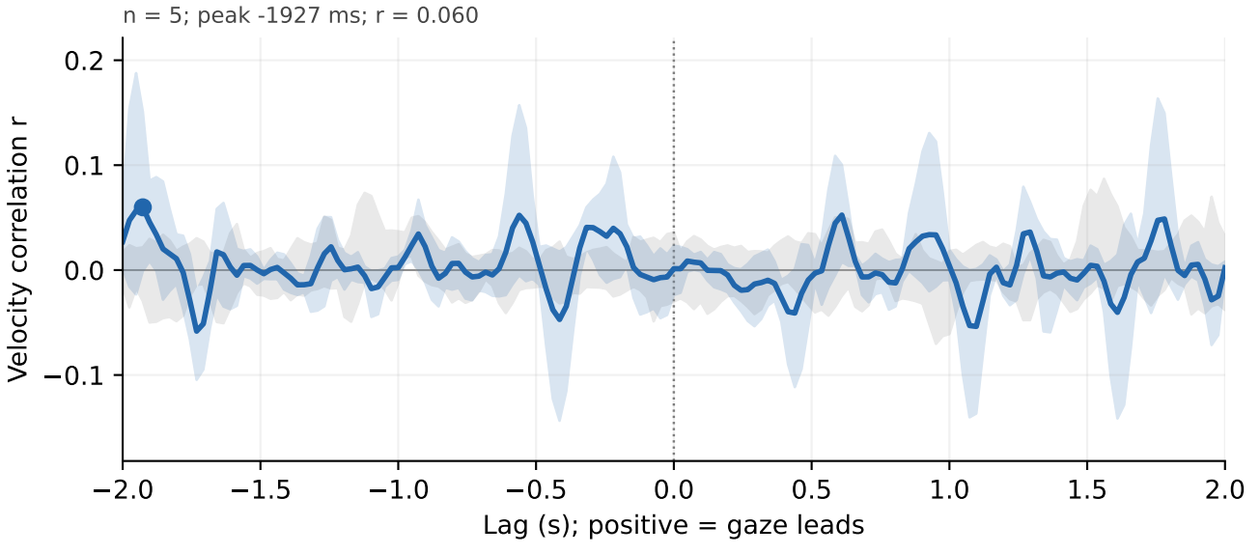
pull through and cinch / psm_stage



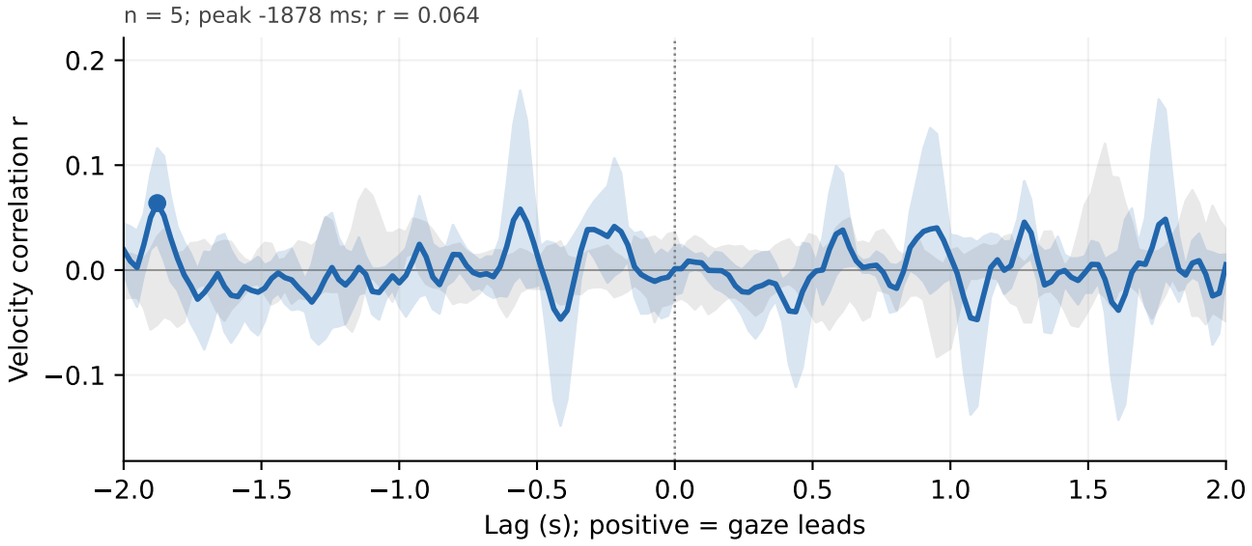
pull through and cinch / within_stage



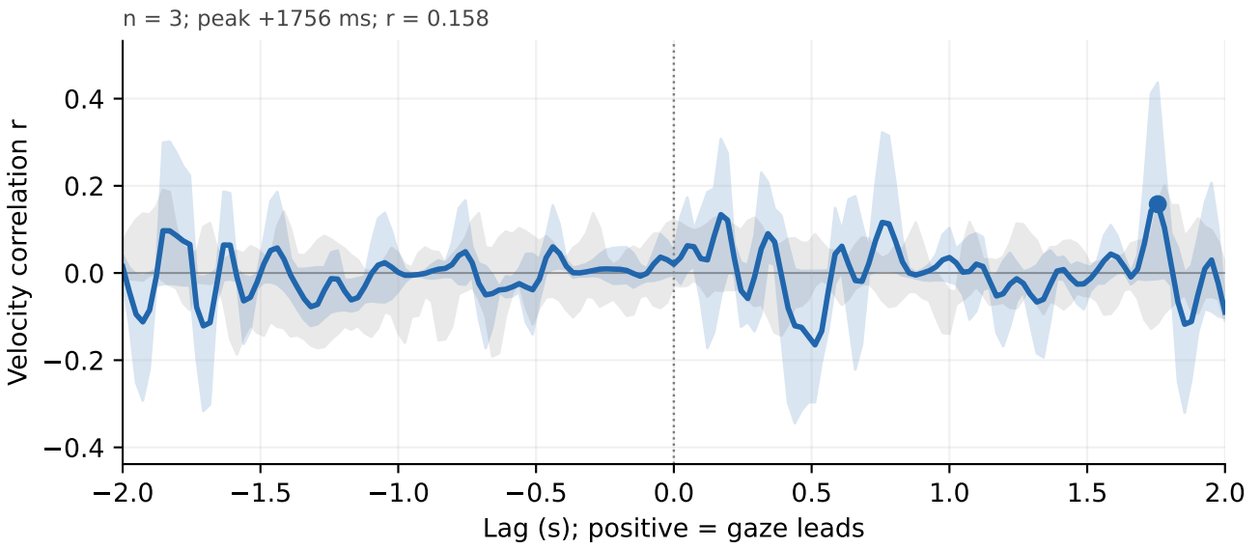
reach string / psm_stage



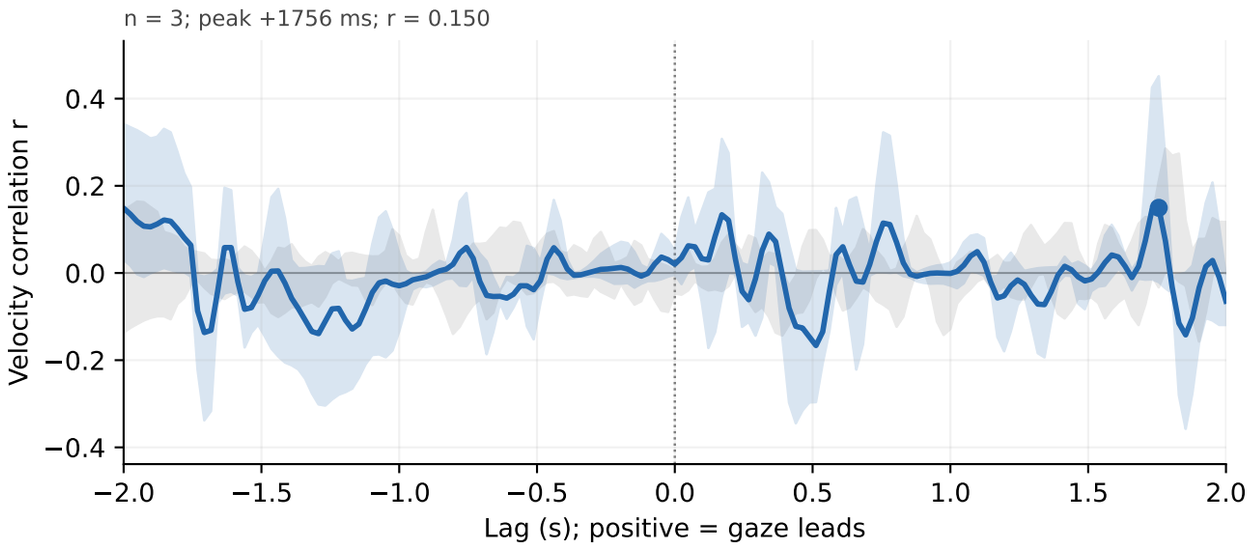
reach string / within_stage



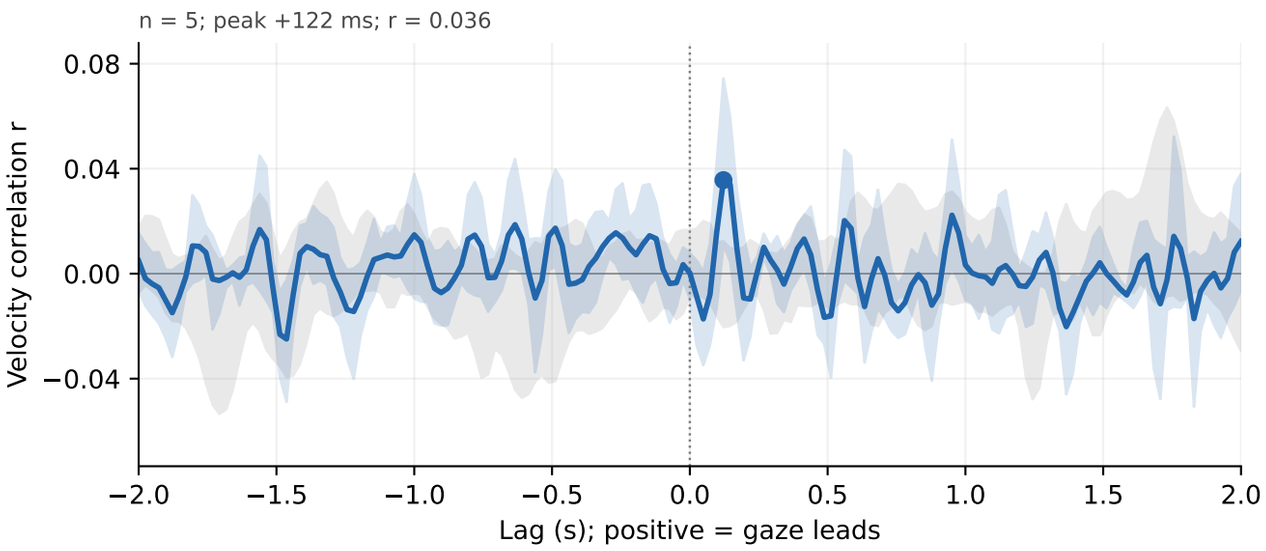
threading string through tower / psm_stage



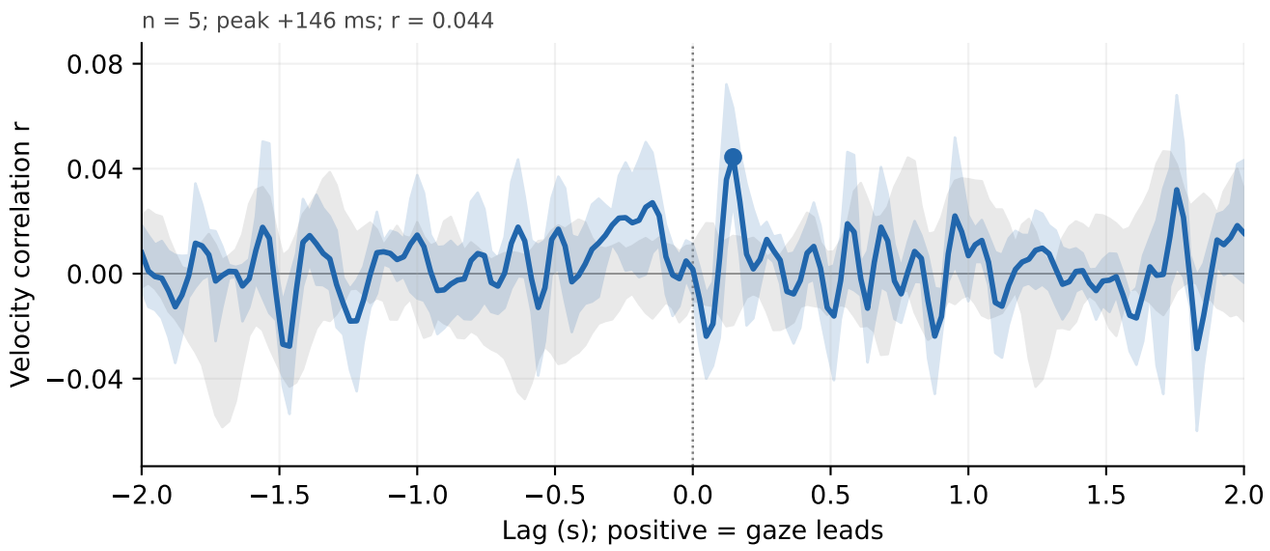
threading string through tower / within_stage



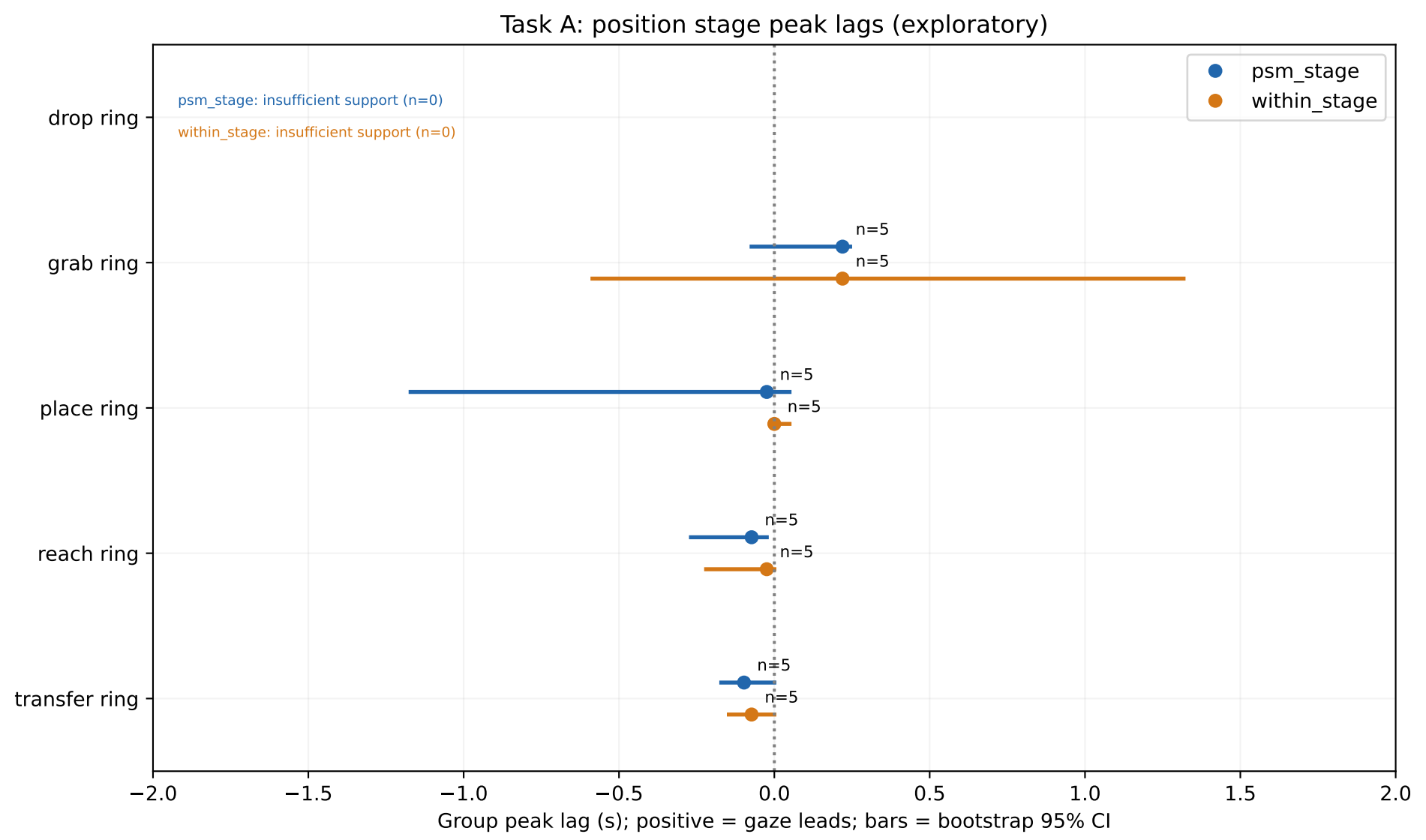
wrap around psm / psm_stage



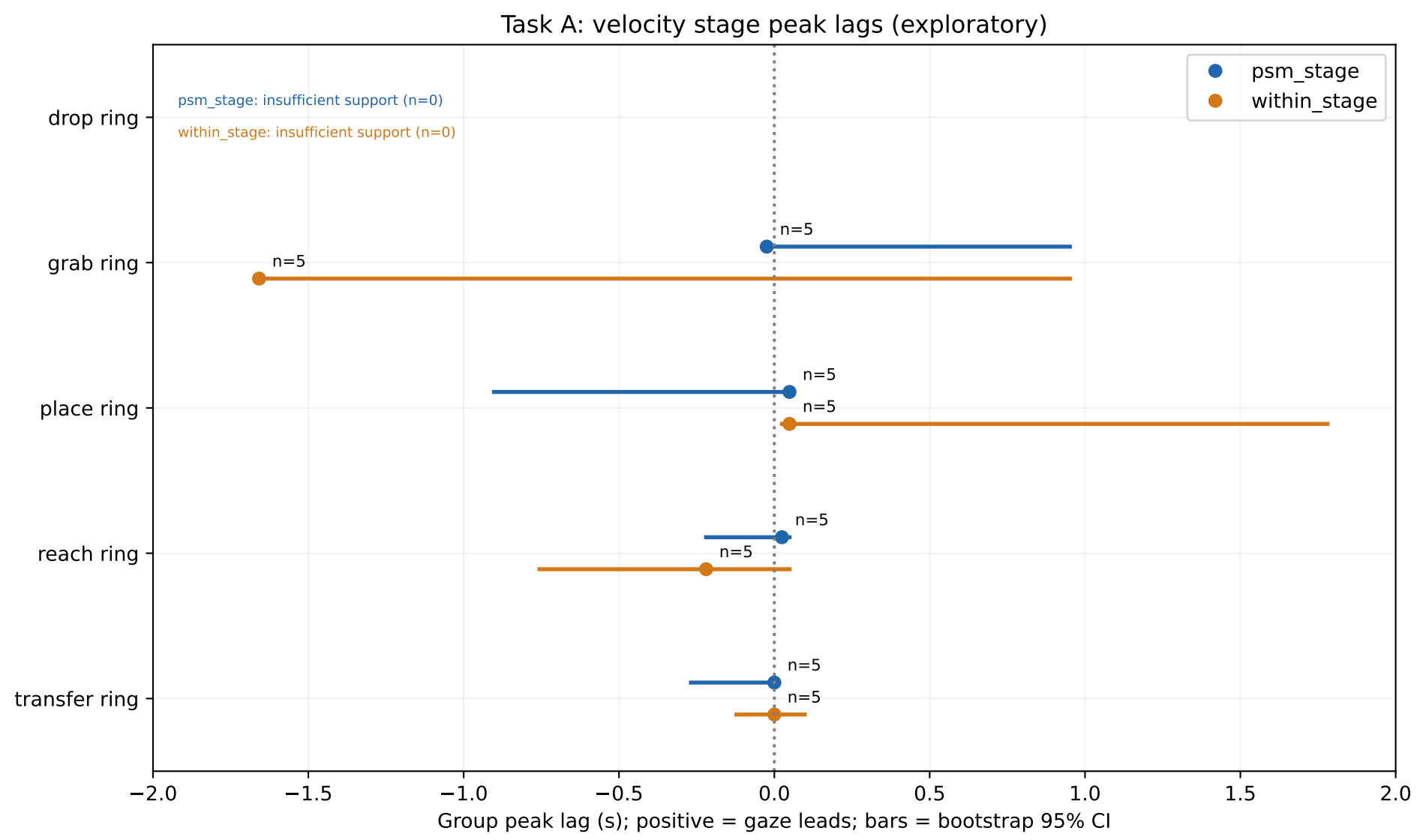
wrap around psm / within_stage



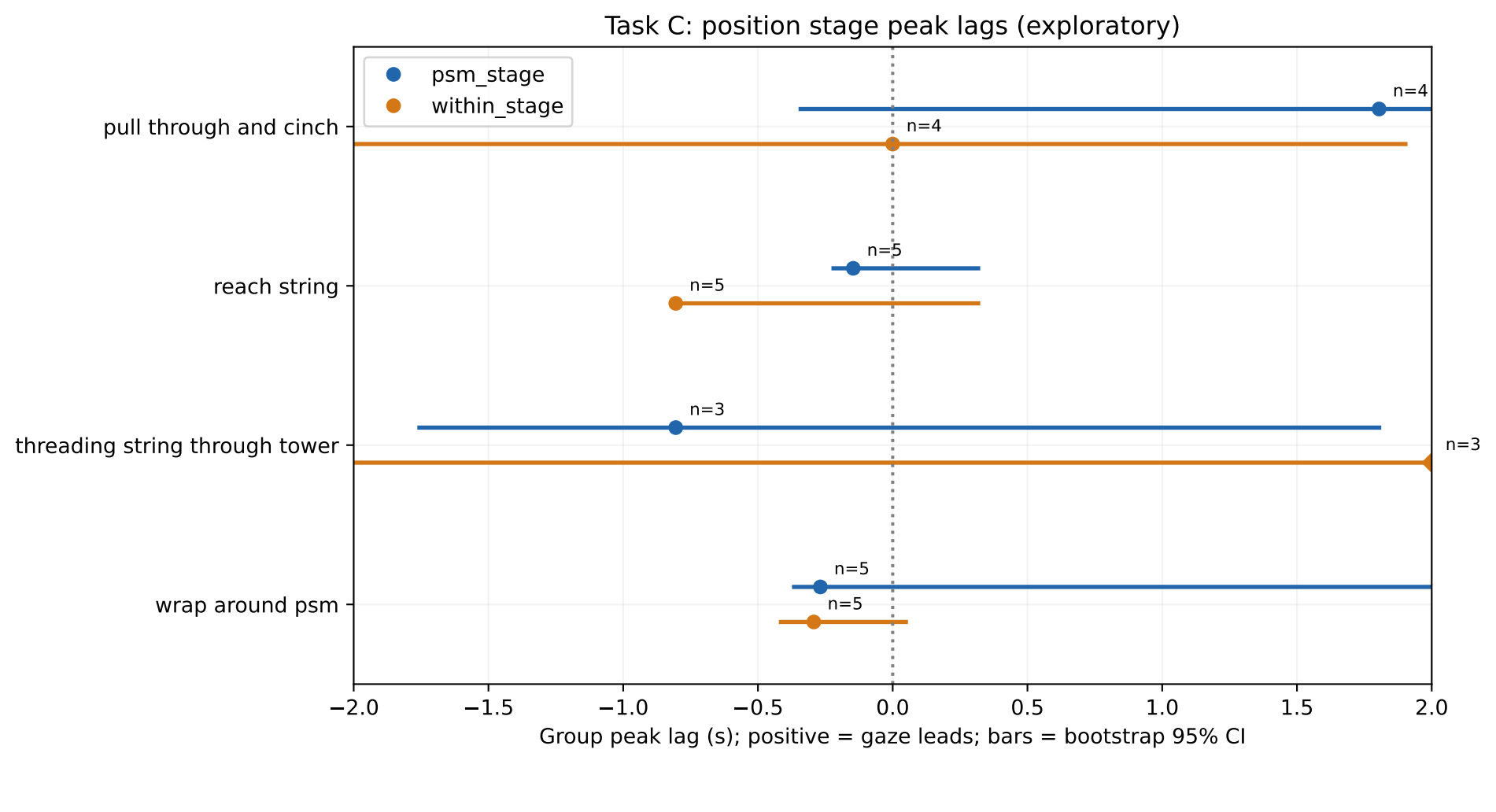
One PSM selected per recording/stage by largest signed xcorr peak; selection does not identify the operated instrument.



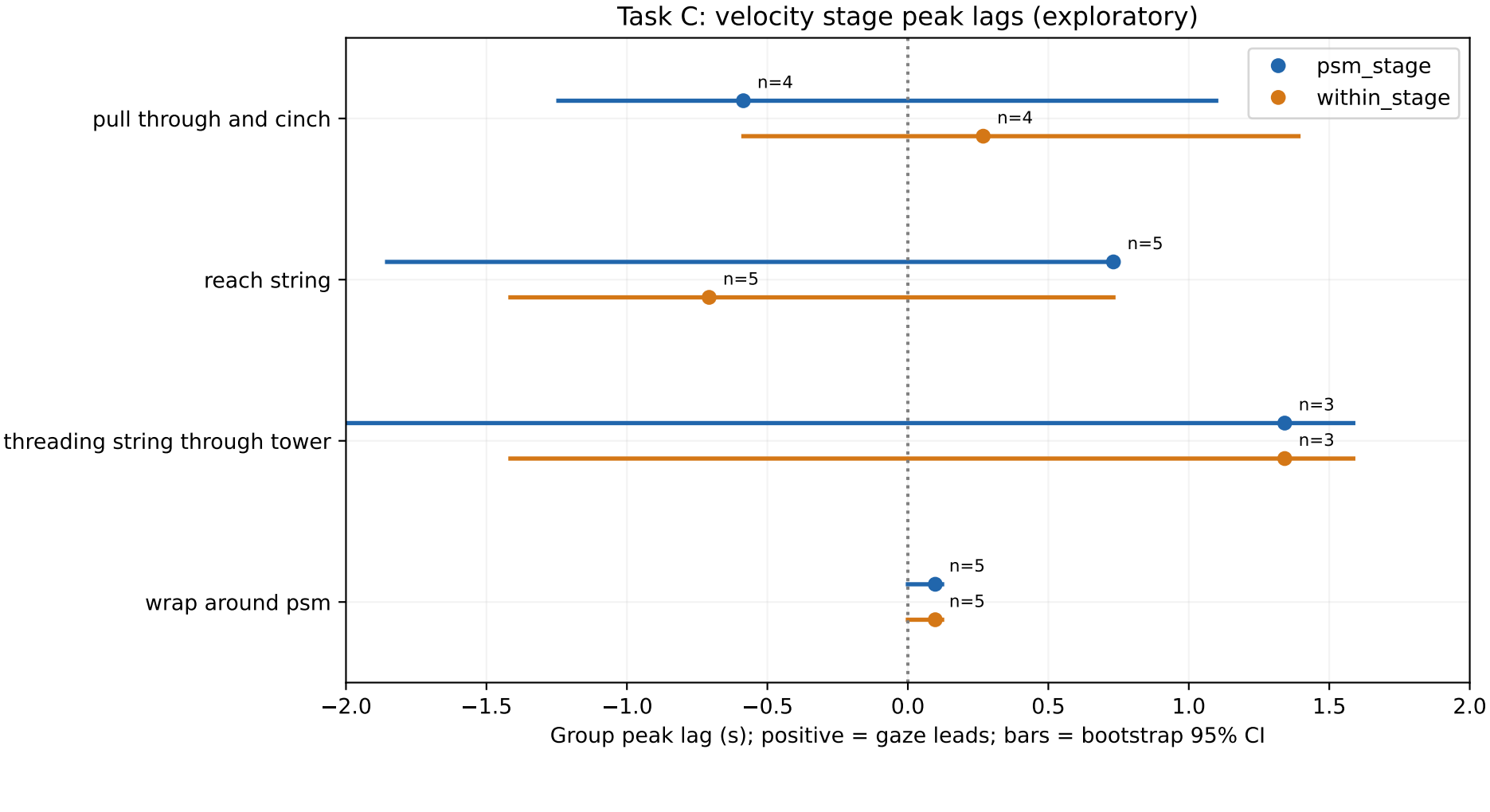
One PSM selected per recording/stage by largest signed xcorr peak; selection does not identify the operated instrument.



One PSM selected per recording/stage by largest signed xcorr peak; selection does not identify the operated instrument.



One PSM selected per recording/stage by largest signed xcorr peak; selection does not identify the operated instrument.



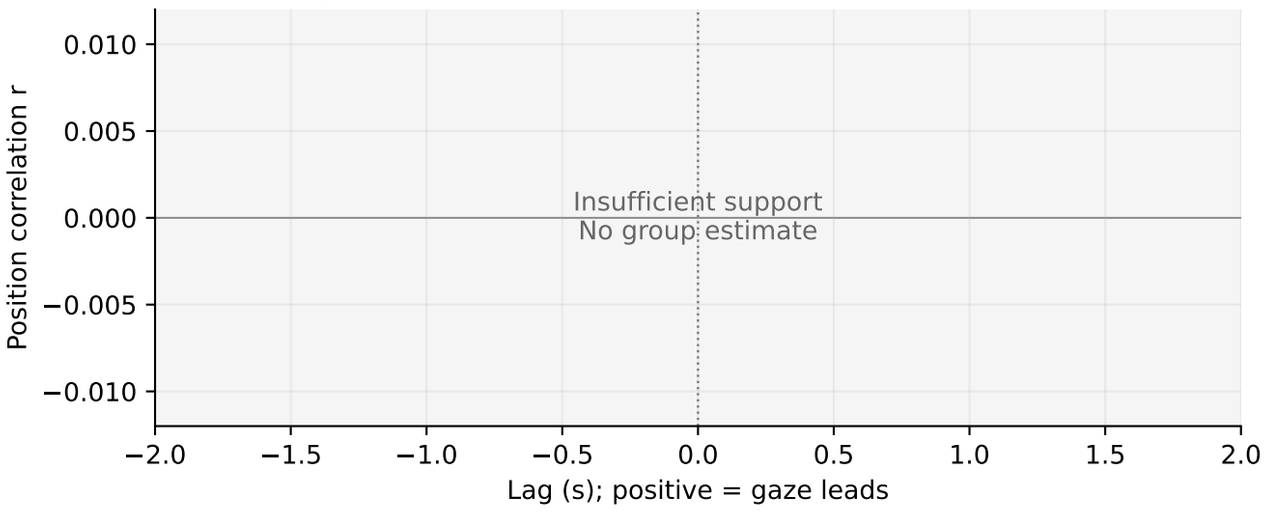
One PSM selected per recording/stage by largest signed xcorr peak; selection does not identify the operated instrument.

Task A: position, maximum-xcorr PSM per stage

Unpaired reference (95%) | Participant bootstrap (95%) | Group mean

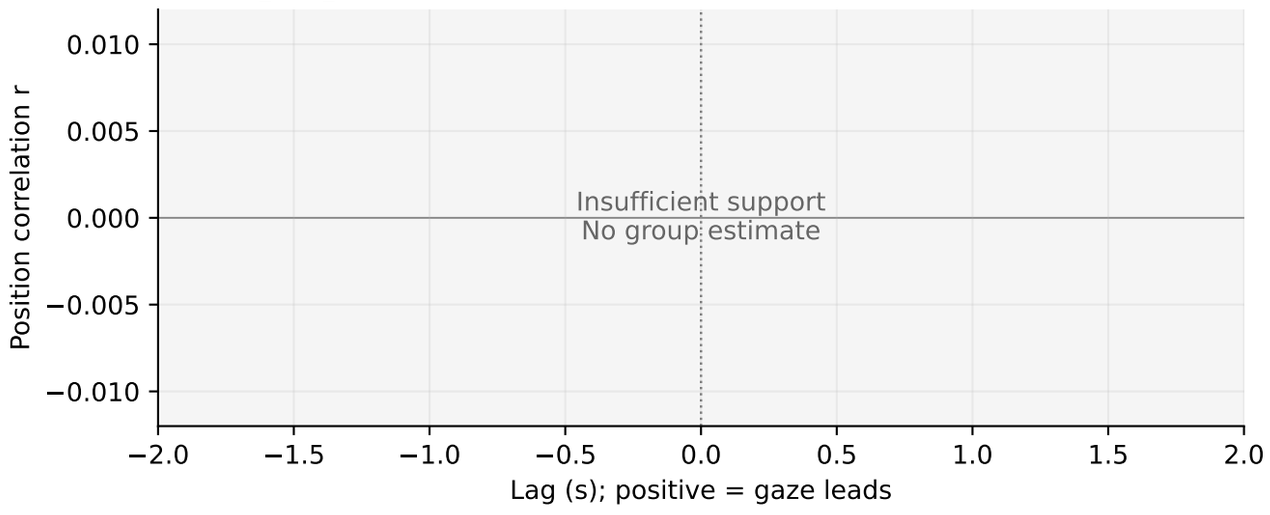
drop ring / psm_stage

n = 0; no_valid_curve



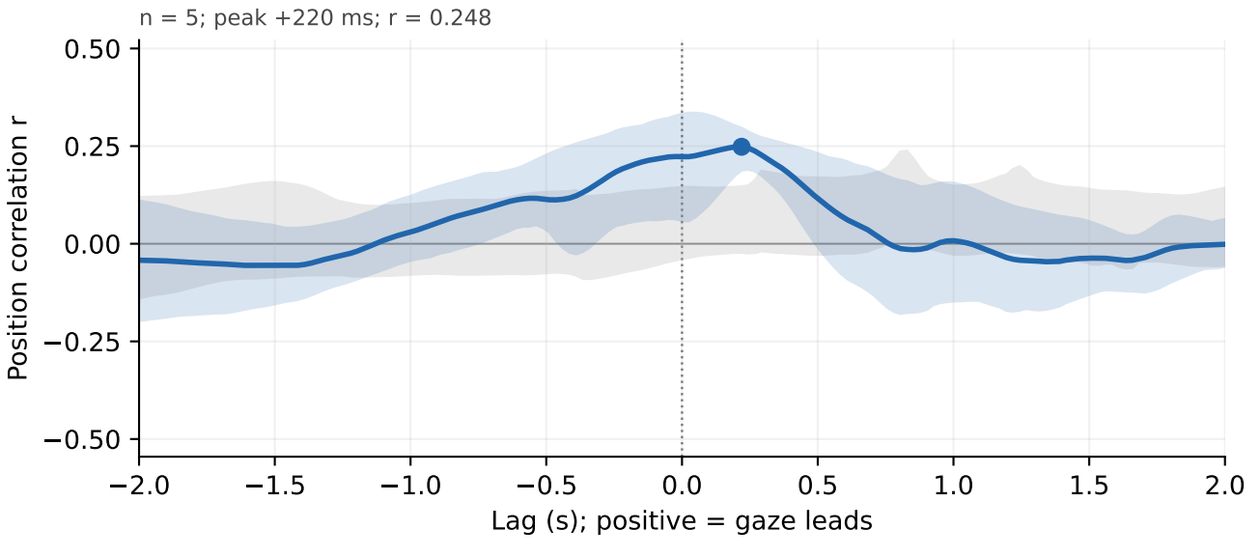
drop ring / within_stage

n = 0; no_valid_curve



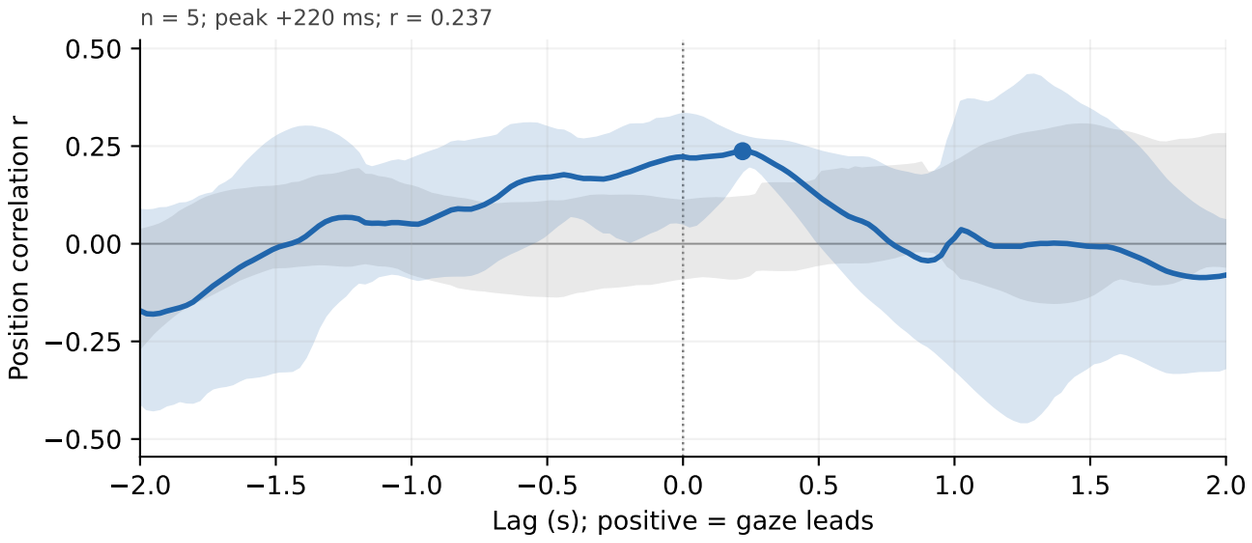
grab ring / psm_stage

n = 5; peak +220 ms; r = 0.248



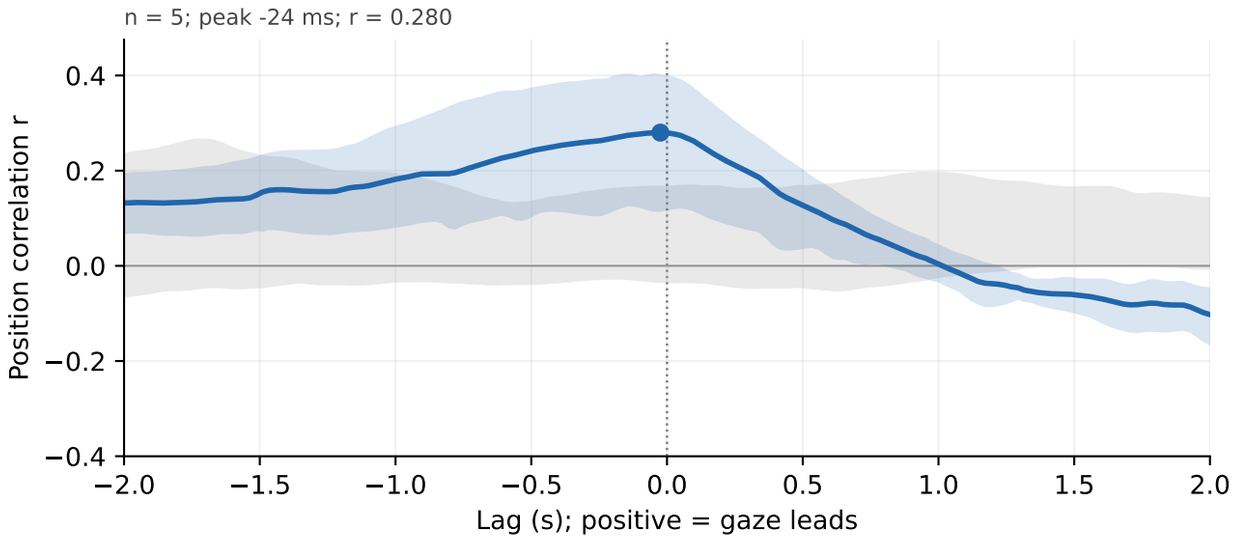
grab ring / within_stage

n = 5; peak +220 ms; r = 0.237



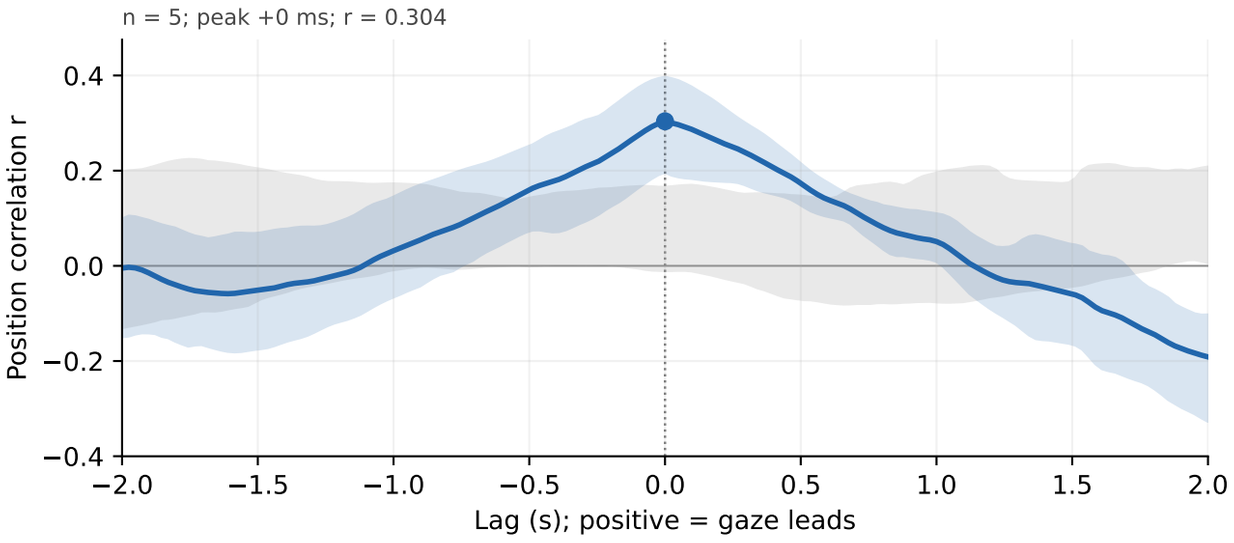
place ring / psm_stage

n = 5; peak -24 ms; r = 0.280



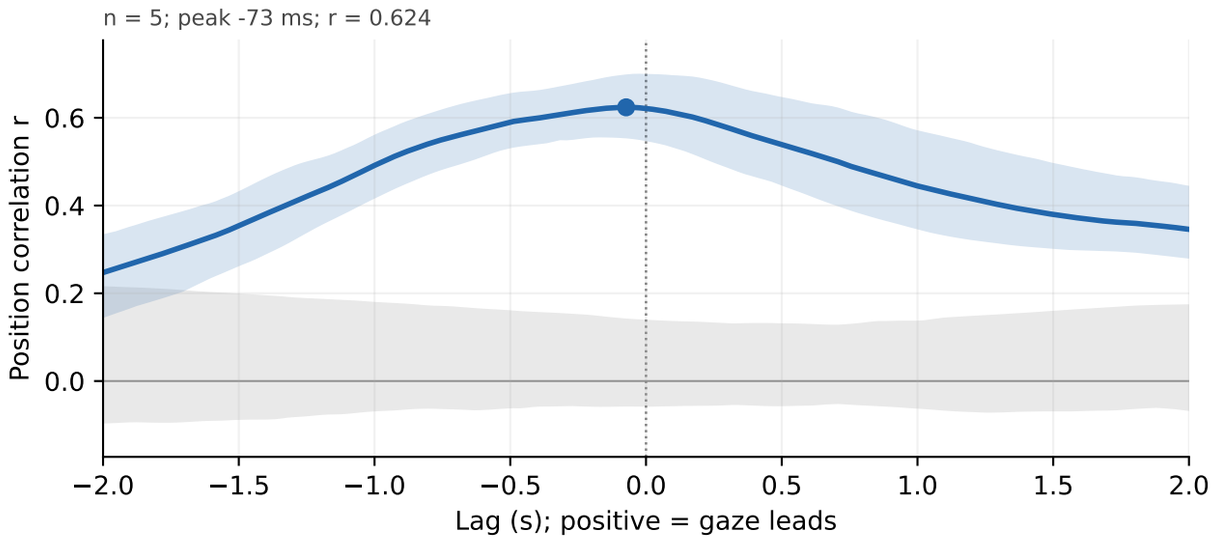
place ring / within_stage

n = 5; peak +0 ms; r = 0.304



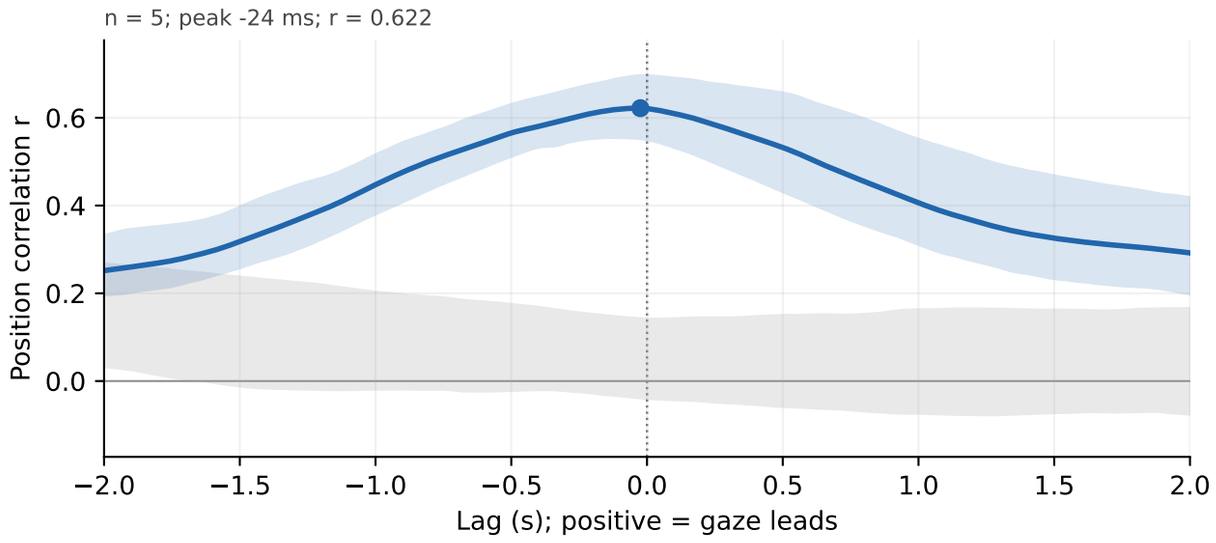
reach ring / psm_stage

n = 5; peak -73 ms; r = 0.624



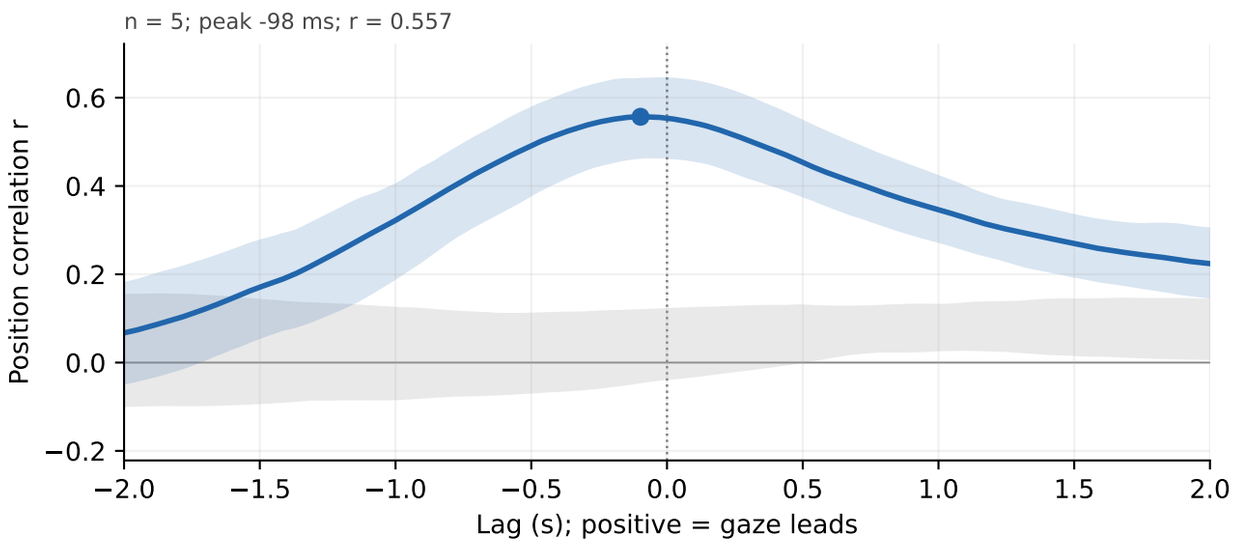
reach ring / within_stage

n = 5; peak -24 ms; r = 0.622



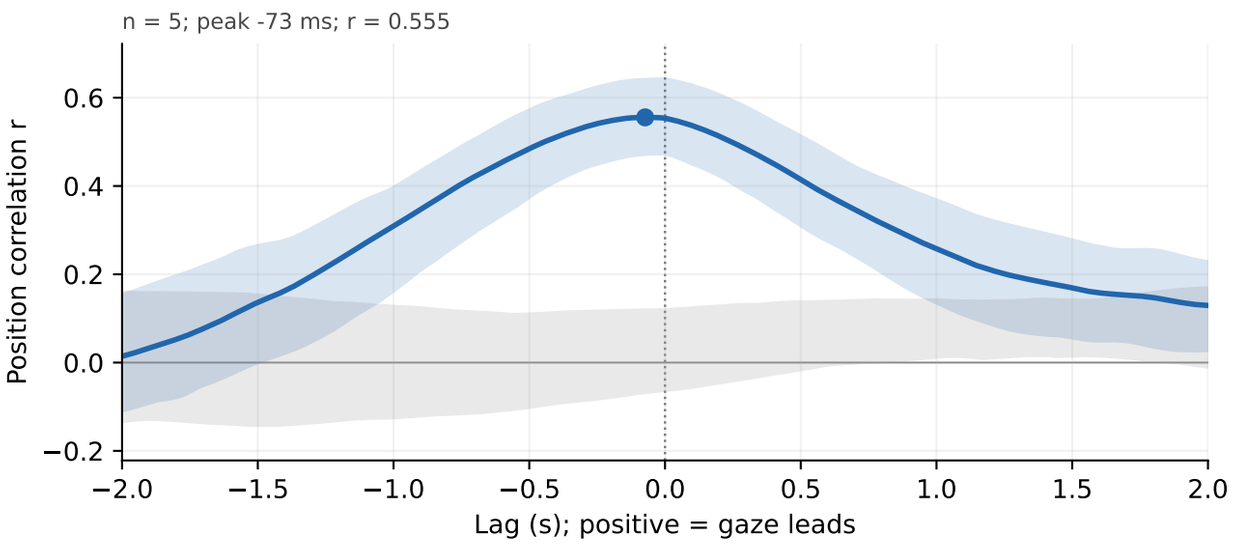
transfer ring / psm_stage

n = 5; peak -98 ms; r = 0.557



transfer ring / within_stage

n = 5; peak -73 ms; r = 0.555

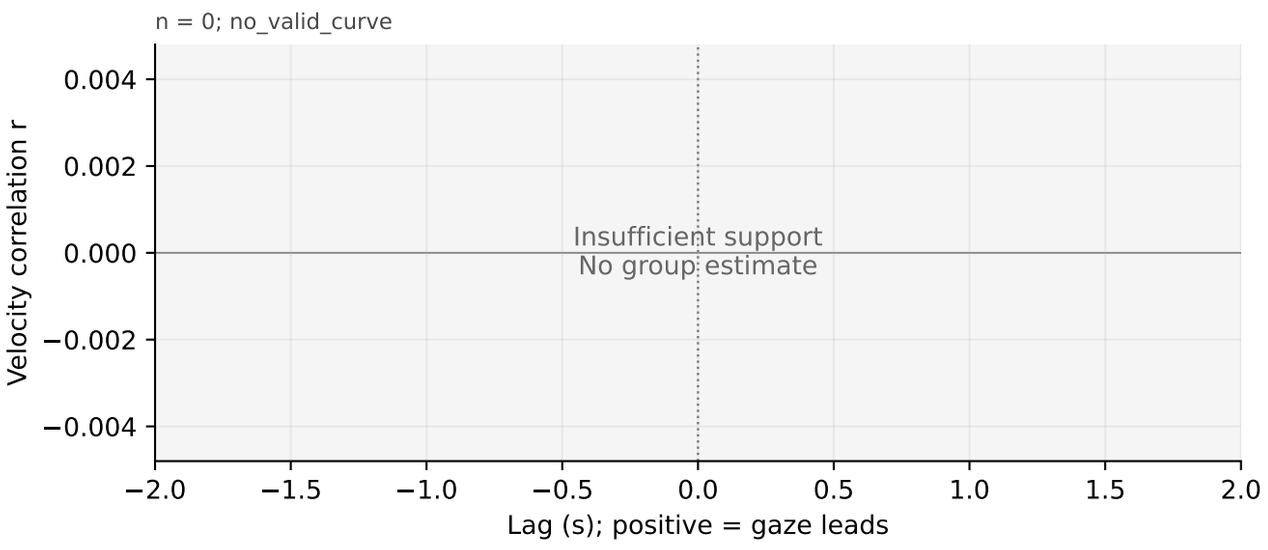


One PSM selected per recording/stage by largest signed xcorr peak; selection does not identify the operated instrument.

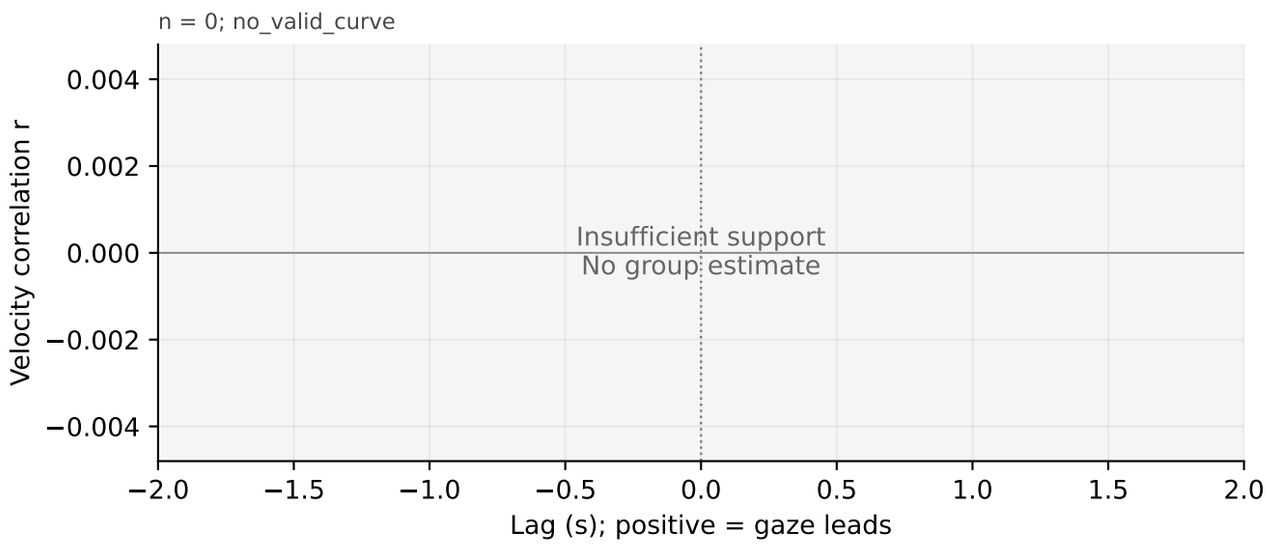
Task A: velocity, maximum-xcorr PSM per stage

Unpaired reference (95%) | Participant bootstrap (95%) | Group mean

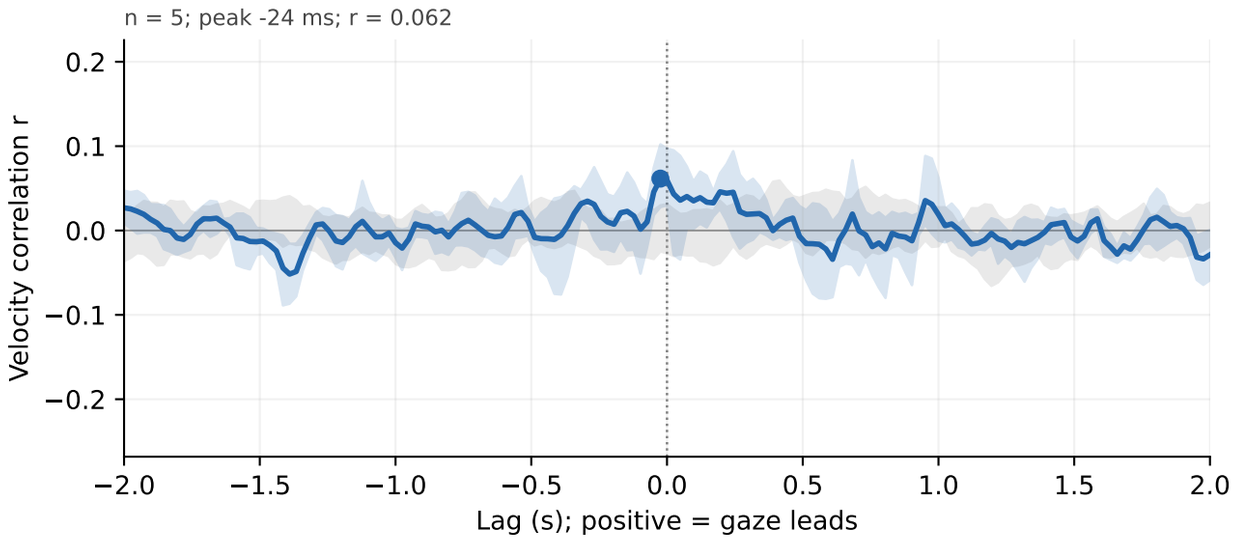
drop ring / psm_stage



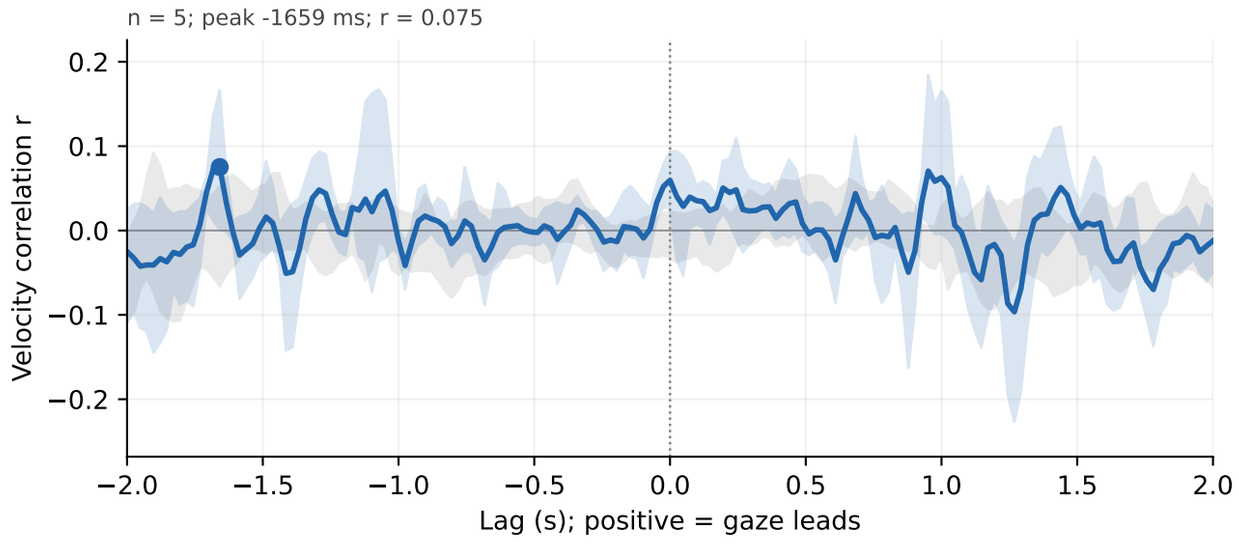
drop ring / within_stage



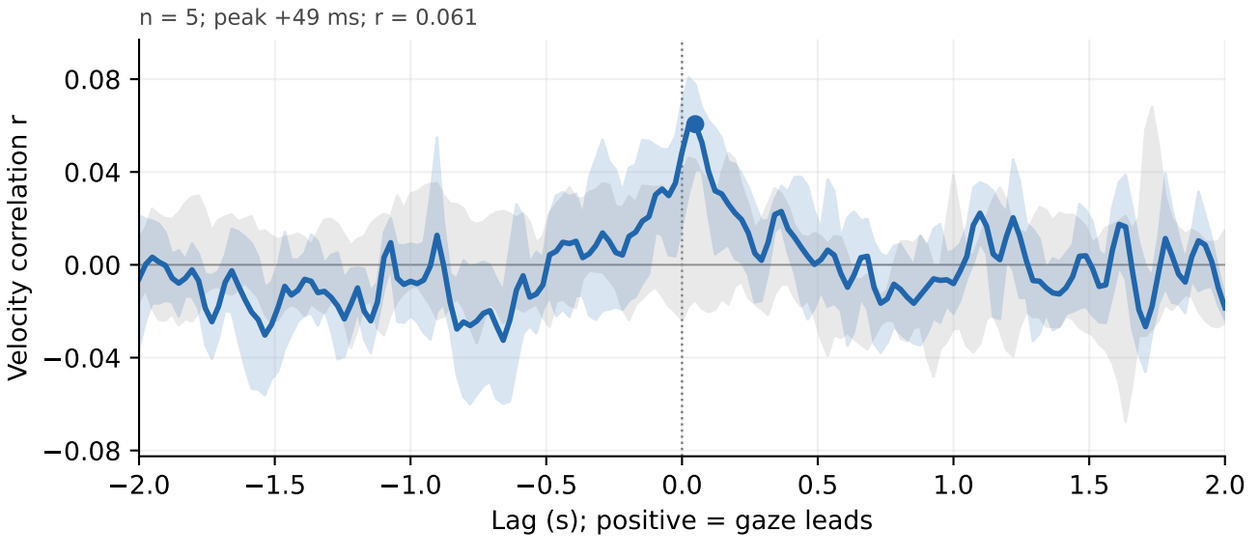
grab ring / psm_stage



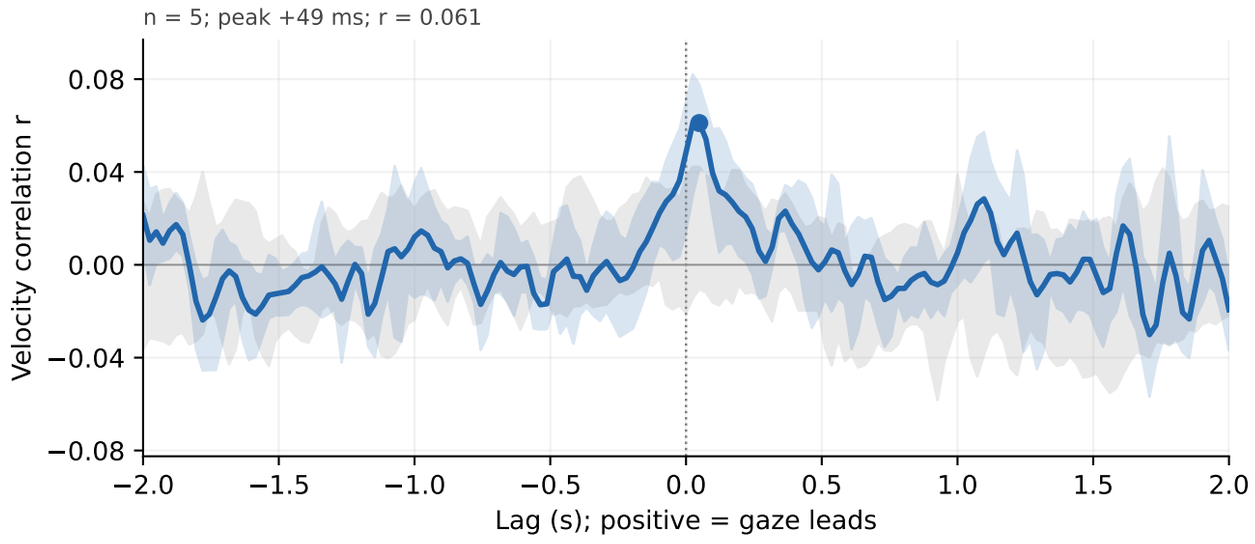
grab ring / within_stage



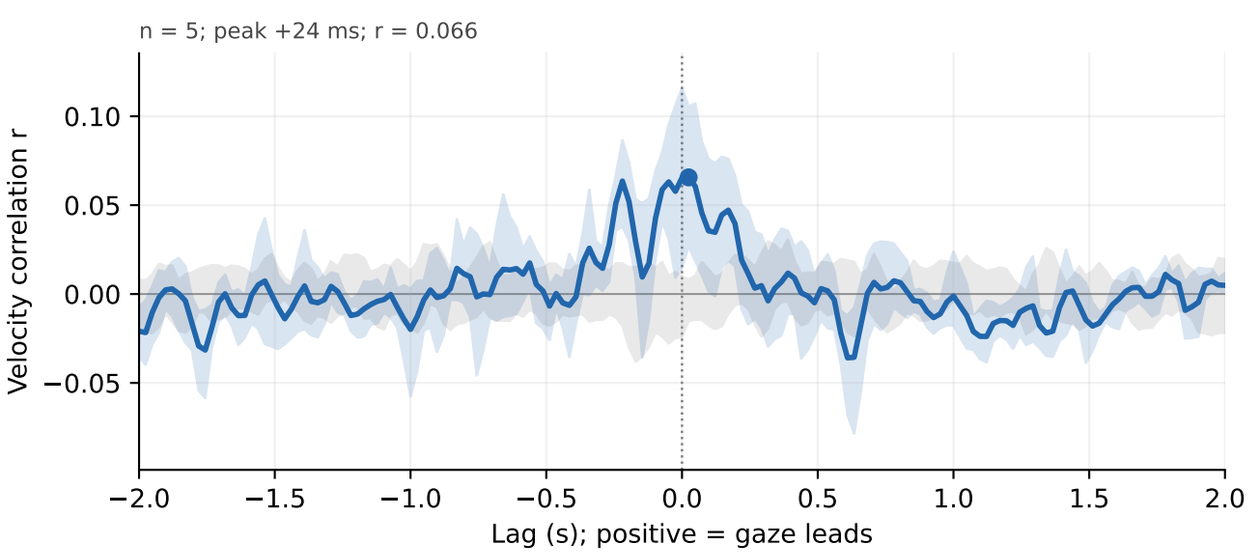
place ring / psm_stage



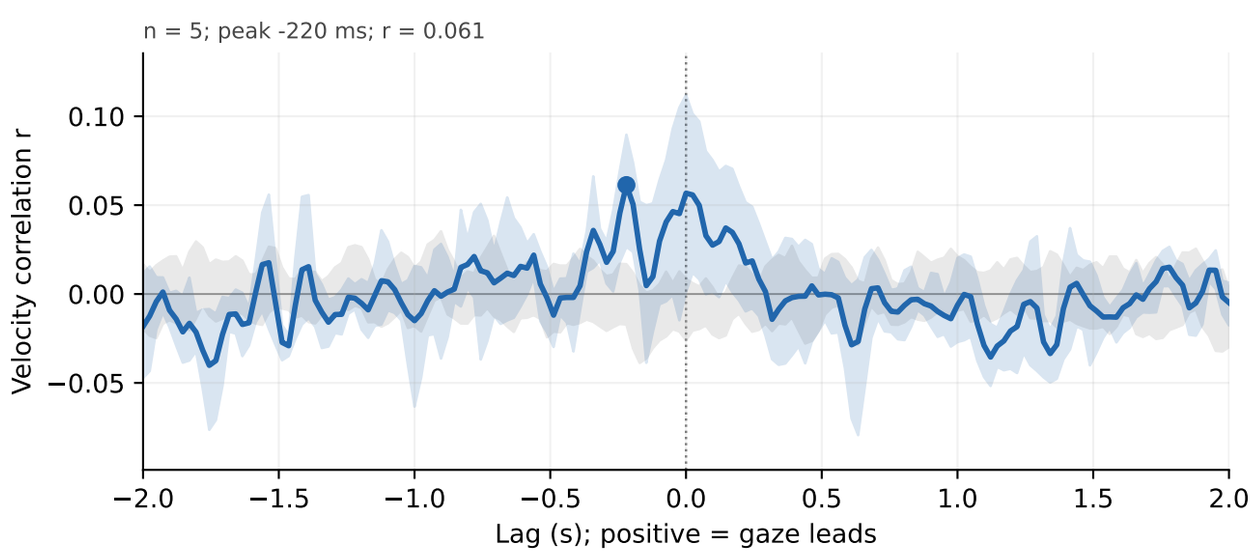
place ring / within_stage



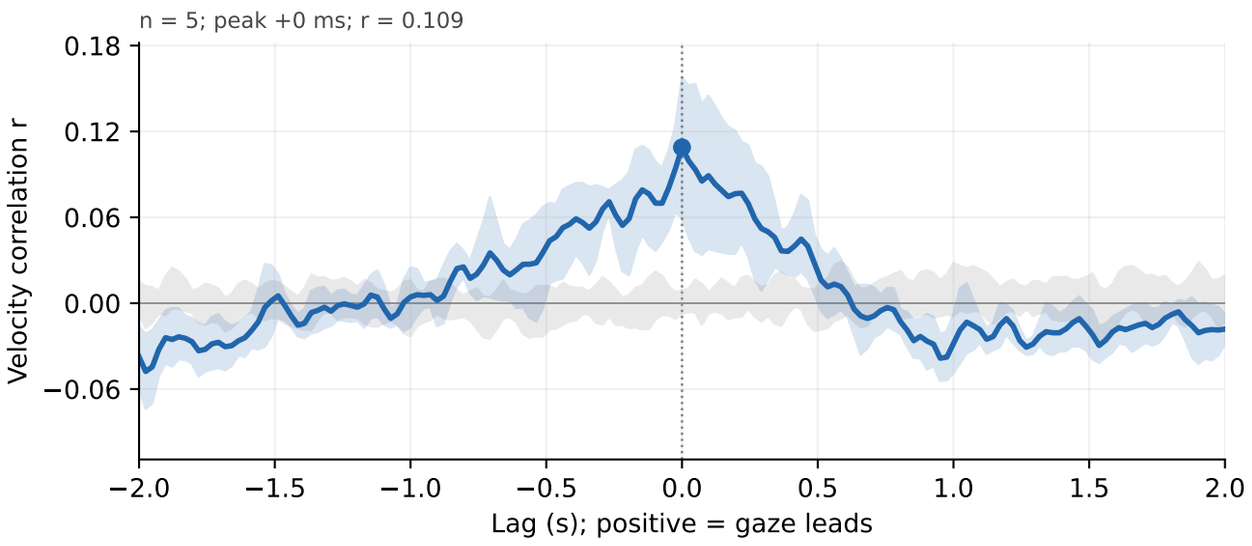
reach ring / psm_stage



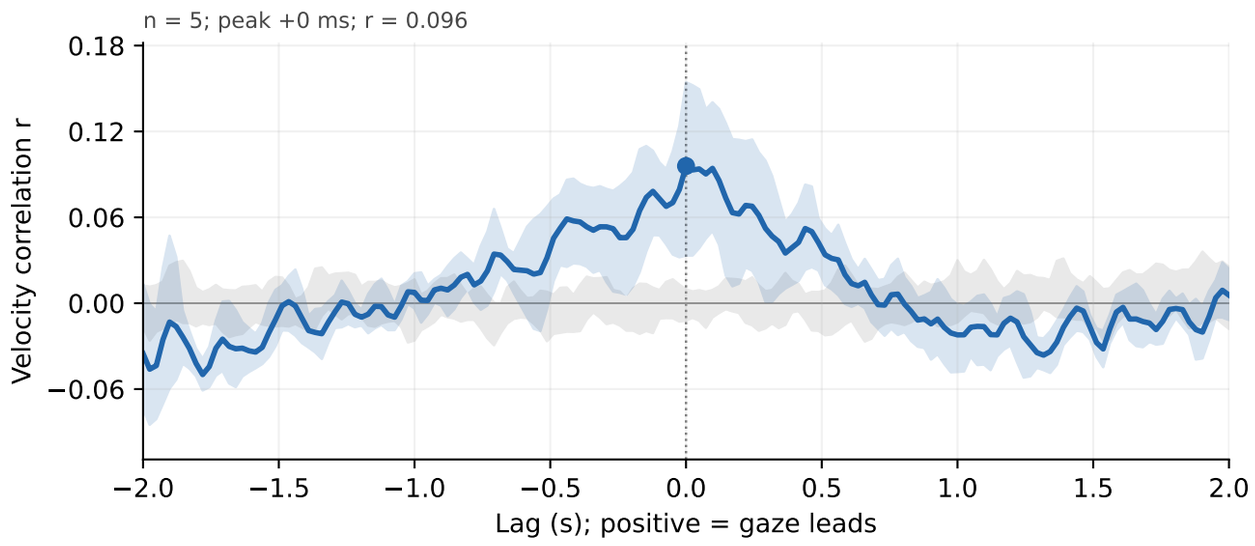
reach ring / within_stage



transfer ring / psm_stage



transfer ring / within_stage

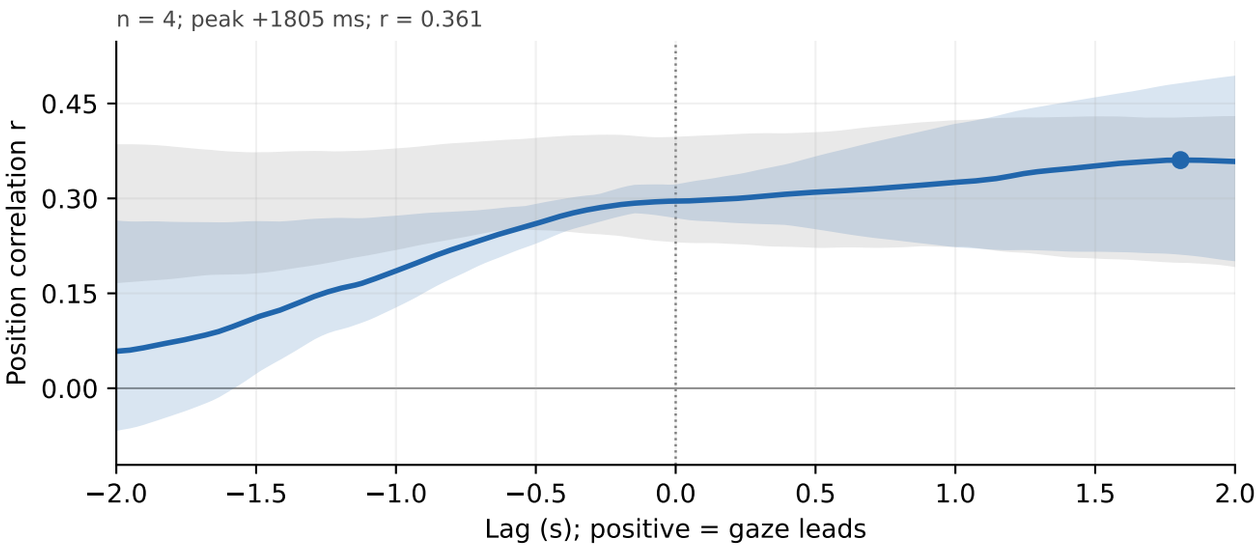


One PSM selected per recording/stage by largest signed xcorr peak; selection does not identify the operated instrument.

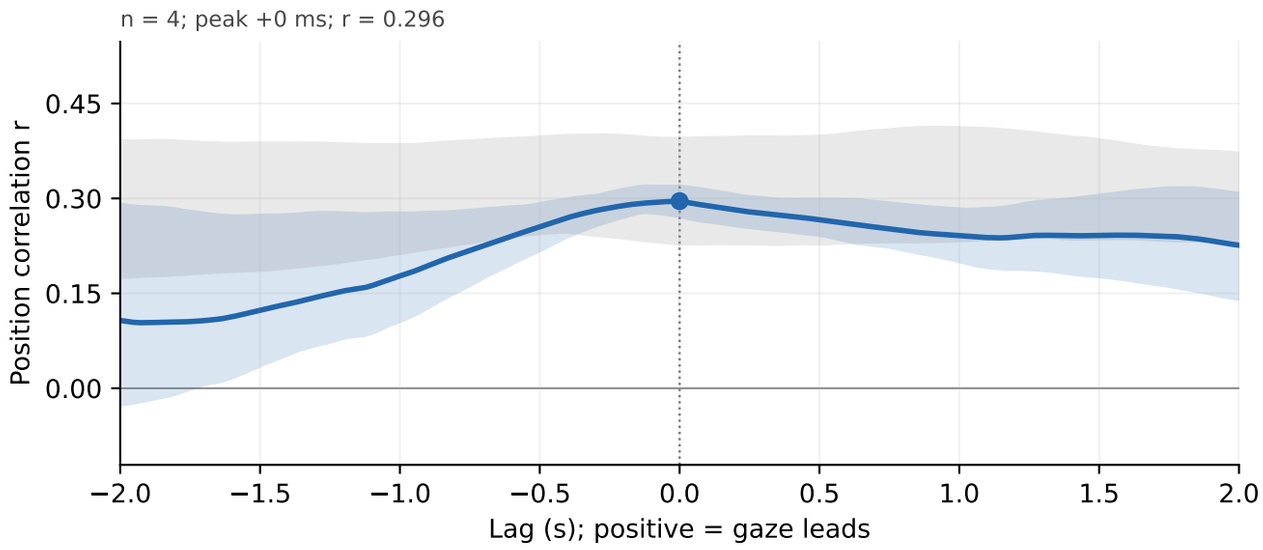
Task C: position, maximum-xcorr PSM per stage

Unpaired reference (95%) | Participant bootstrap (95%) | Group mean

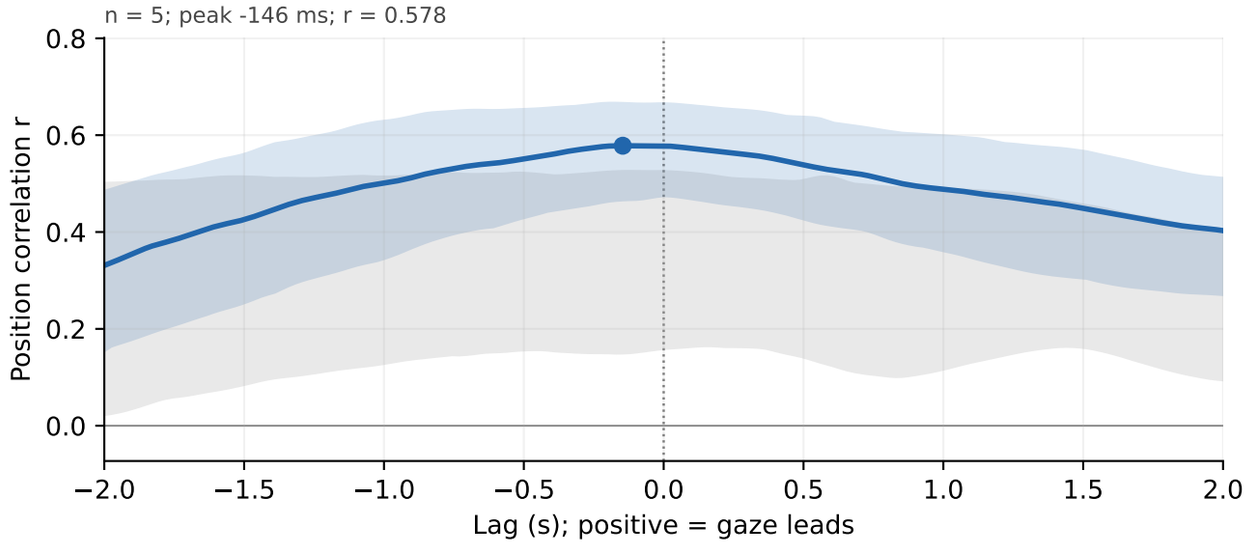
pull through and cinch / psm_stage



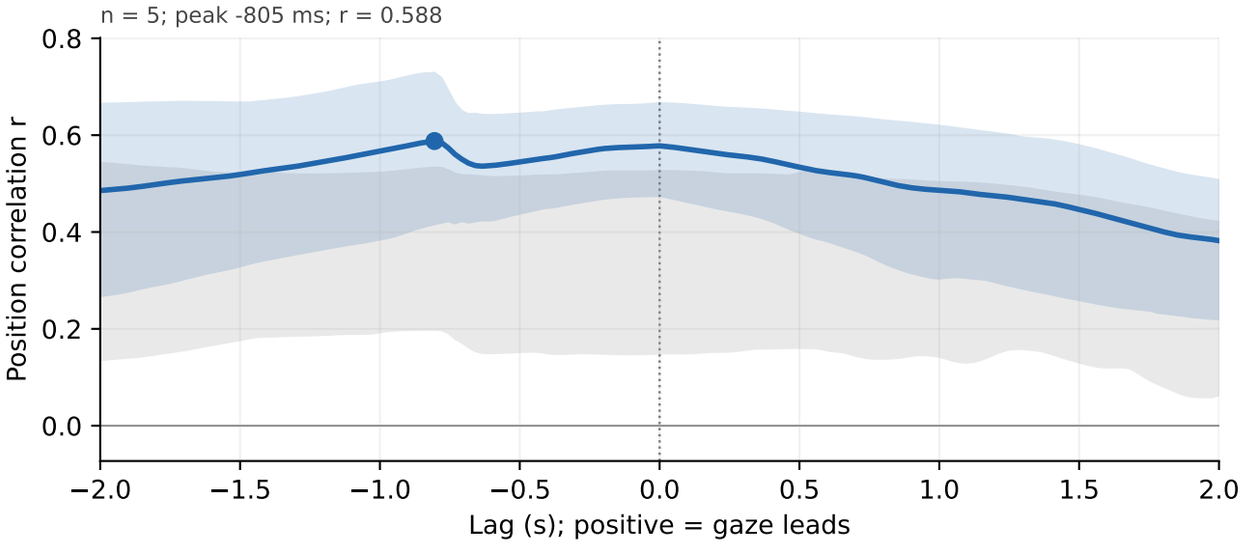
pull through and cinch / within_stage



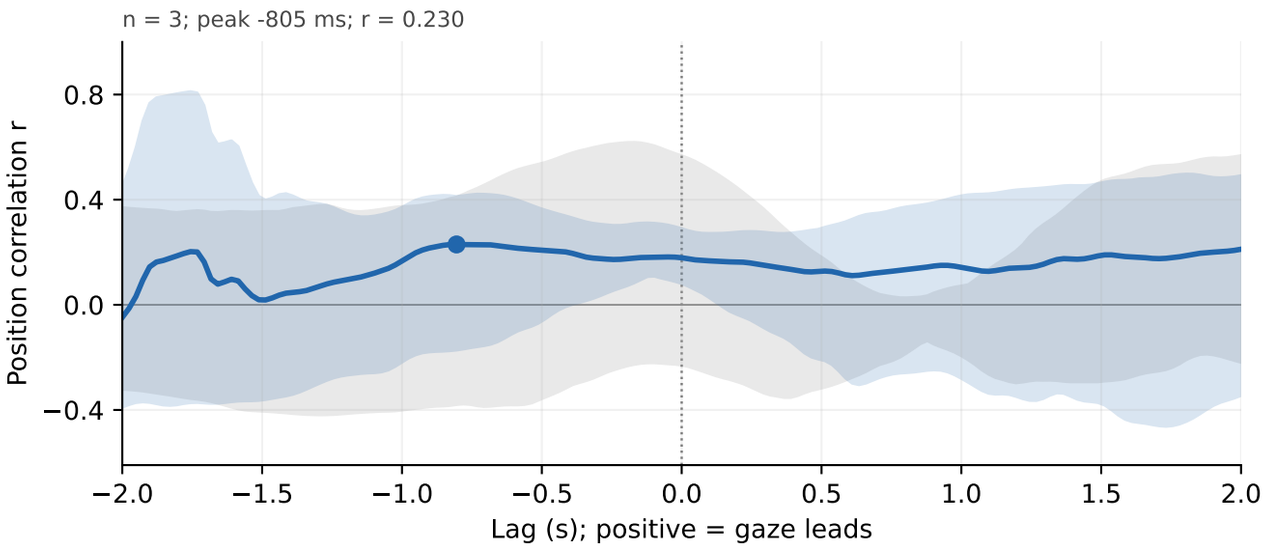
reach string / psm_stage



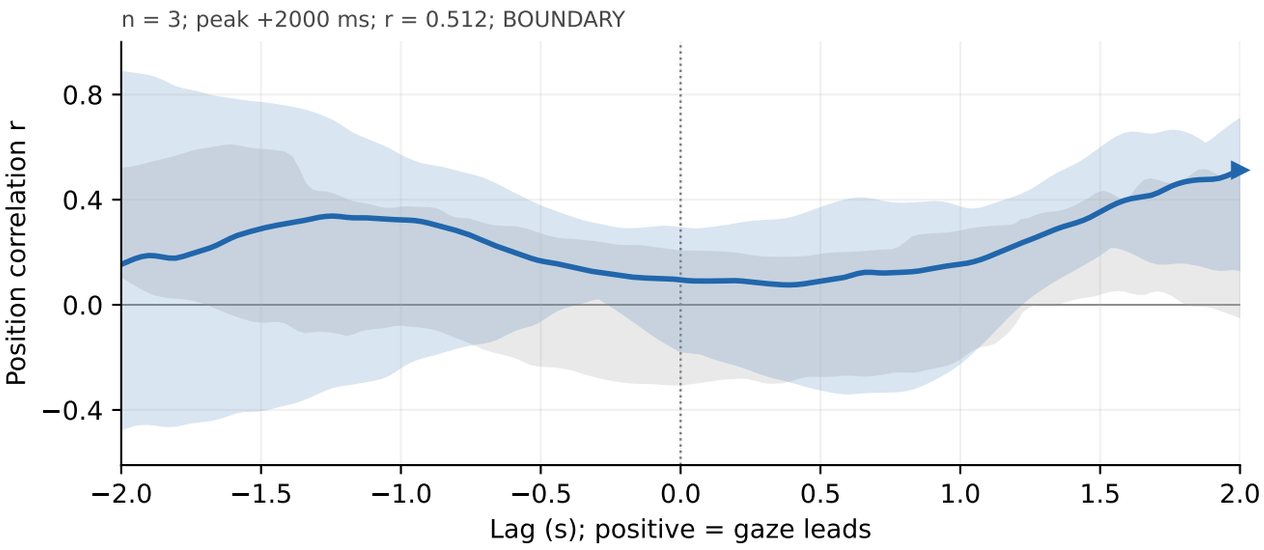
reach string / within_stage



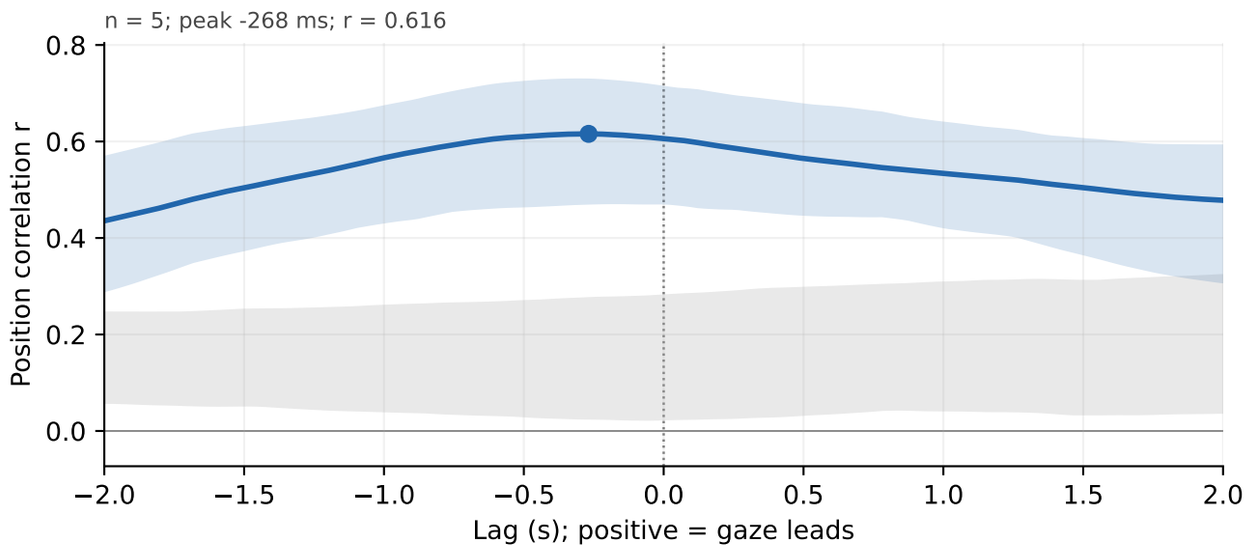
threading string through tower / psm_stage



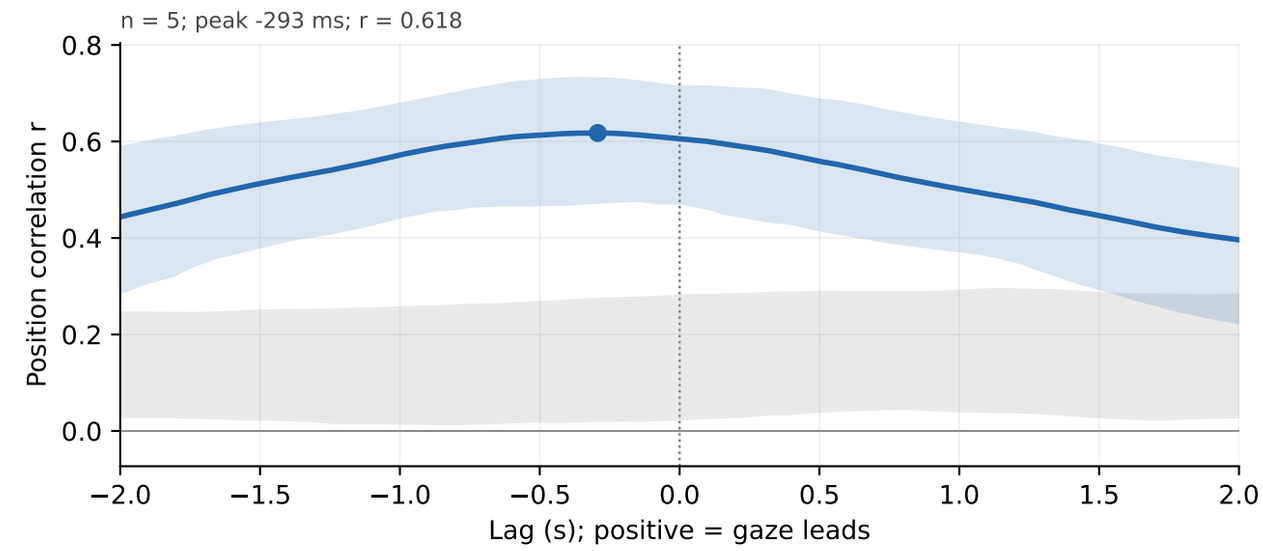
threading string through tower / within_stage



wrap around psm / psm_stage



wrap around psm / within_stage

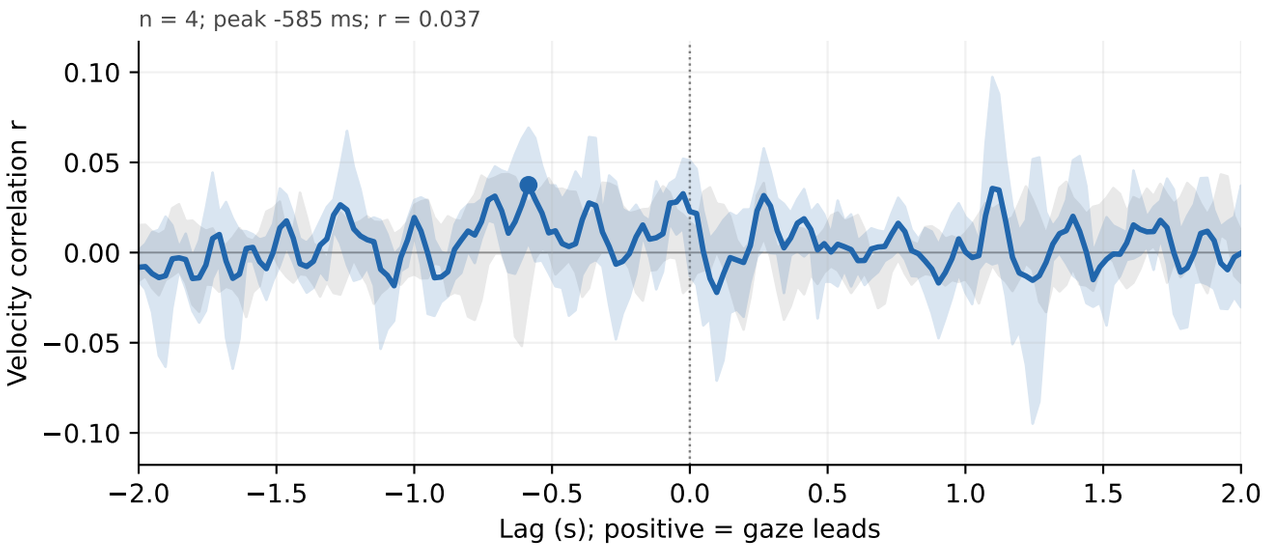


One PSM selected per recording/stage by largest signed xcorr peak; selection does not identify the operated instrument.

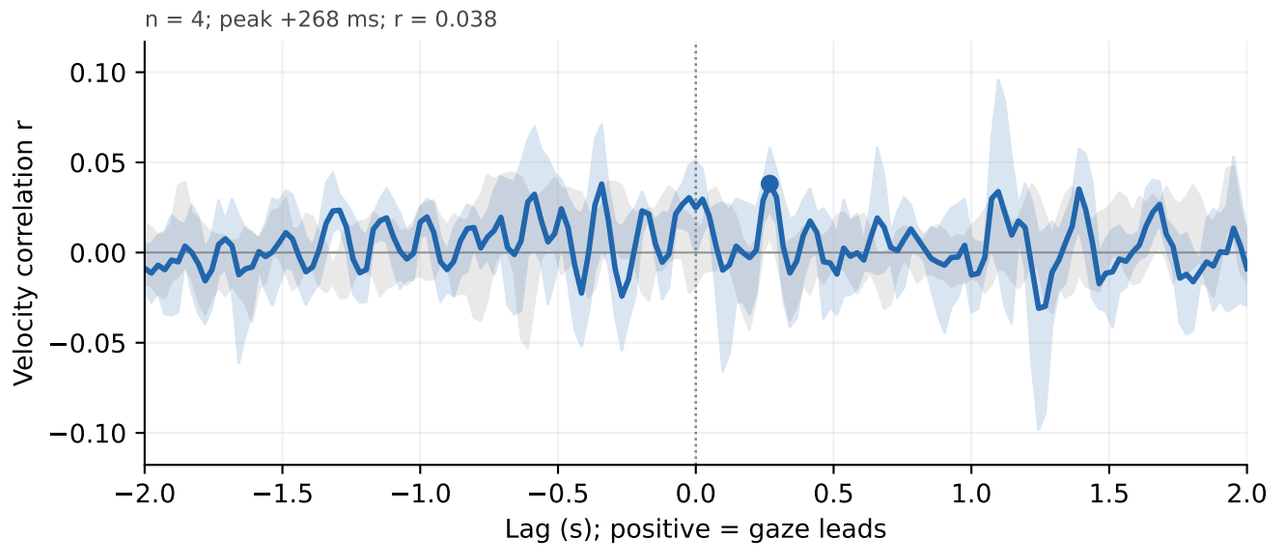
Task C: velocity, maximum-xcorr PSM per stage

Unpaired reference (95%) | Participant bootstrap (95%) | Group mean

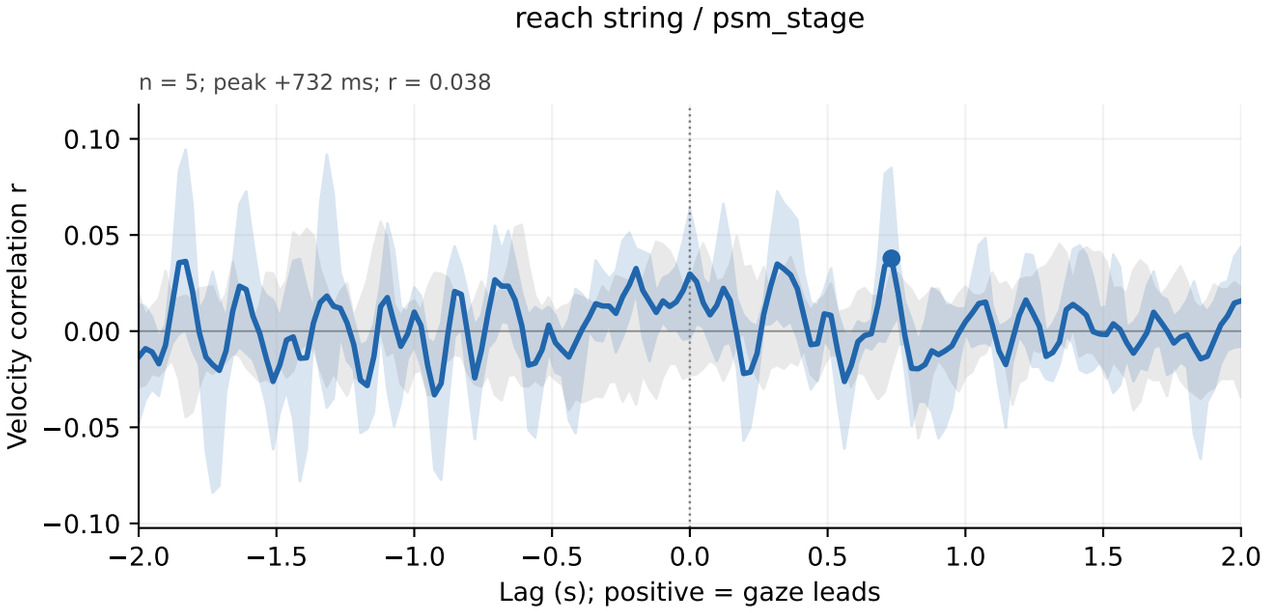
pull through and cinch / psm_stage



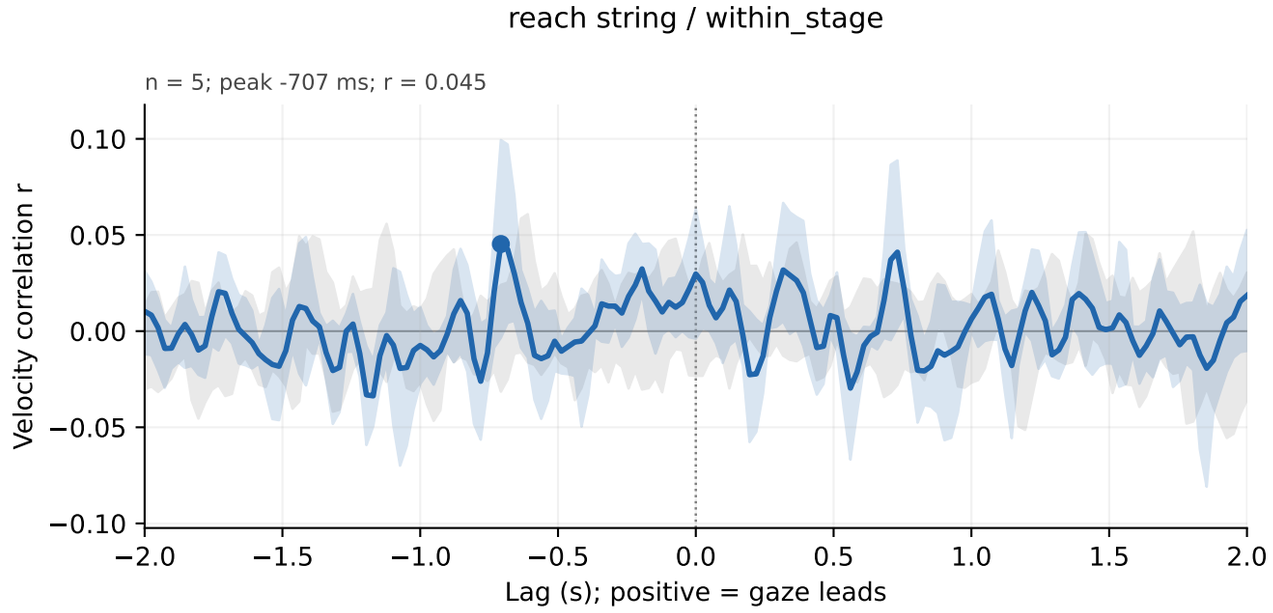
pull through and cinch / within_stage



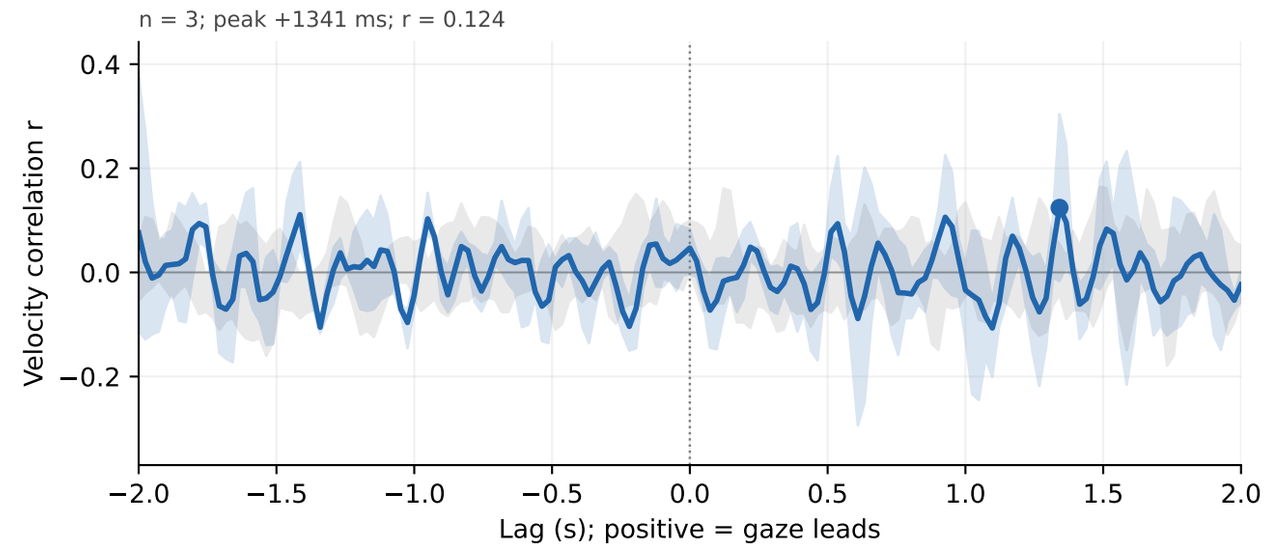
reach string / psm_stage



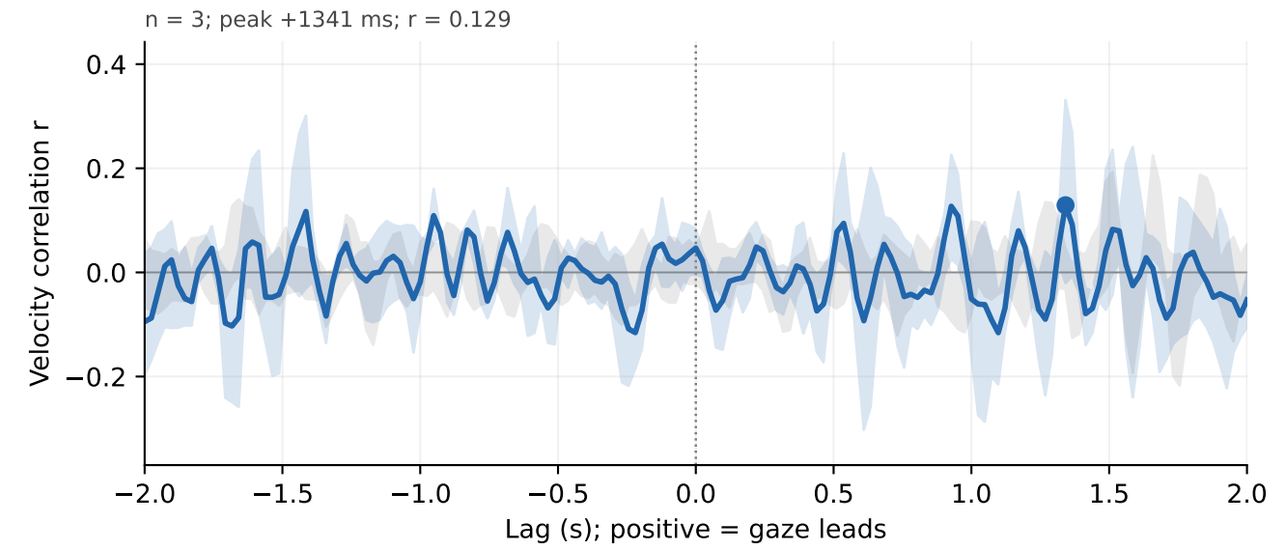
reach string / within_stage



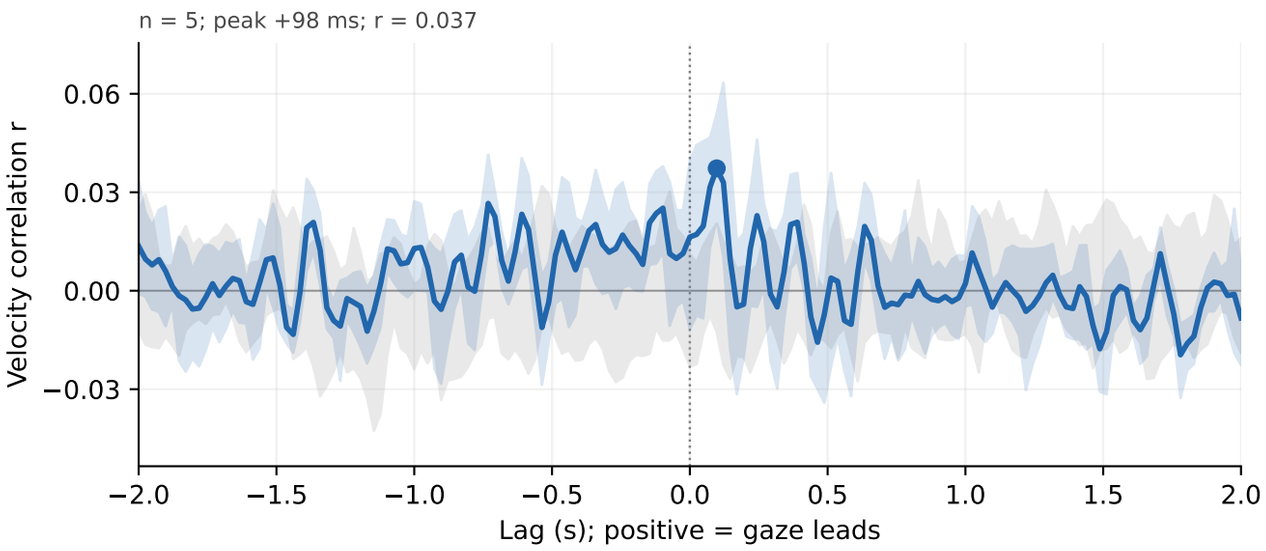
threading string through tower / psm_stage



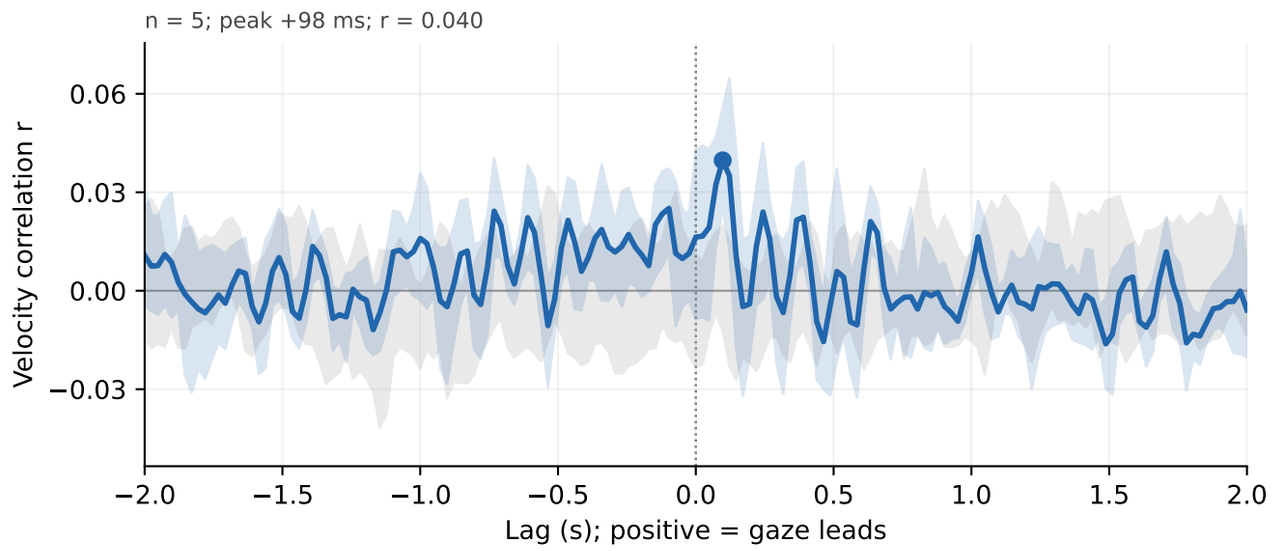
threading string through tower / within_stage



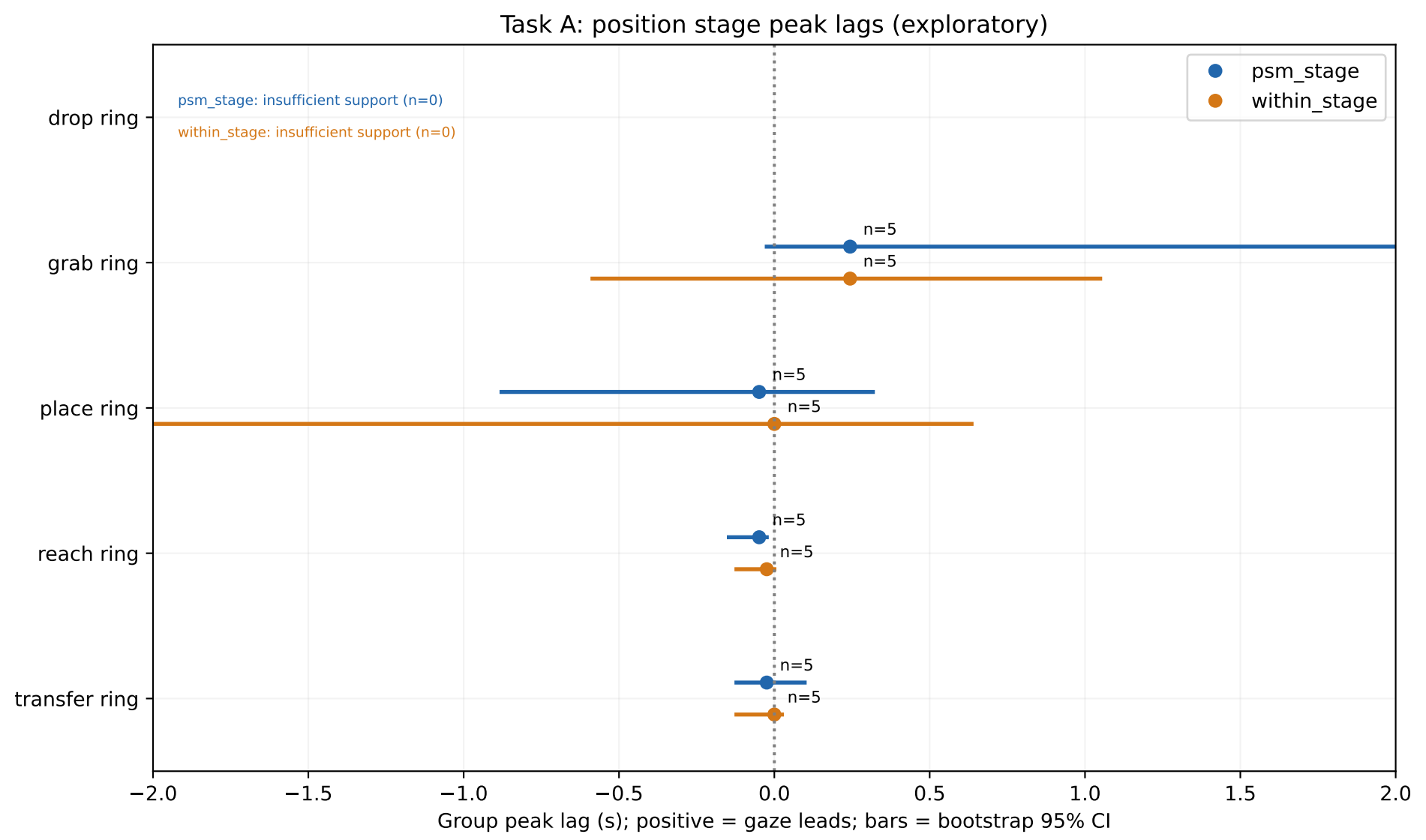
wrap around psm / psm_stage



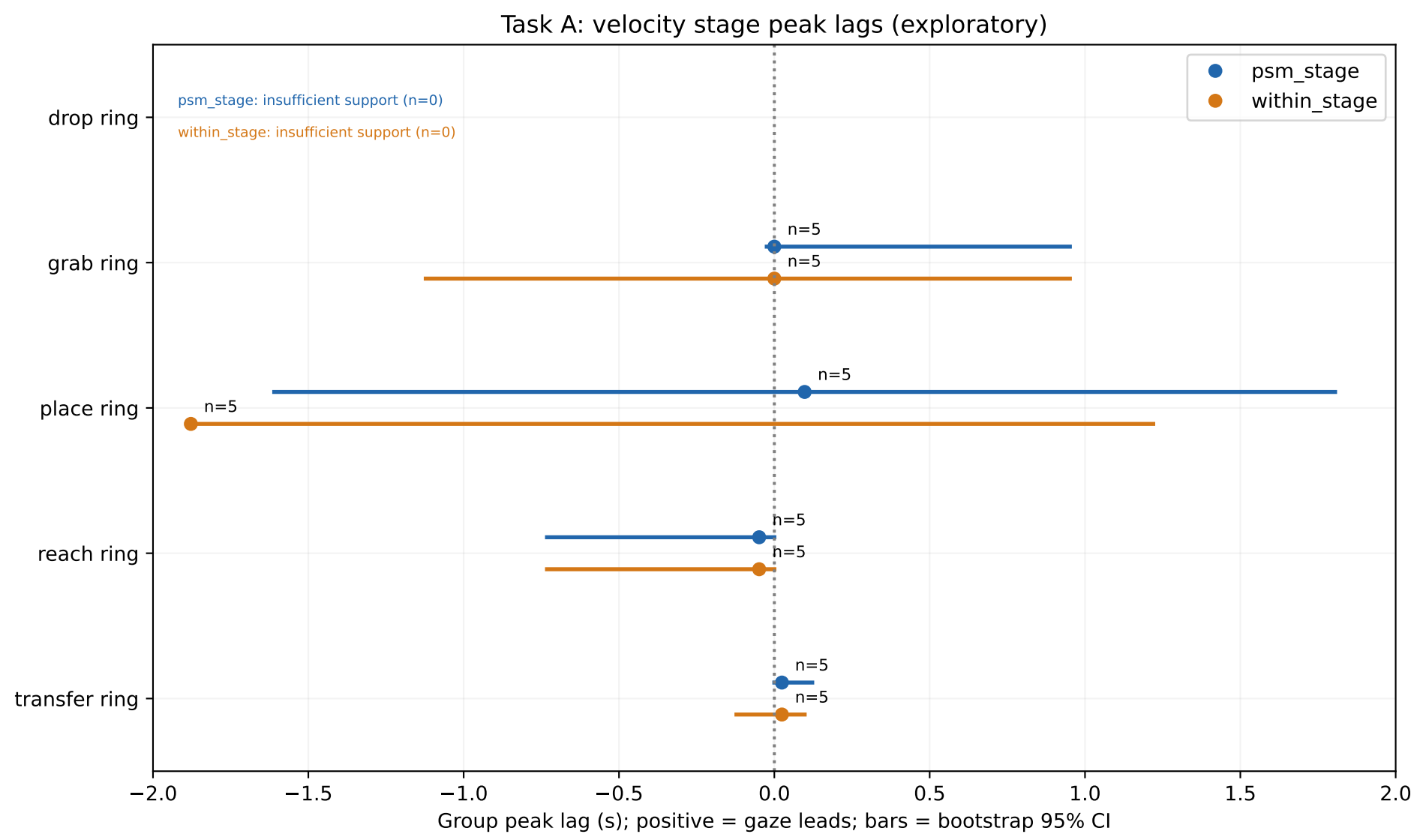
wrap around psm / within_stage



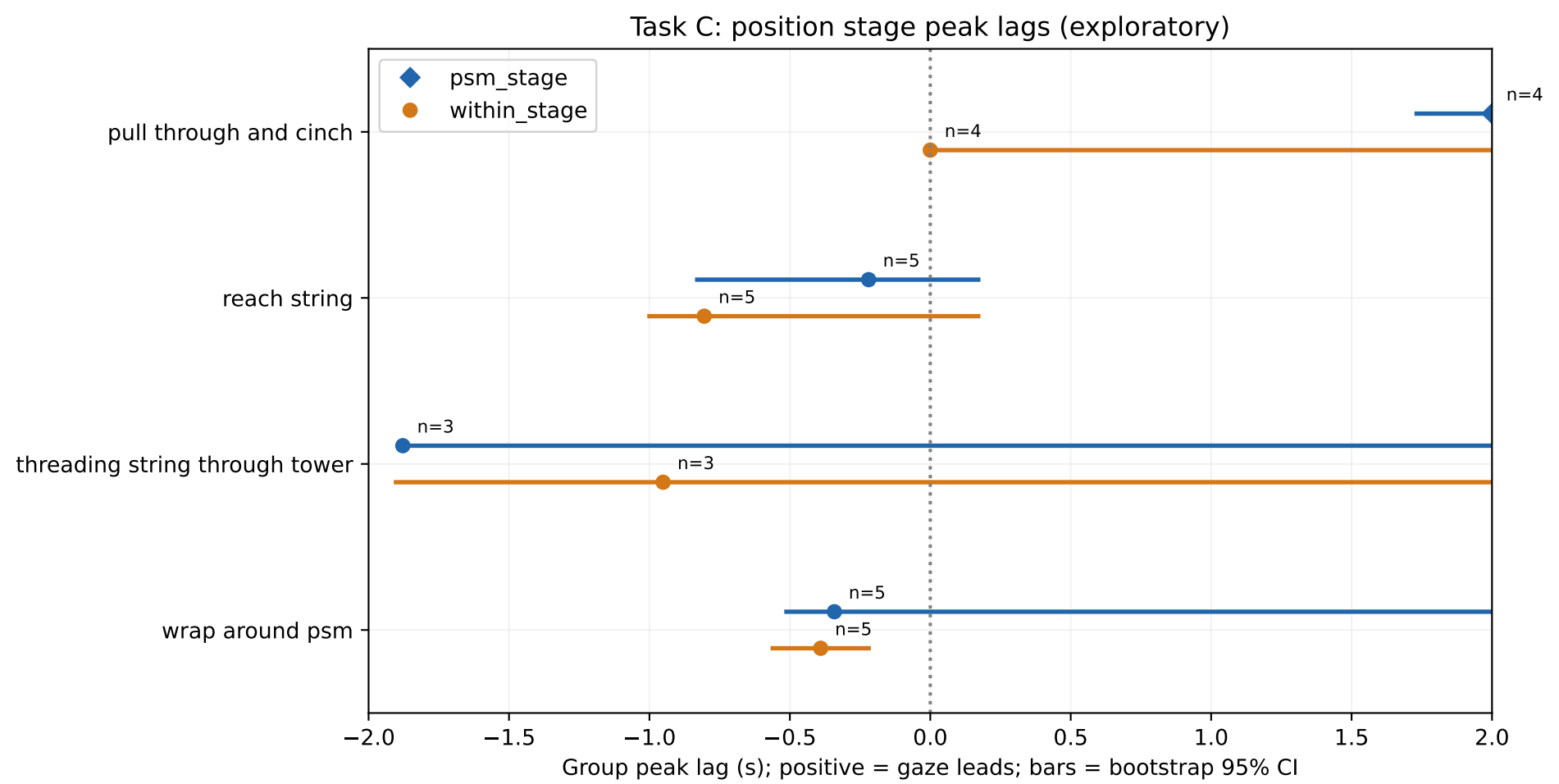
One PSM selected per recording/stage by largest signed xcorr peak; selection does not identify the operated instrument.



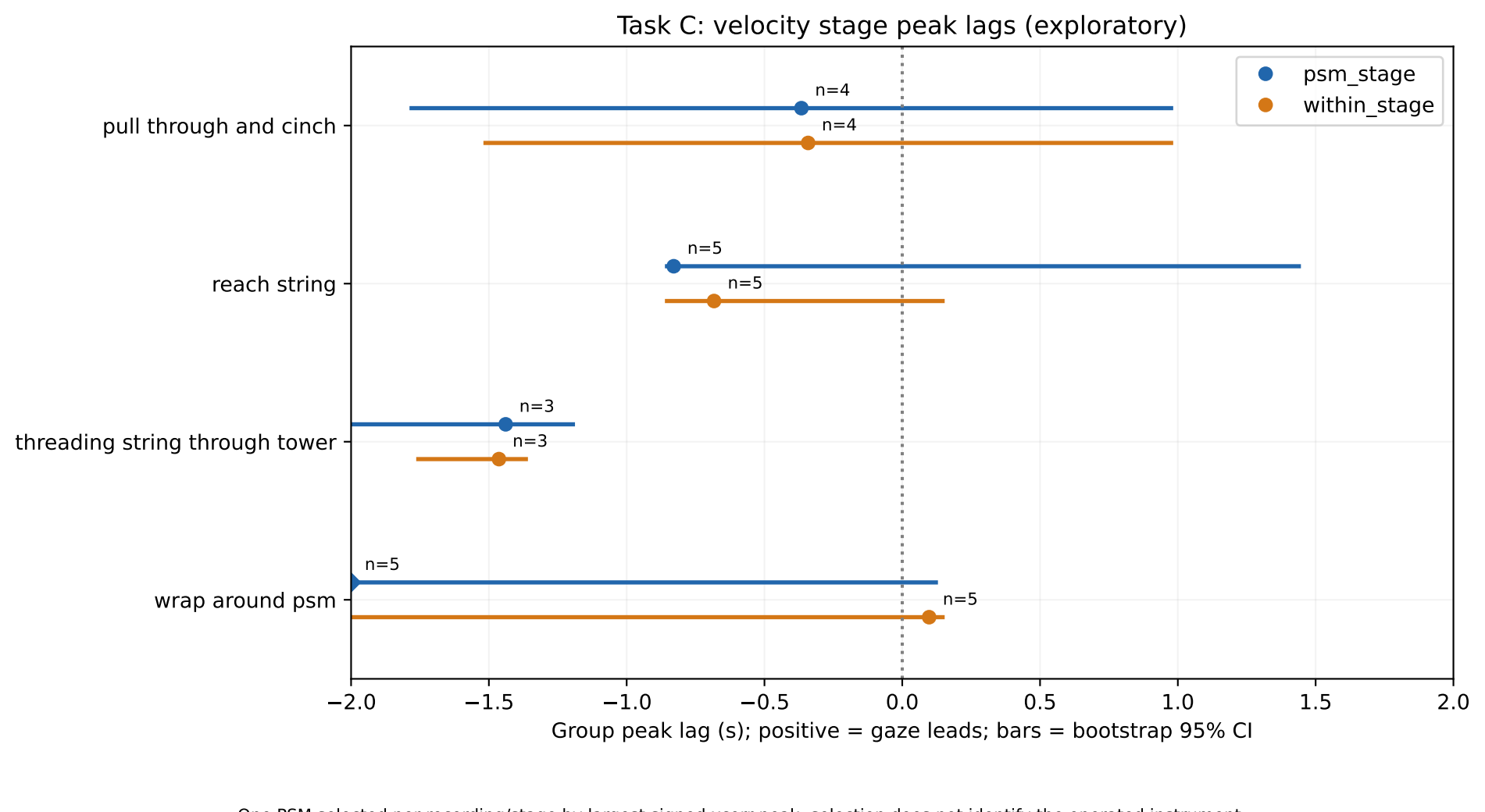
One PSM selected per recording/stage by largest signed xcorr peak; selection does not identify the operated instrument.



One PSM selected per recording/stage by largest signed xcorr peak; selection does not identify the operated instrument.



One PSM selected per recording/stage by largest signed xcorr peak; selection does not identify the operated instrument.



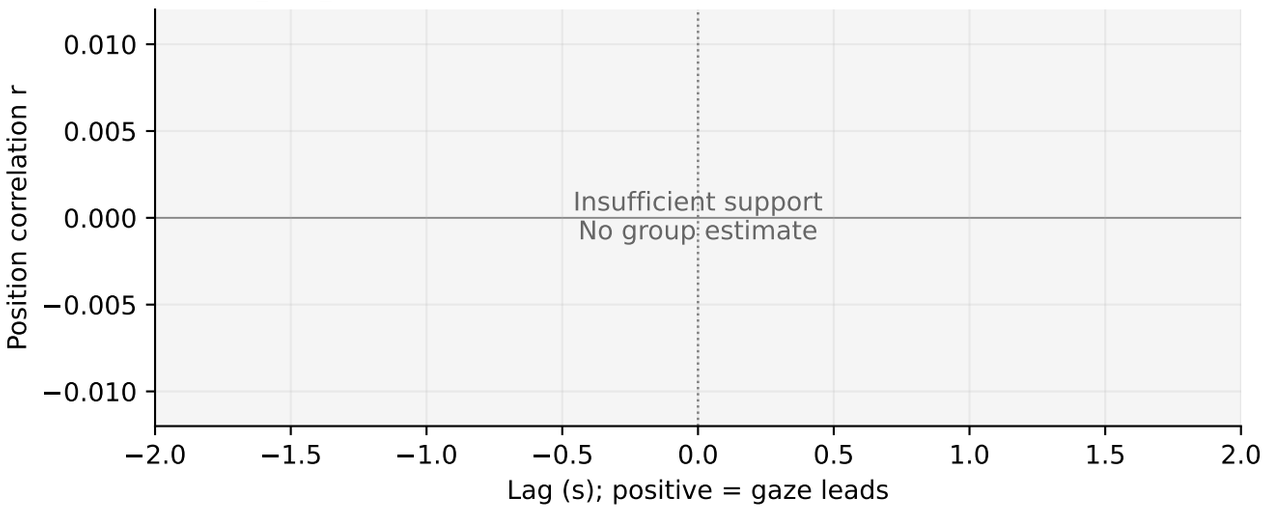
One PSM selected per recording/stage by largest signed xcorr peak; selection does not identify the operated instrument.

Task A: position, maximum-xcorr PSM per stage

Unpaired reference (95%) | Participant bootstrap (95%) | Group mean

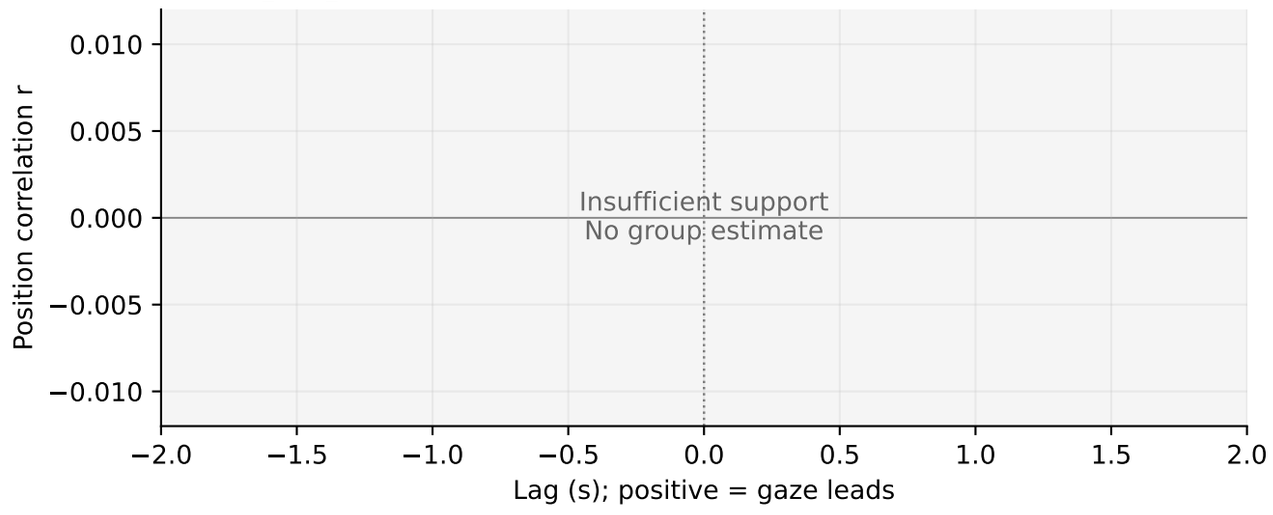
drop ring / psm_stage

n = 0; no_valid_curve



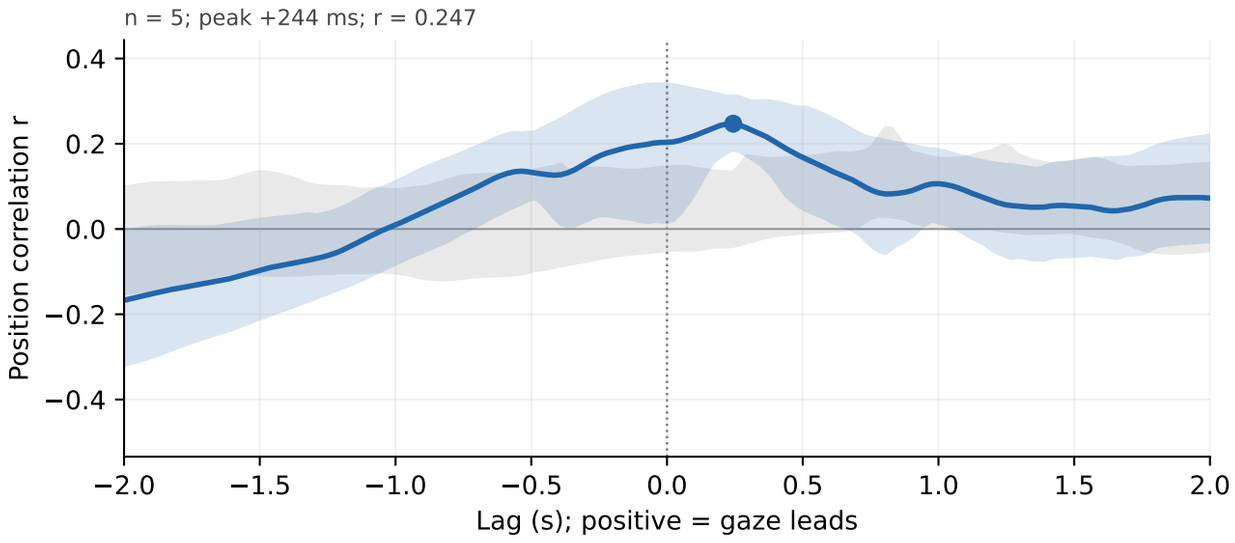
drop ring / within_stage

n = 0; no_valid_curve



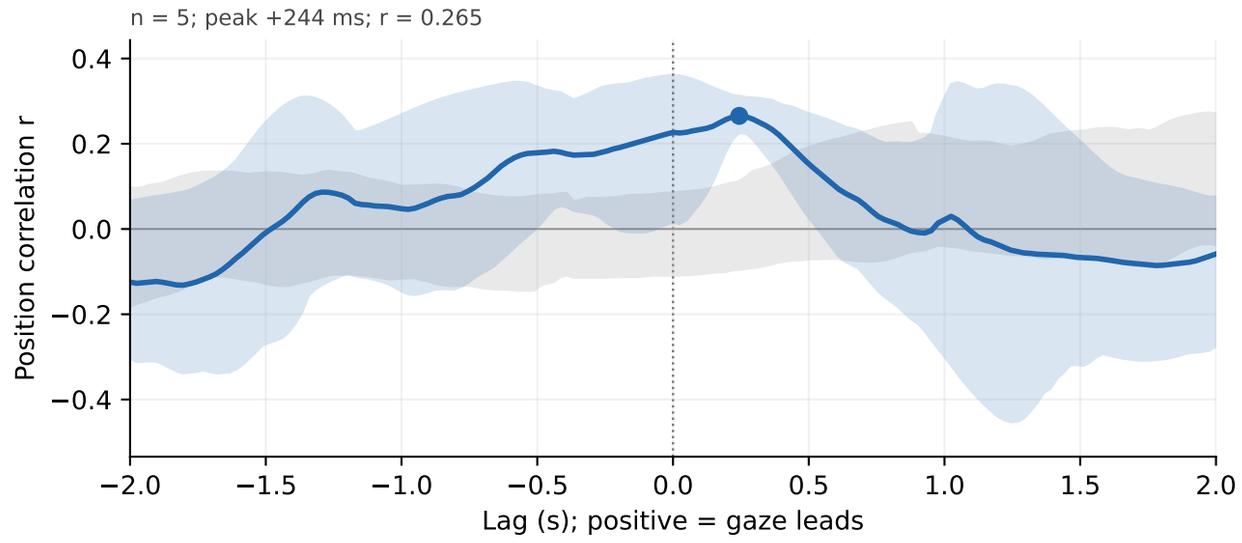
grab ring / psm_stage

n = 5; peak +244 ms; r = 0.247



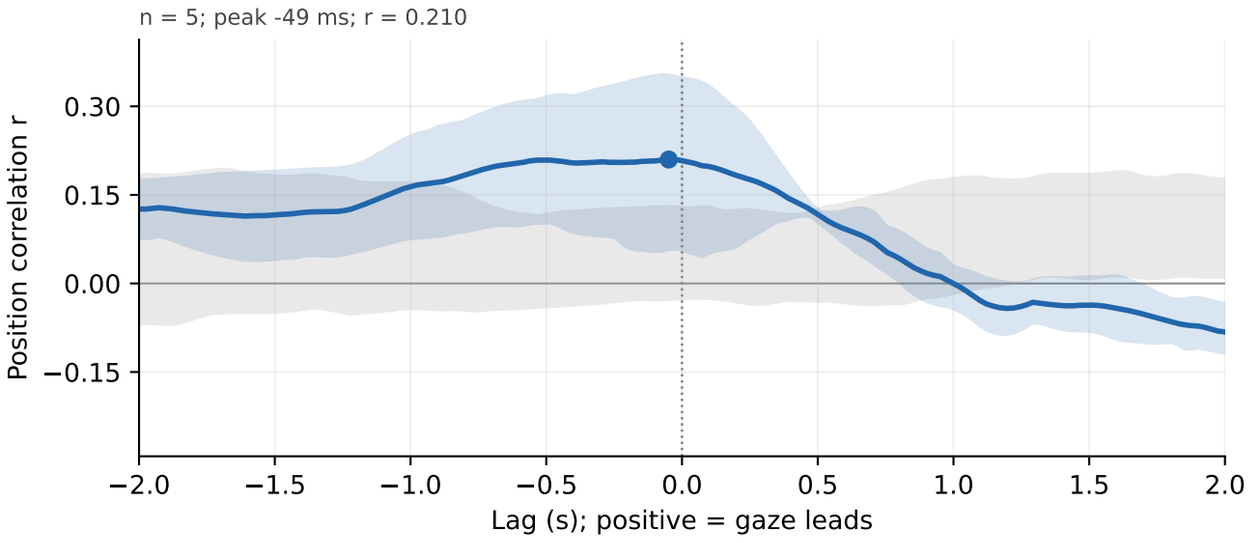
grab ring / within_stage

n = 5; peak +244 ms; r = 0.265



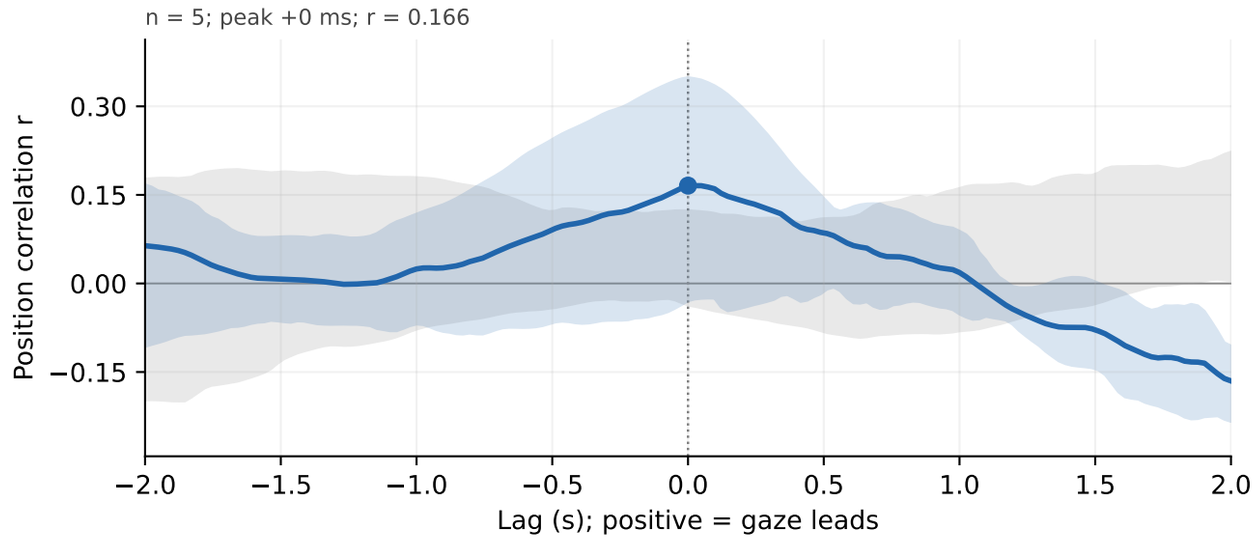
place ring / psm_stage

n = 5; peak -49 ms; r = 0.210



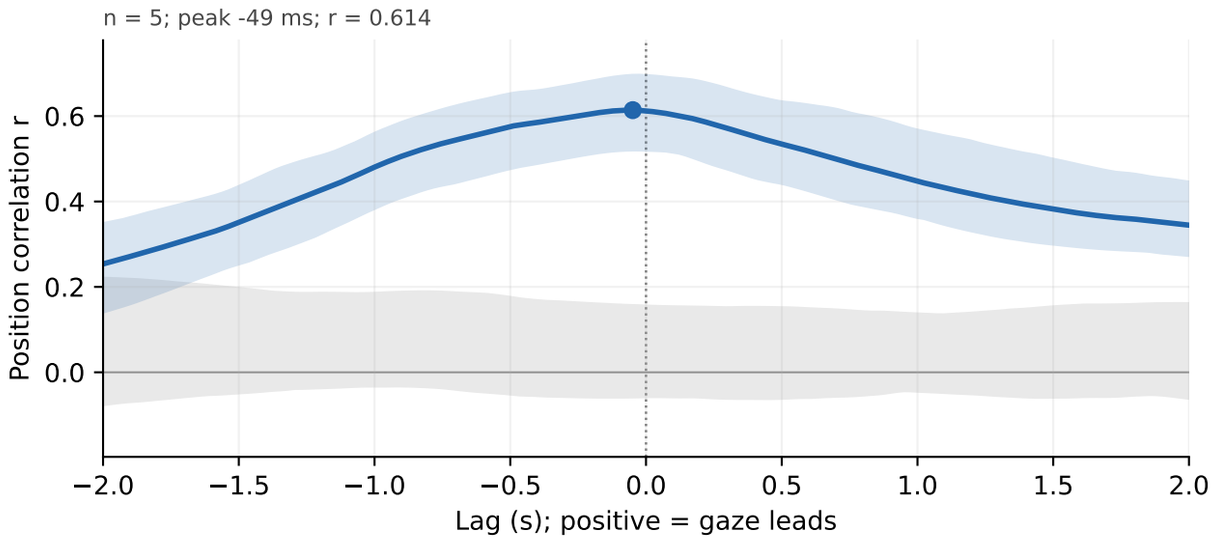
place ring / within_stage

n = 5; peak +0 ms; r = 0.166



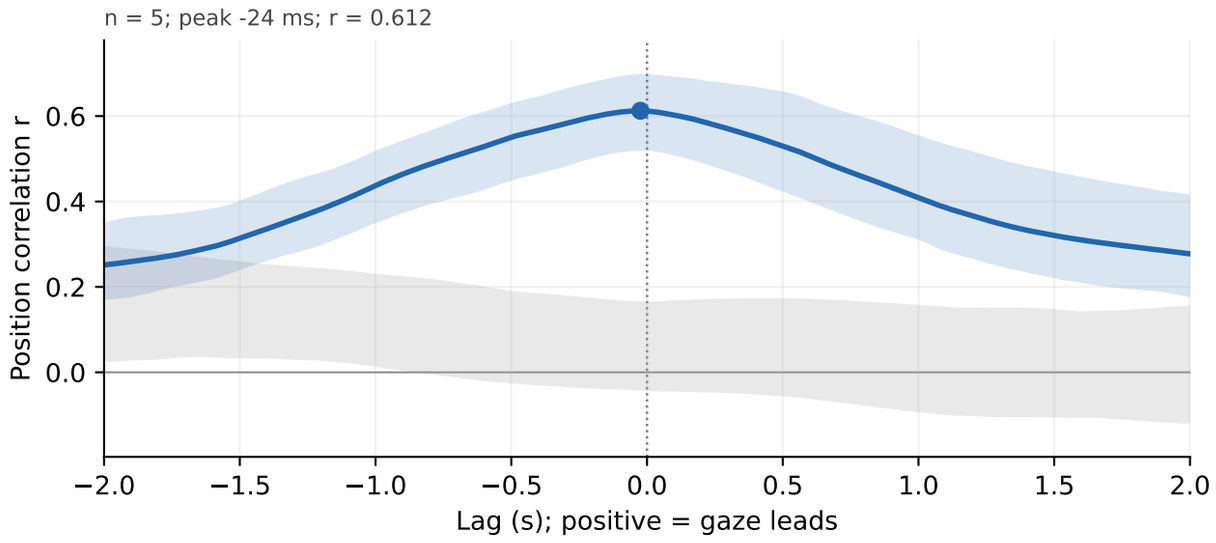
reach ring / psm_stage

n = 5; peak -49 ms; r = 0.614



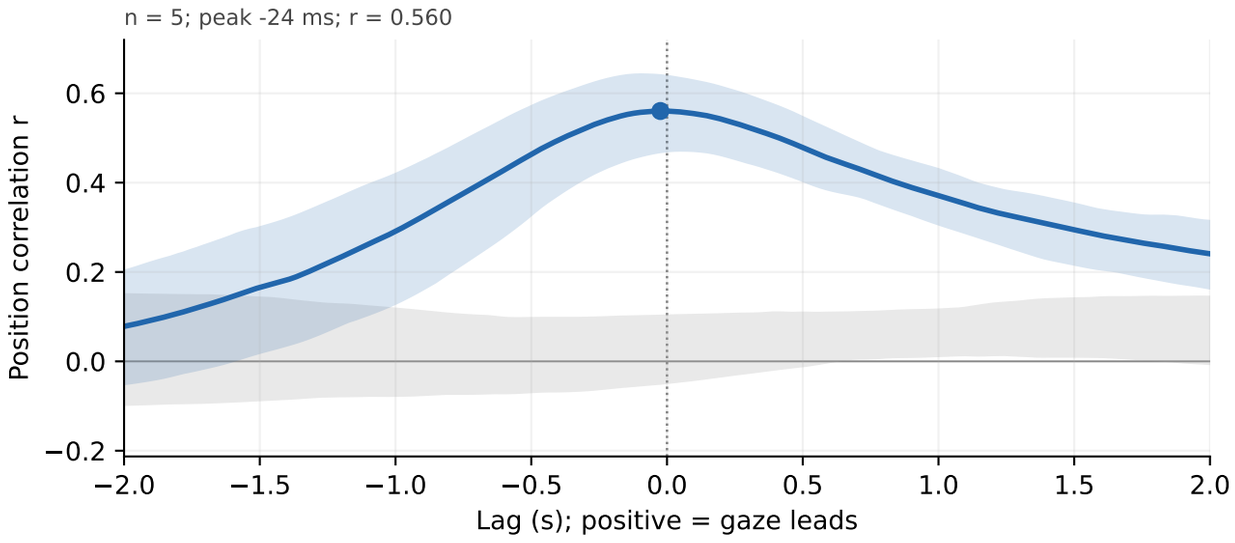
reach ring / within_stage

n = 5; peak -24 ms; r = 0.612



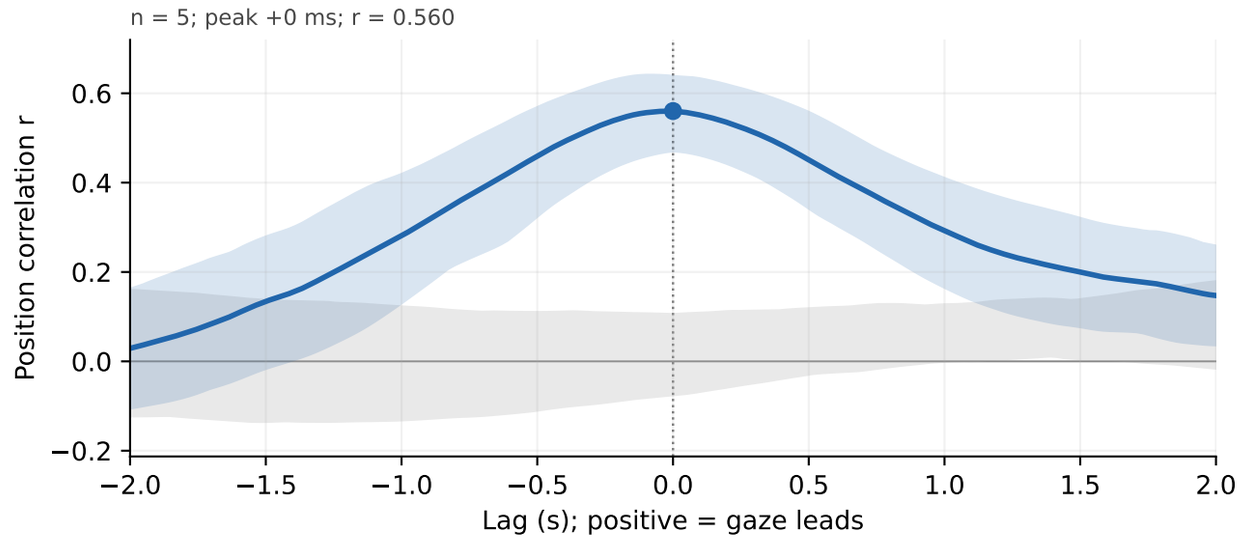
transfer ring / psm_stage

n = 5; peak -24 ms; r = 0.560



transfer ring / within_stage

n = 5; peak +0 ms; r = 0.560

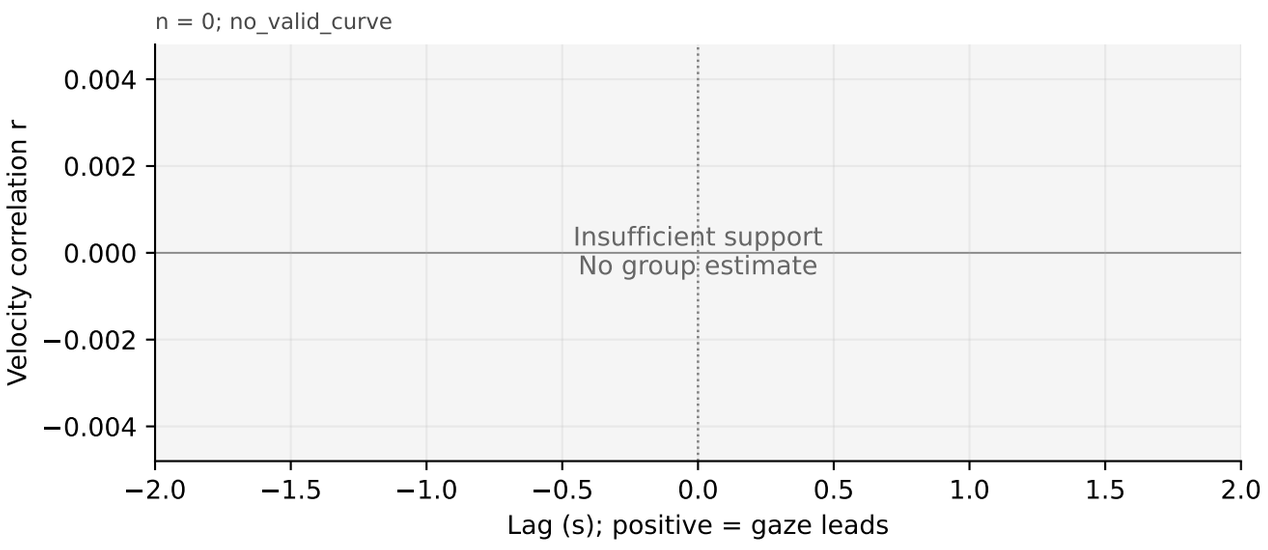


One PSM selected per recording/stage by largest signed xcorr peak; selection does not identify the operated instrument.

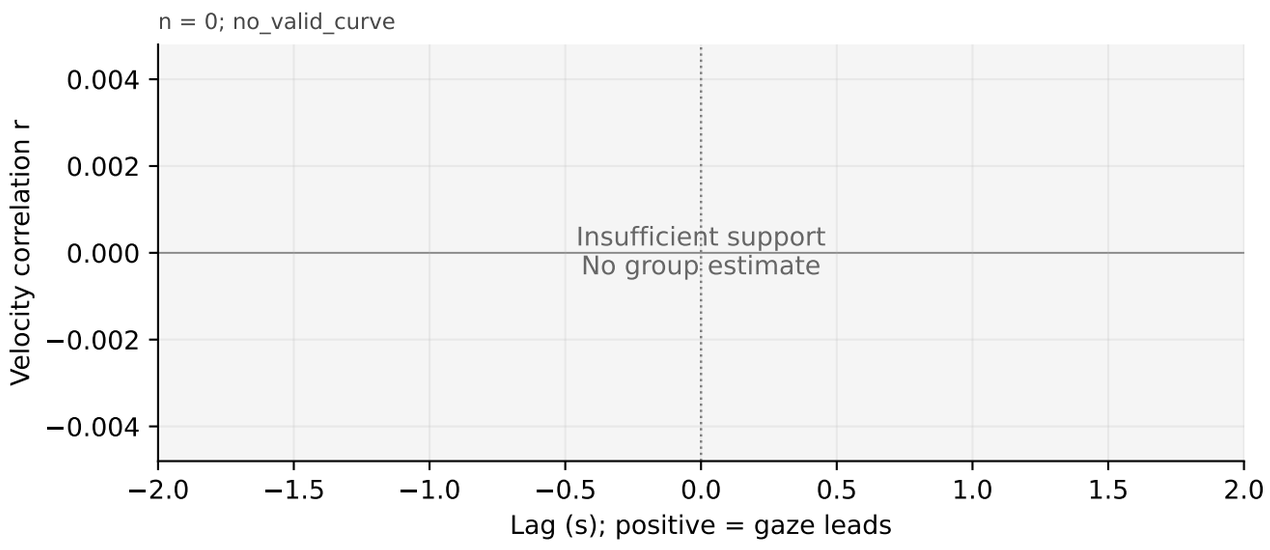
Task A: velocity, maximum-xcorr PSM per stage

Unpaired reference (95%) | Participant bootstrap (95%) | Group mean

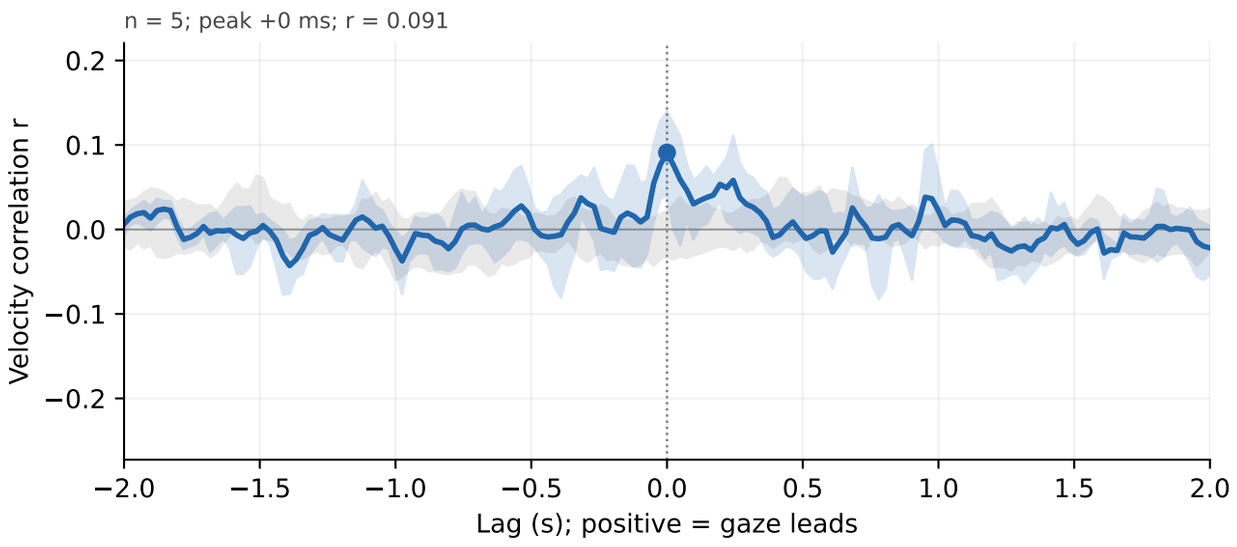
drop ring / psm_stage



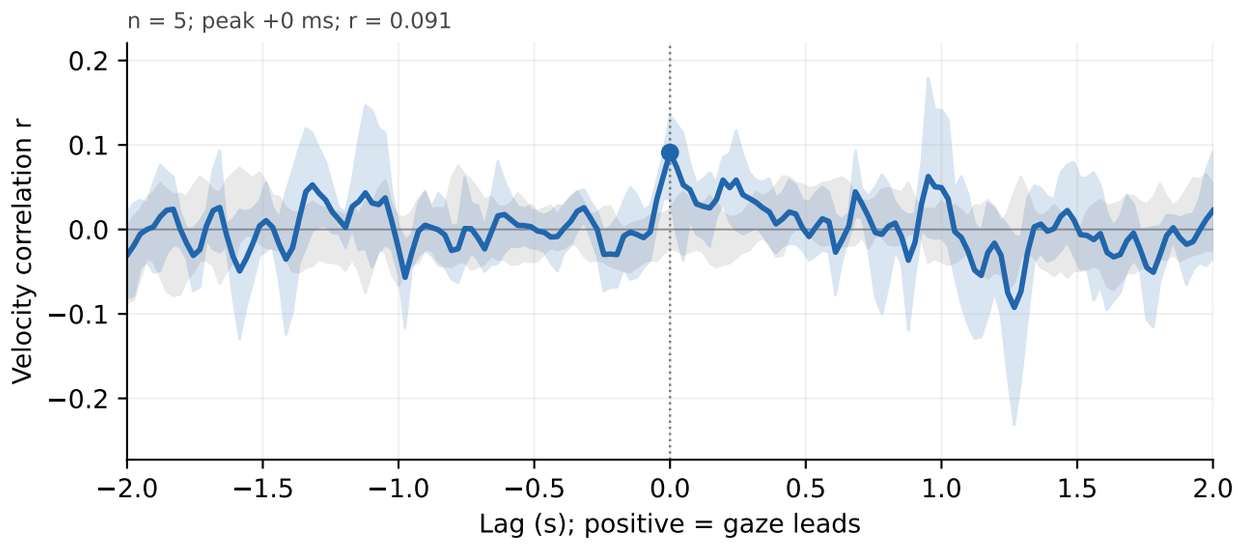
drop ring / within_stage



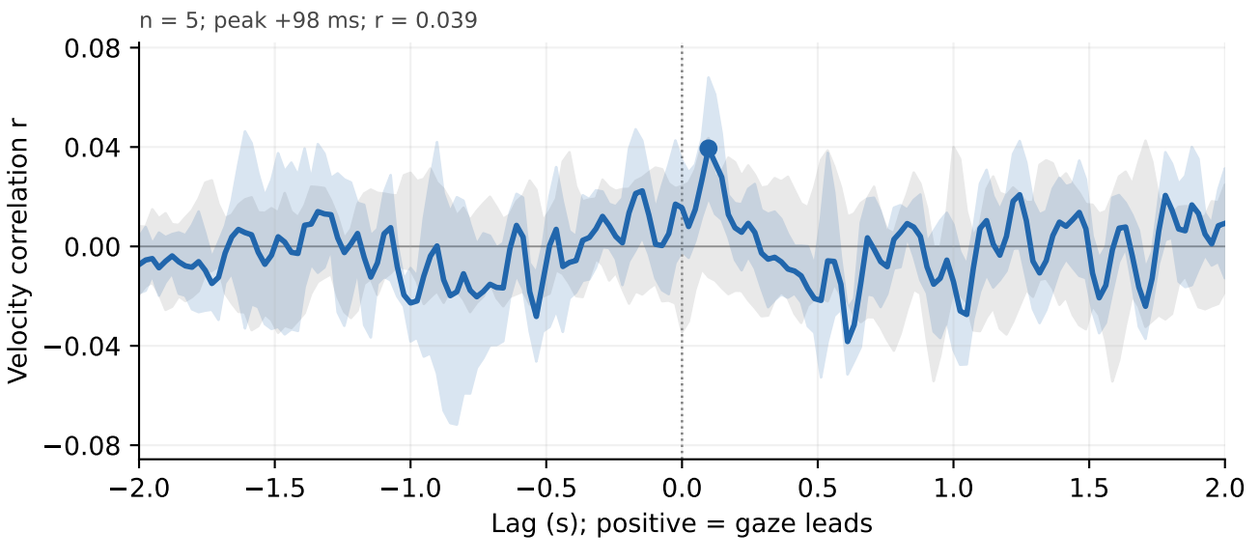
grab ring / psm_stage



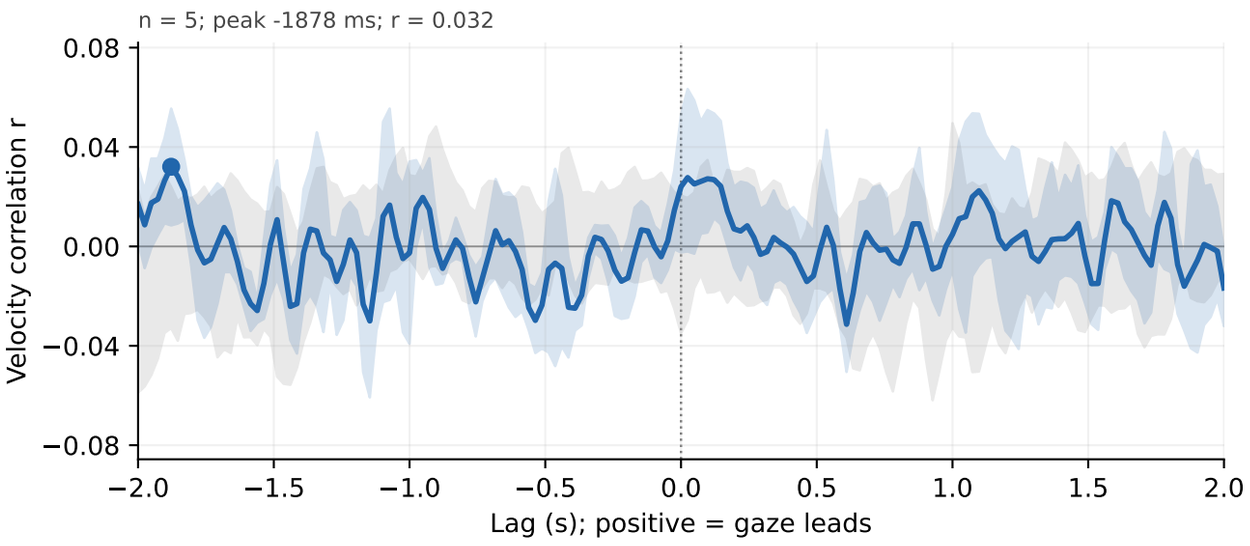
grab ring / within_stage



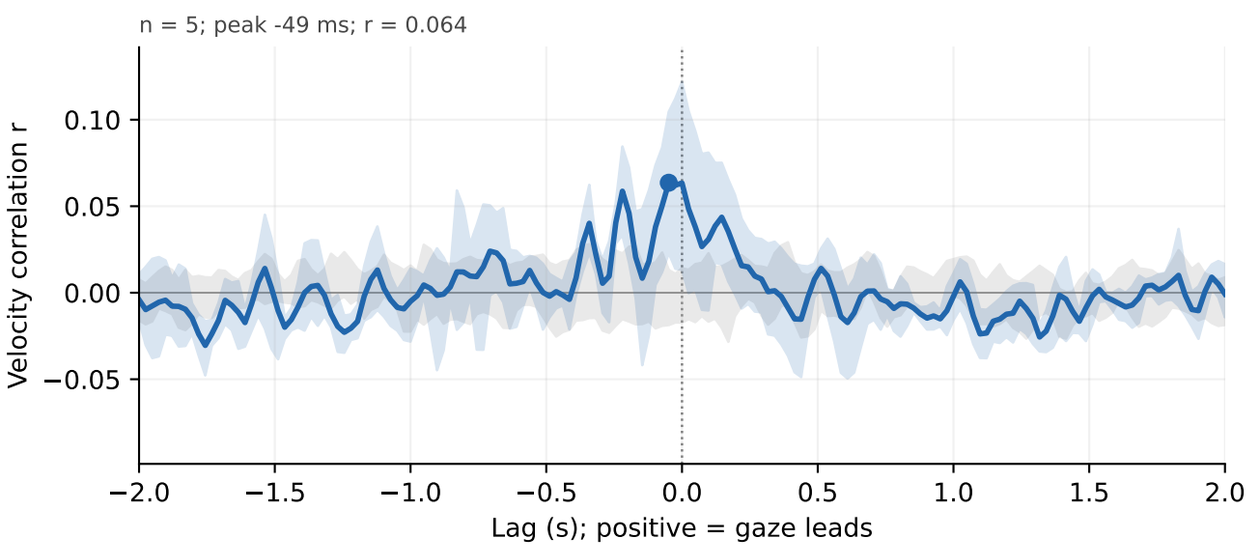
place ring / psm_stage



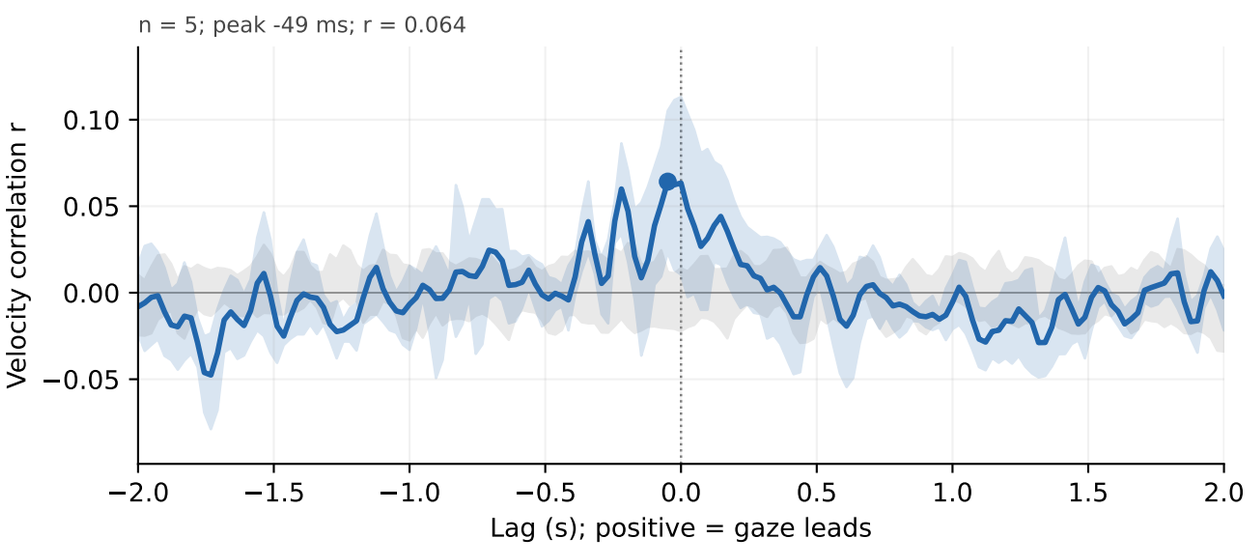
place ring / within_stage



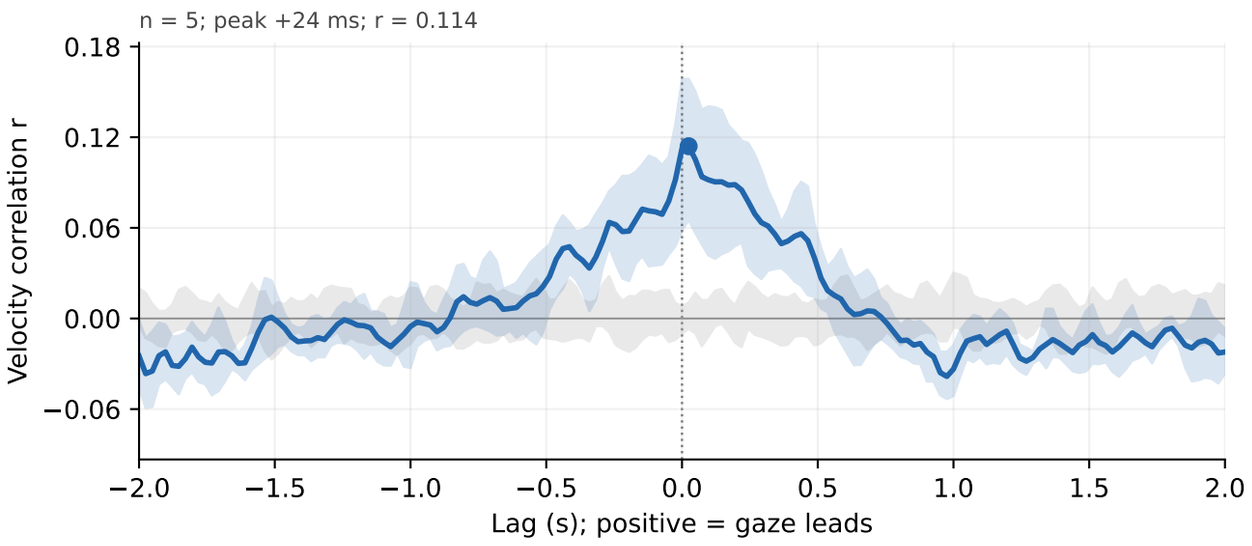
reach ring / psm_stage



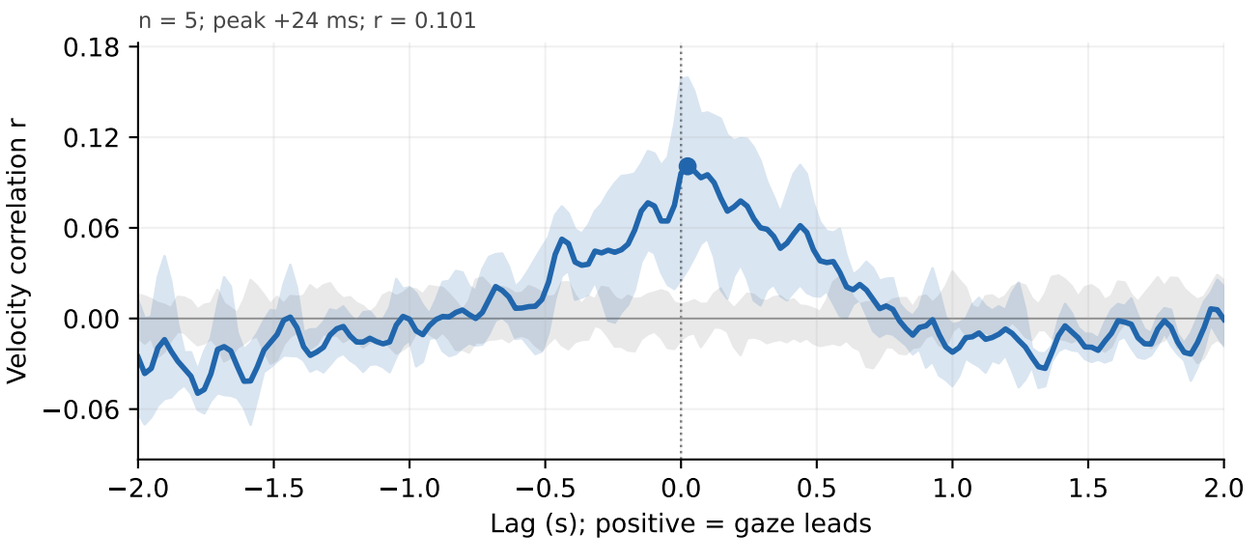
reach ring / within_stage



transfer ring / psm_stage



transfer ring / within_stage

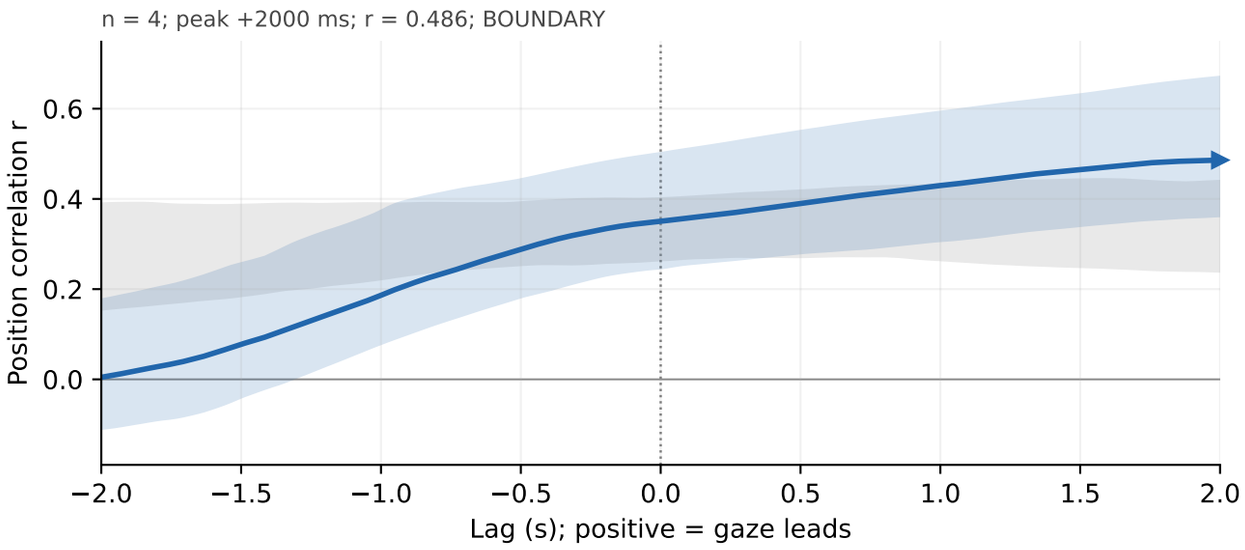


One PSM selected per recording/stage by largest signed xcorr peak; selection does not identify the operated instrument.

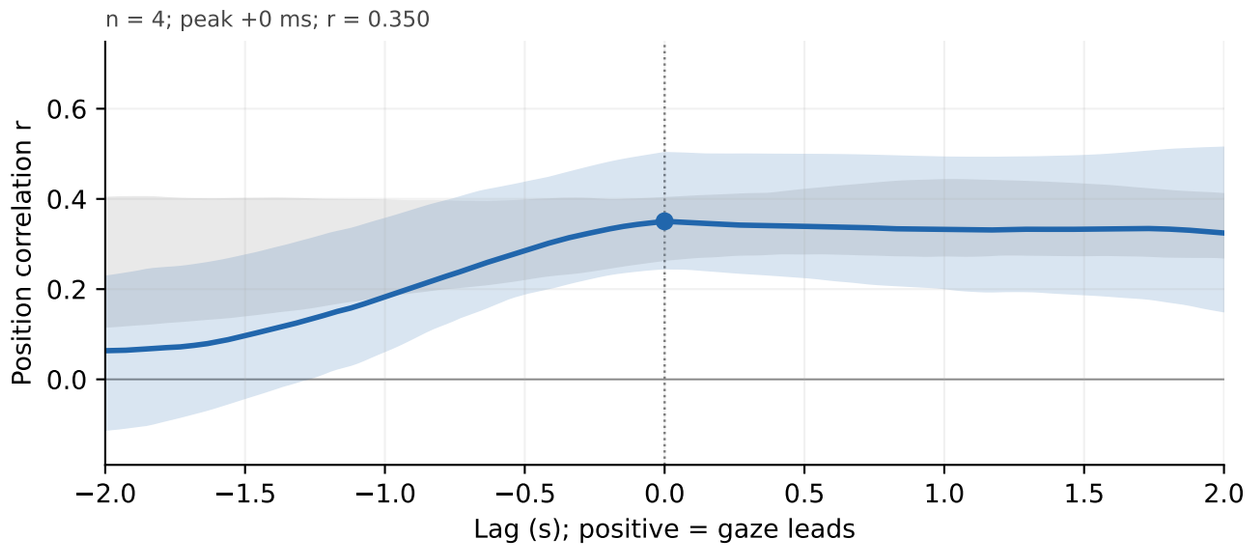
Task C: position, maximum-xcorr PSM per stage

Unpaired reference (95%) | Participant bootstrap (95%) | Group mean

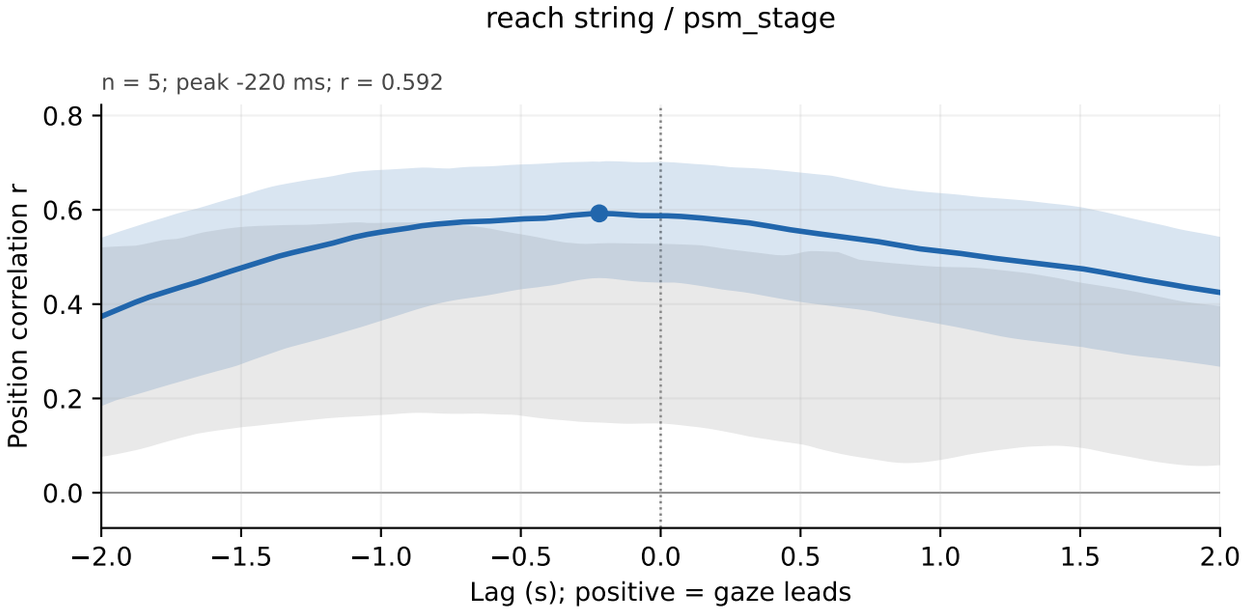
pull through and cinch / psm_stage



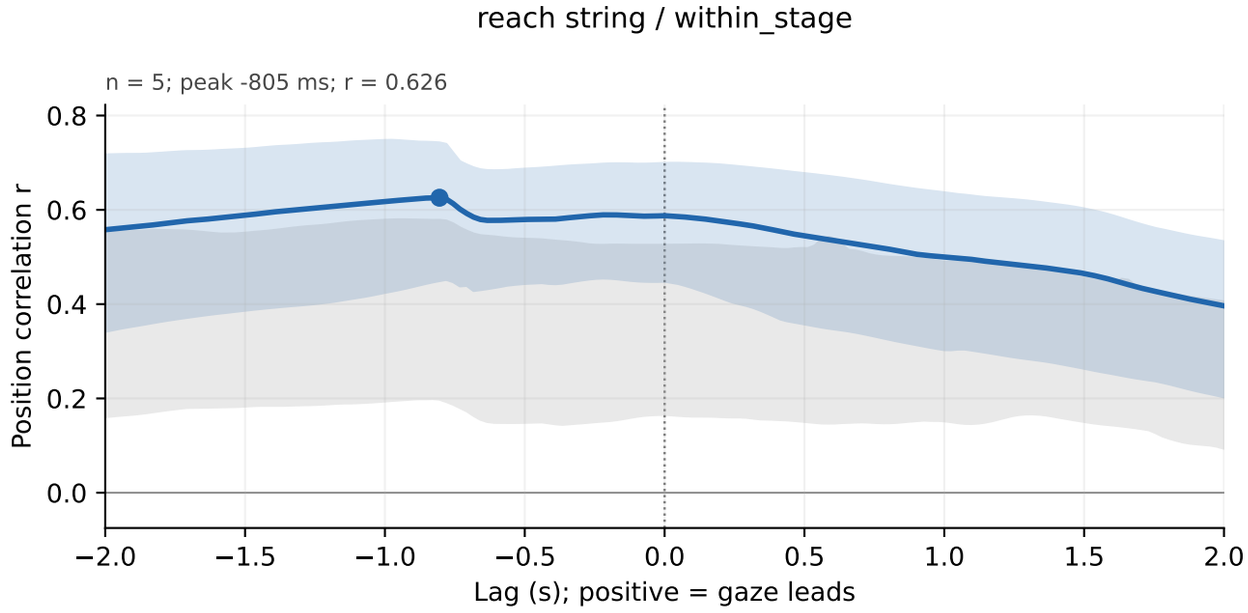
pull through and cinch / within_stage



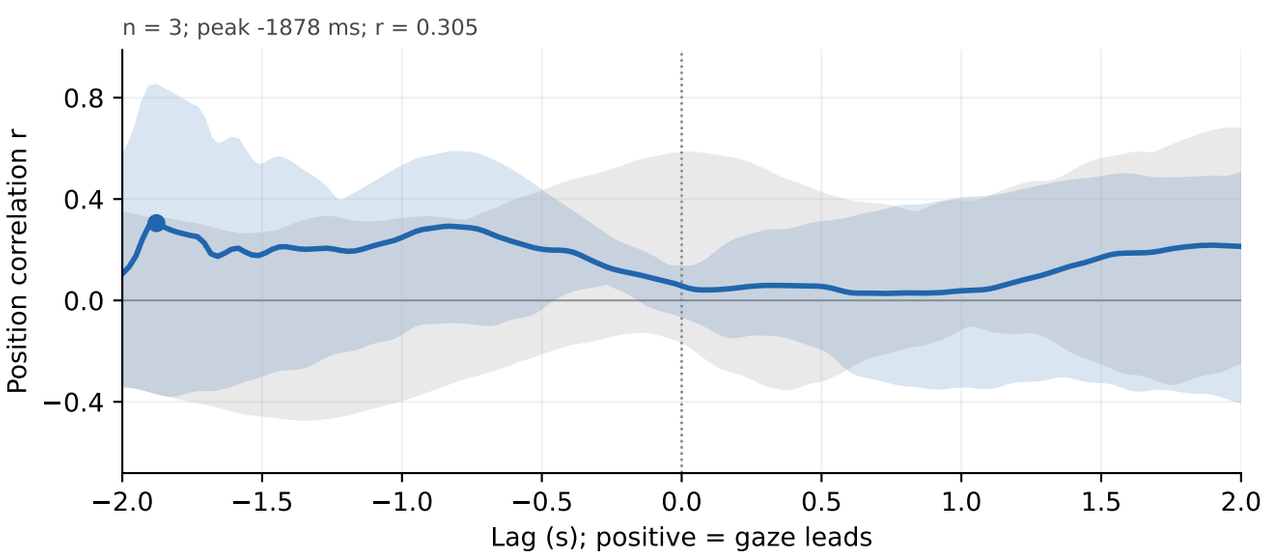
reach string / psm_stage



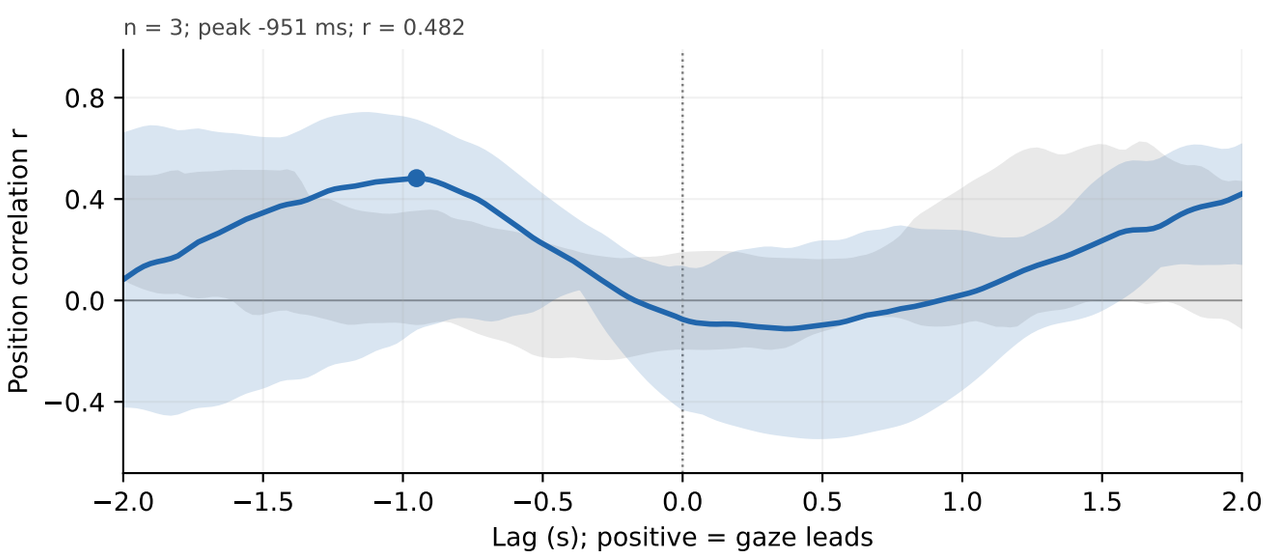
reach string / within_stage



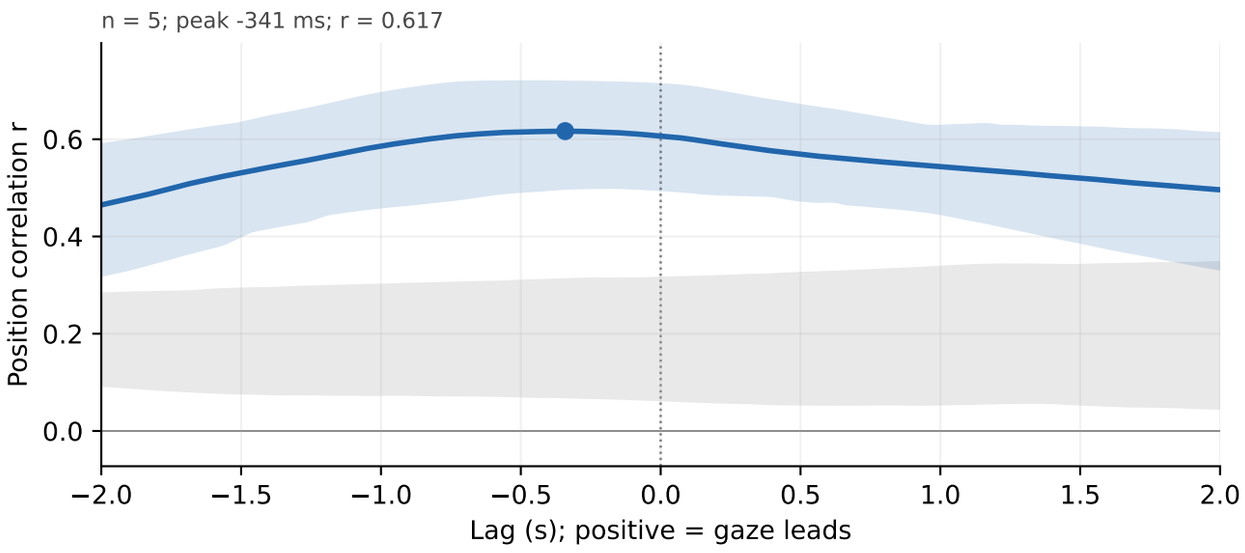
threading string through tower / psm_stage



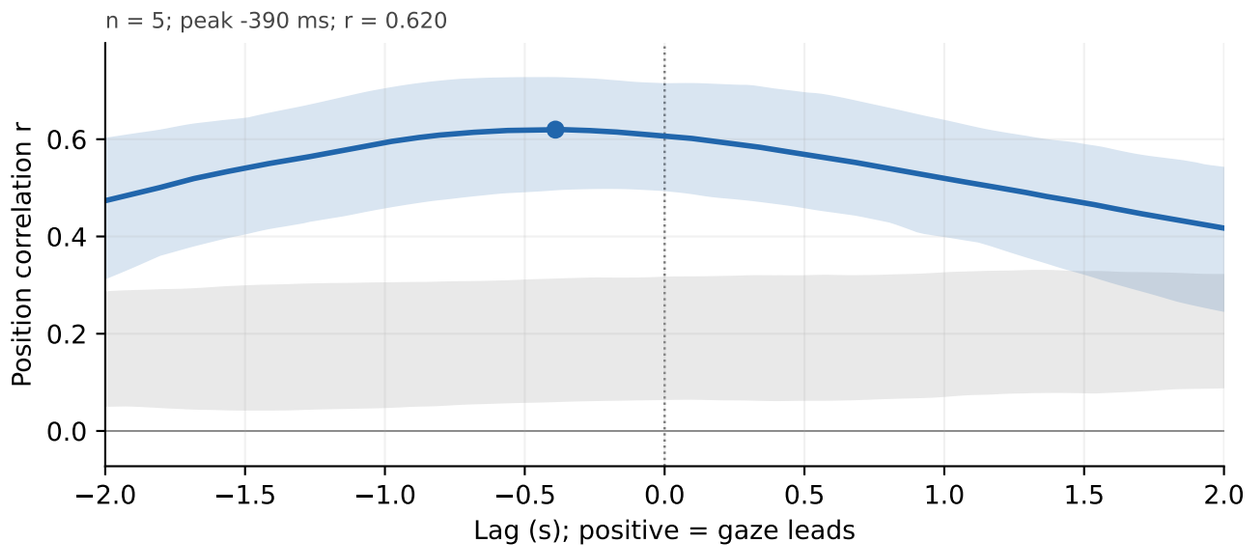
threading string through tower / within_stage



wrap around psm / psm_stage



wrap around psm / within_stage

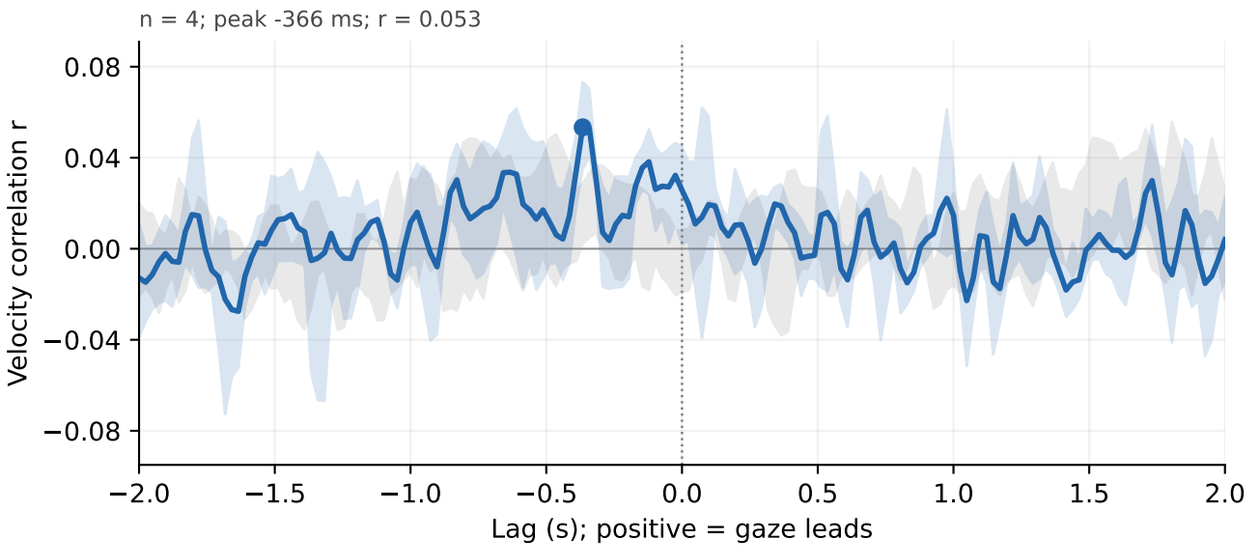


One PSM selected per recording/stage by largest signed xcorr peak; selection does not identify the operated instrument.

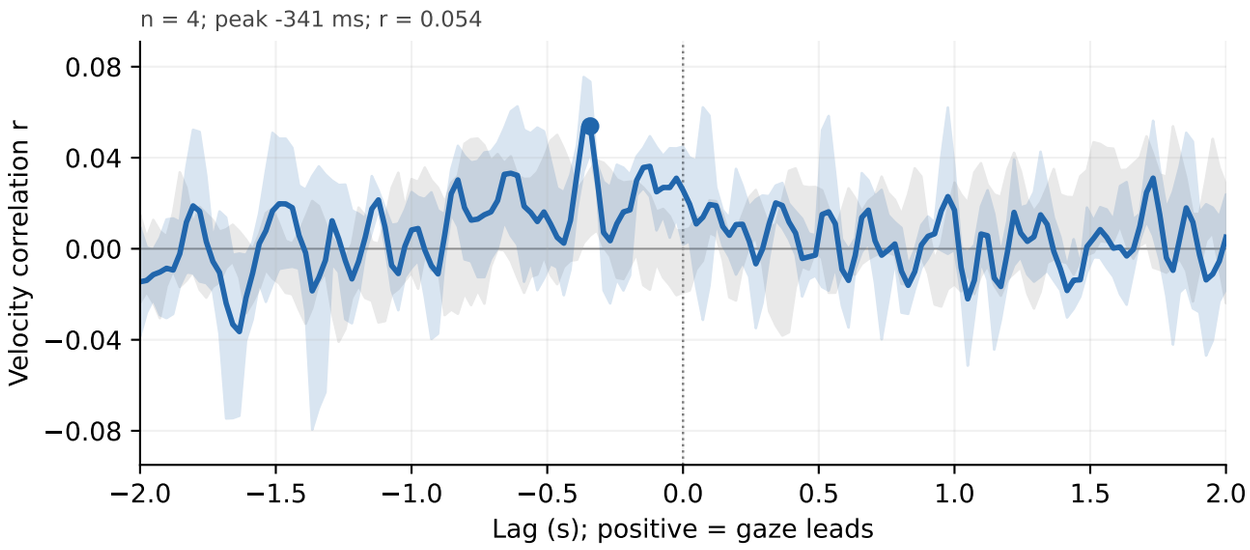
Task C: velocity, maximum-xcorr PSM per stage

Unpaired reference (95%) | Participant bootstrap (95%) | Group mean

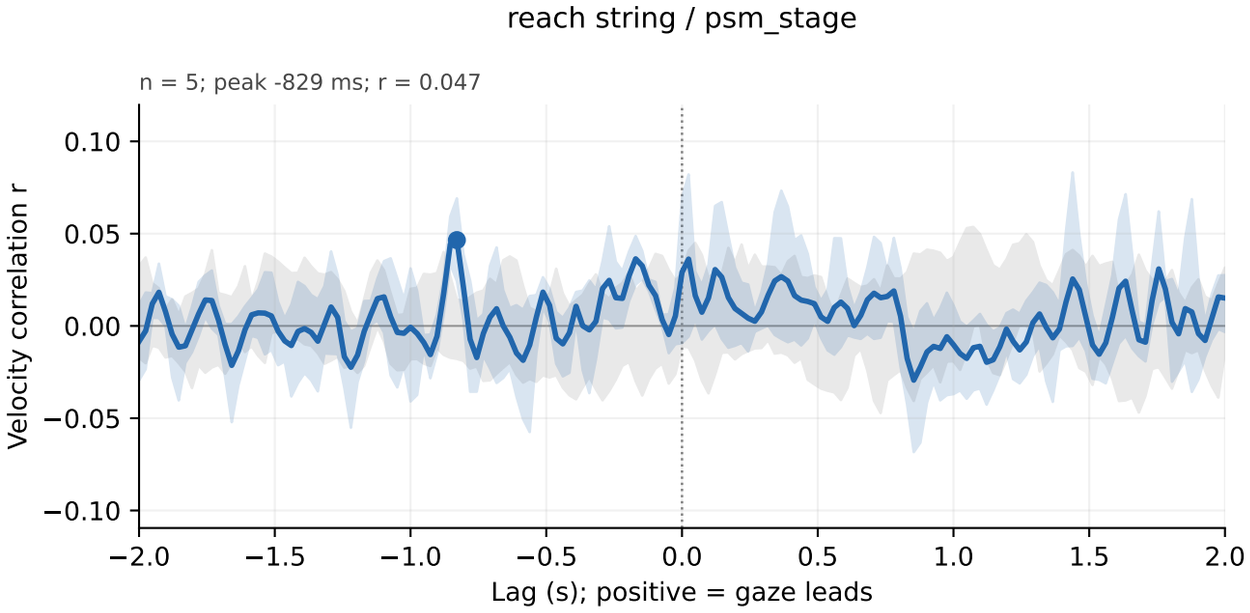
pull through and cinch / psm_stage



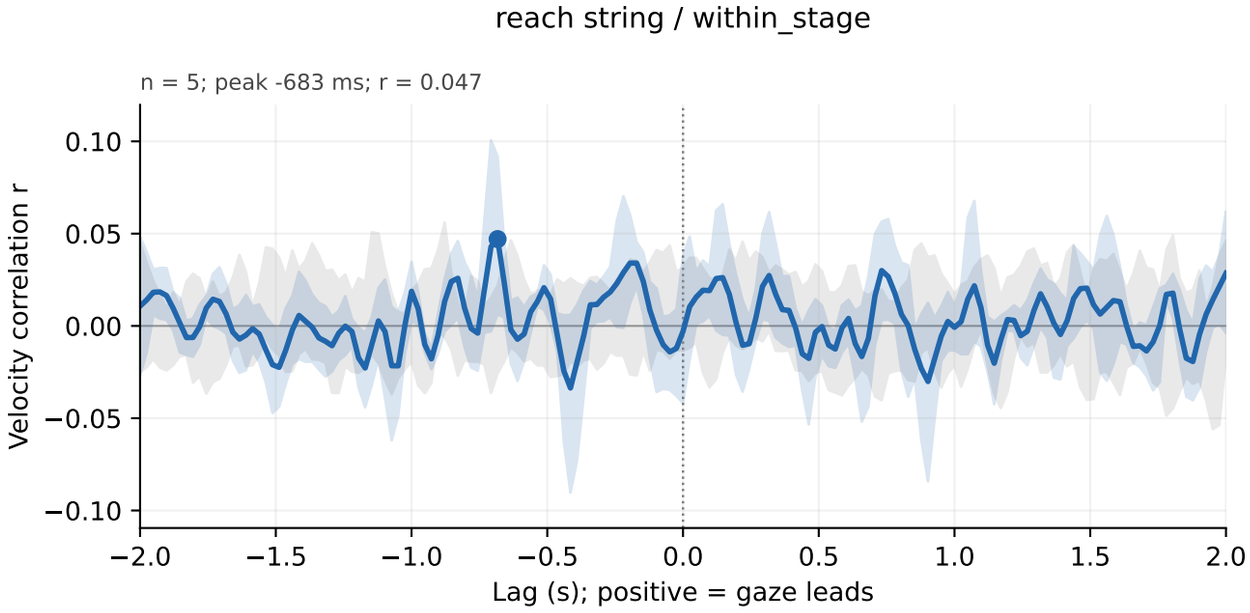
pull through and cinch / within_stage



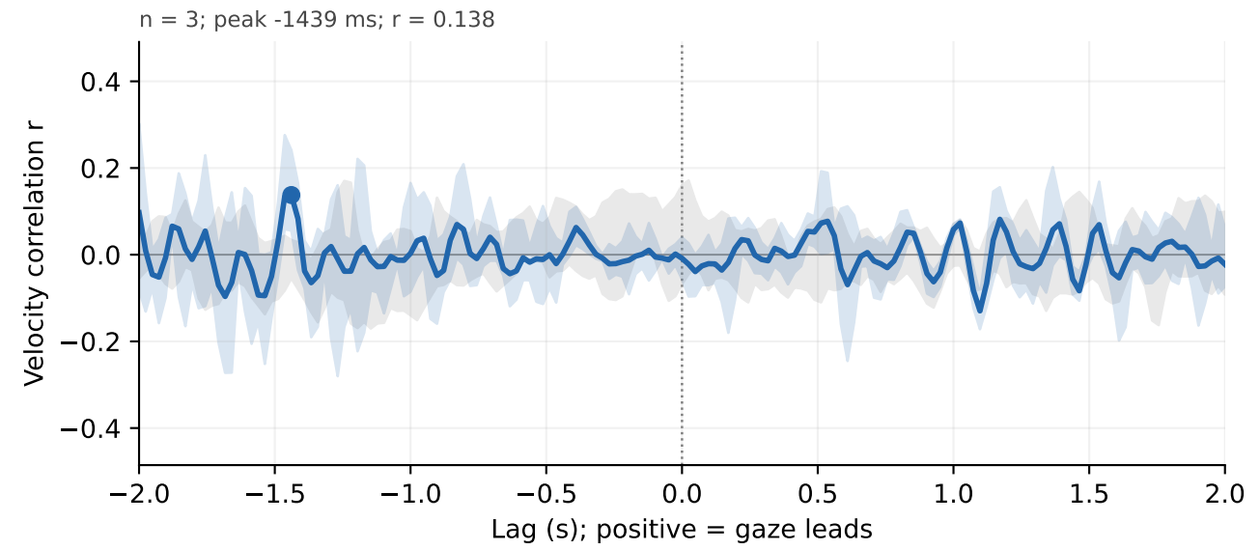
reach string / psm_stage



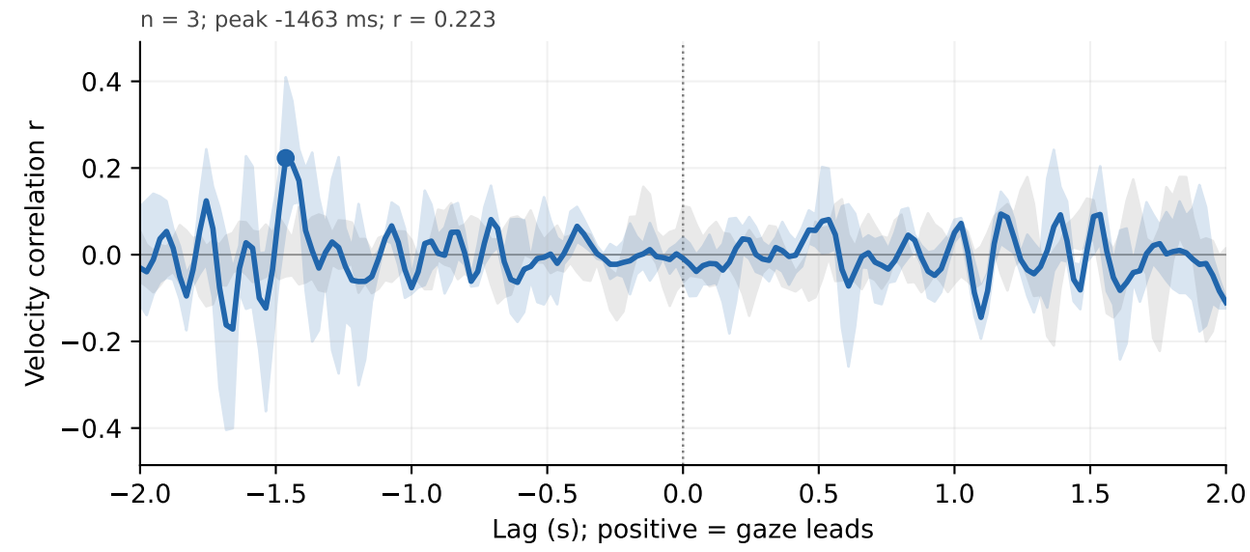
reach string / within_stage



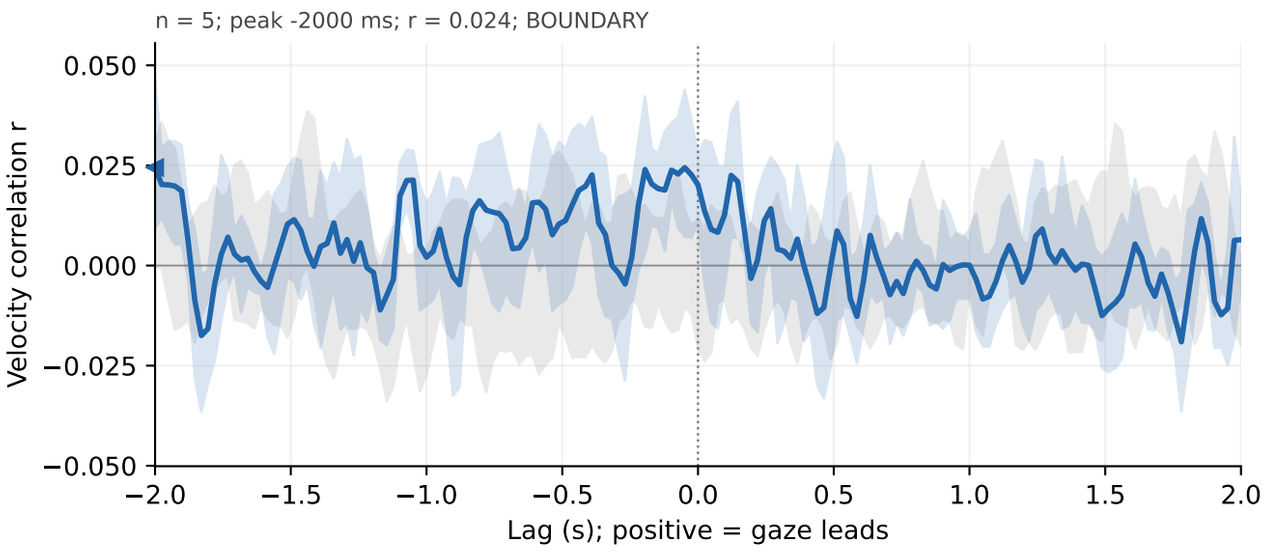
threading string through tower / psm_stage



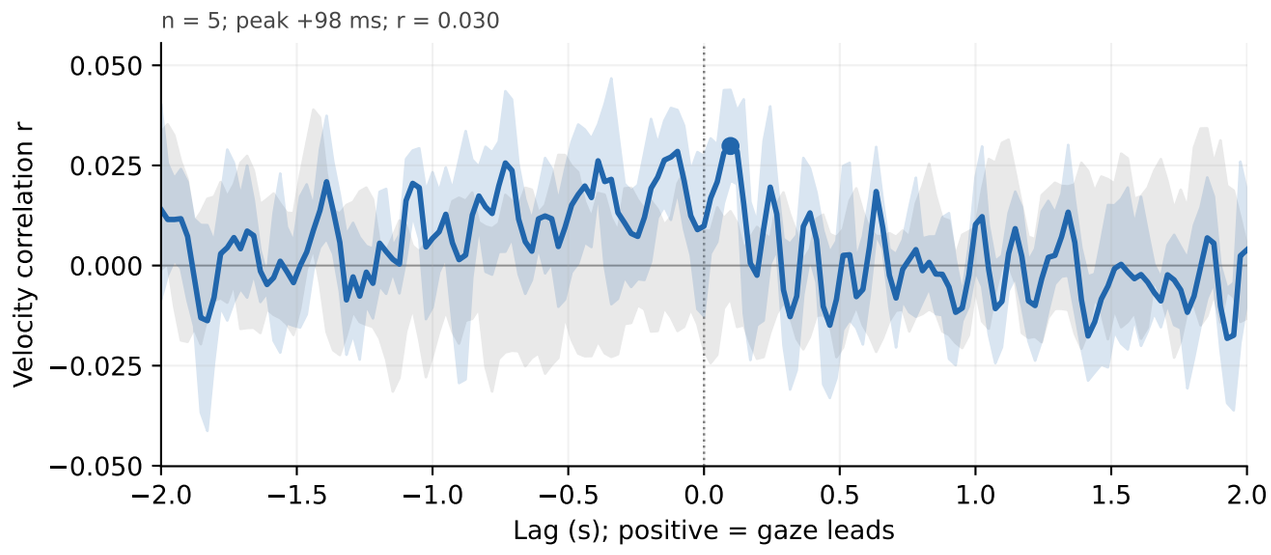
threading string through tower / within_stage



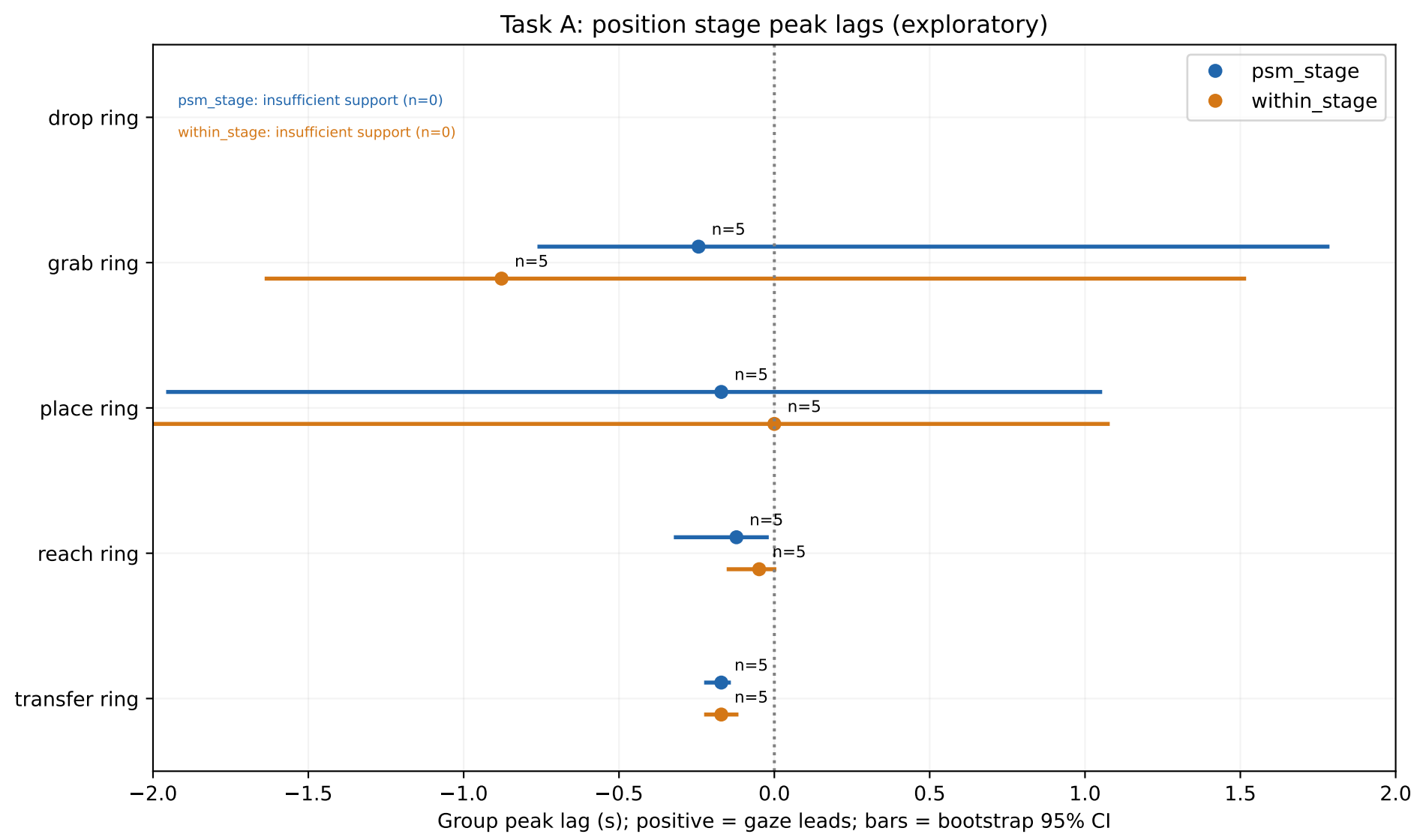
wrap around psm / psm_stage



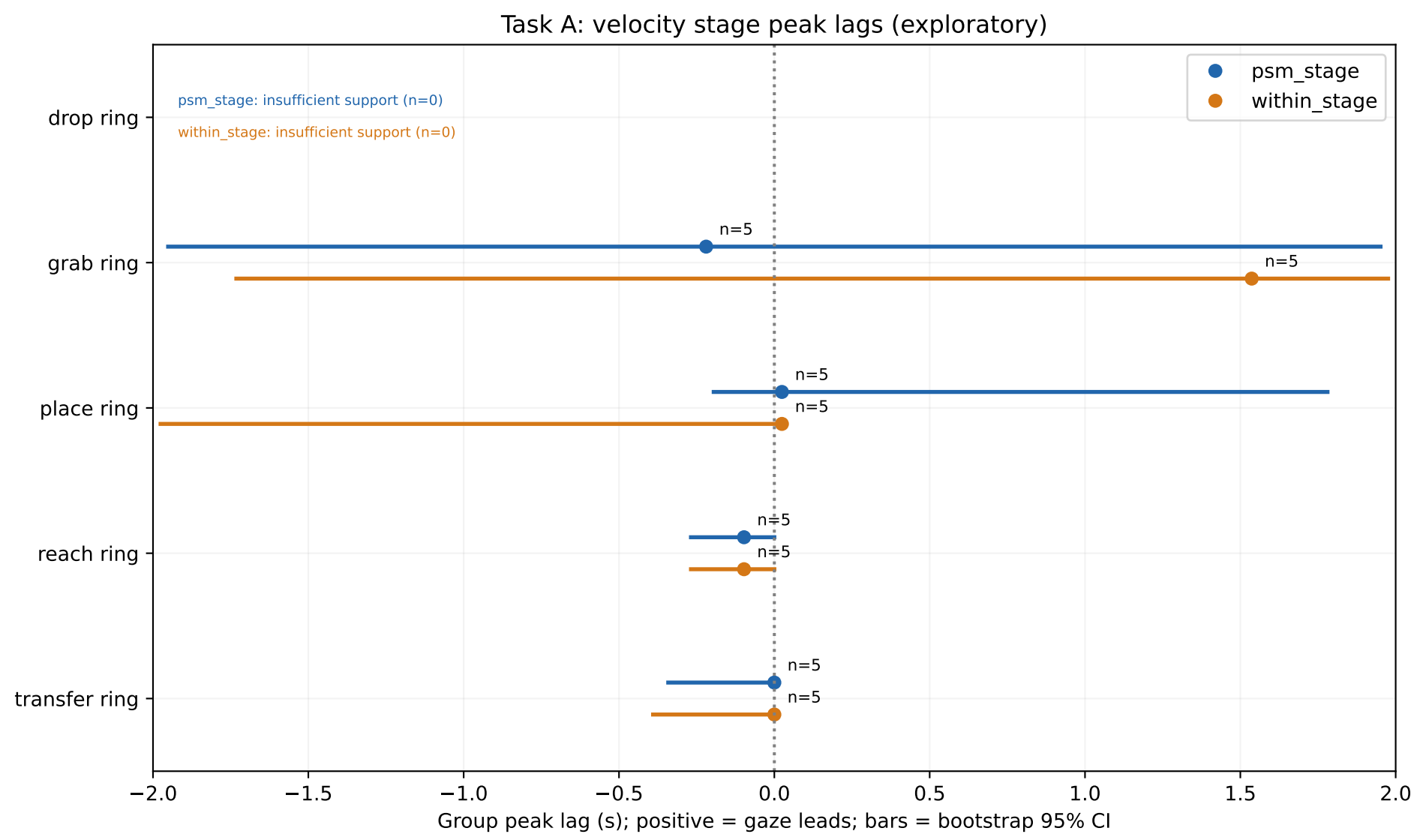
wrap around psm / within_stage



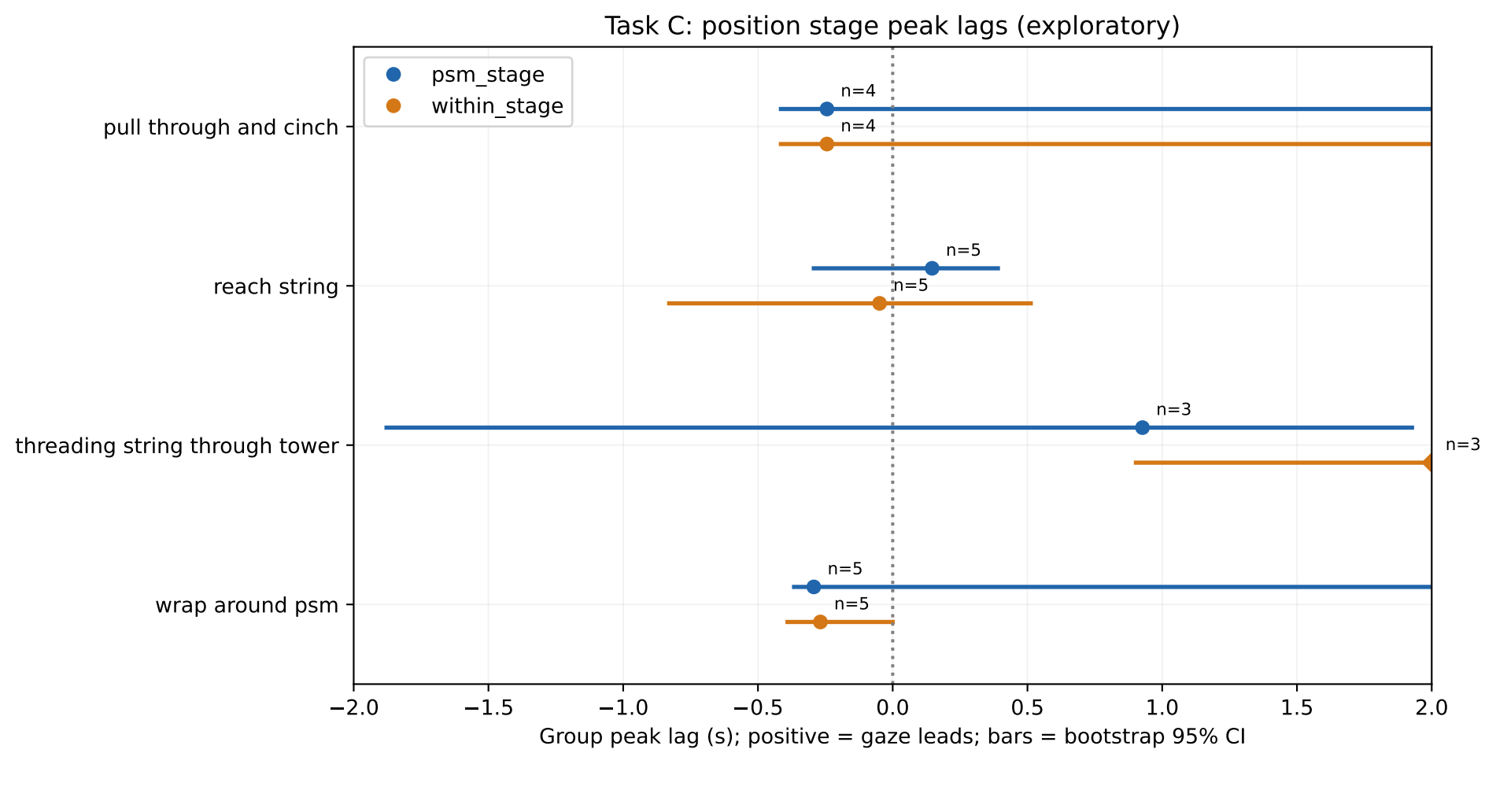
One PSM selected per recording/stage by largest signed xcorr peak; selection does not identify the operated instrument.



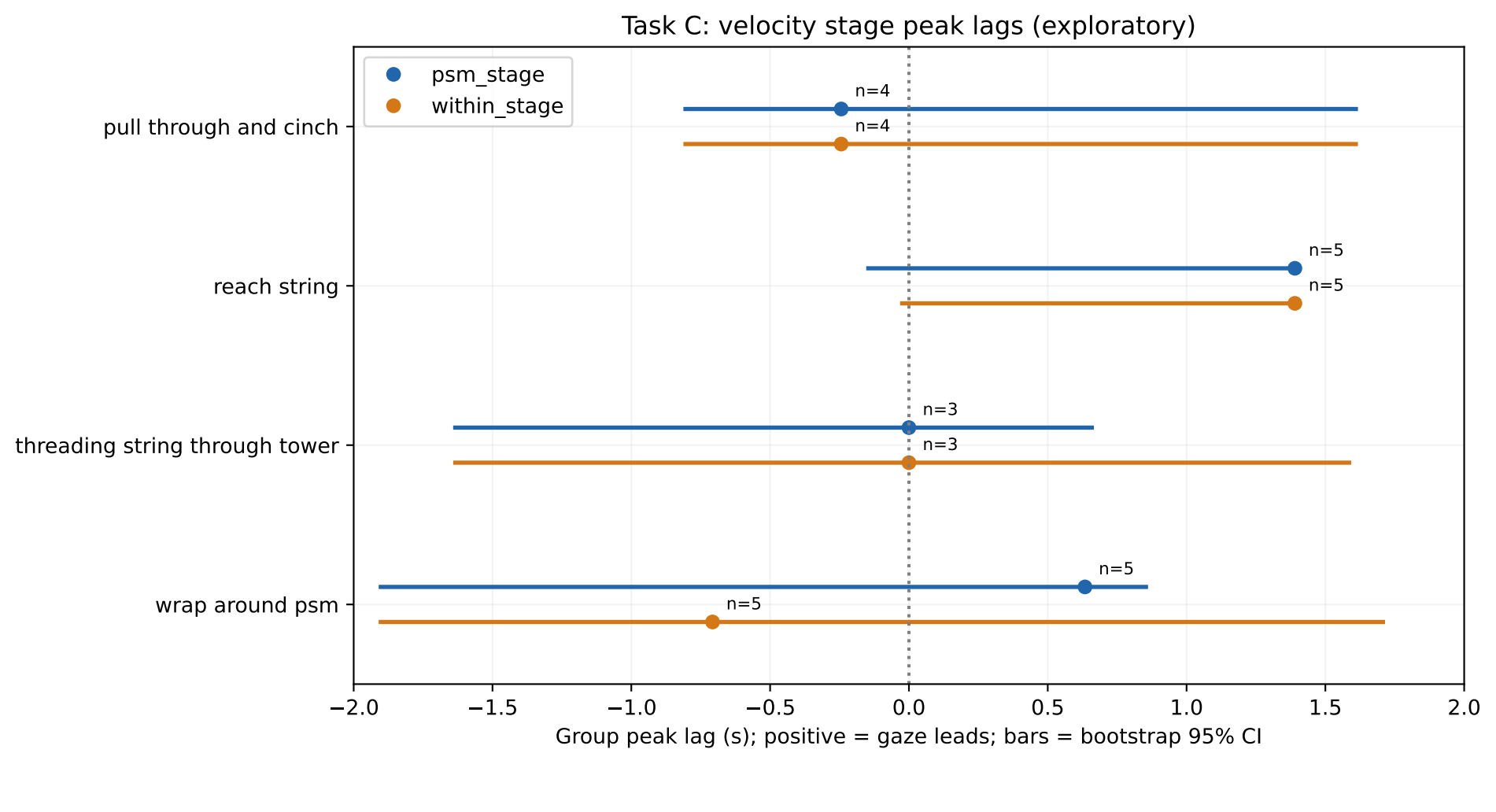
One PSM selected per recording/stage by largest signed xcorr peak; selection does not identify the operated instrument.



One PSM selected per recording/stage by largest signed xcorr peak; selection does not identify the operated instrument.



One PSM selected per recording/stage by largest signed xcorr peak; selection does not identify the operated instrument.



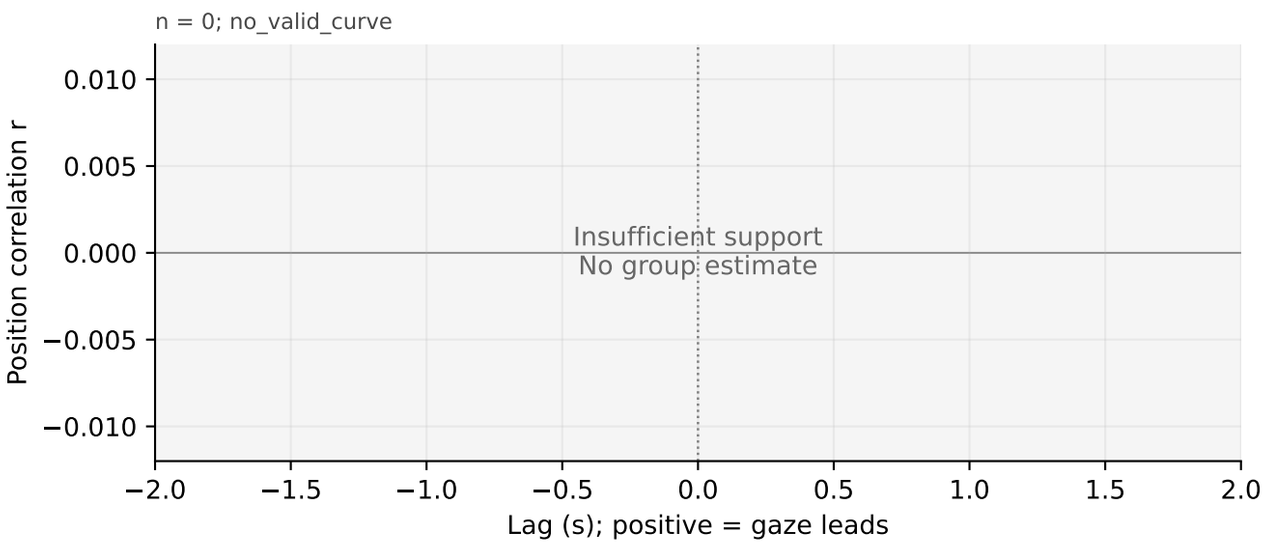
One PSM selected per recording/stage by largest signed xcorr peak; selection does not identify the operated instrument.

Task A: position, maximum-xcorr PSM per stage

Unpaired reference (95%) · Participant bootstrap (95%) · Group mean

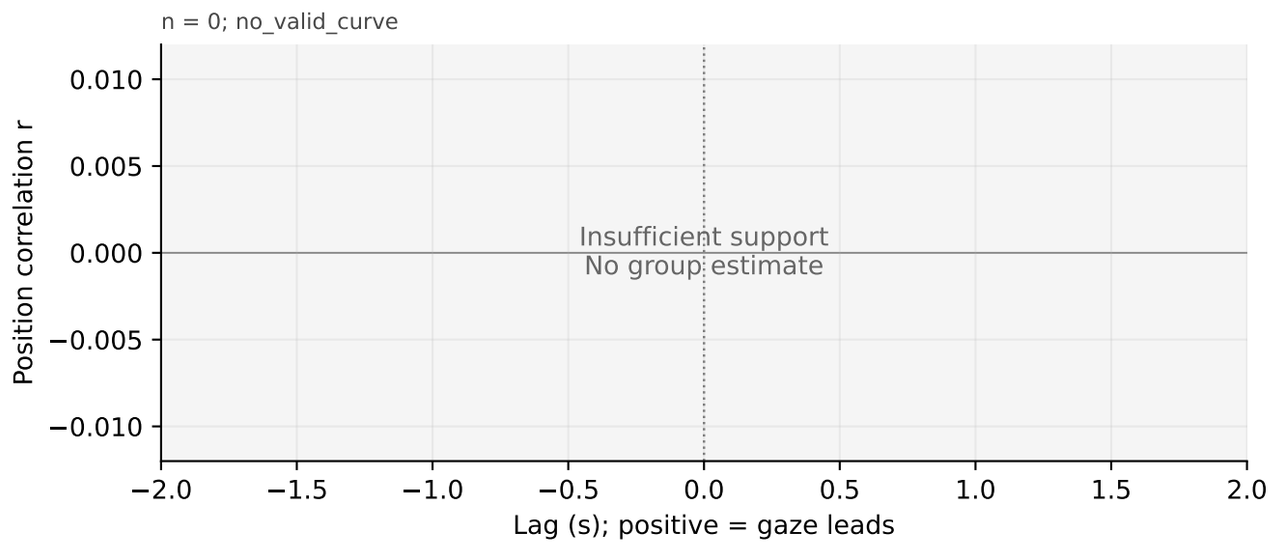
drop ring / psm_stage

n = 0; no_valid_curve



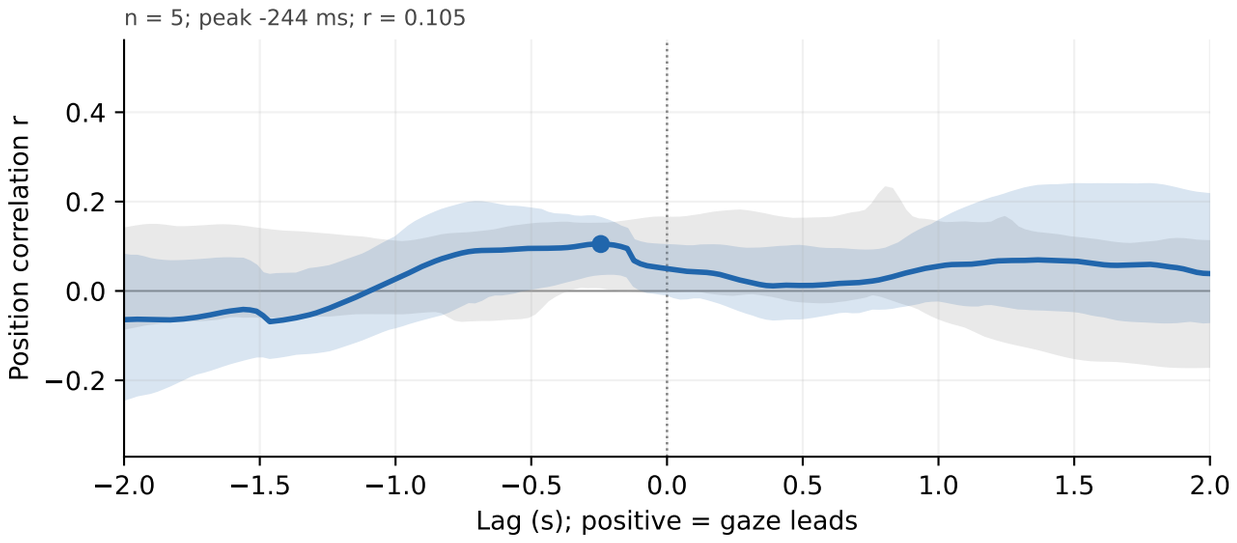
drop ring / within_stage

n = 0; no_valid_curve



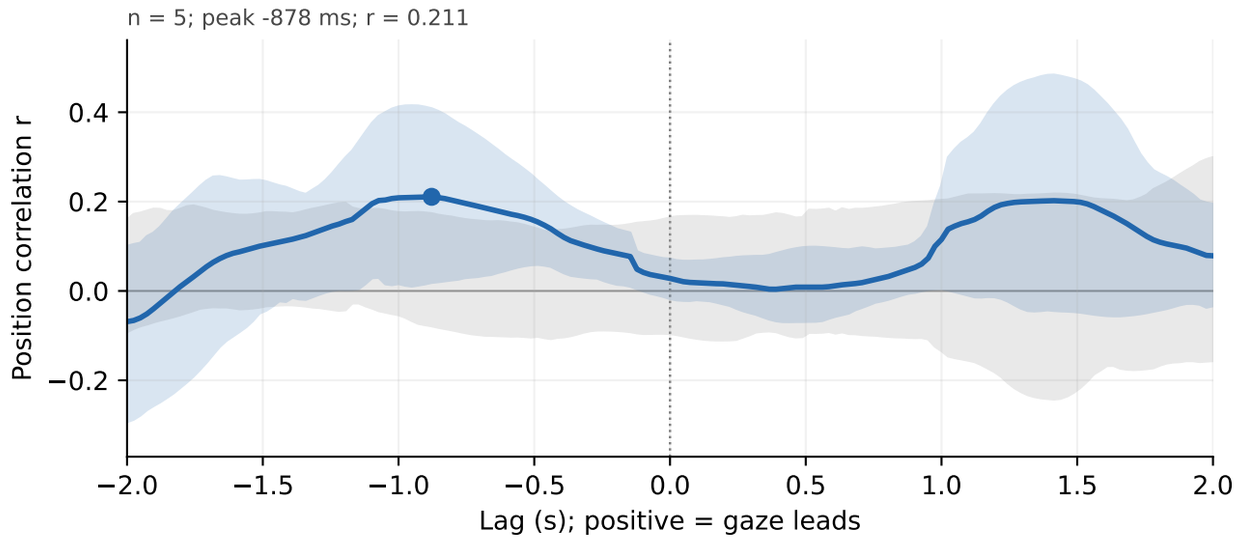
grab ring / psm_stage

n = 5; peak -244 ms; r = 0.105



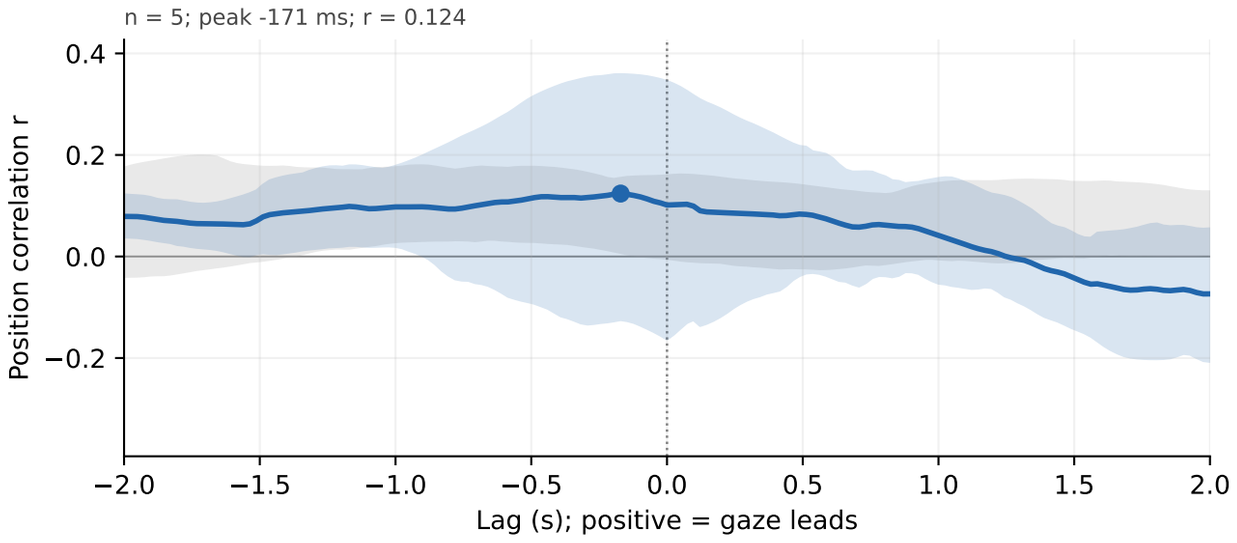
grab ring / within_stage

n = 5; peak -878 ms; r = 0.211



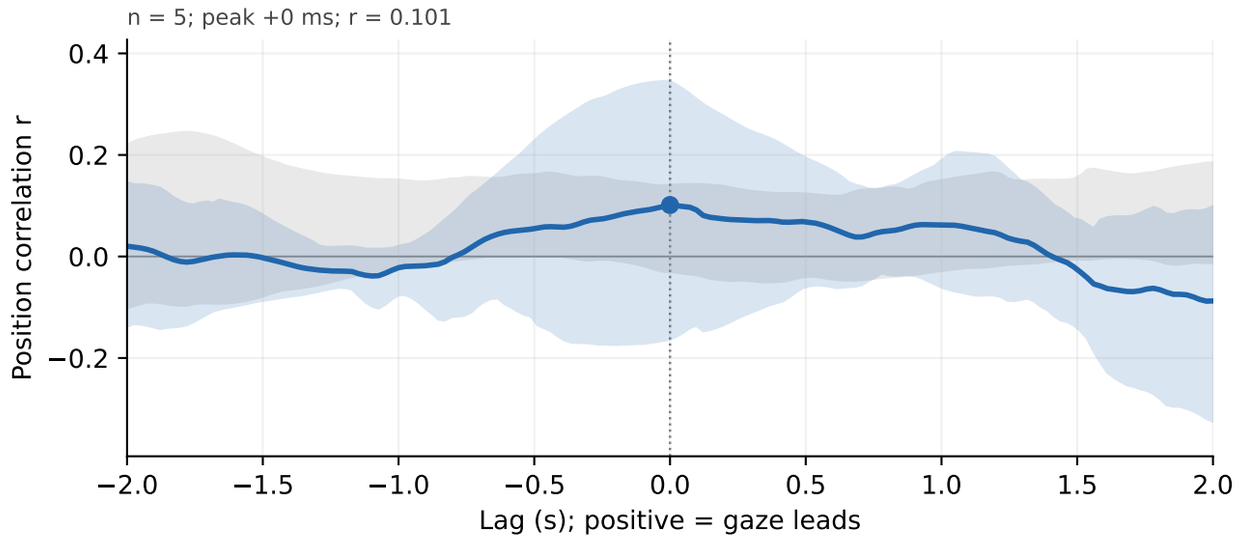
place ring / psm_stage

n = 5; peak -171 ms; r = 0.124



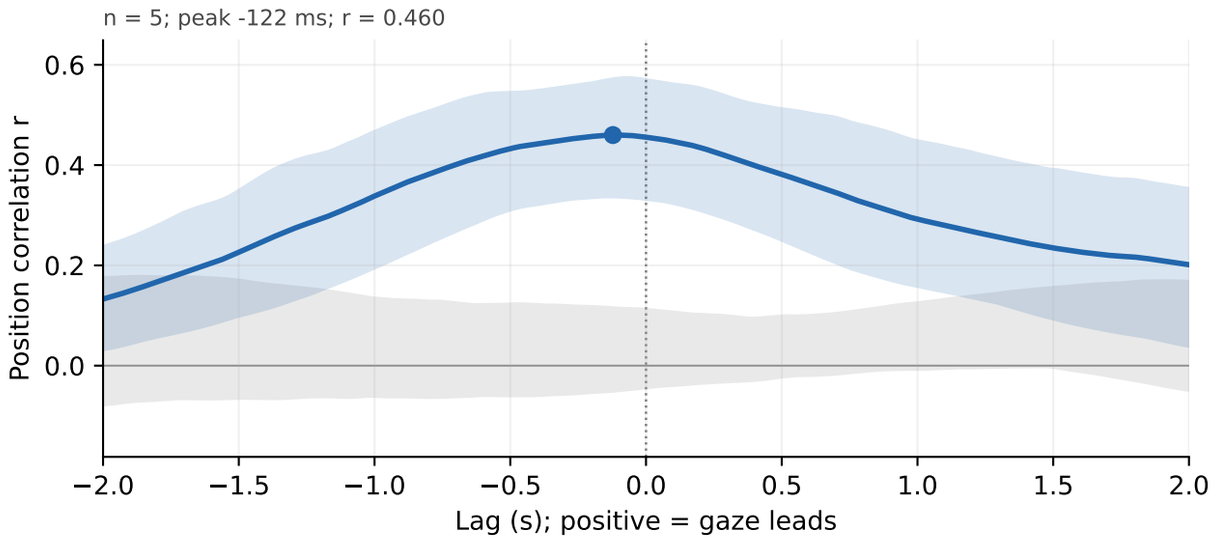
place ring / within_stage

n = 5; peak +0 ms; r = 0.101



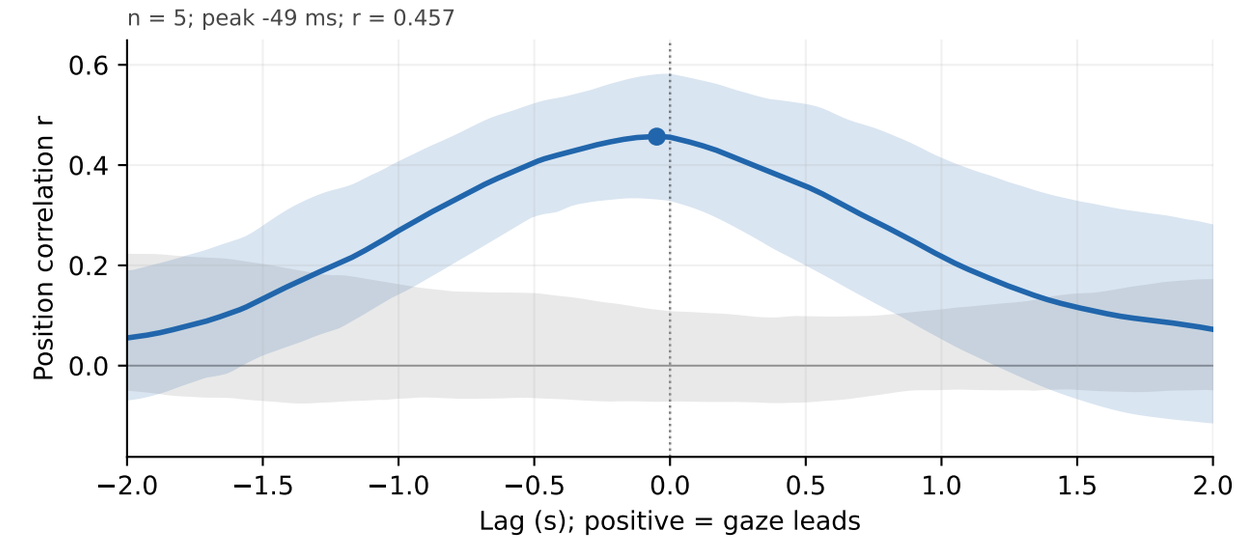
reach ring / psm_stage

n = 5; peak -122 ms; r = 0.460



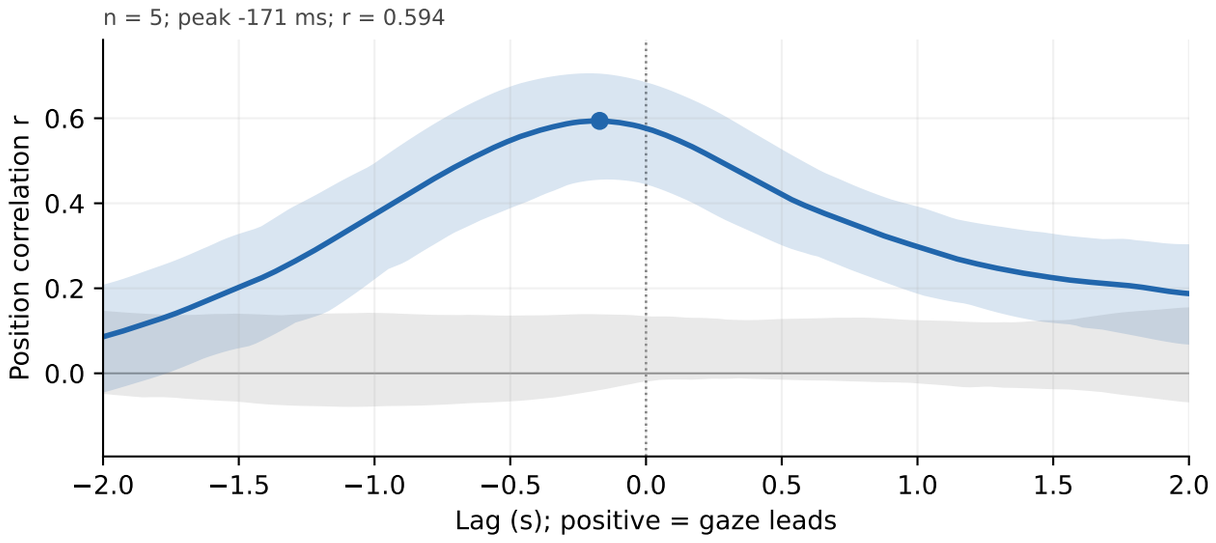
reach ring / within_stage

n = 5; peak -49 ms; r = 0.457



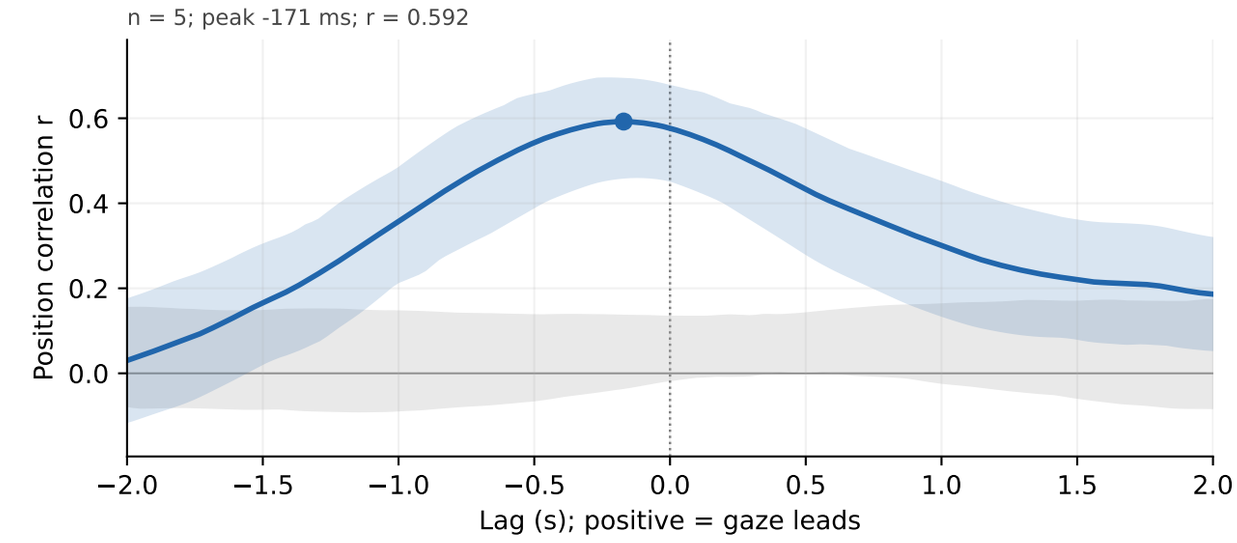
transfer ring / psm_stage

n = 5; peak -171 ms; r = 0.594



transfer ring / within_stage

n = 5; peak -171 ms; r = 0.592



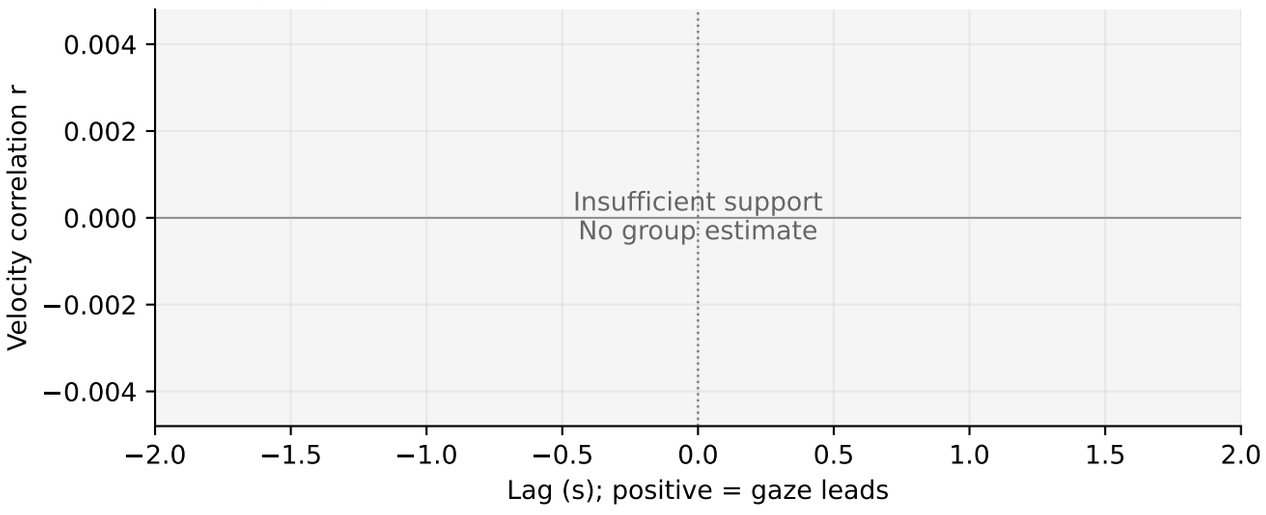
One PSM selected per recording/stage by largest signed xcorr peak; selection does not identify the operated instrument.

Task A: velocity, maximum-xcorr PSM per stage

Unpaired reference (95%) | Participant bootstrap (95%) | Group mean

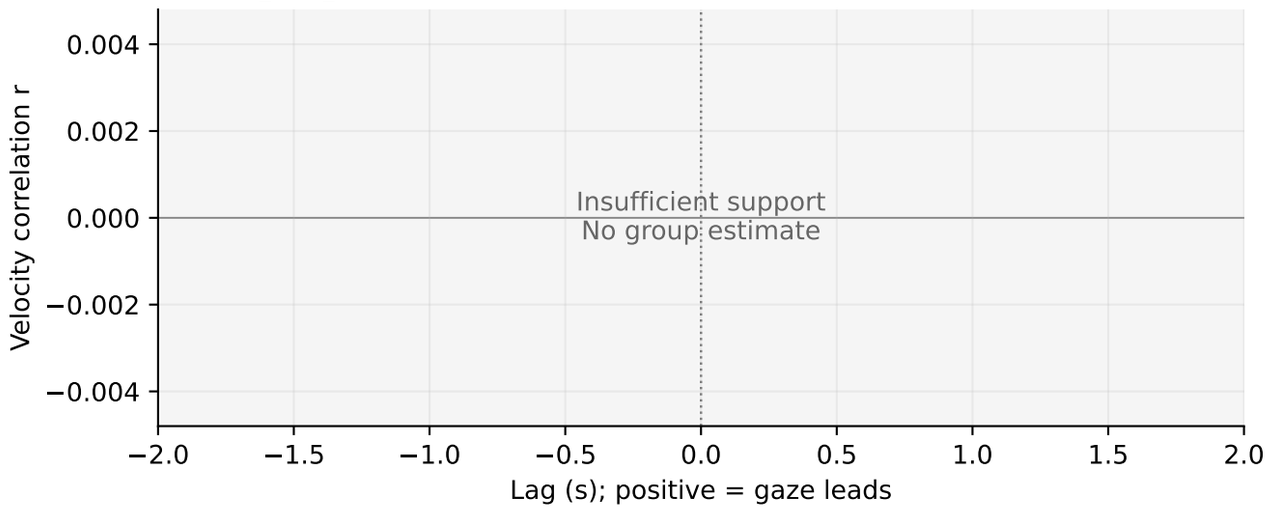
drop ring / psm_stage

n = 0; no_valid_curve



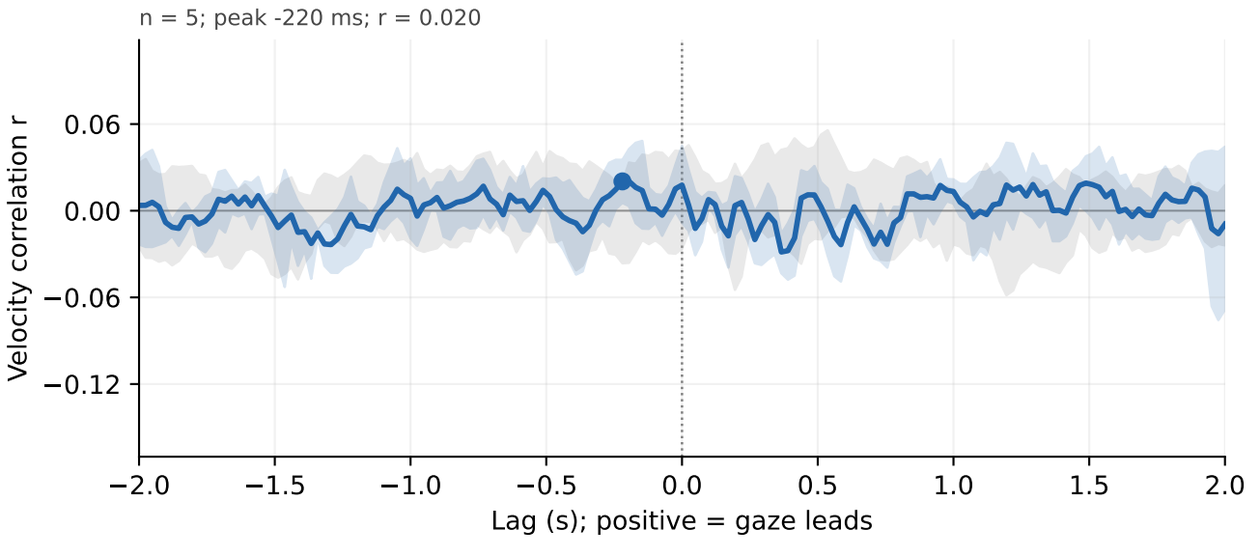
drop ring / within_stage

n = 0; no_valid_curve



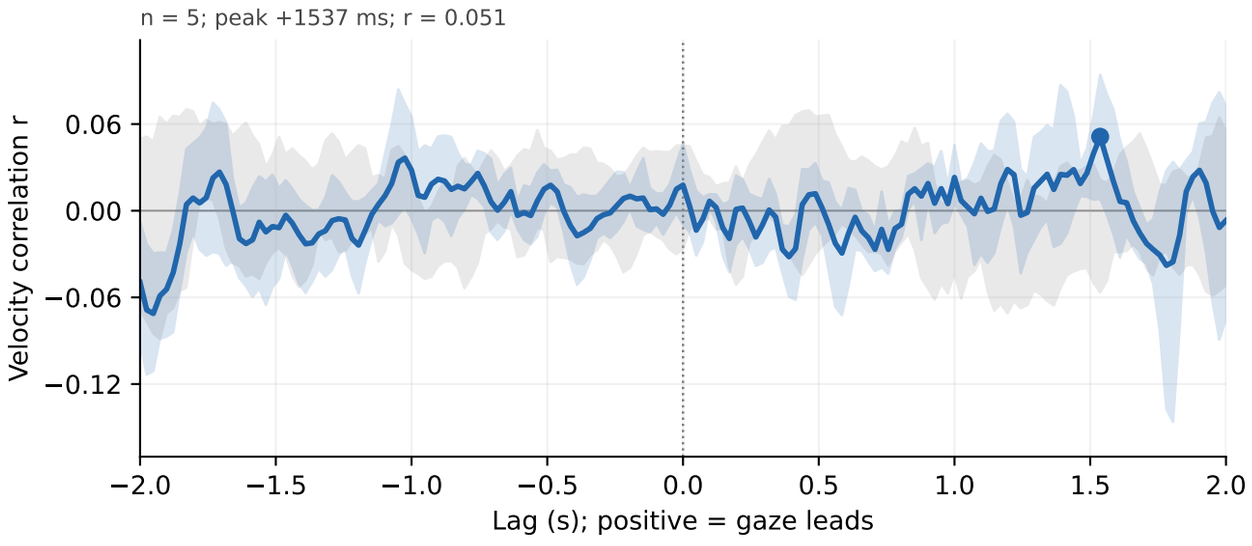
grab ring / psm_stage

n = 5; peak -220 ms; r = 0.020



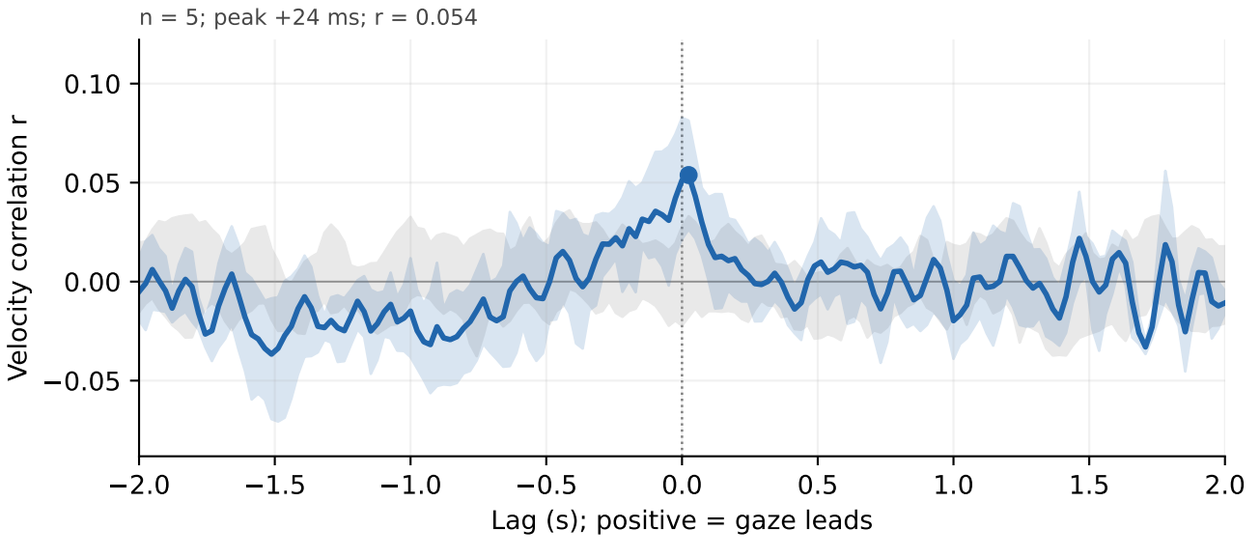
grab ring / within_stage

n = 5; peak +1537 ms; r = 0.051



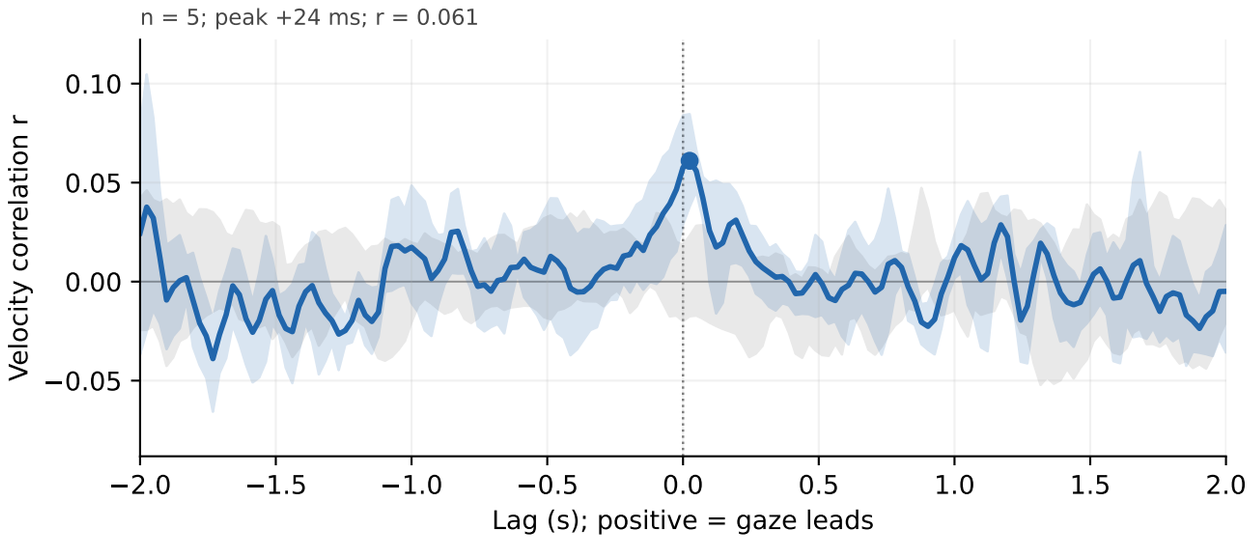
place ring / psm_stage

n = 5; peak +24 ms; r = 0.054



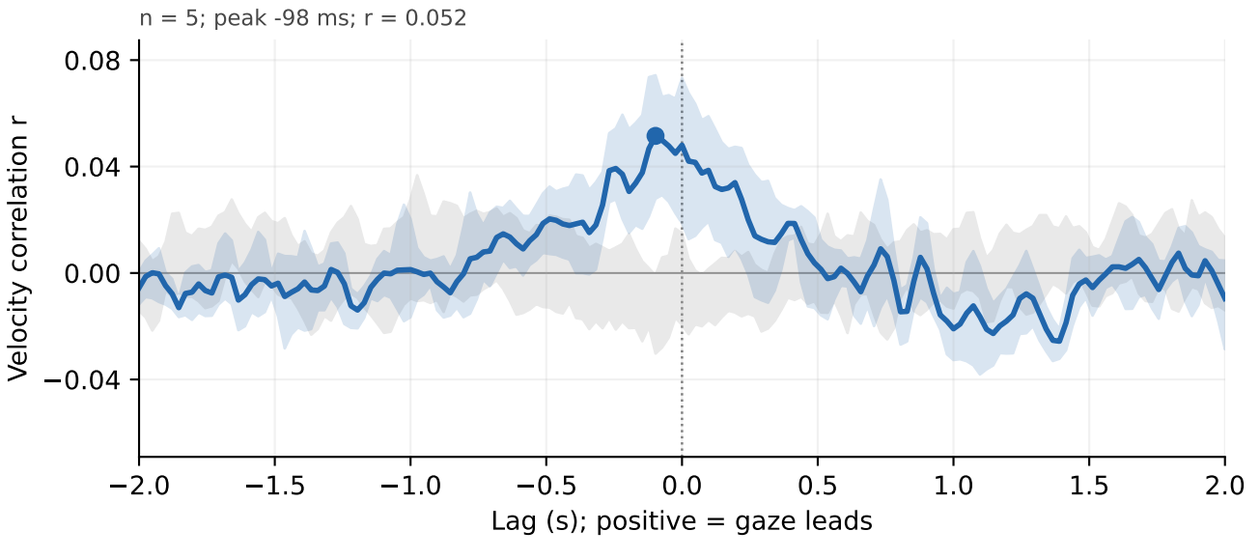
place ring / within_stage

n = 5; peak +24 ms; r = 0.061



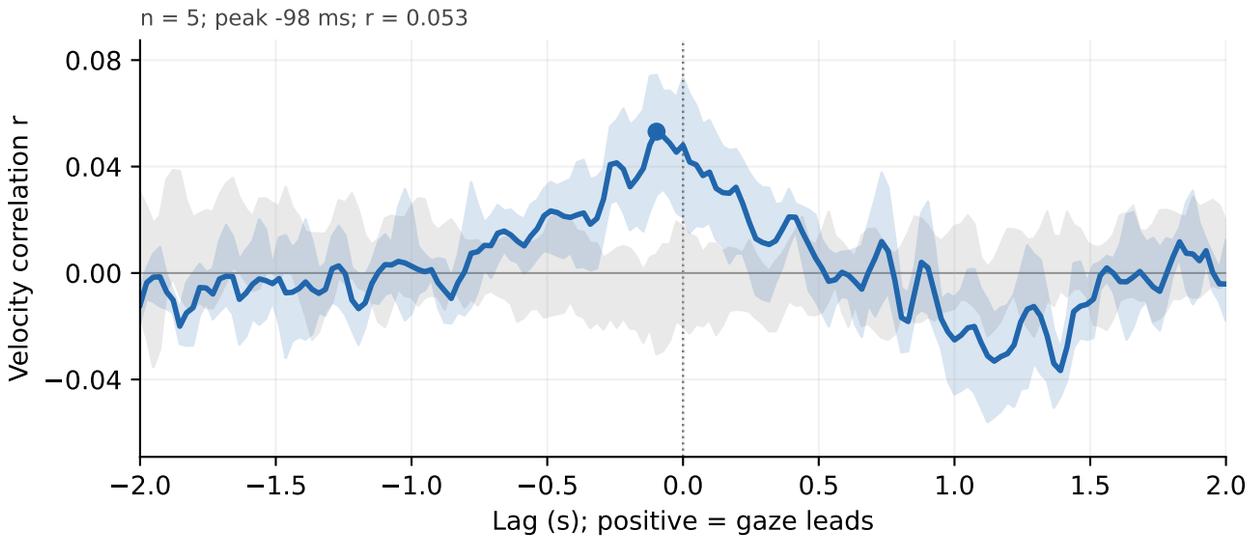
reach ring / psm_stage

n = 5; peak -98 ms; r = 0.052



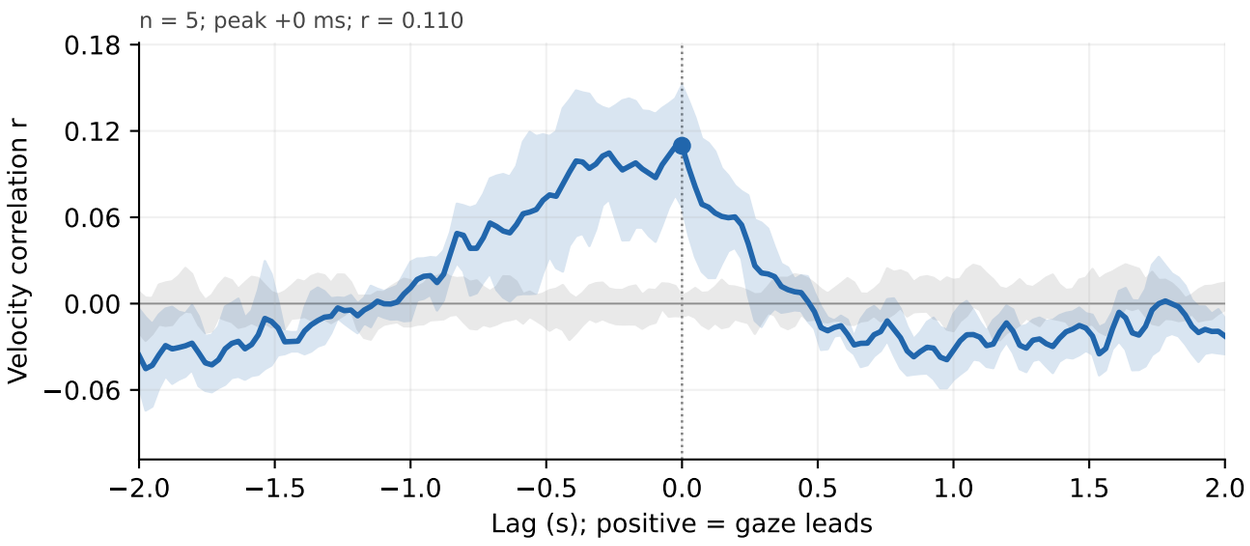
reach ring / within_stage

n = 5; peak -98 ms; r = 0.053



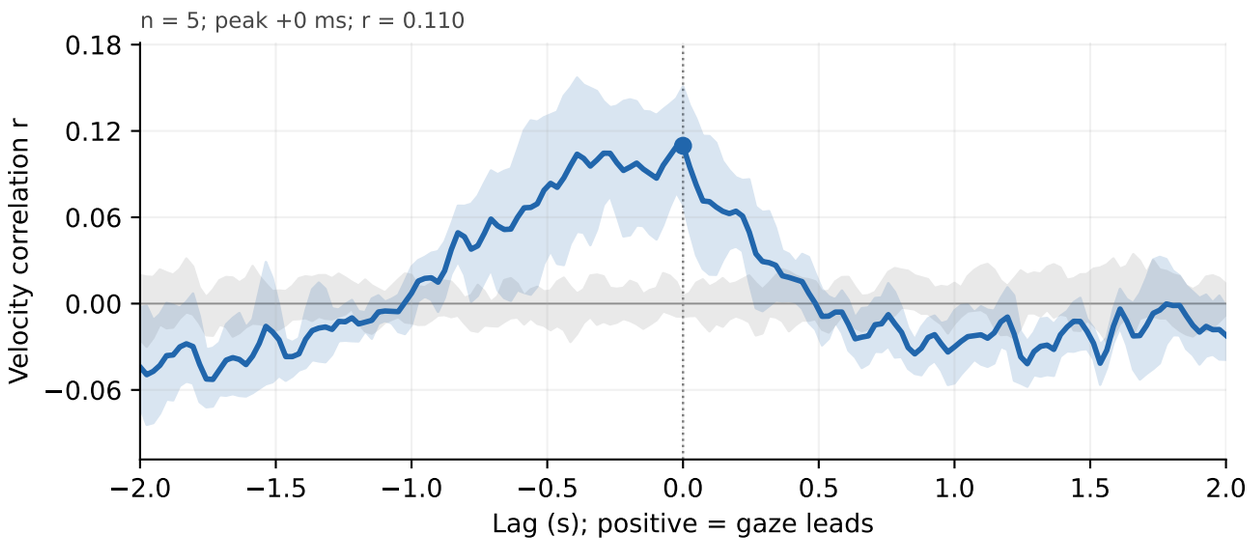
transfer ring / psm_stage

n = 5; peak +0 ms; r = 0.110



transfer ring / within_stage

n = 5; peak +0 ms; r = 0.110

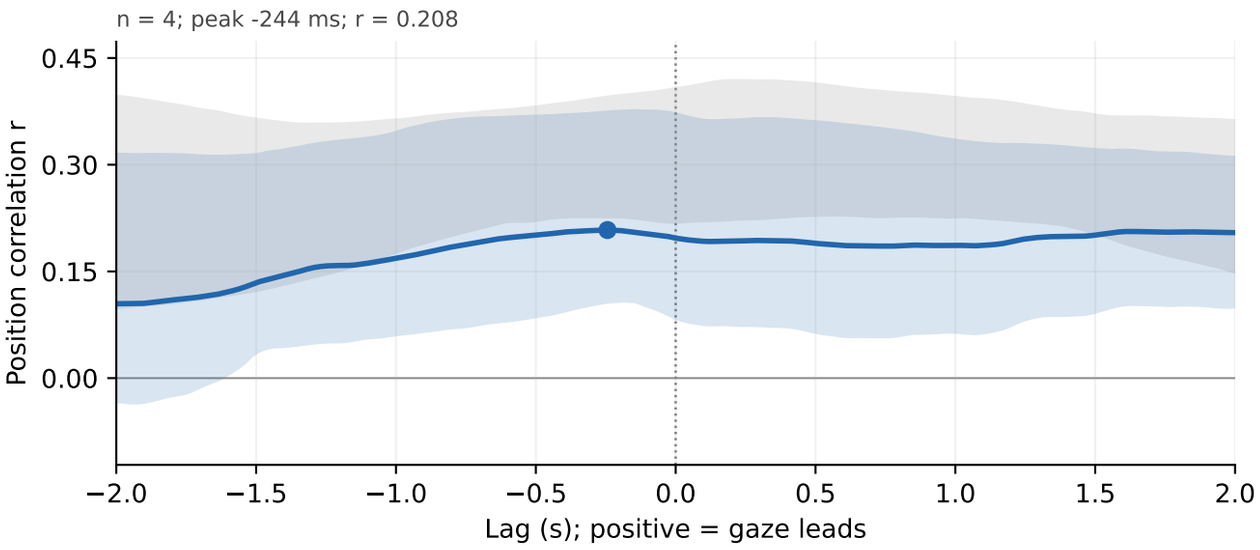


One PSM selected per recording/stage by largest signed xcorr peak; selection does not identify the operated instrument.

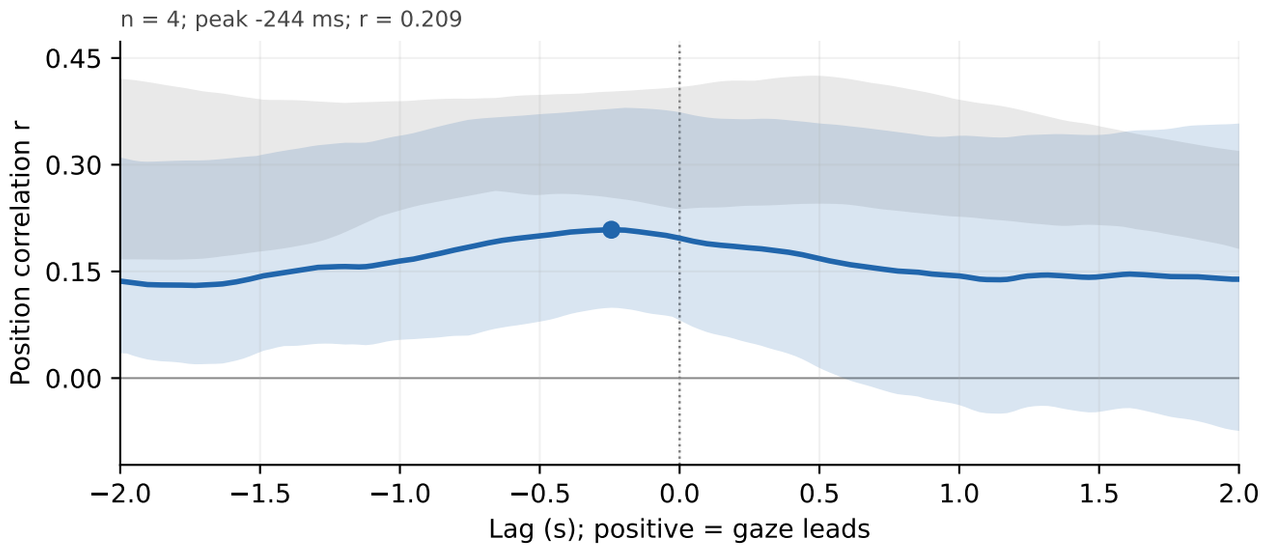
Task C: position, maximum-xcorr PSM per stage

Unpaired reference (95%) | Participant bootstrap (95%) | Group mean

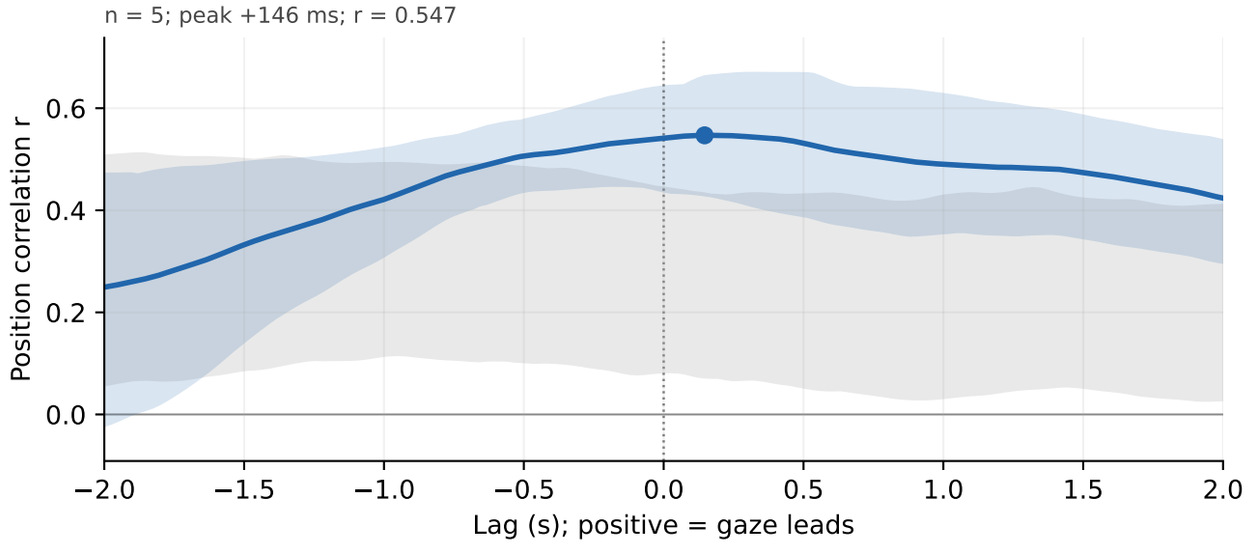
pull through and cinch / psm_stage



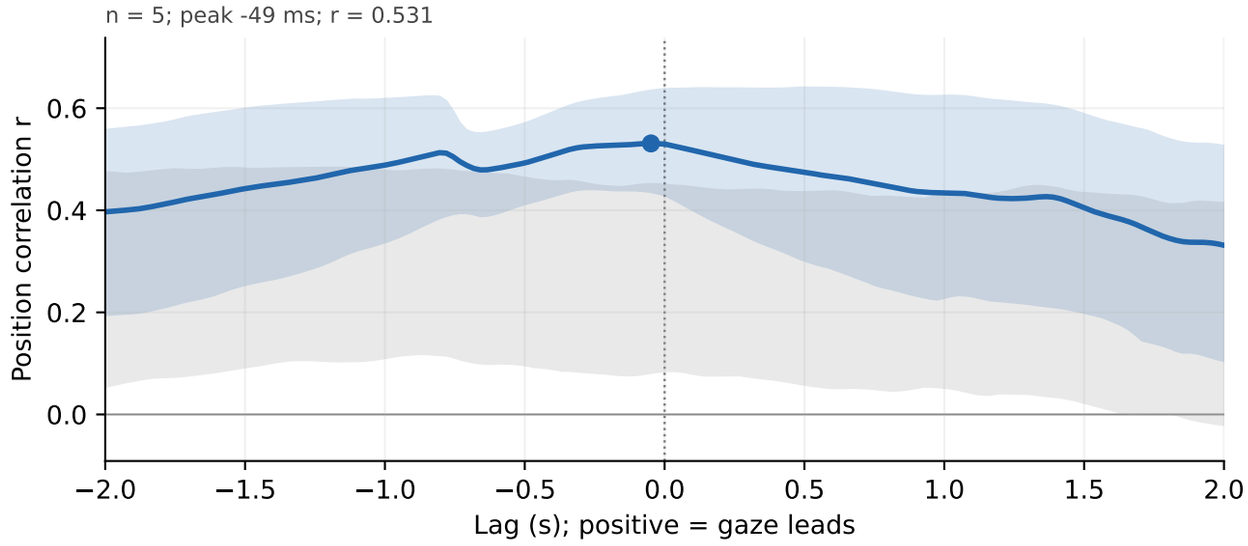
pull through and cinch / within_stage



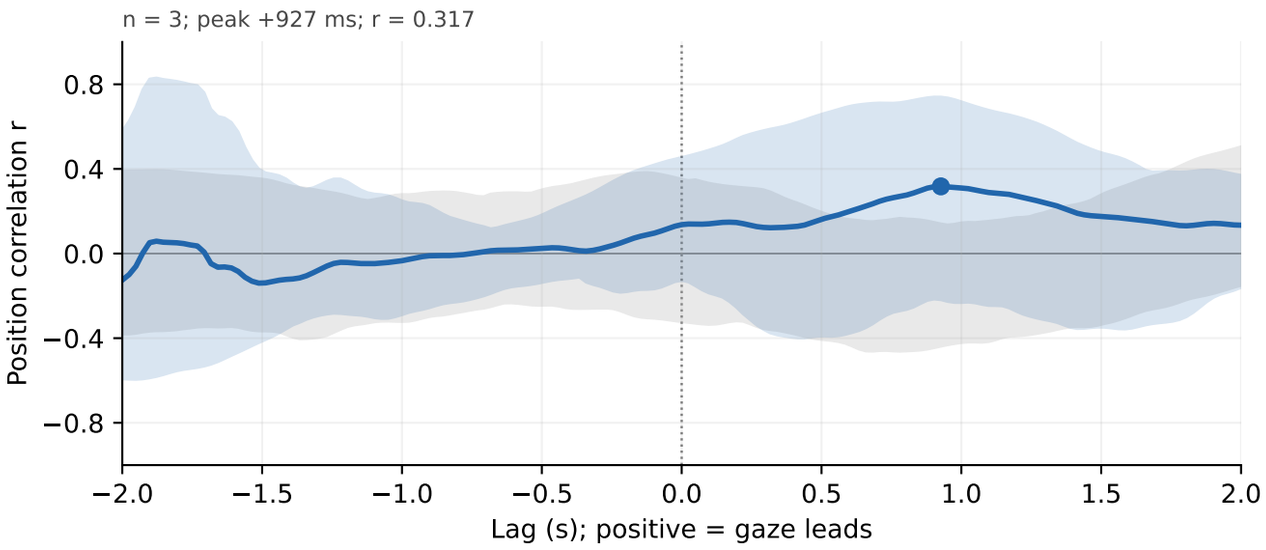
reach string / psm_stage



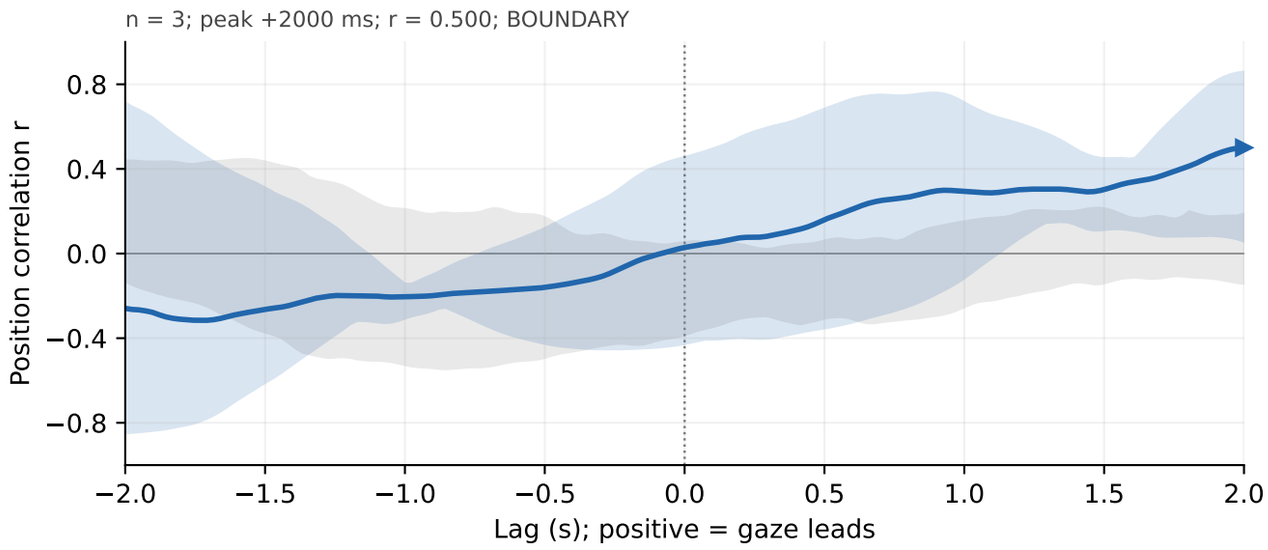
reach string / within_stage



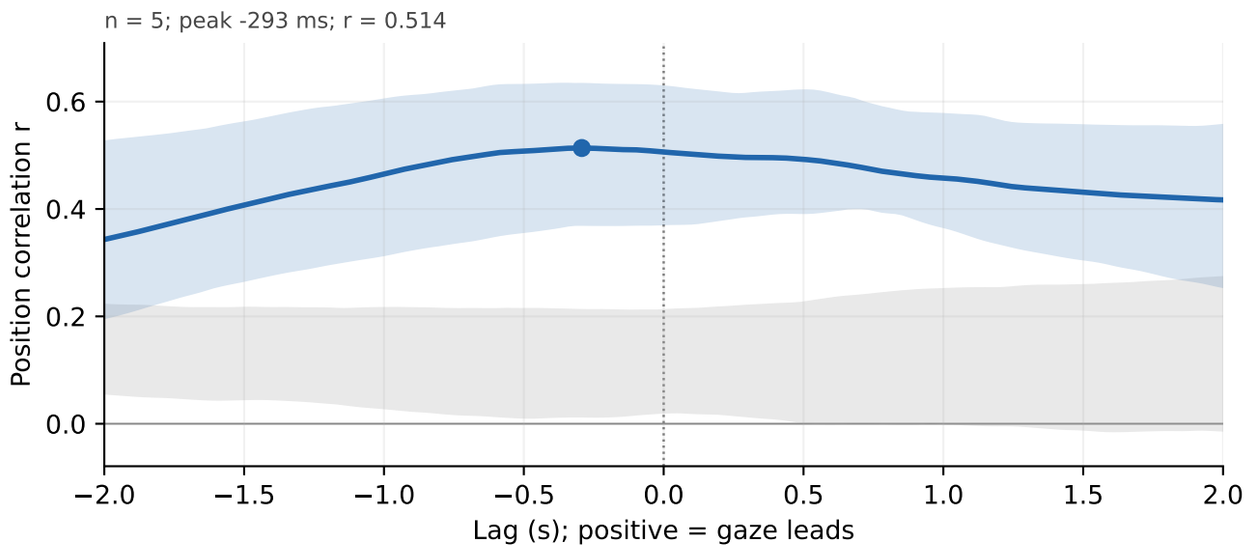
threading string through tower / psm_stage



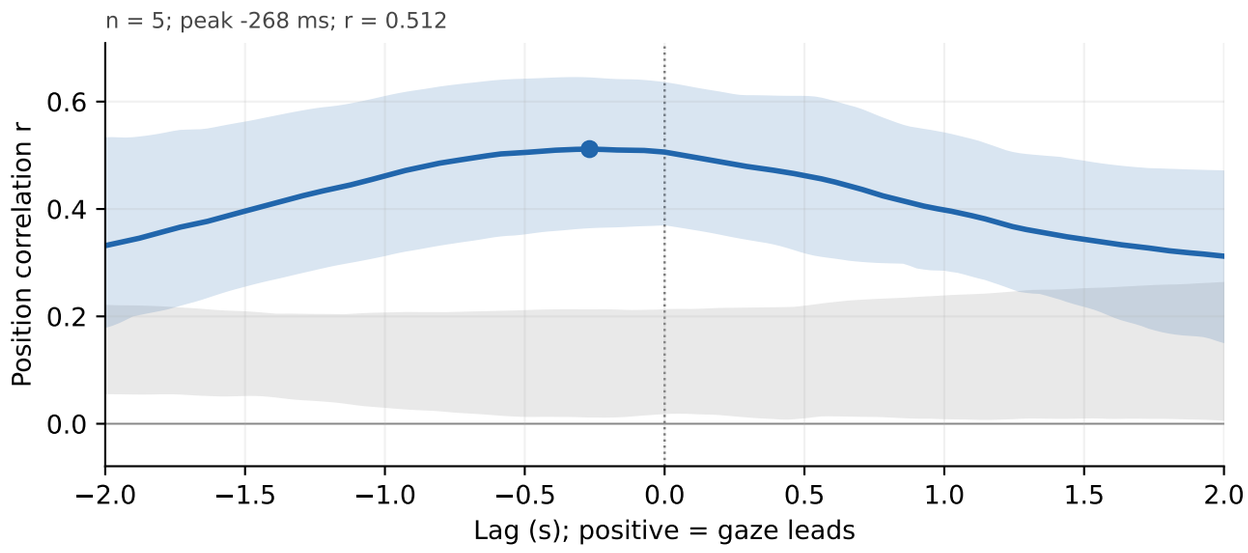
threading string through tower / within_stage



wrap around psm / psm_stage



wrap around psm / within_stage

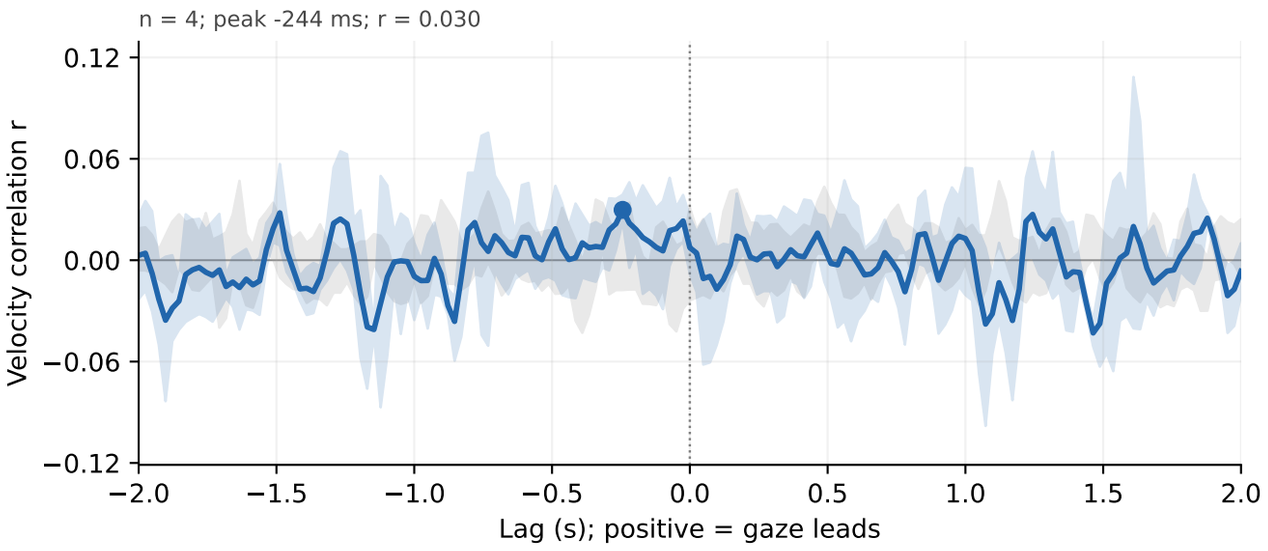


One PSM selected per recording/stage by largest signed xcorr peak; selection does not identify the operated instrument.

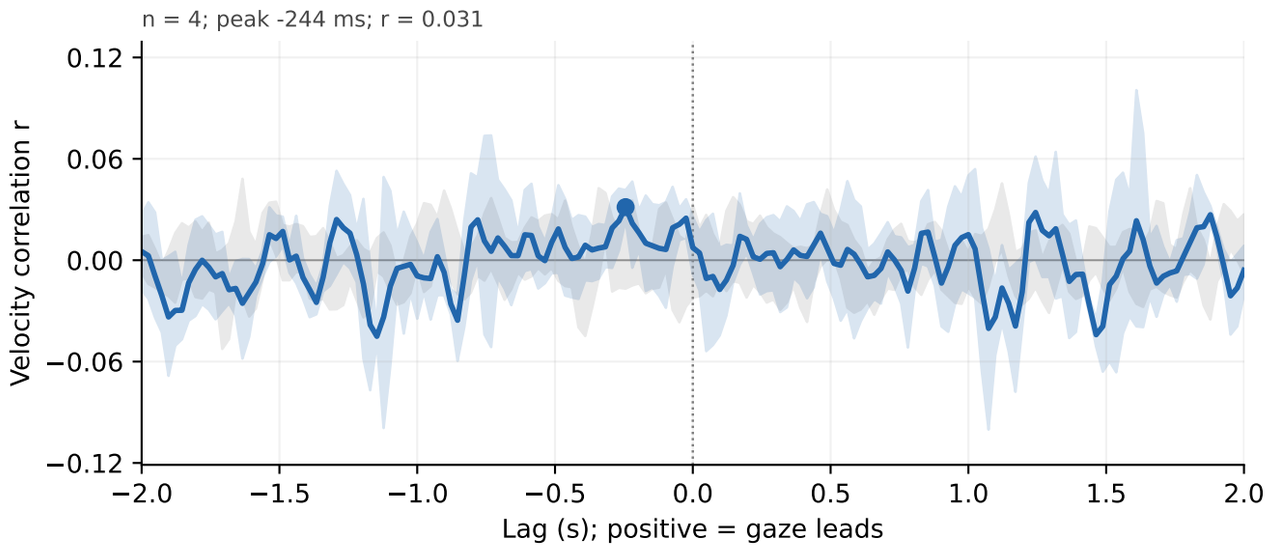
Task C: velocity, maximum-xcorr PSM per stage

Unpaired reference (95%) | Participant bootstrap (95%) | Group mean

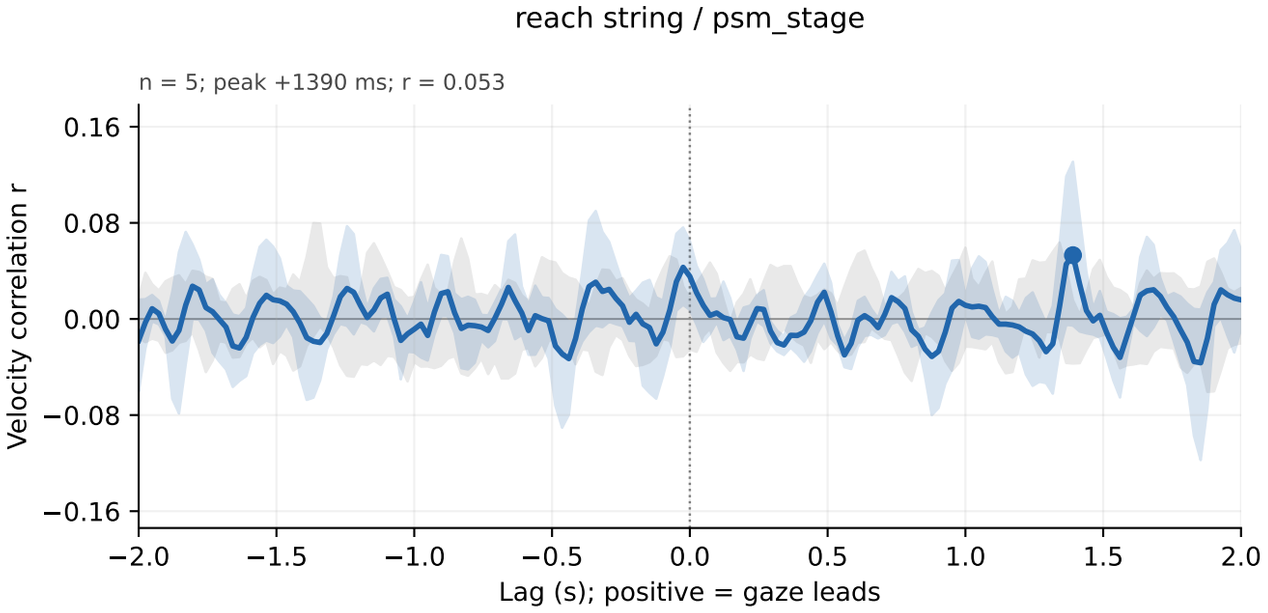
pull through and cinch / psm_stage



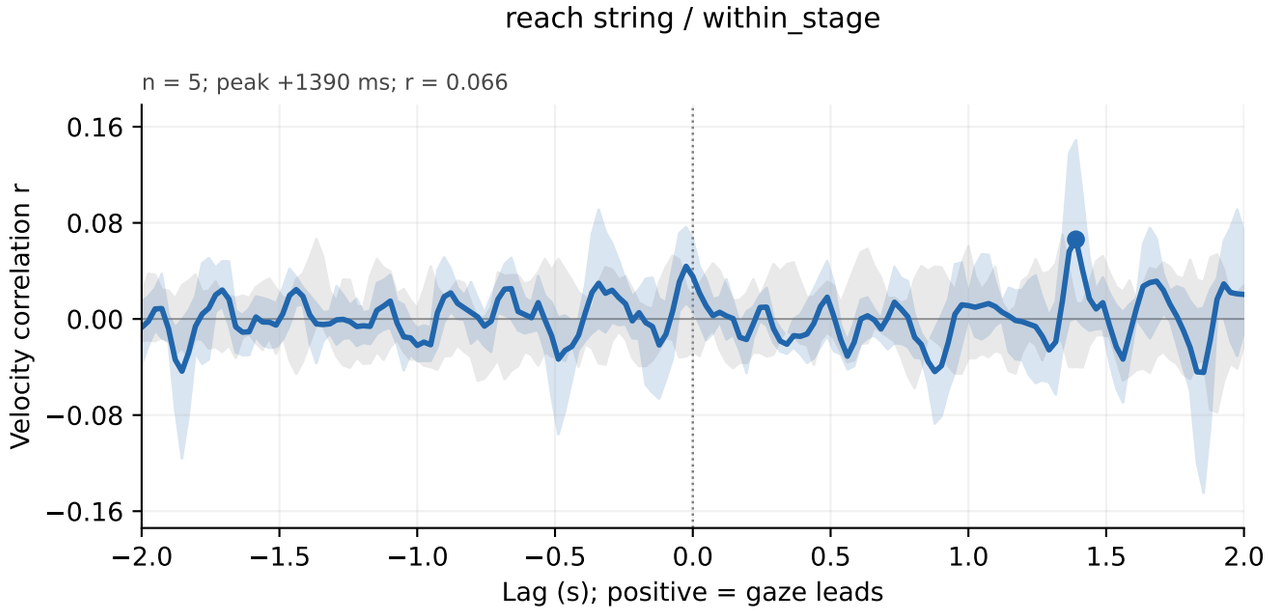
pull through and cinch / within_stage



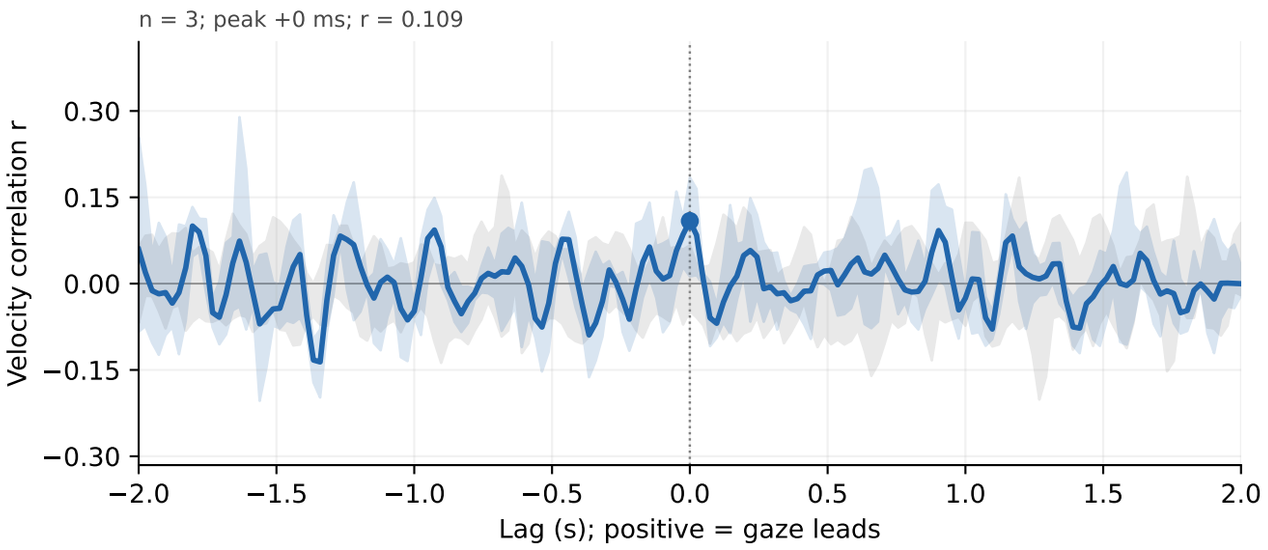
reach string / psm_stage



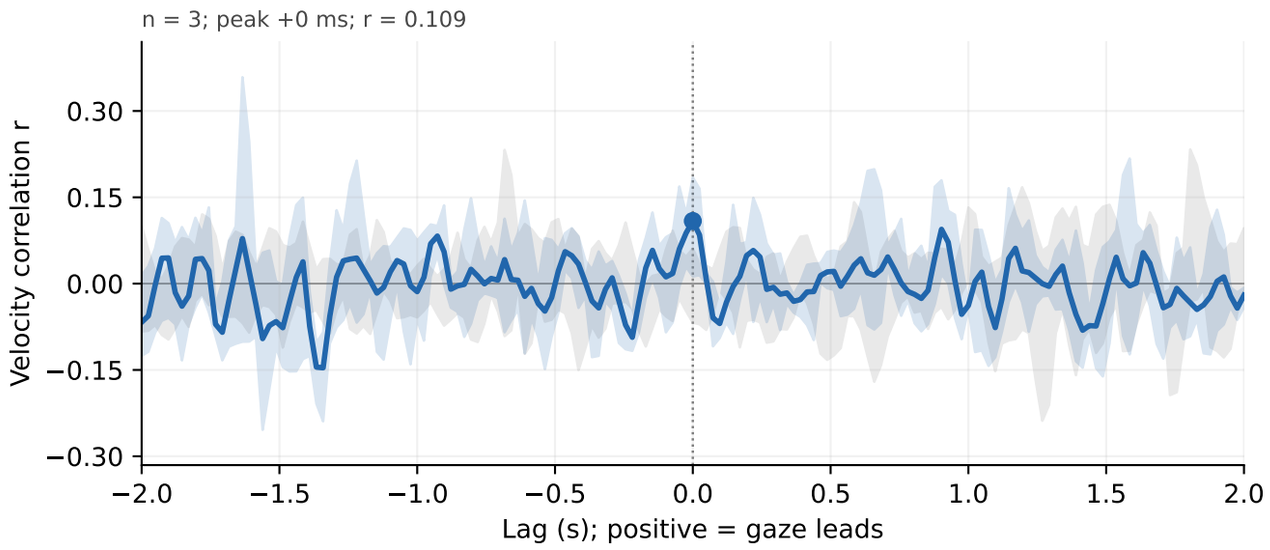
reach string / within_stage



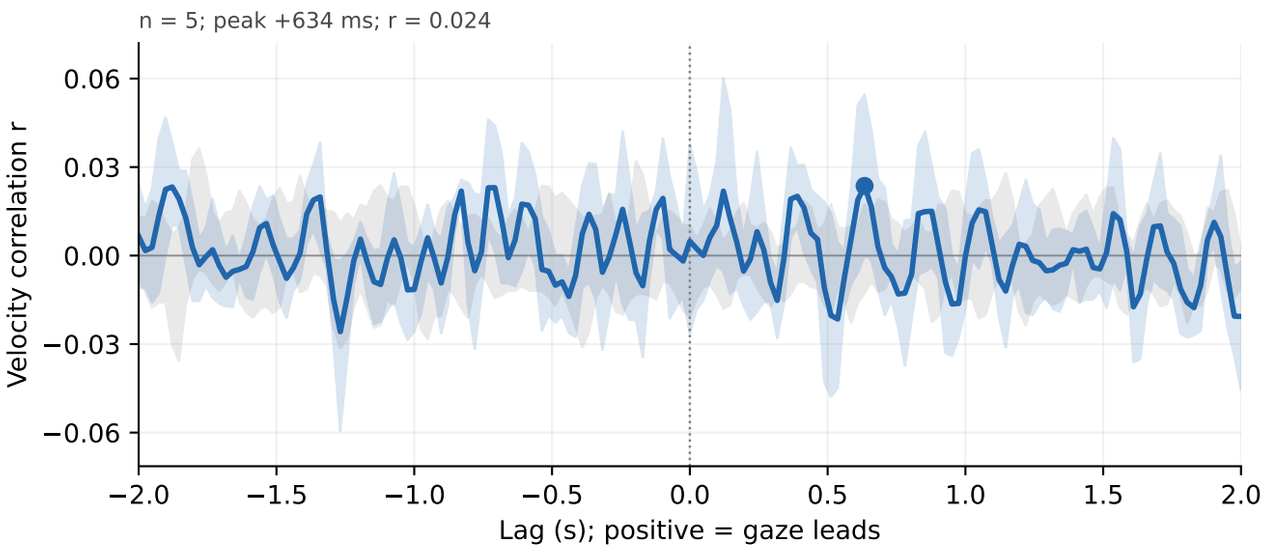
threading string through tower / psm_stage



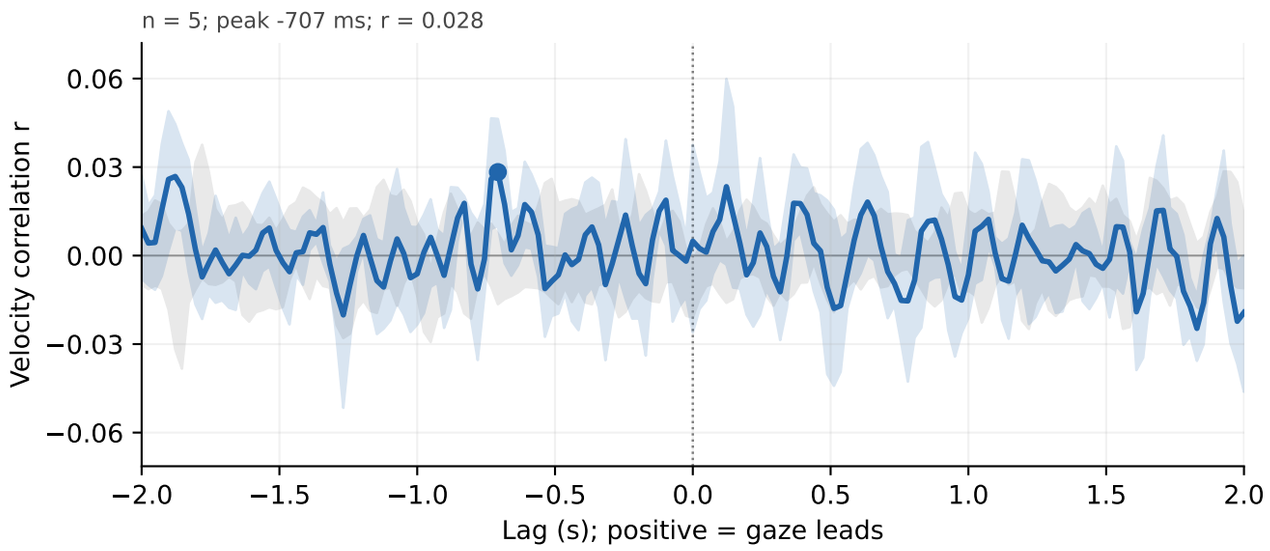
threading string through tower / within_stage



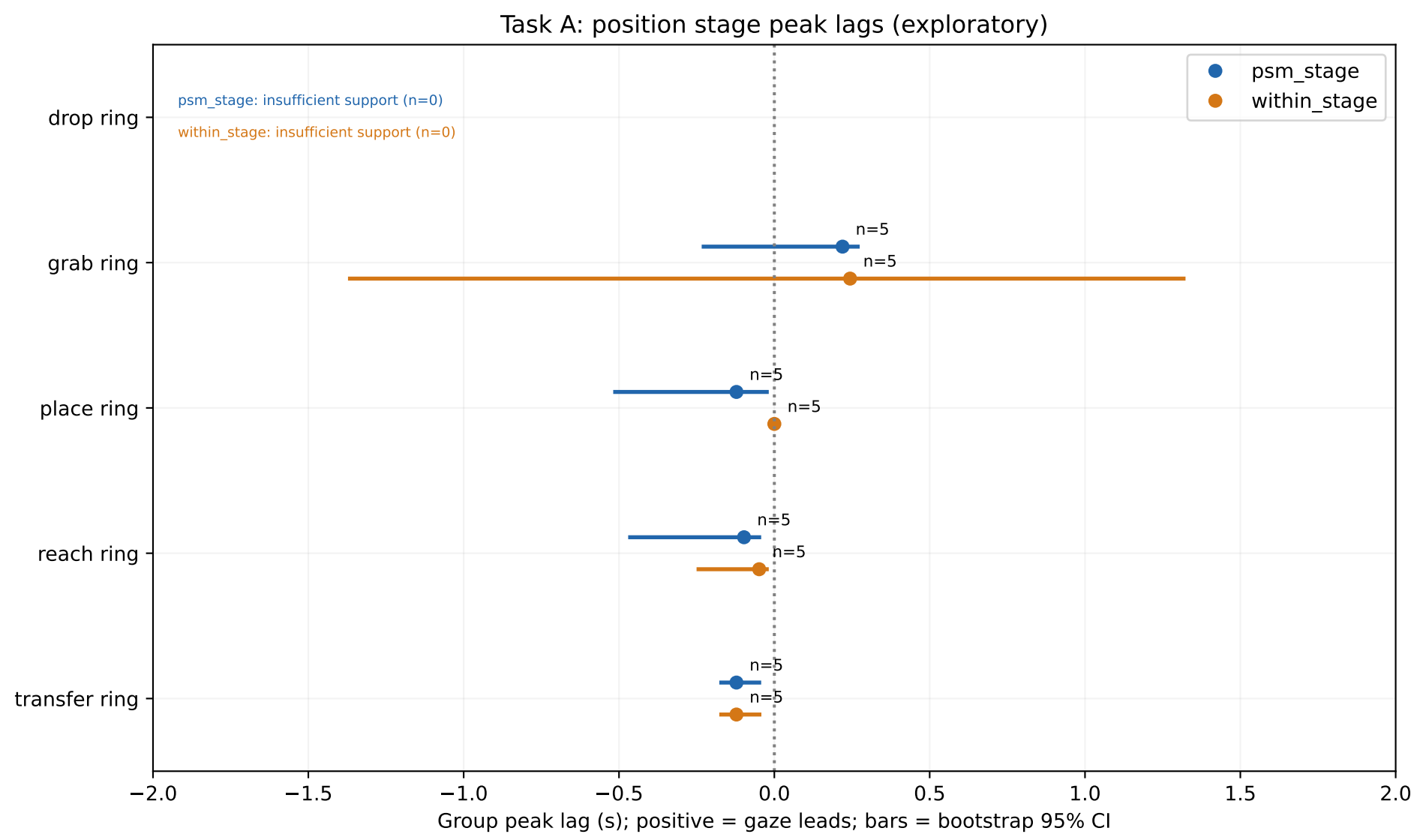
wrap around psm / psm_stage



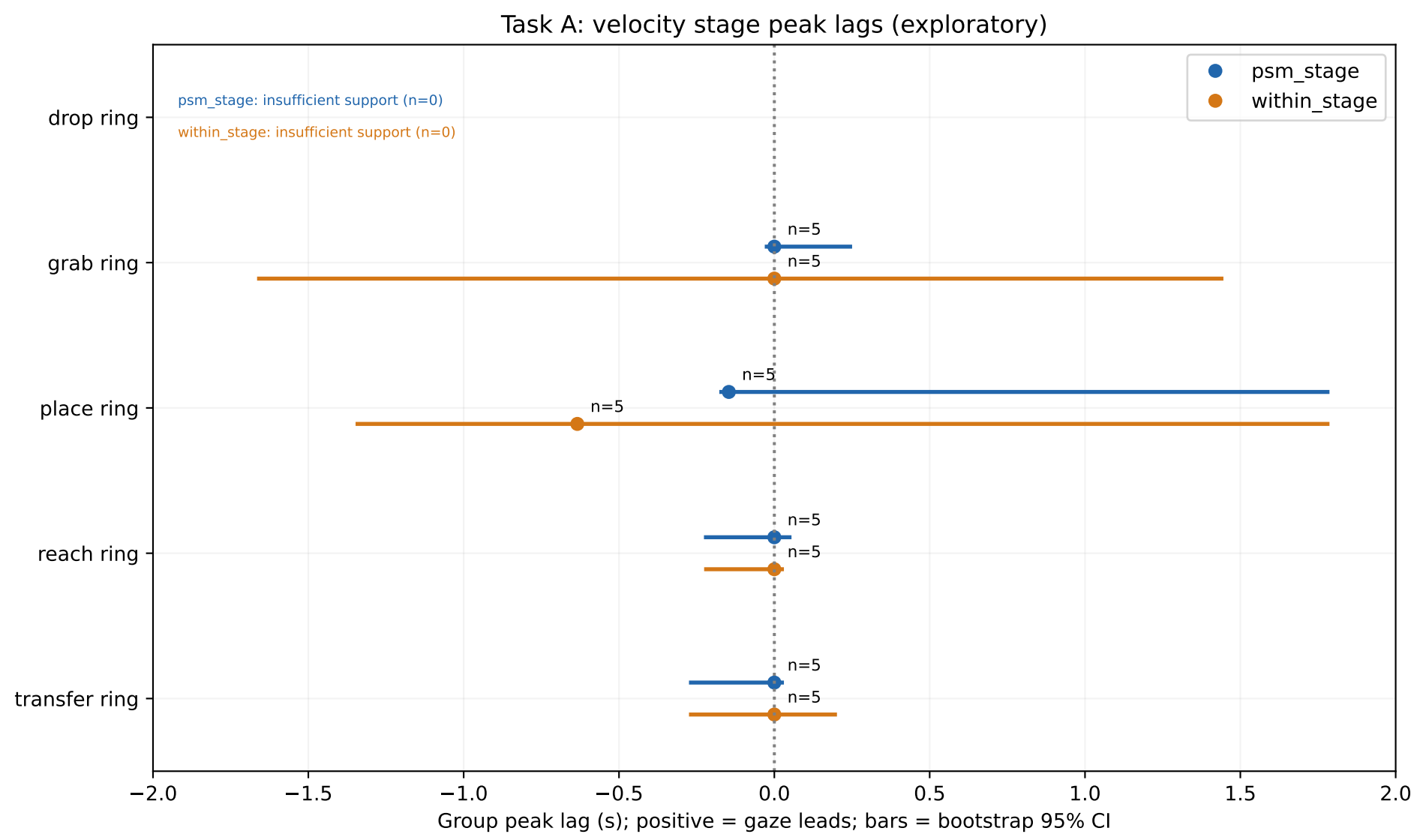
wrap around psm / within_stage



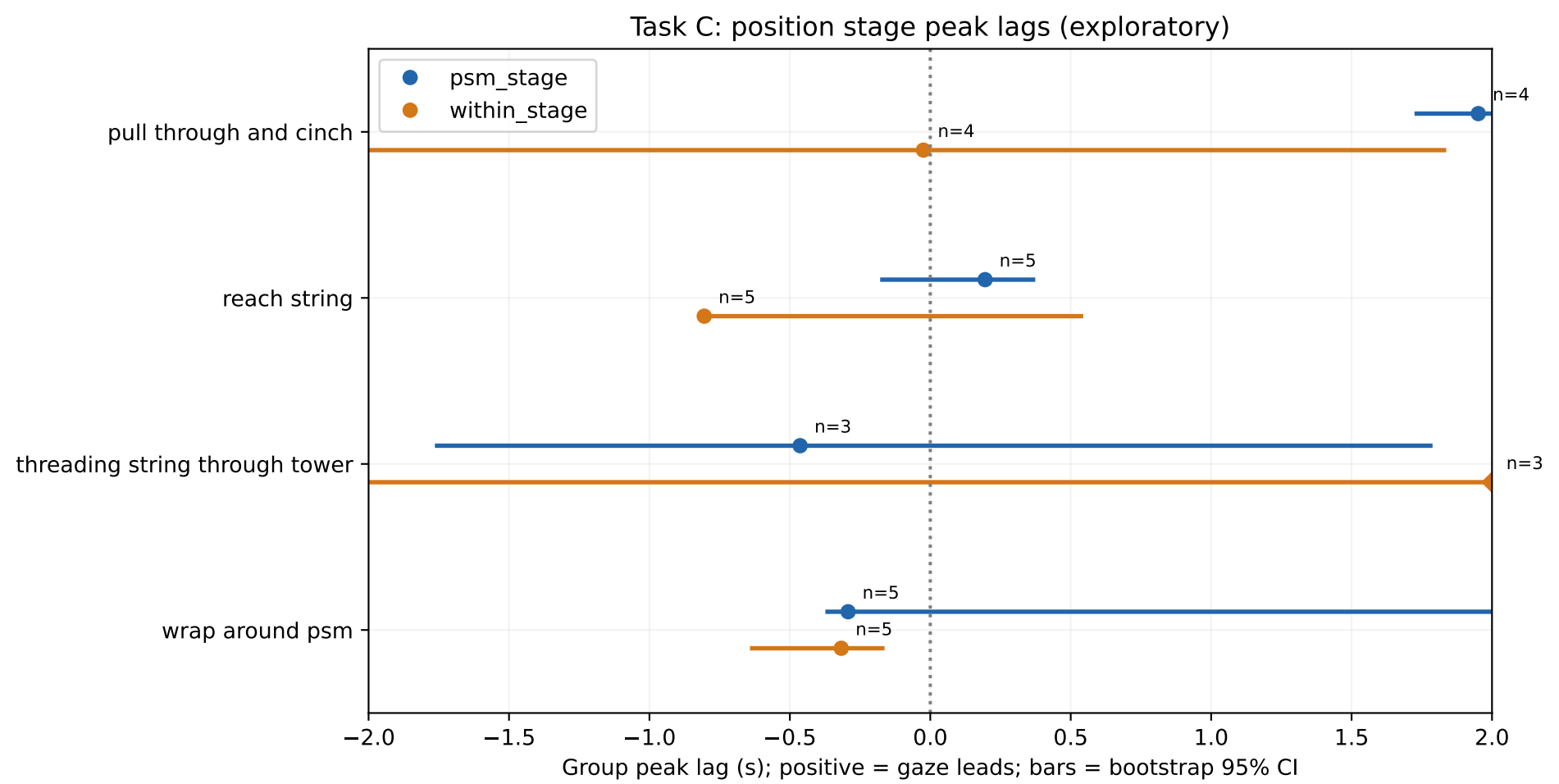
One PSM selected per recording/stage by largest signed xcorr peak; selection does not identify the operated instrument.



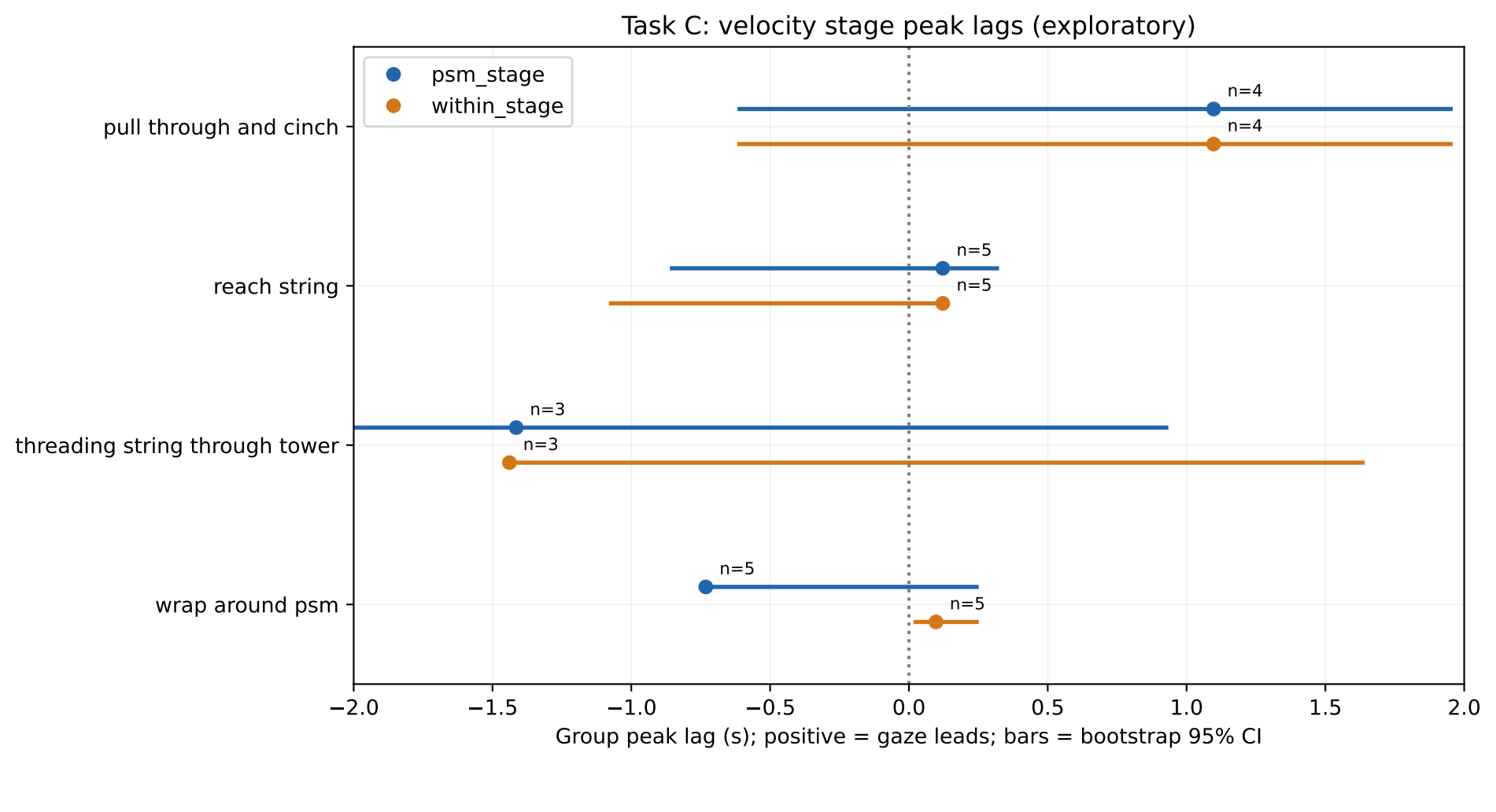
One PSM selected per recording/stage by largest signed xcorr peak; selection does not identify the operated instrument.



One PSM selected per recording/stage by largest signed xcorr peak; selection does not identify the operated instrument.



One PSM selected per recording/stage by largest signed xcorr peak; selection does not identify the operated instrument.



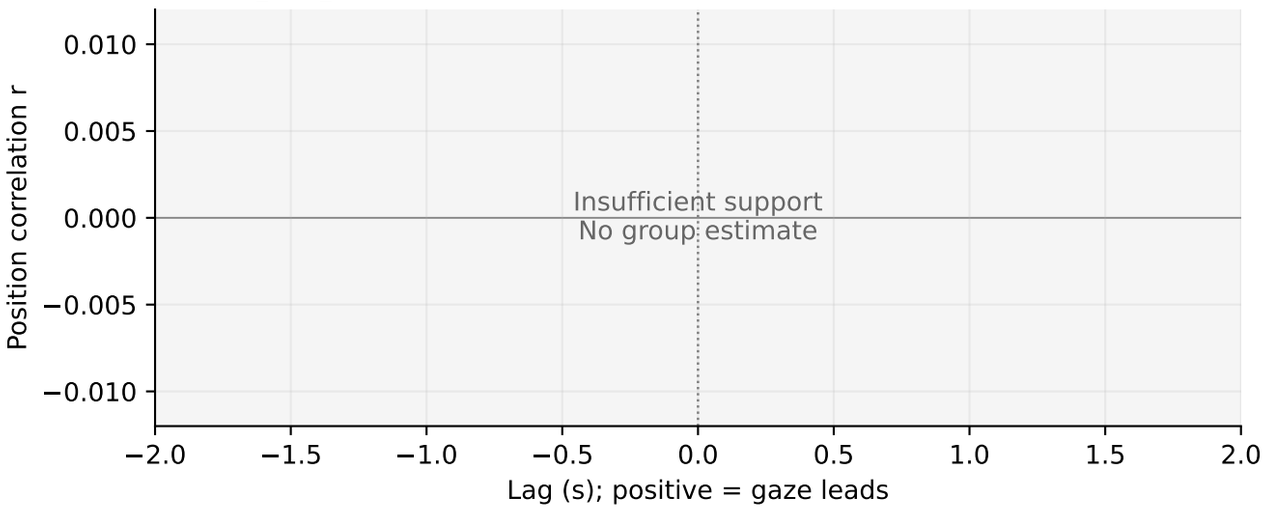
One PSM selected per recording/stage by largest signed xcorr peak; selection does not identify the operated instrument.

Task A: position, maximum-xcorr PSM per stage

Unpaired reference (95%) | Participant bootstrap (95%) | Group mean

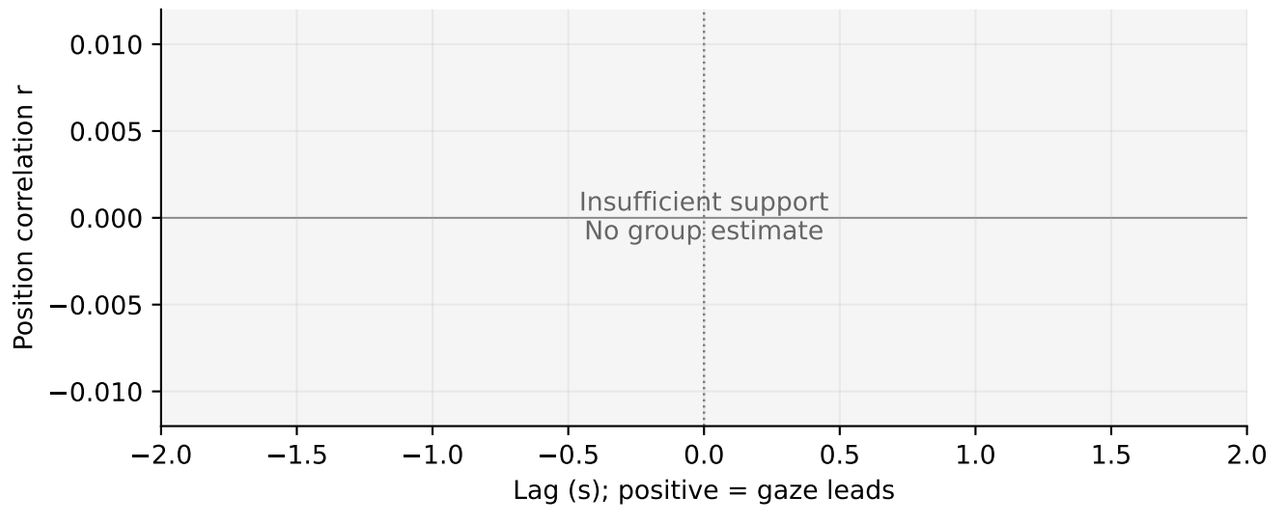
drop ring / psm_stage

n = 0; no_valid_curve



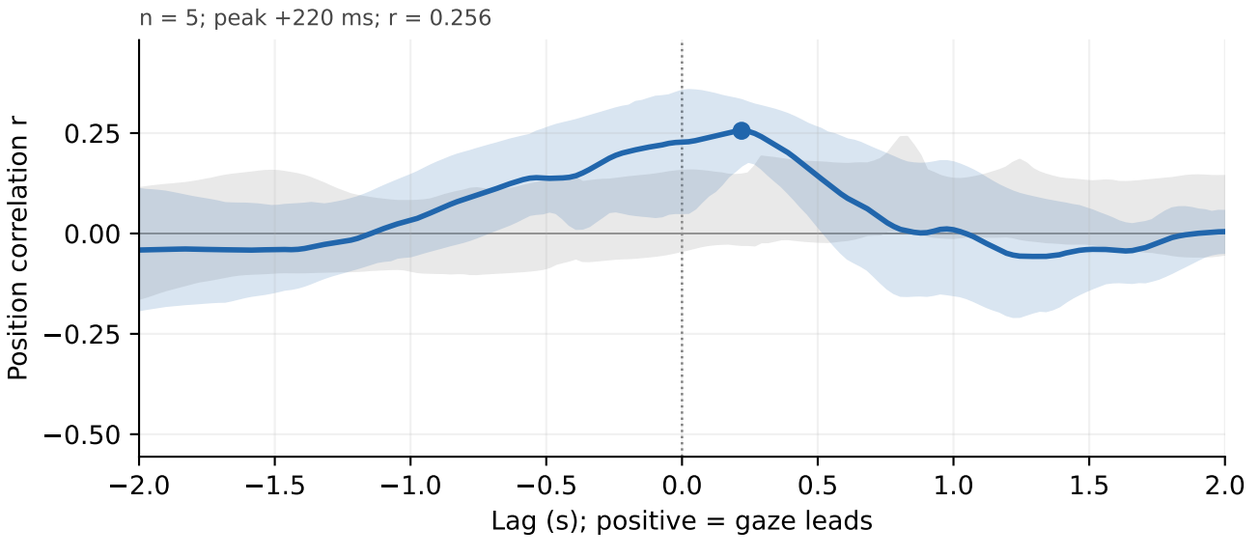
drop ring / within_stage

n = 0; no_valid_curve



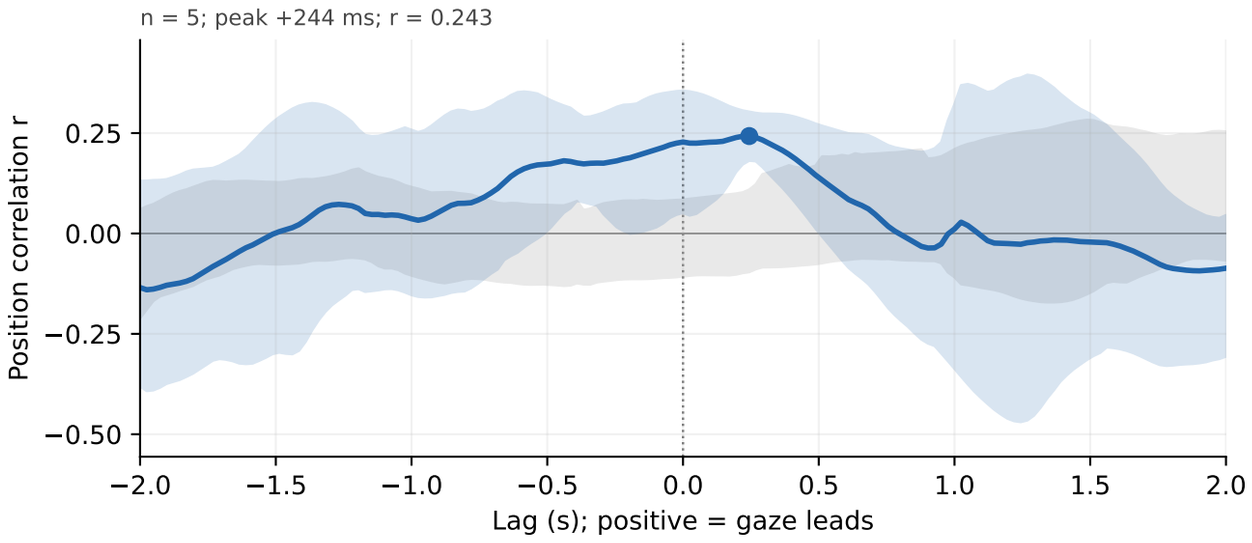
grab ring / psm_stage

n = 5; peak +220 ms; r = 0.256



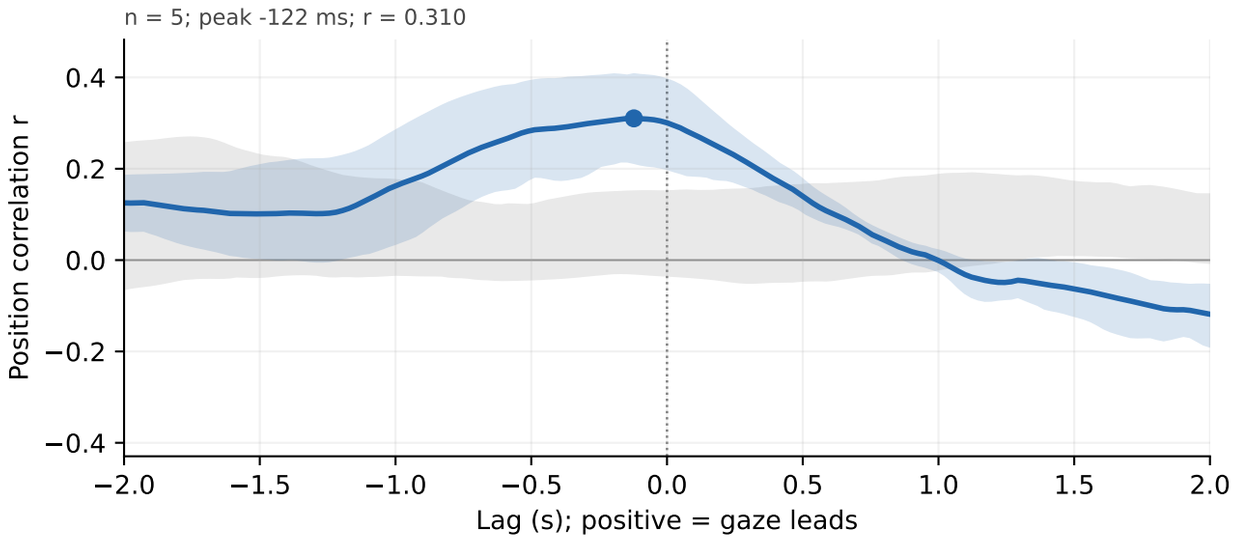
grab ring / within_stage

n = 5; peak +244 ms; r = 0.243



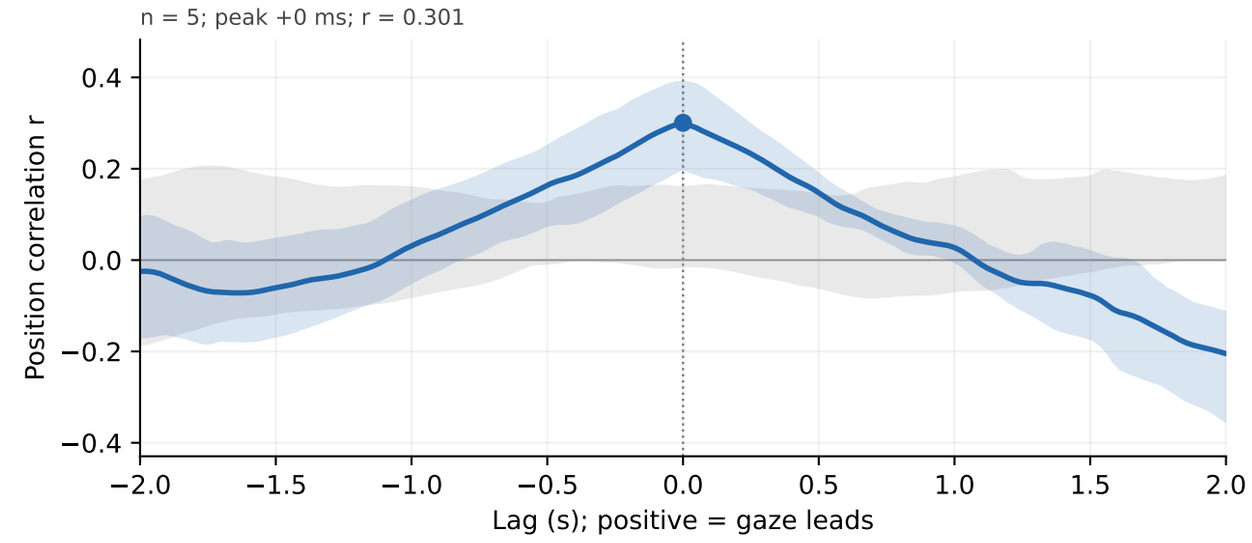
place ring / psm_stage

n = 5; peak -122 ms; r = 0.310



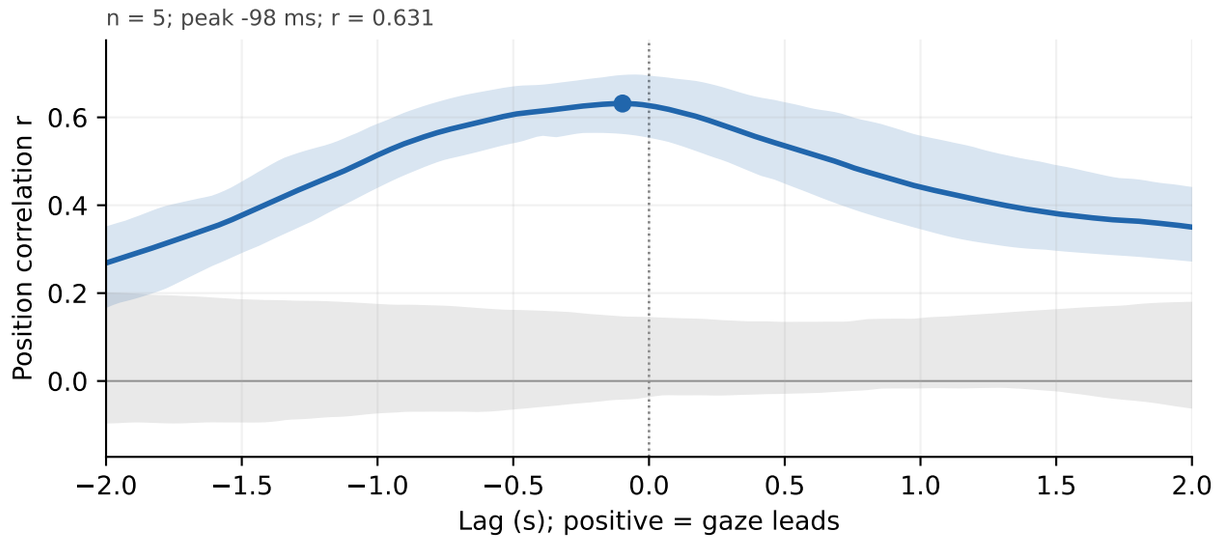
place ring / within_stage

n = 5; peak +0 ms; r = 0.301



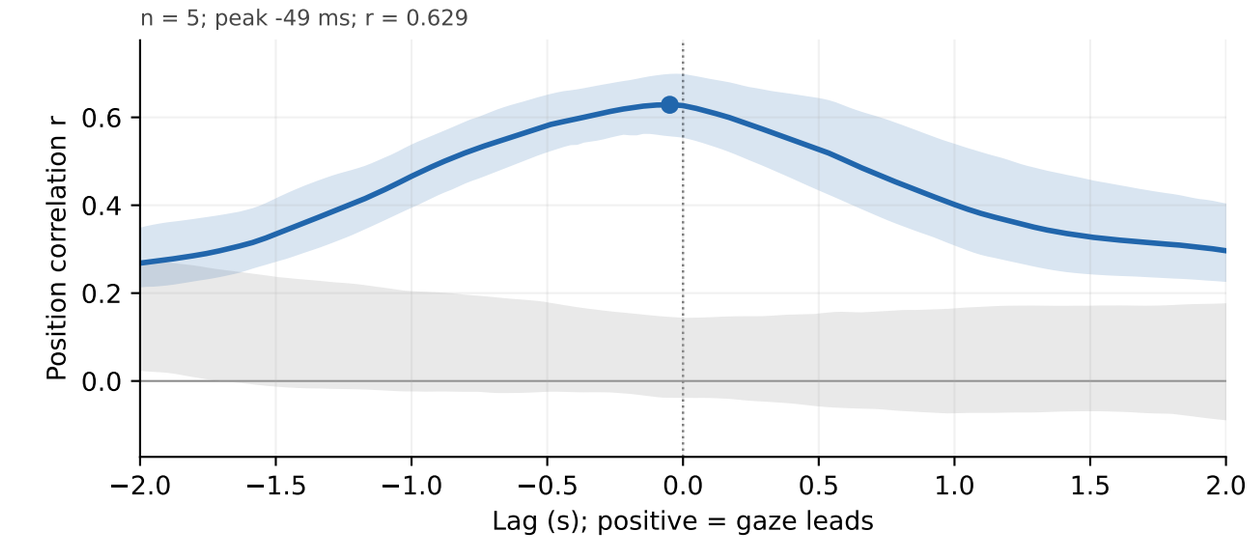
reach ring / psm_stage

n = 5; peak -98 ms; r = 0.631



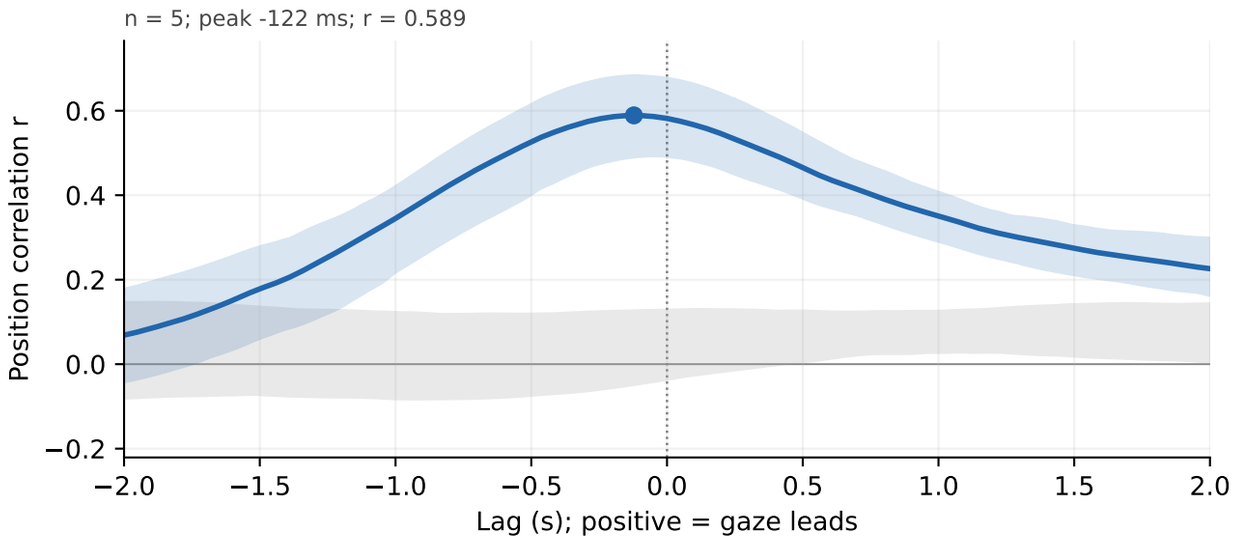
reach ring / within_stage

n = 5; peak -49 ms; r = 0.629



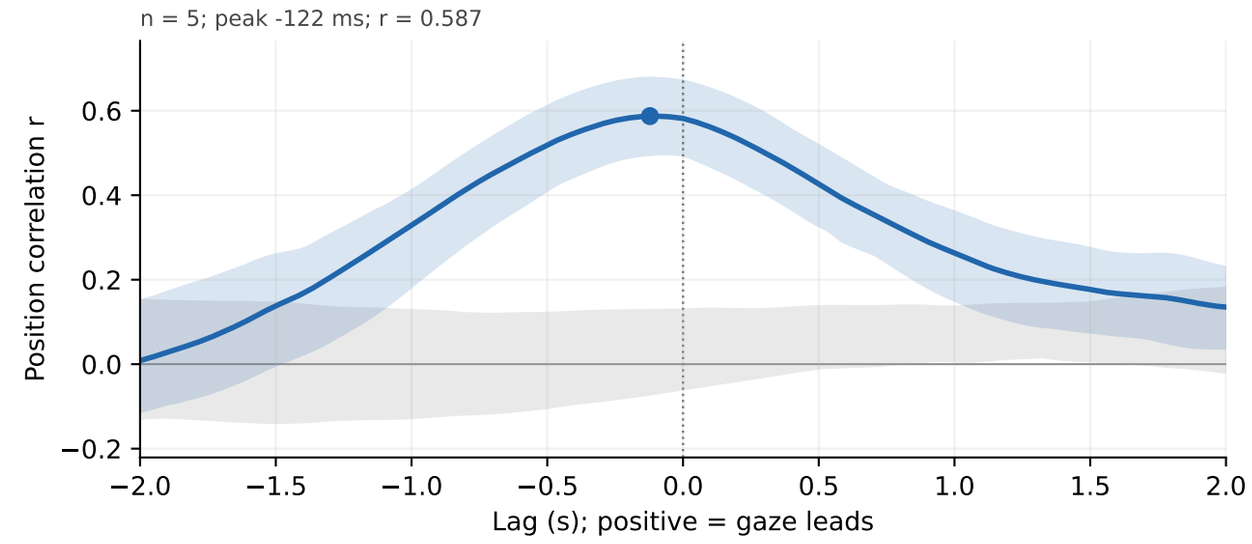
transfer ring / psm_stage

n = 5; peak -122 ms; r = 0.589



transfer ring / within_stage

n = 5; peak -122 ms; r = 0.587

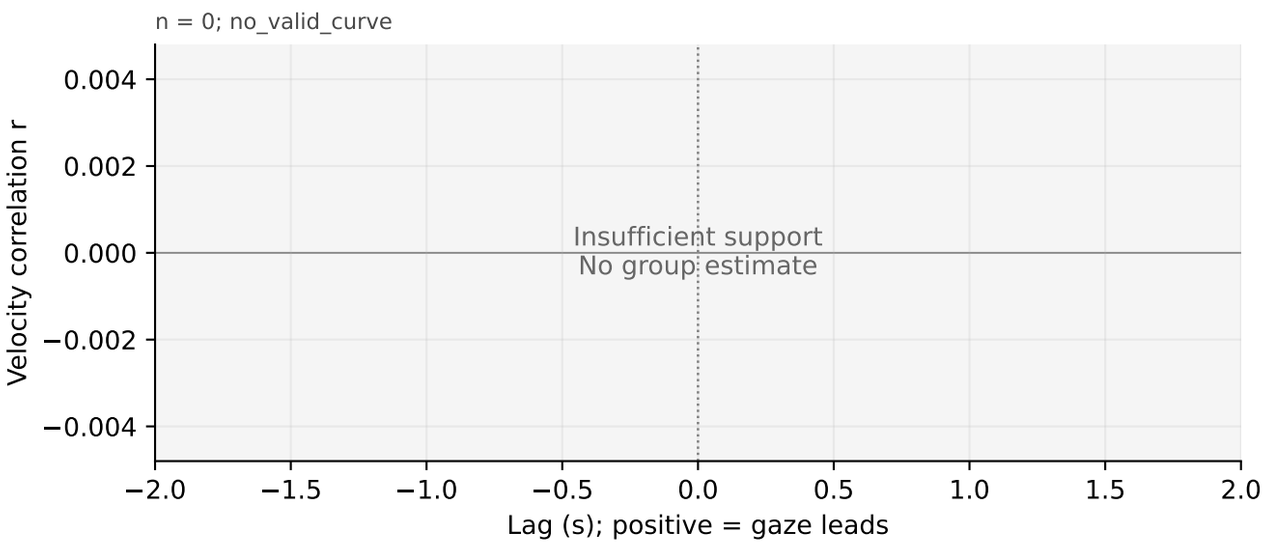


One PSM selected per recording/stage by largest signed xcorr peak; selection does not identify the operated instrument.

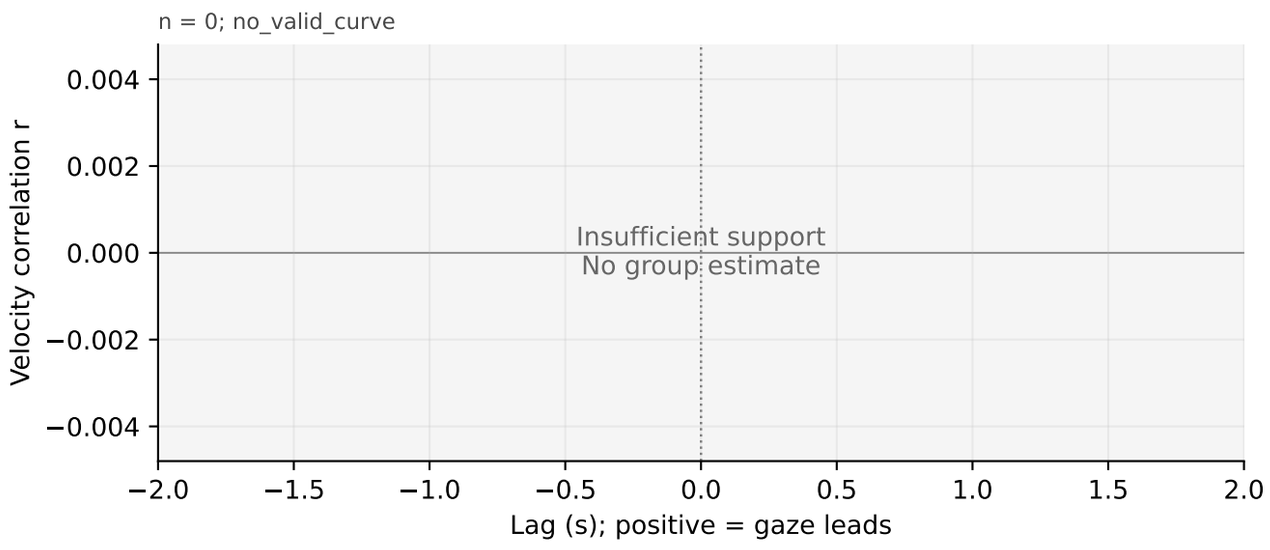
Task A: velocity, maximum-xcorr PSM per stage

Unpaired reference (95%) | Participant bootstrap (95%) | Group mean

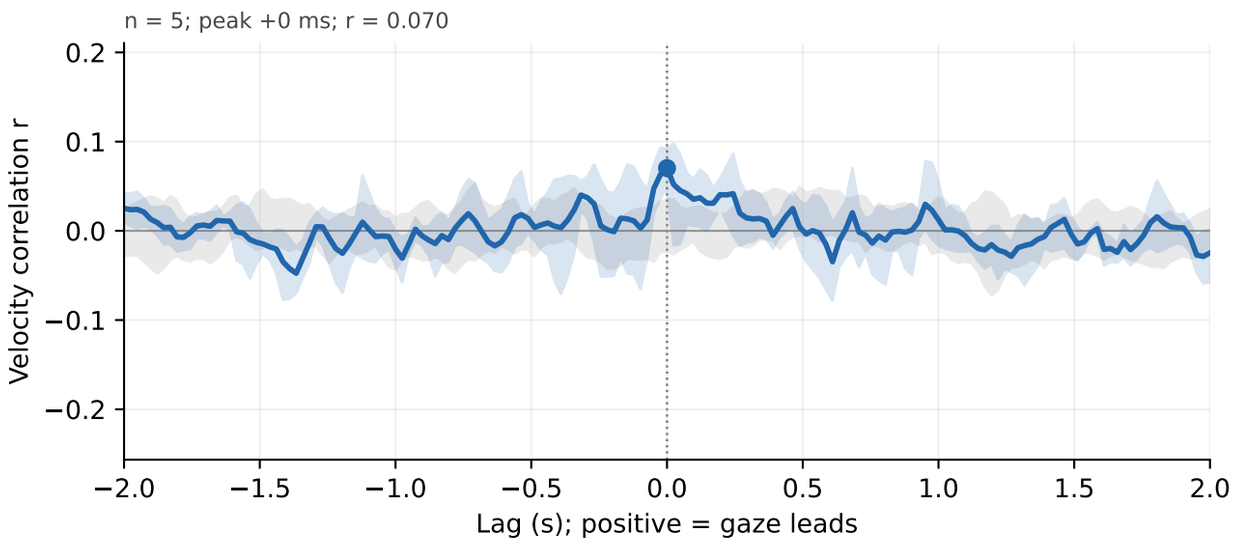
drop ring / psm_stage



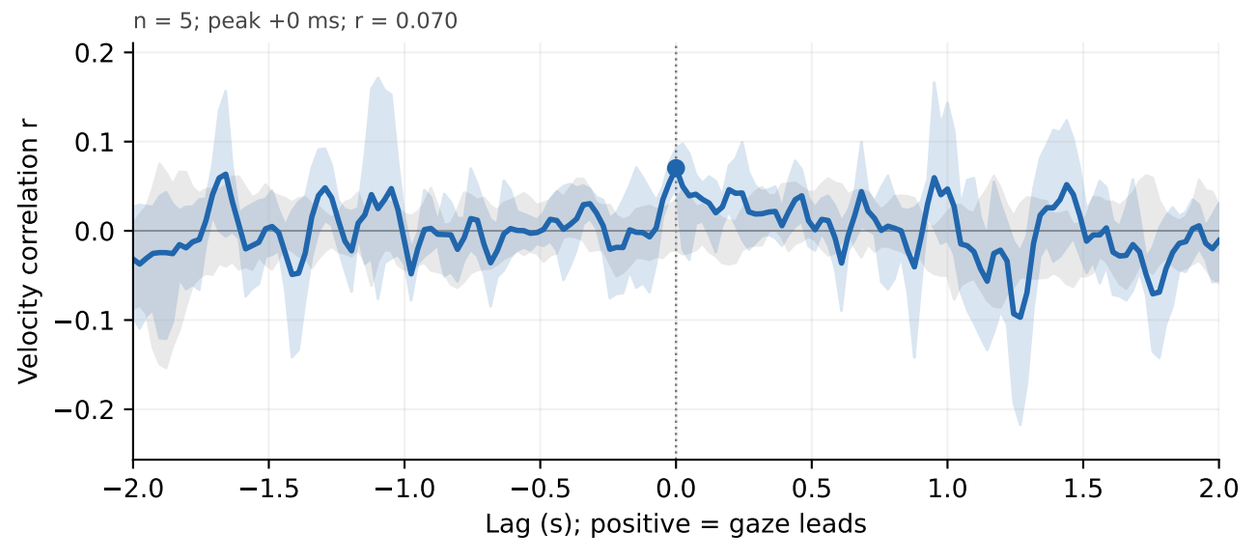
drop ring / within_stage



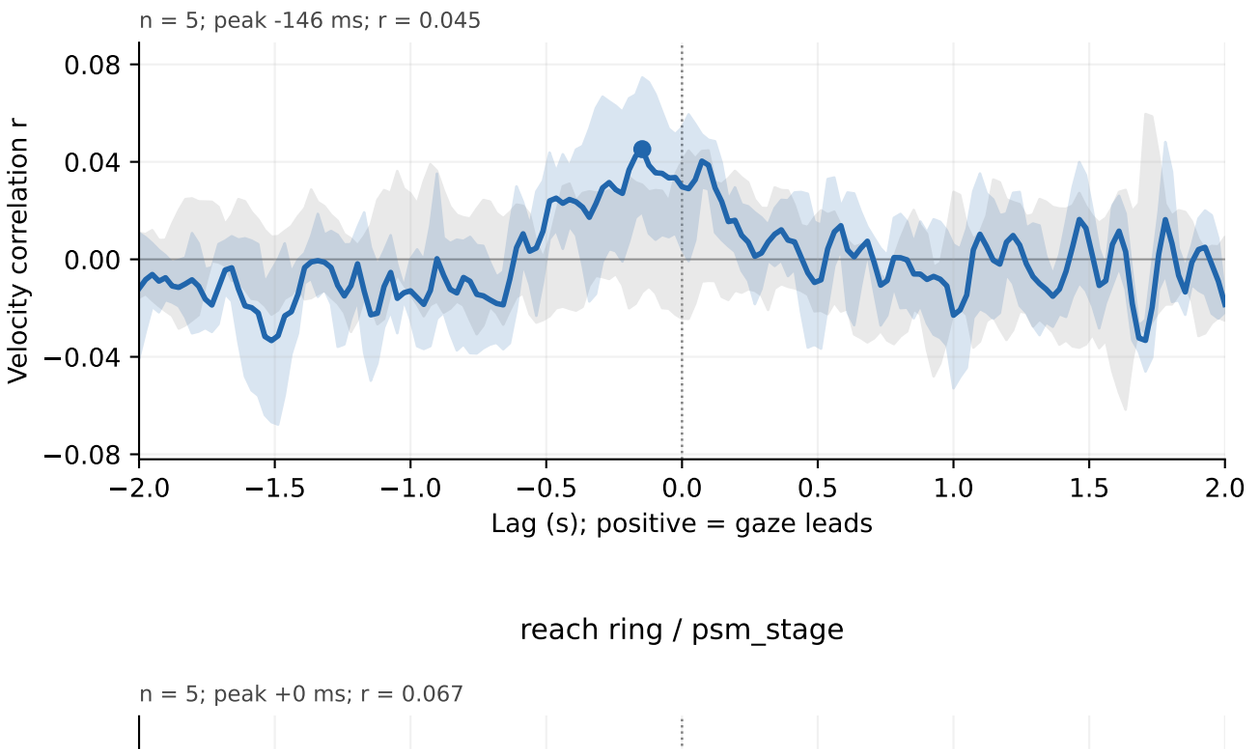
grab ring / psm_stage



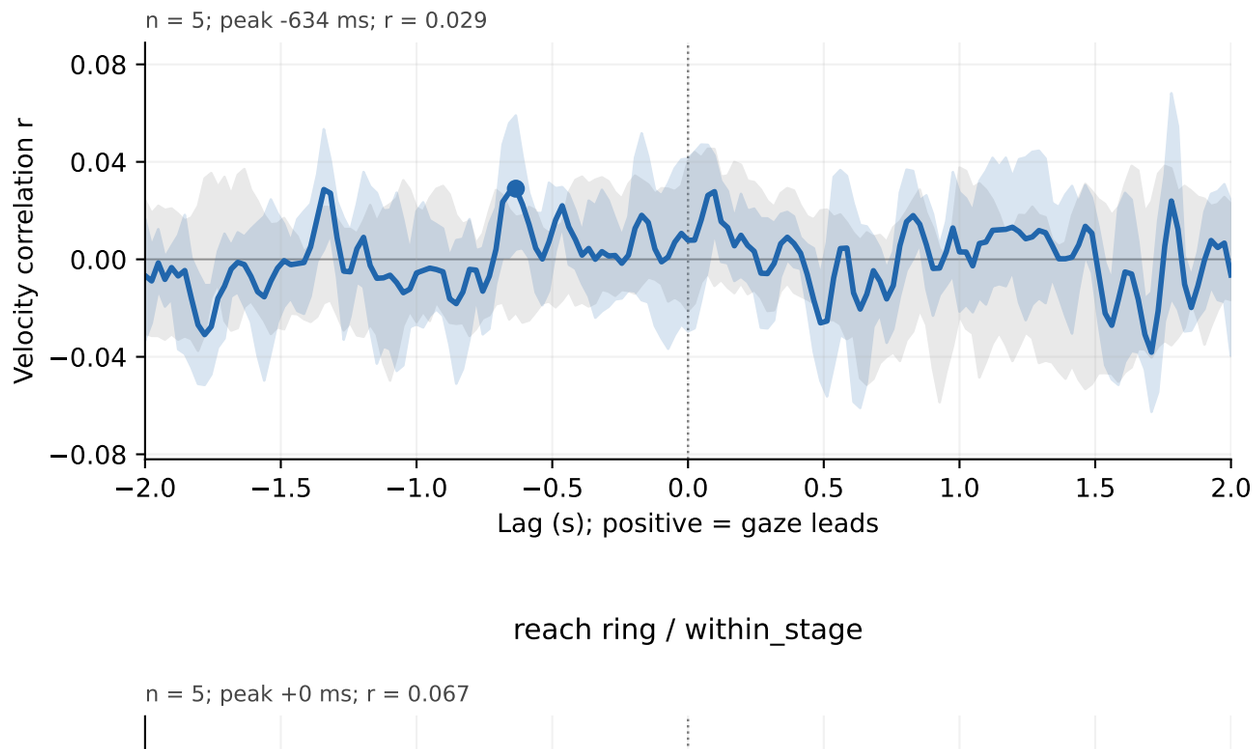
grab ring / within_stage



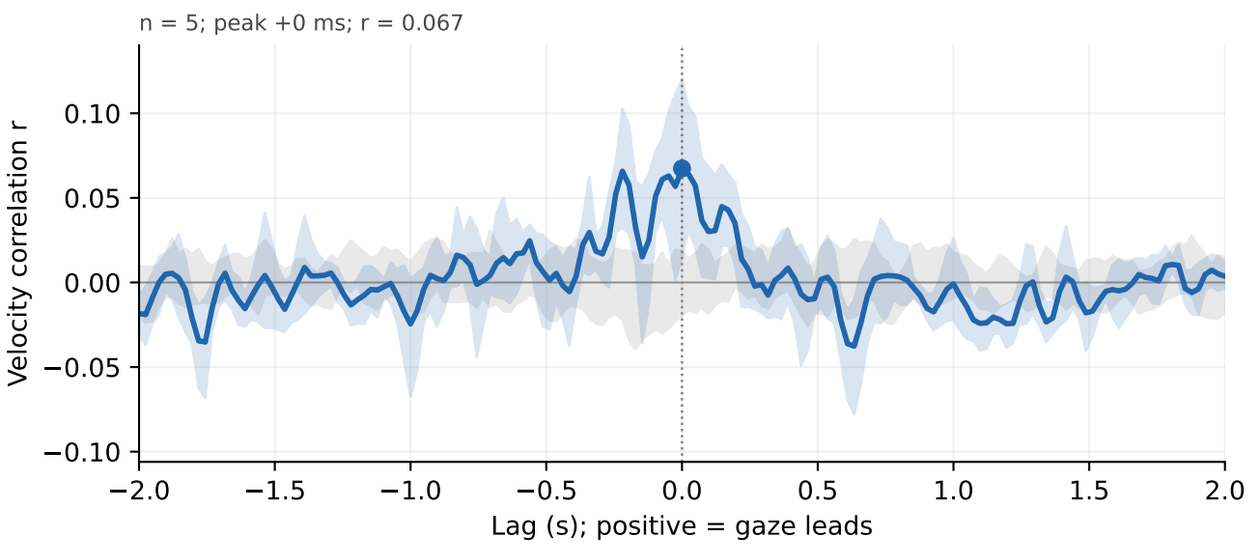
place ring / psm_stage



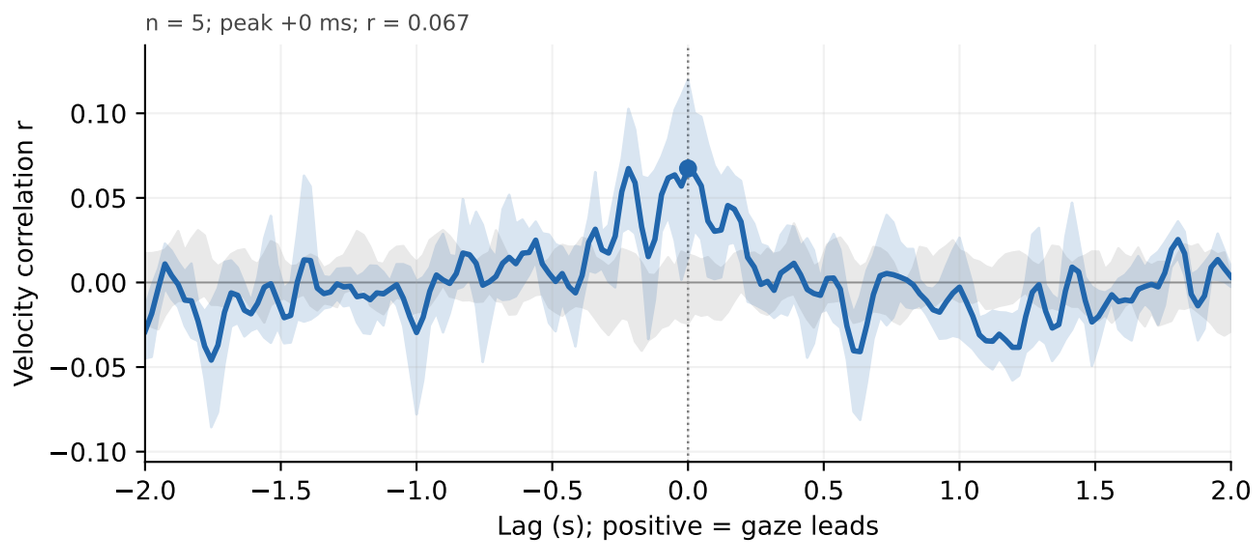
place ring / within_stage



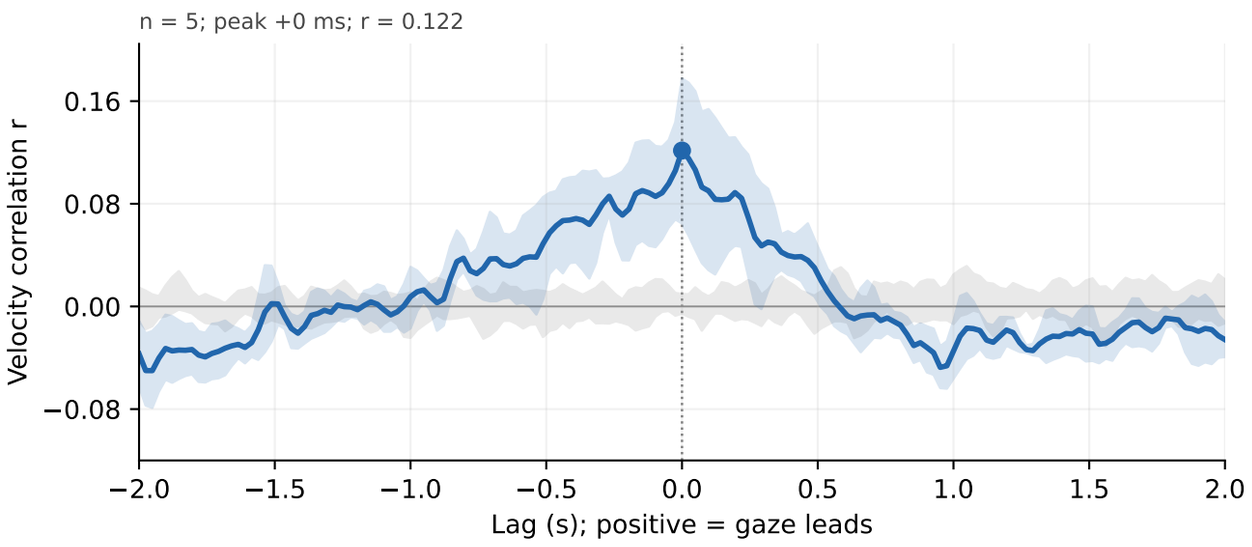
reach ring / psm_stage



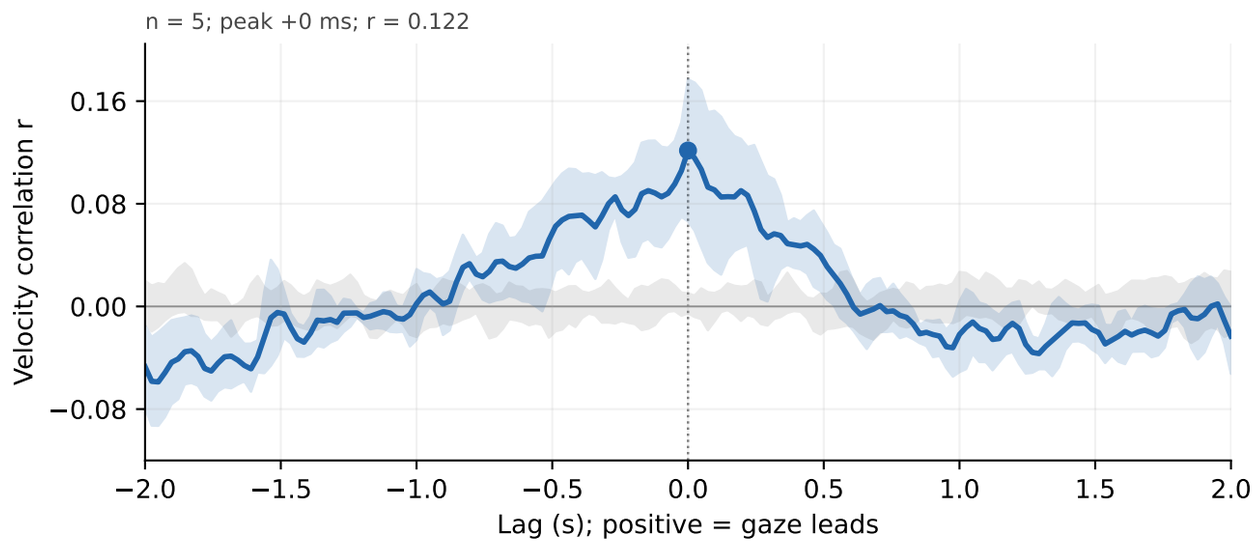
reach ring / within_stage



transfer ring / psm_stage



transfer ring / within_stage

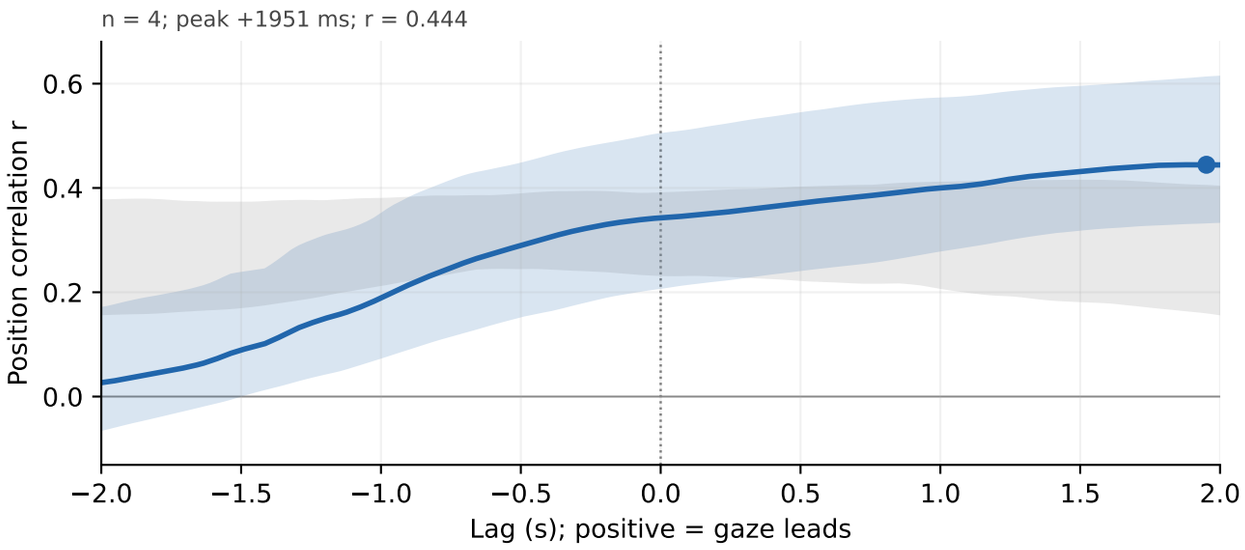


One PSM selected per recording/stage by largest signed xcorr peak; selection does not identify the operated instrument.

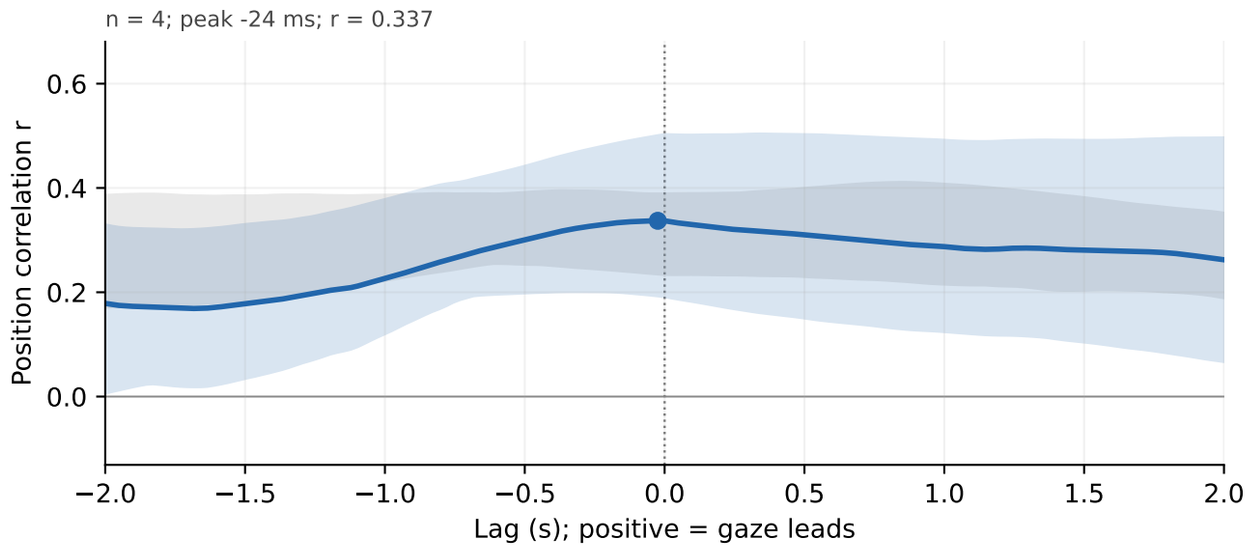
Task C: position, maximum-xcorr PSM per stage

Unpaired reference (95%) | Participant bootstrap (95%) | Group mean

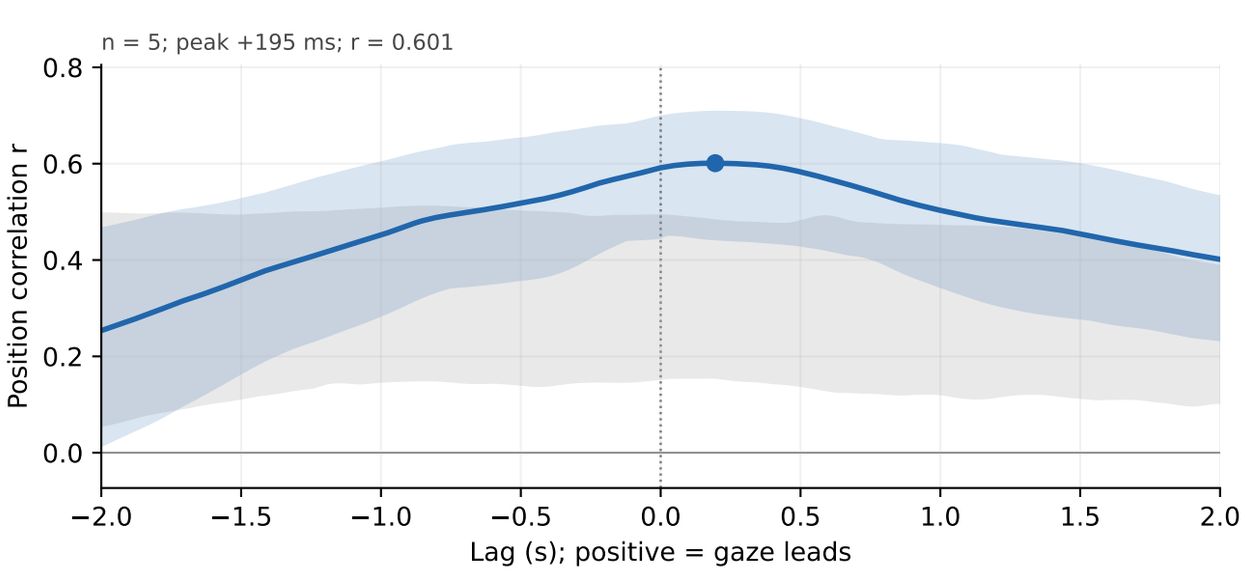
pull through and cinch / psm_stage



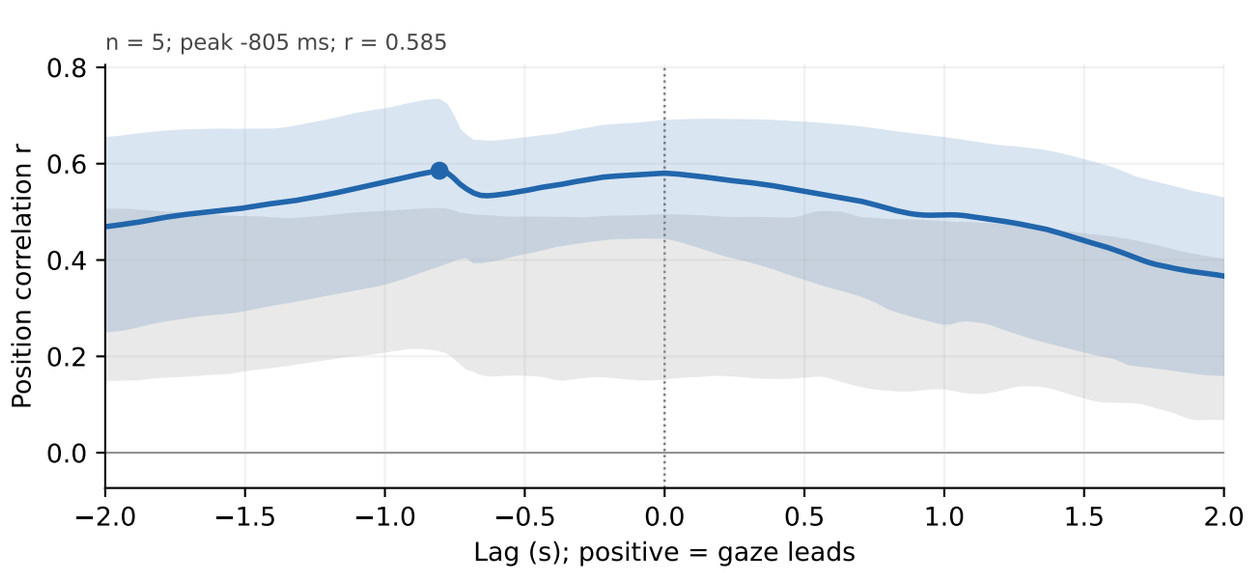
pull through and cinch / within_stage



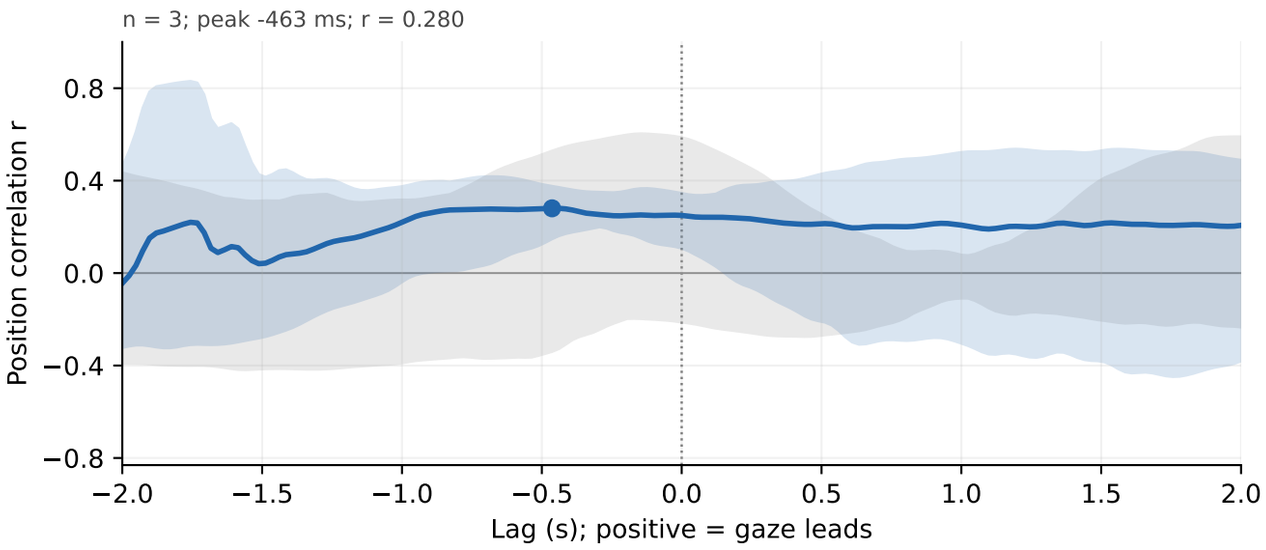
reach string / psm_stage



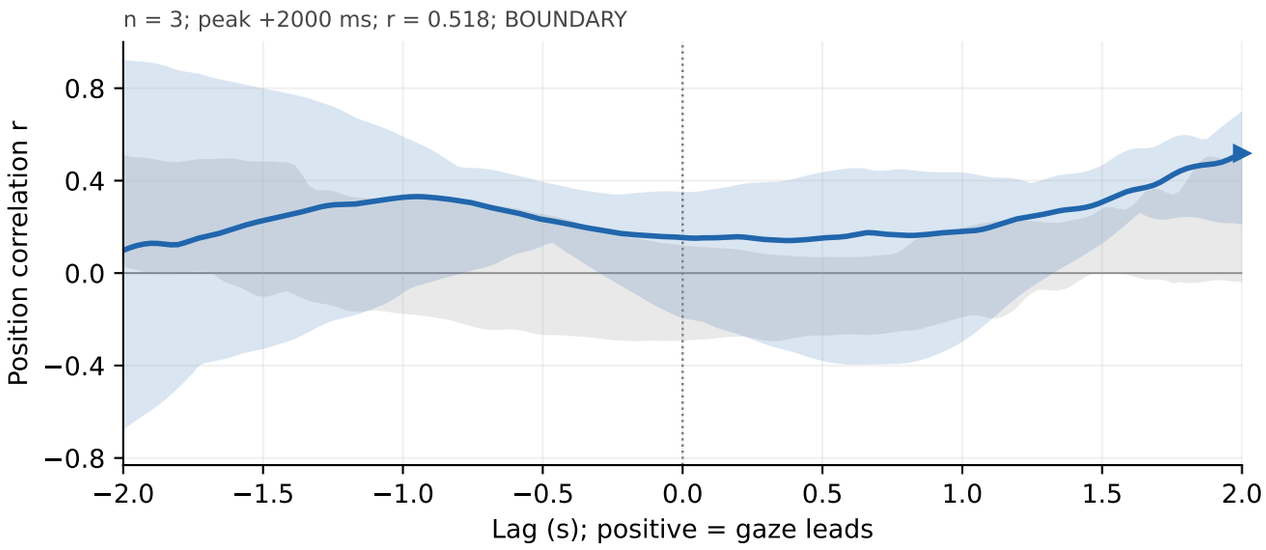
reach string / within_stage



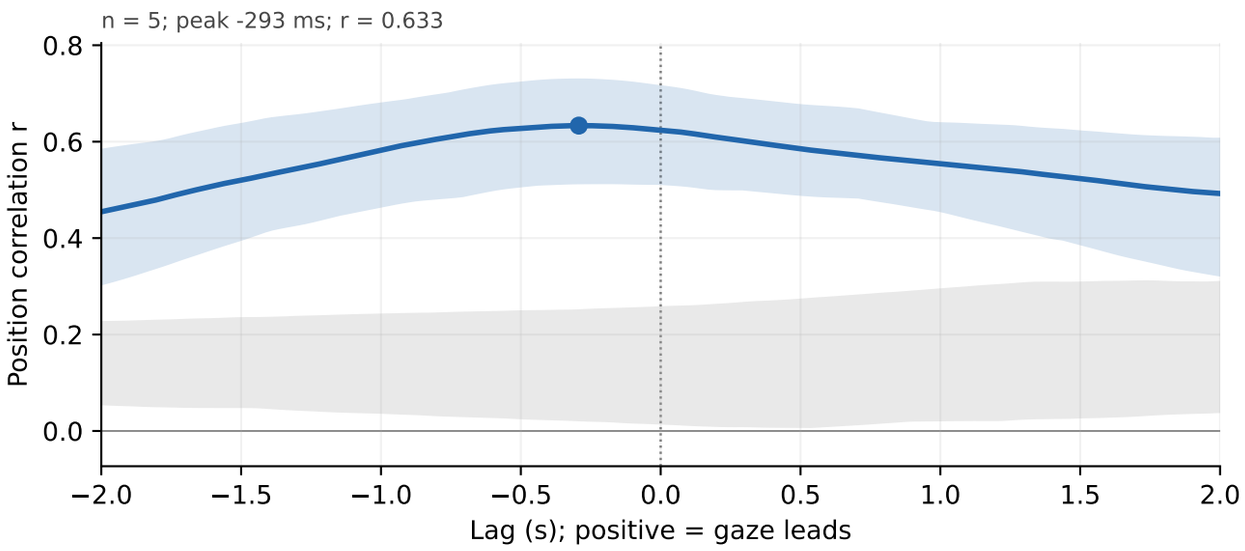
threading string through tower / psm_stage



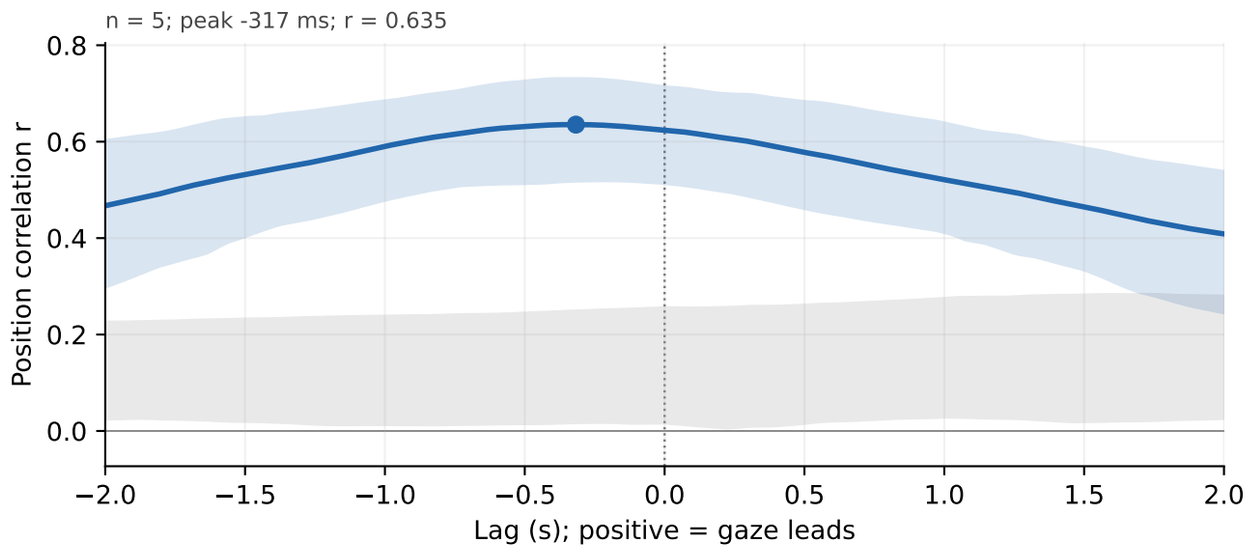
threading string through tower / within_stage



wrap around psm / psm_stage



wrap around psm / within_stage

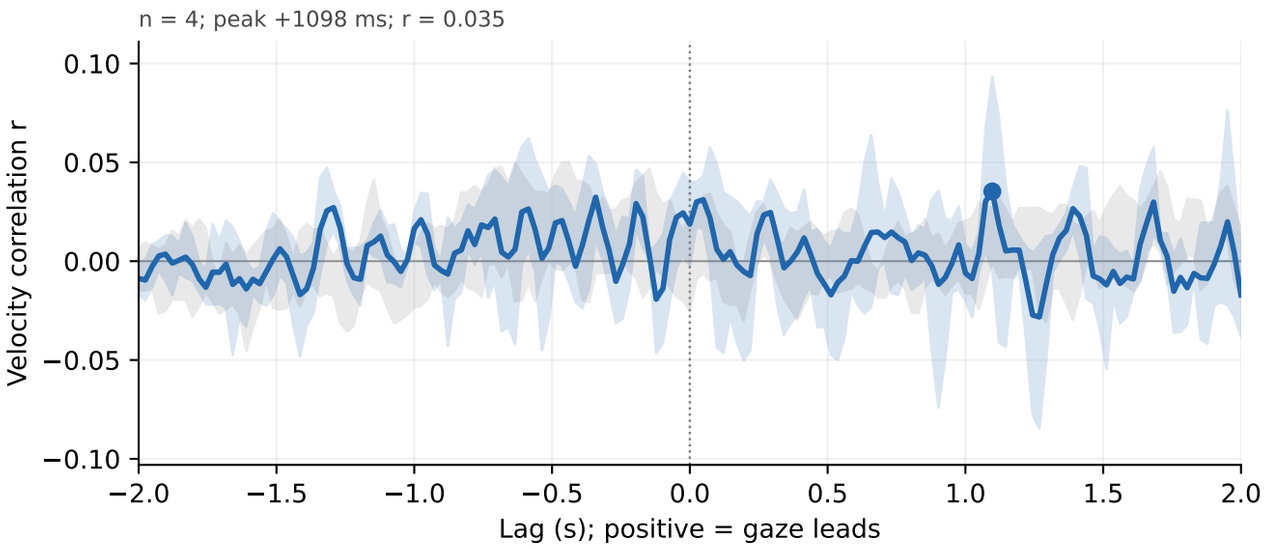


One PSM selected per recording/stage by largest signed xcorr peak; selection does not identify the operated instrument.

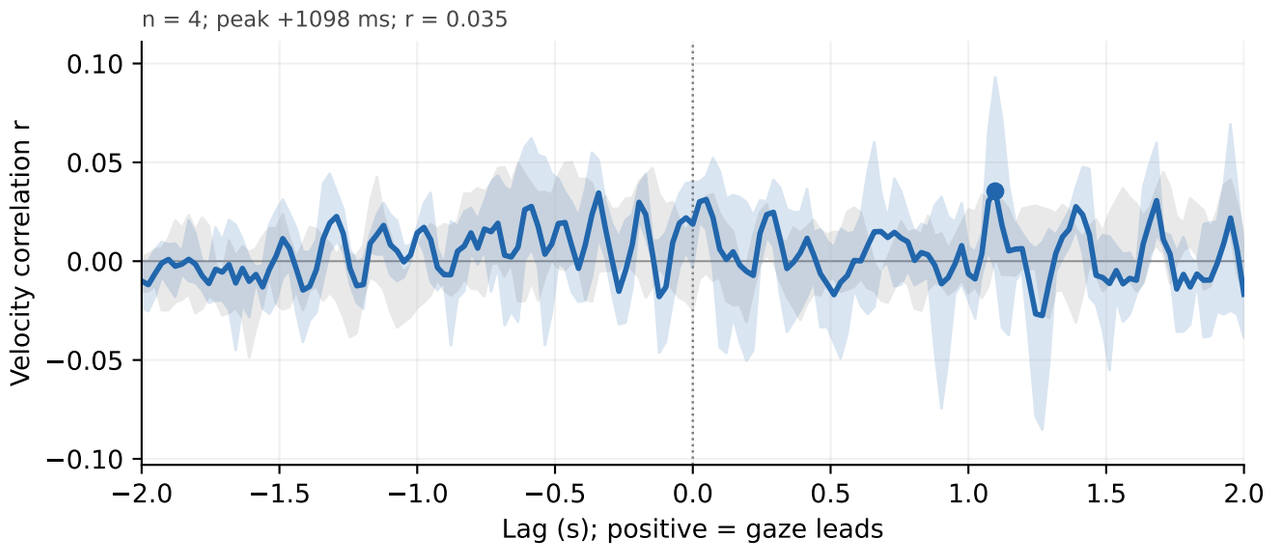
Task C: velocity, maximum-xcorr PSM per stage

Unpaired reference (95%) | Participant bootstrap (95%) | Group mean

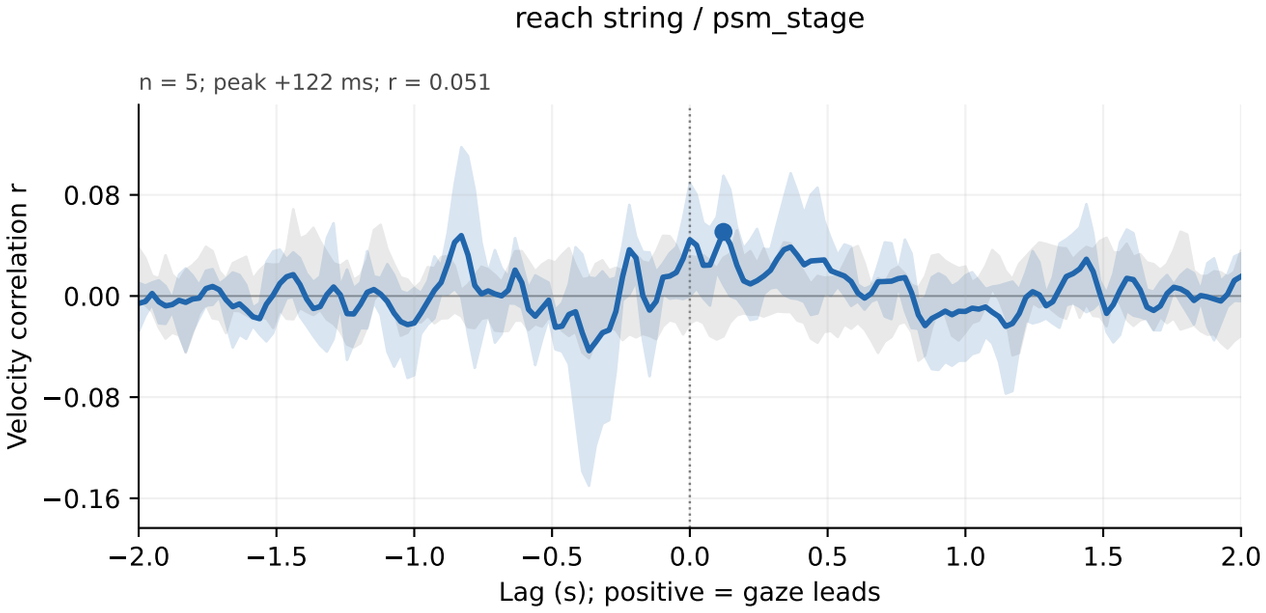
pull through and cinch / psm_stage



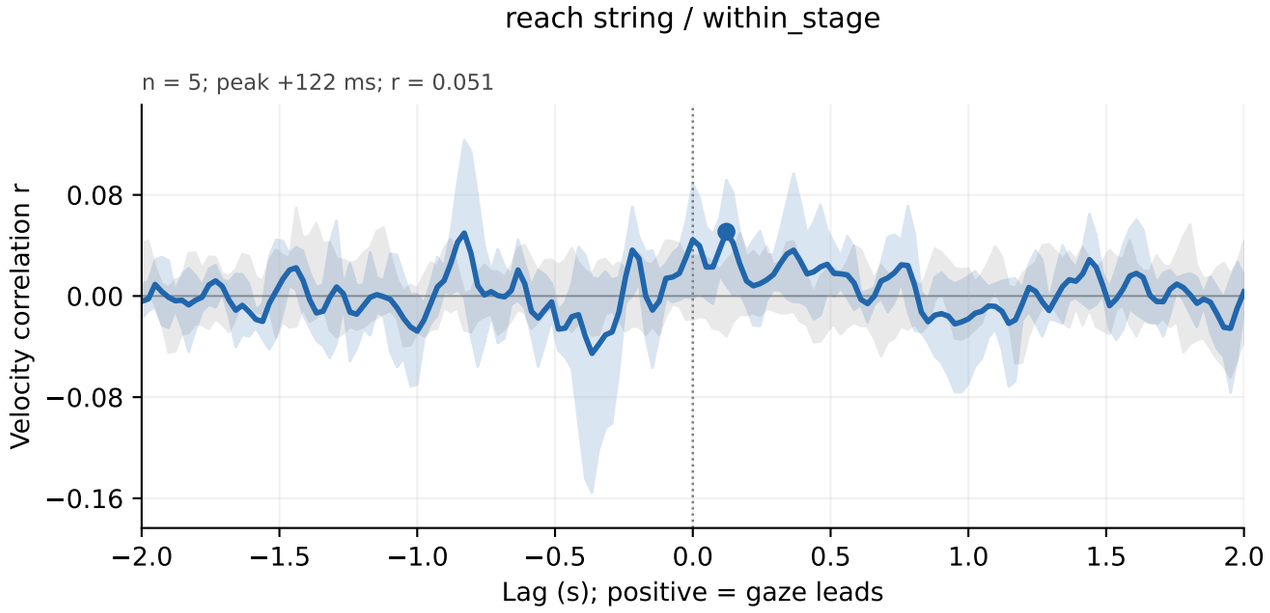
pull through and cinch / within_stage



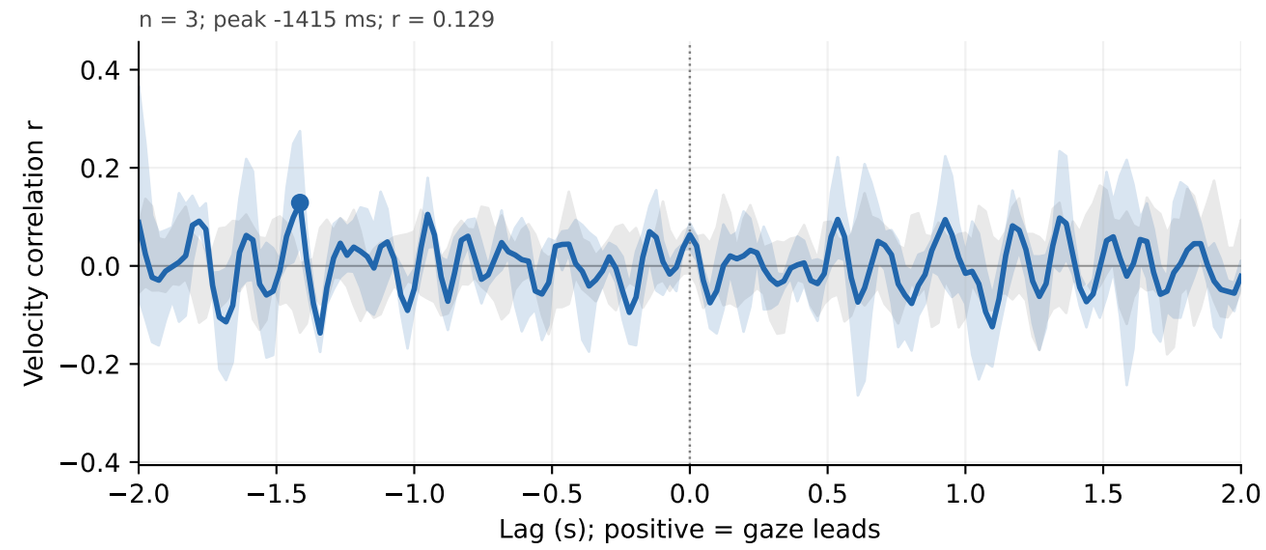
reach string / psm_stage



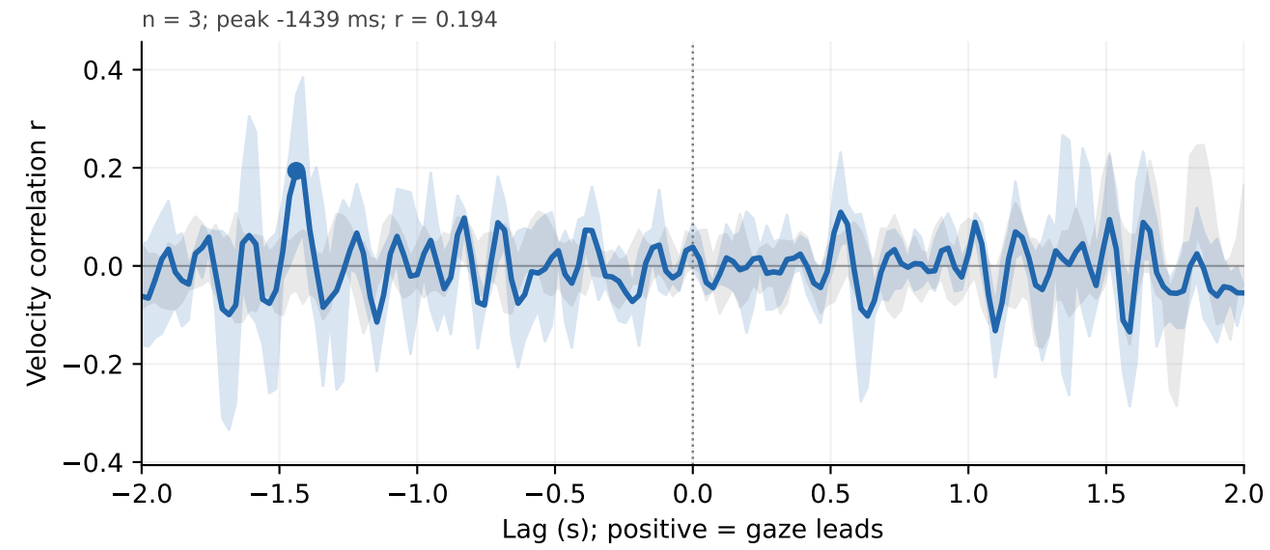
reach string / within_stage



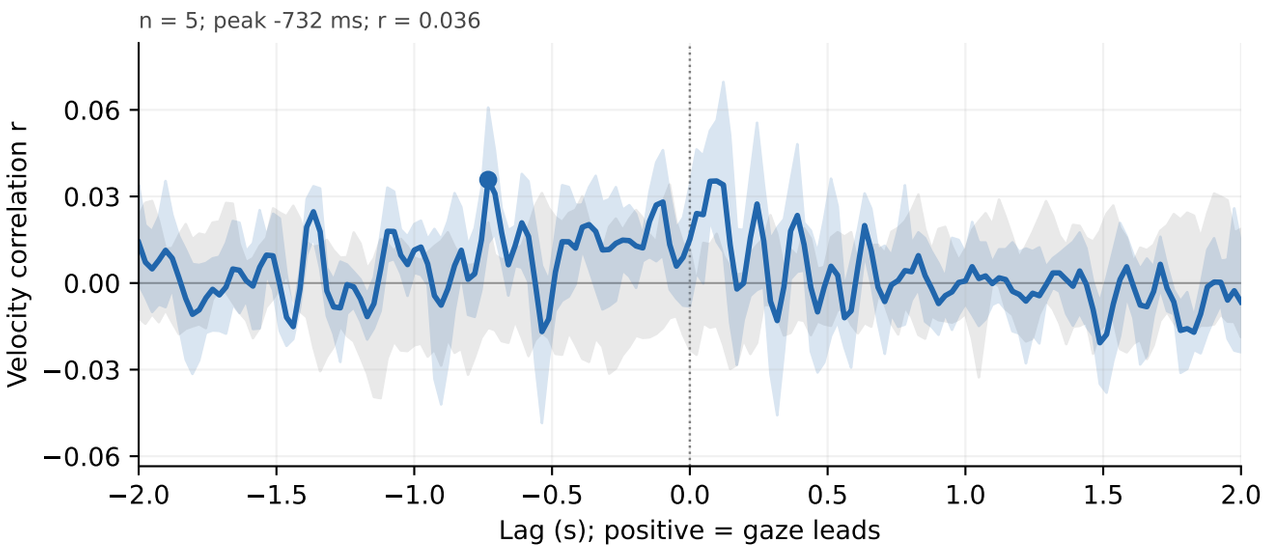
threading string through tower / psm_stage



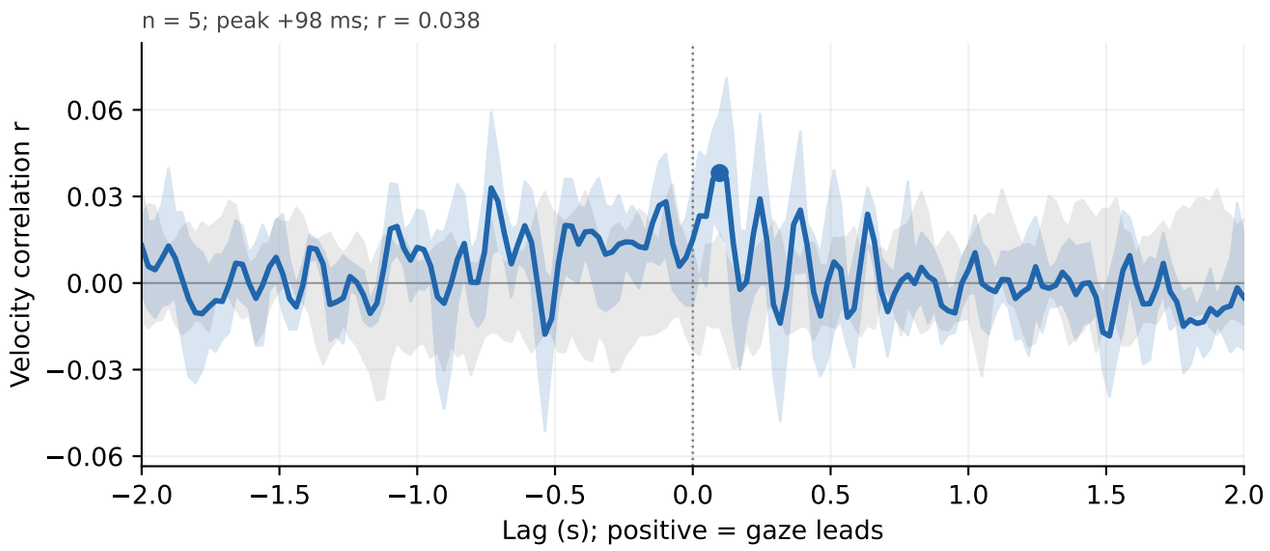
threading string through tower / within_stage



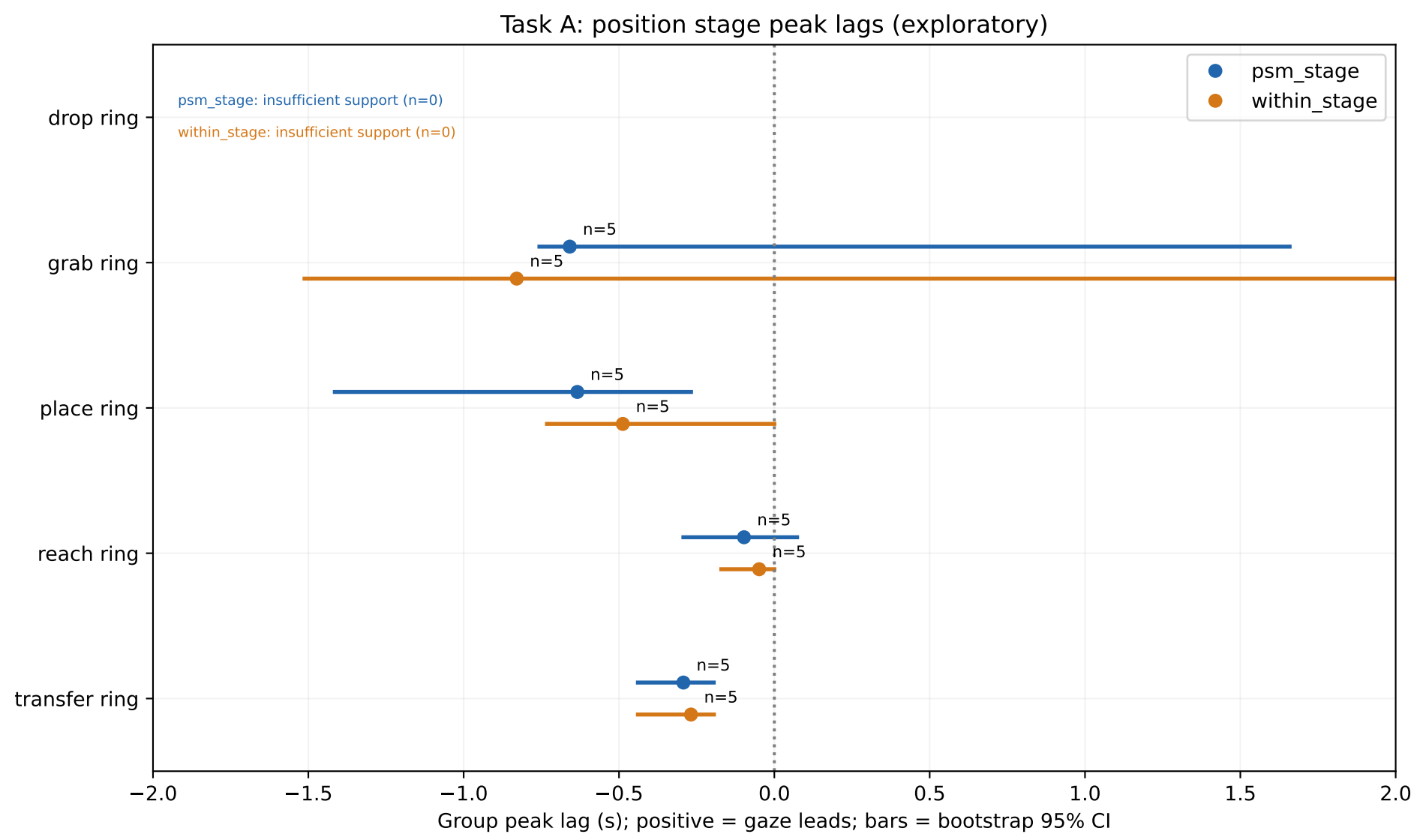
wrap around psm / psm_stage



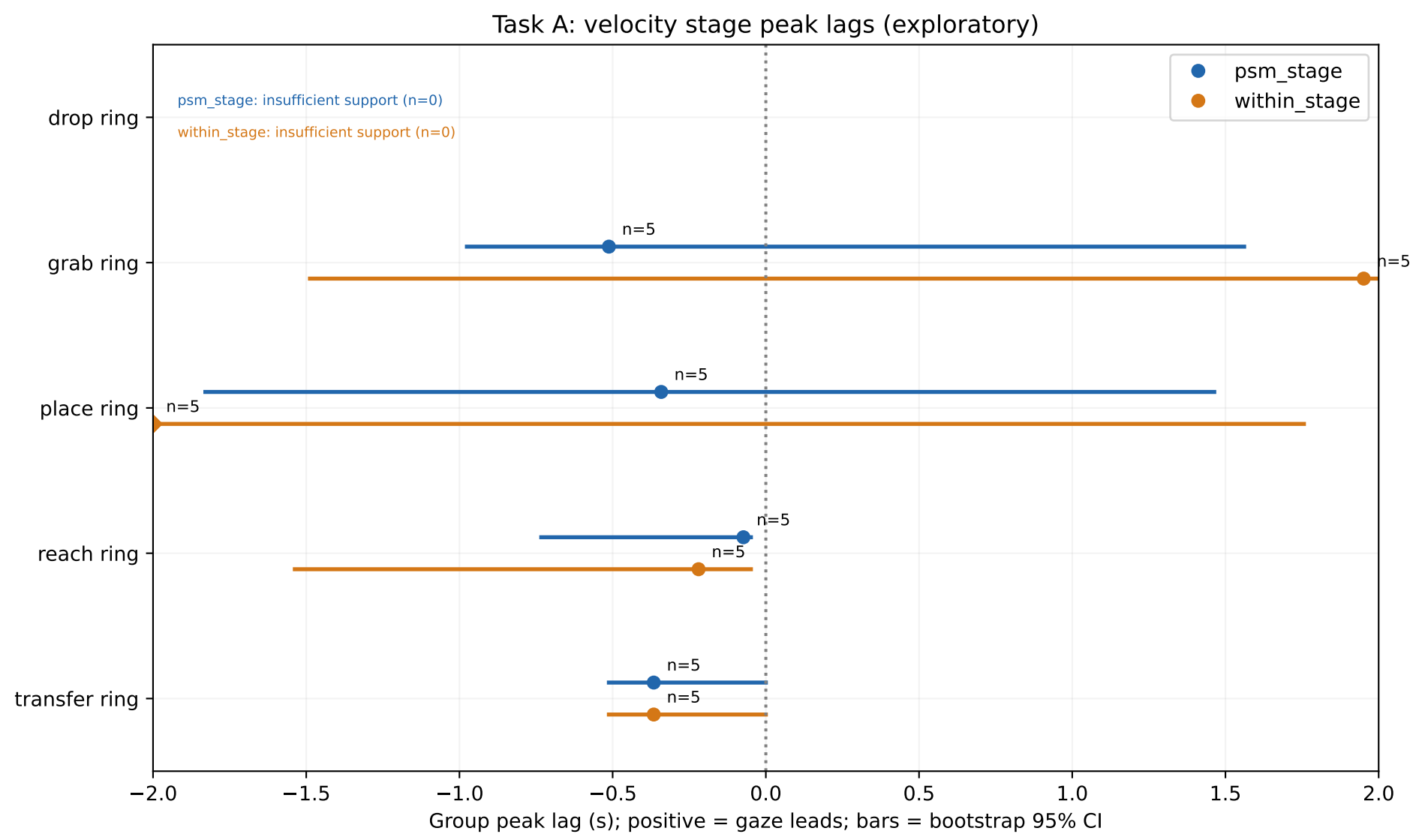
wrap around psm / within_stage



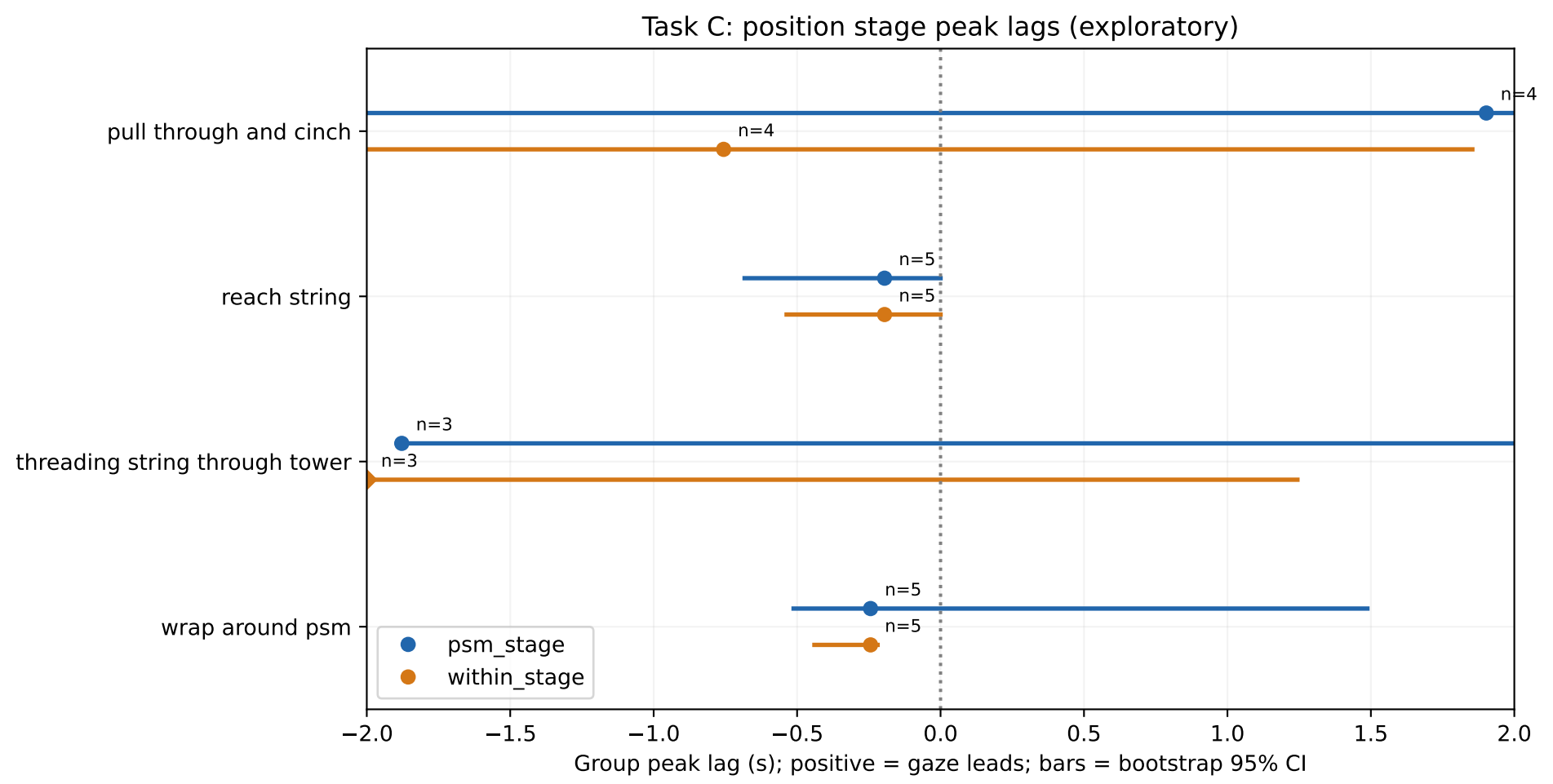
One PSM selected per recording/stage by largest signed xcorr peak; selection does not identify the operated instrument.



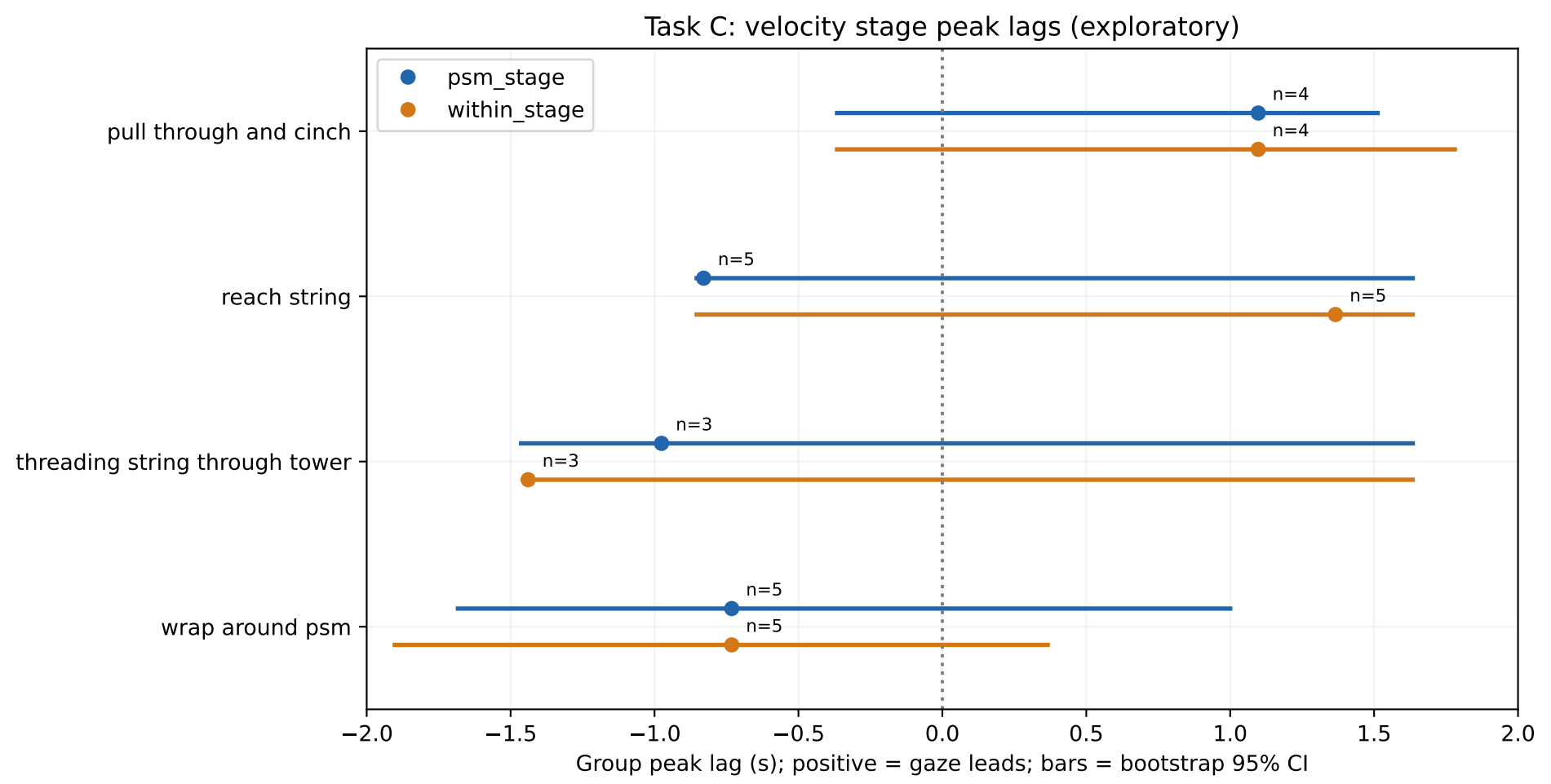
One PSM selected per recording/stage by largest signed xcorr peak; selection does not identify the operated instrument.



One PSM selected per recording/stage by largest signed xcorr peak; selection does not identify the operated instrument.



One PSM selected per recording/stage by largest signed xcorr peak; selection does not identify the operated instrument.



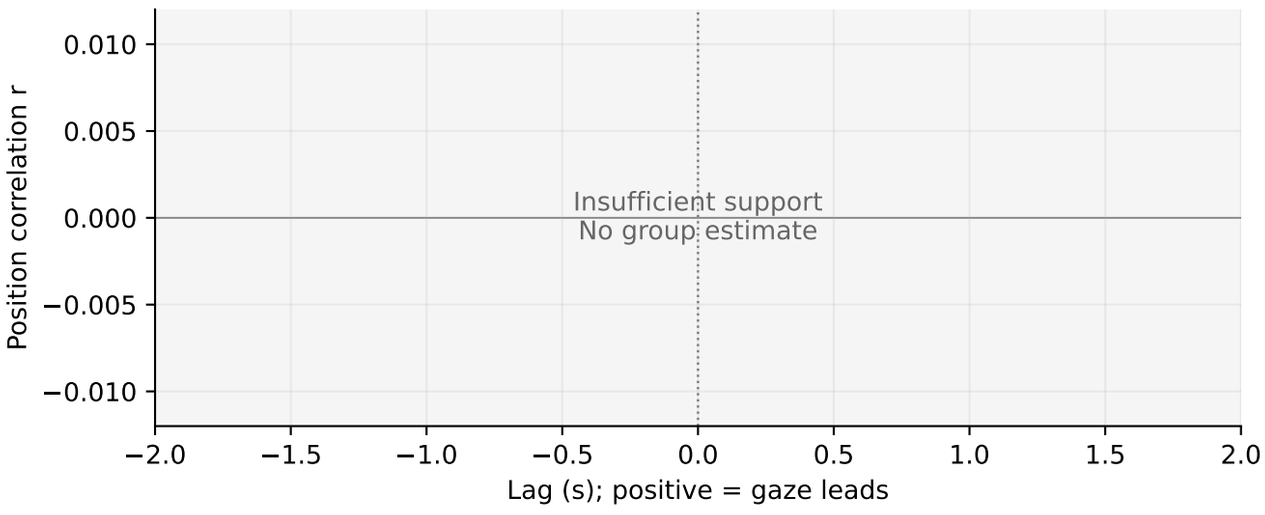
One PSM selected per recording/stage by largest signed xcorr peak; selection does not identify the operated instrument.

Task A: position, maximum-xcorr PSM per stage

Unpaired reference (95%) | Participant bootstrap (95%) | Group mean

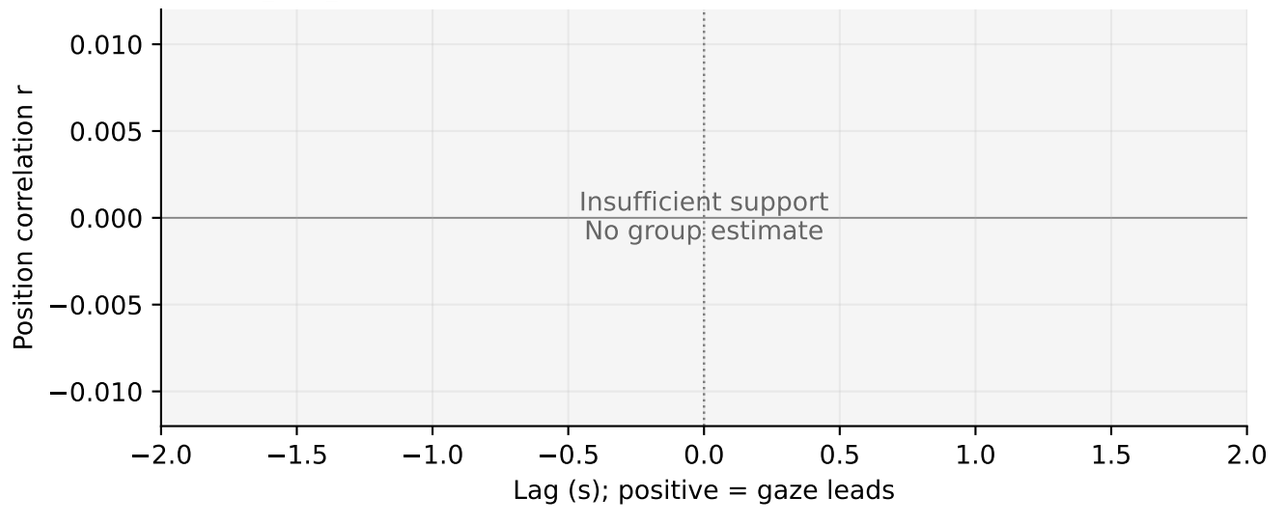
drop ring / psm_stage

n = 0; no_valid_curve



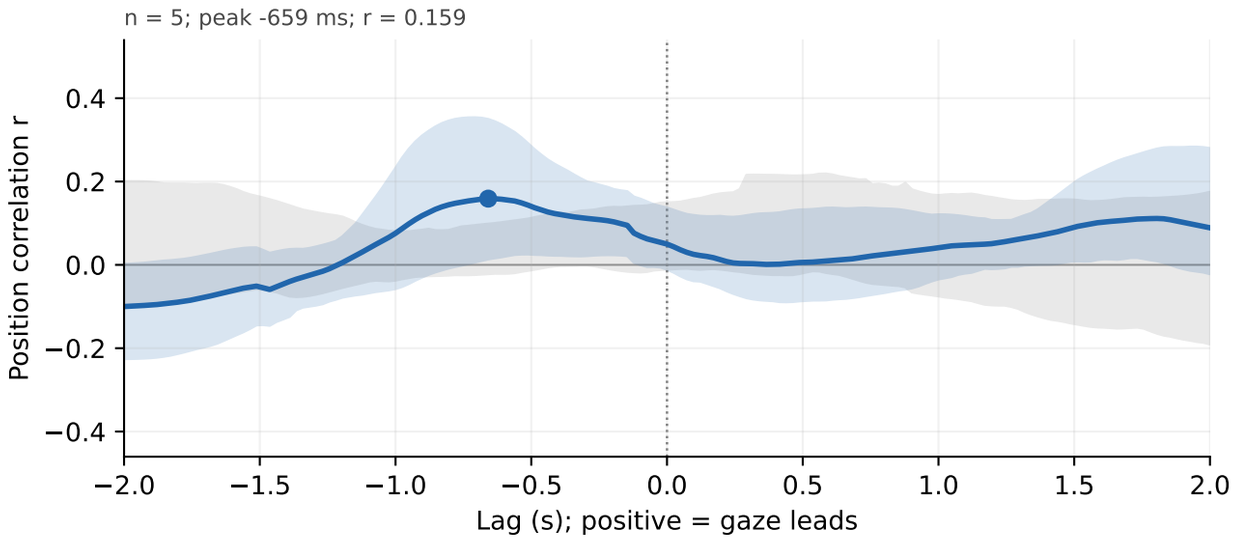
drop ring / within_stage

n = 0; no_valid_curve



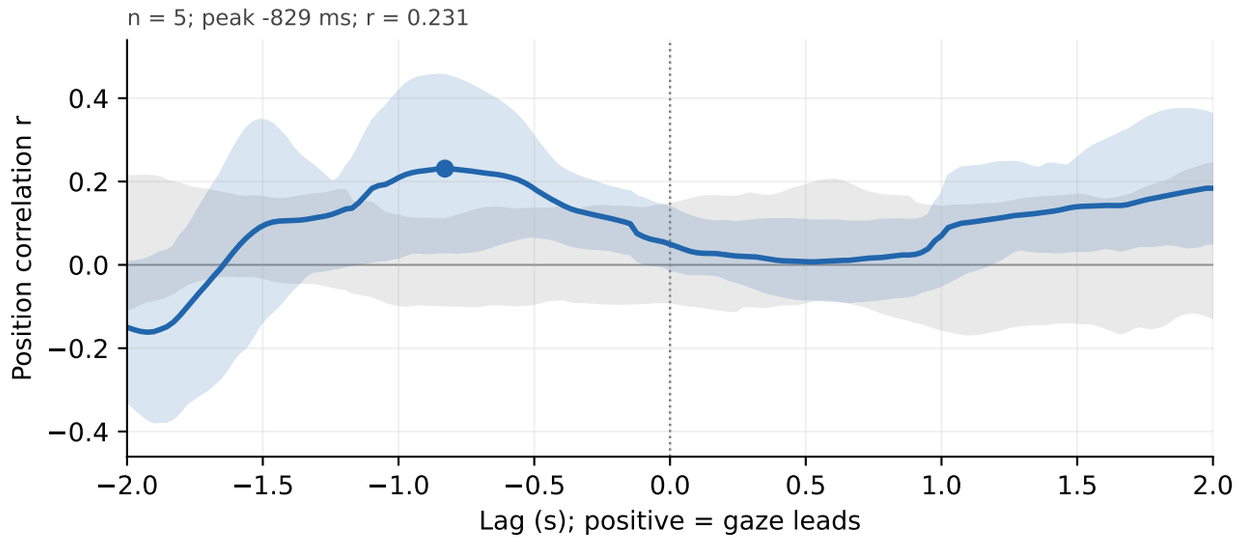
grab ring / psm_stage

n = 5; peak -659 ms; r = 0.159



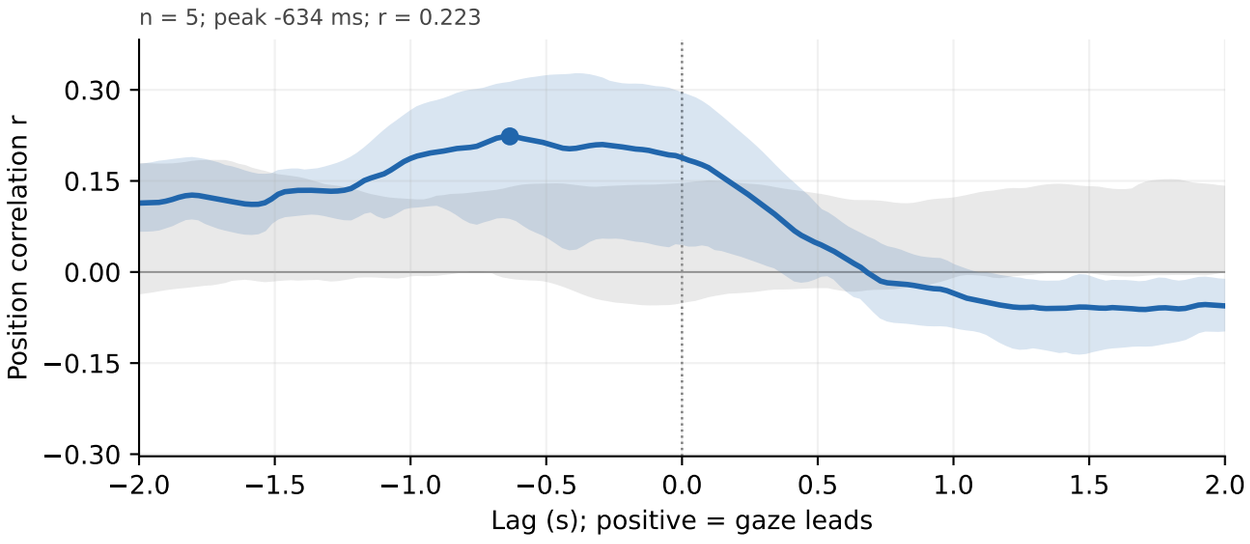
grab ring / within_stage

n = 5; peak -829 ms; r = 0.231



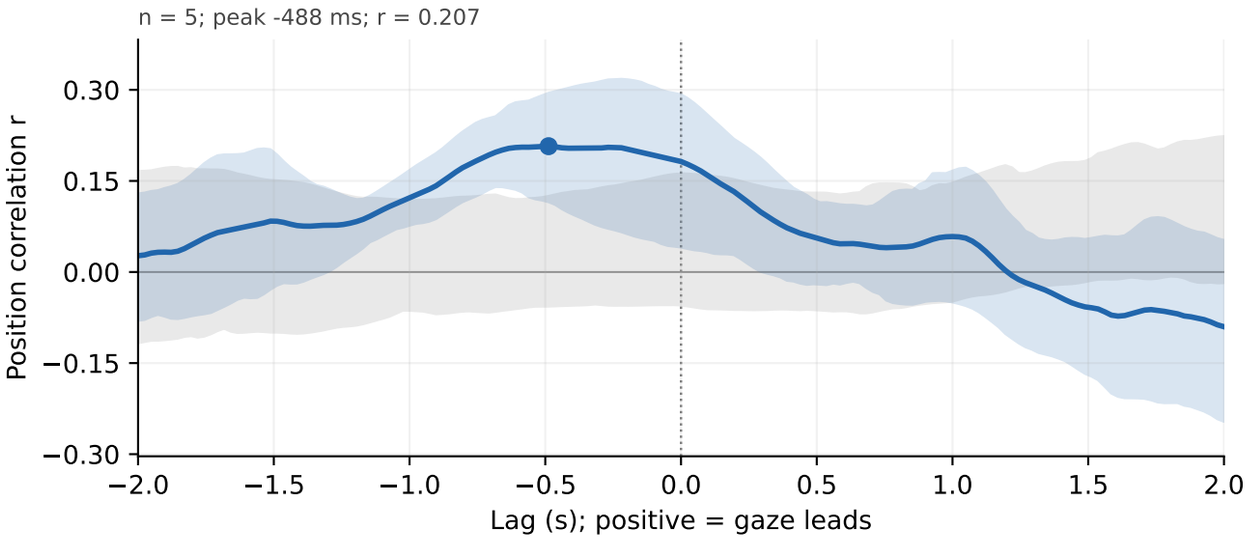
place ring / psm_stage

n = 5; peak -634 ms; r = 0.223



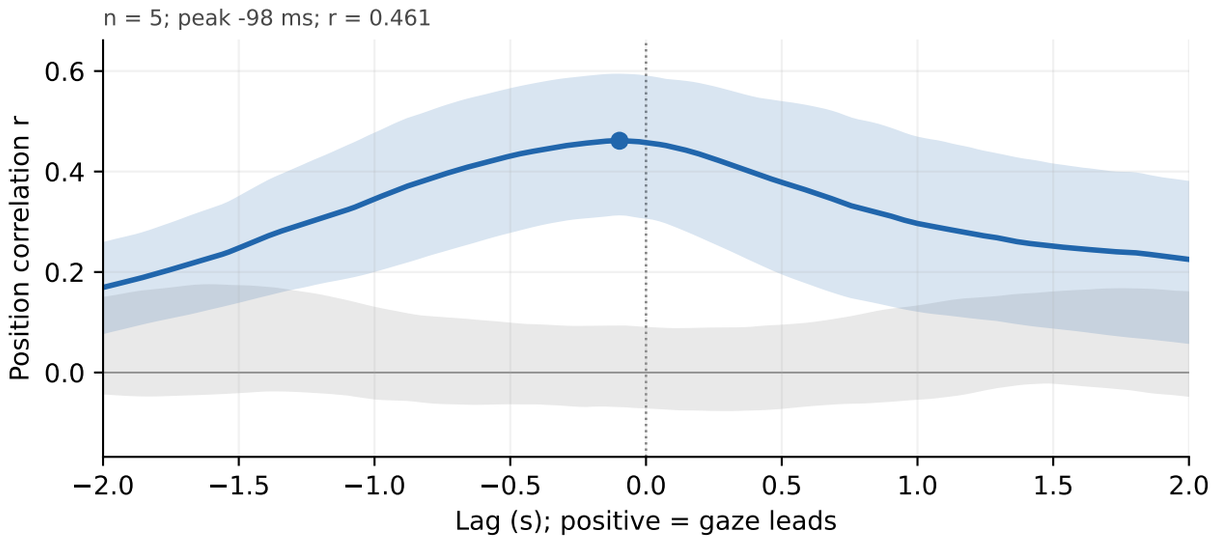
place ring / within_stage

n = 5; peak -488 ms; r = 0.207



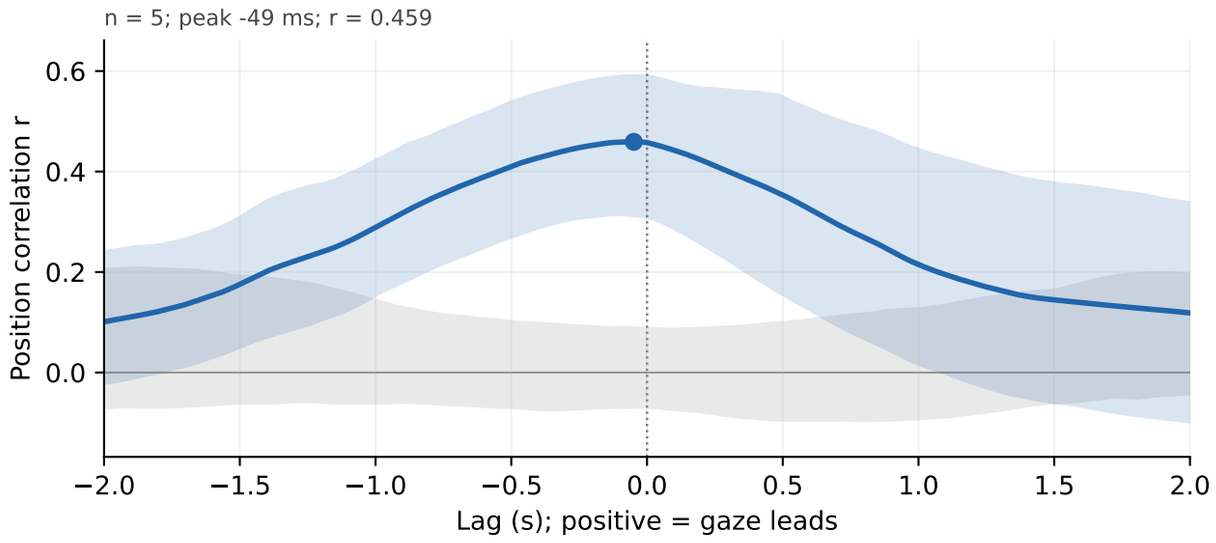
reach ring / psm_stage

n = 5; peak -98 ms; r = 0.461



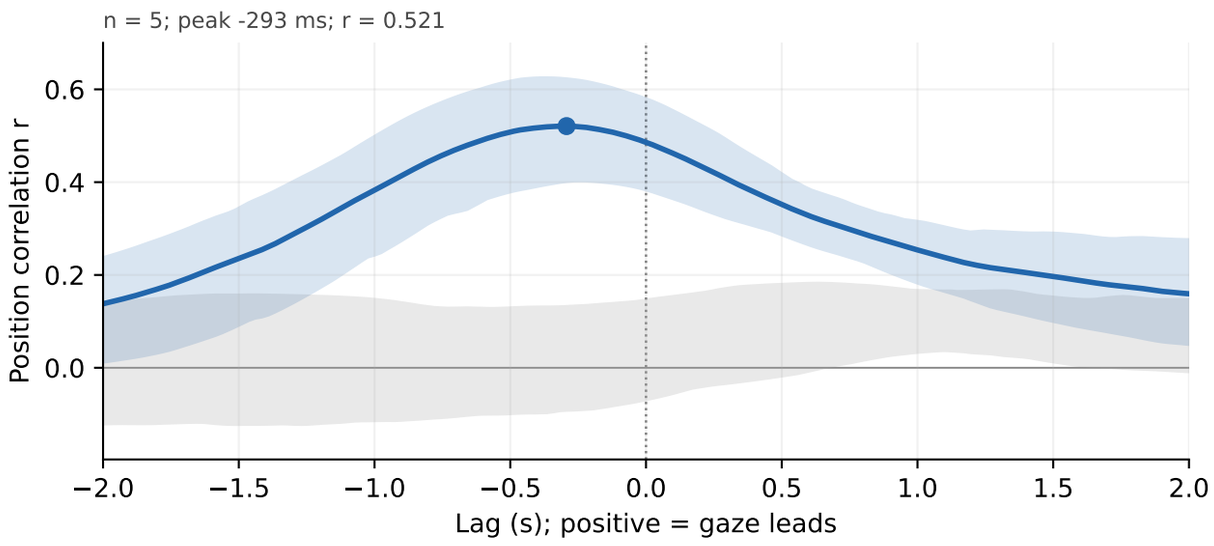
reach ring / within_stage

n = 5; peak -49 ms; r = 0.459



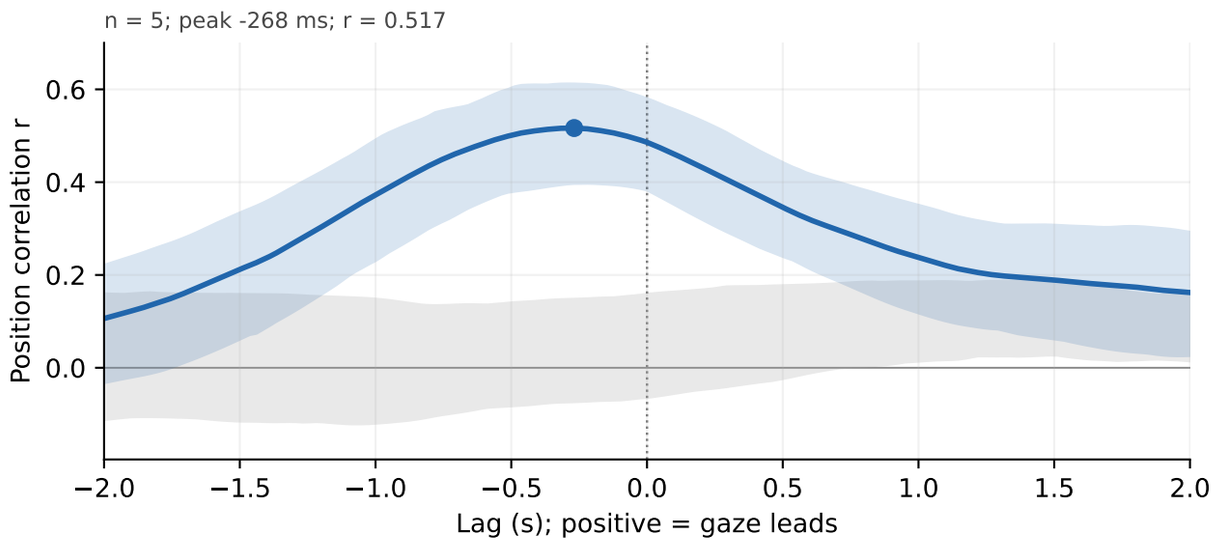
transfer ring / psm_stage

n = 5; peak -293 ms; r = 0.521



transfer ring / within_stage

n = 5; peak -268 ms; r = 0.517

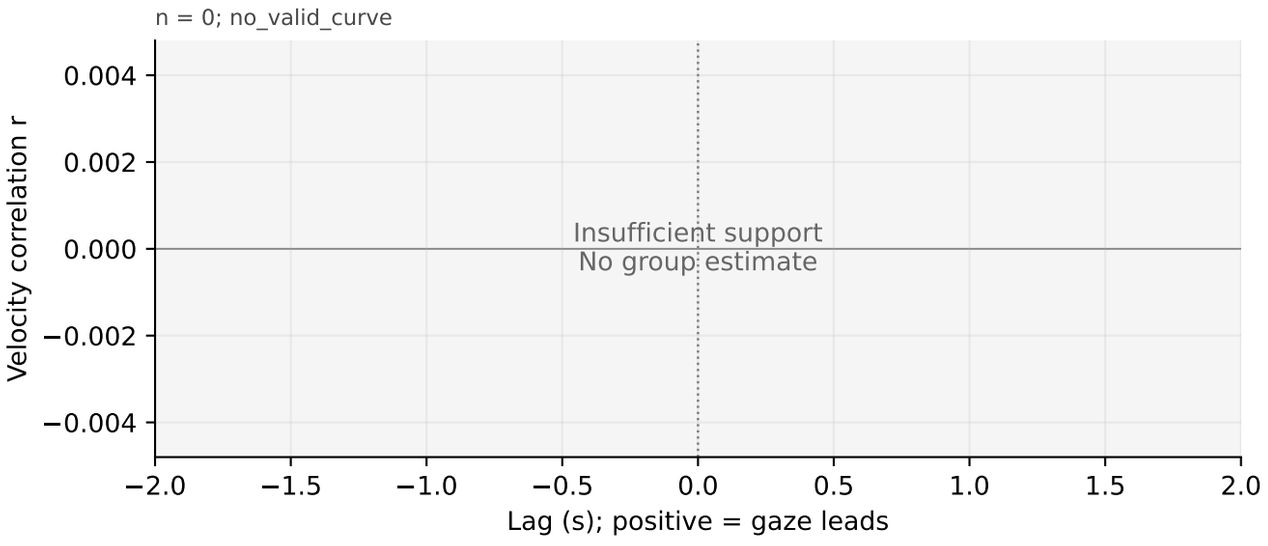


One PSM selected per recording/stage by largest signed xcorr peak; selection does not identify the operated instrument.

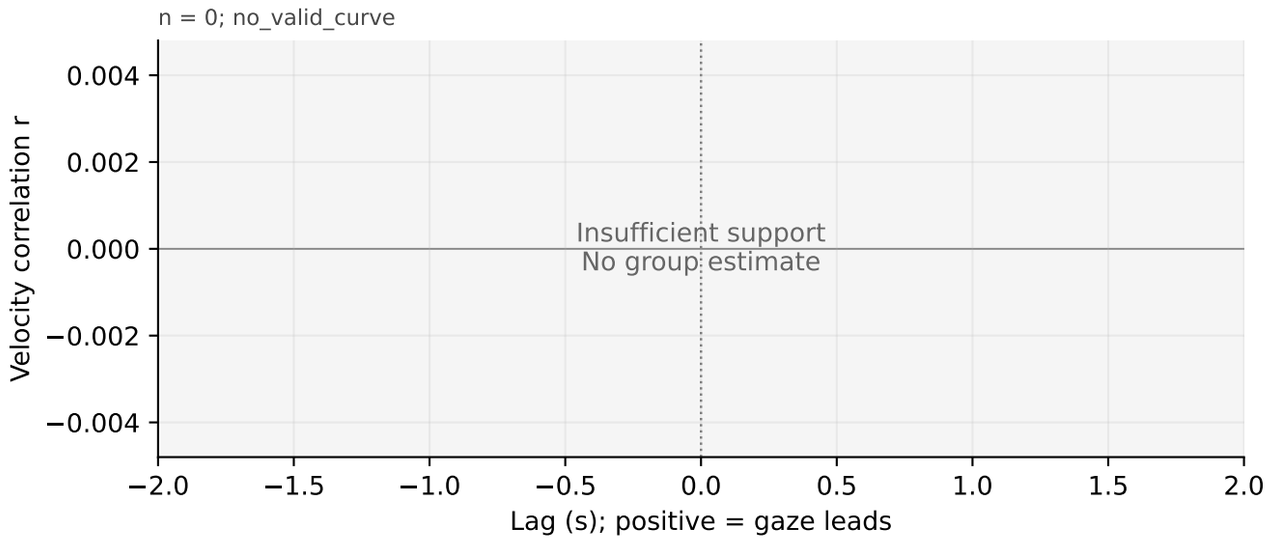
Task A: velocity, maximum-xcorr PSM per stage

Unpaired reference (95%) · Participant bootstrap (95%) · Group mean

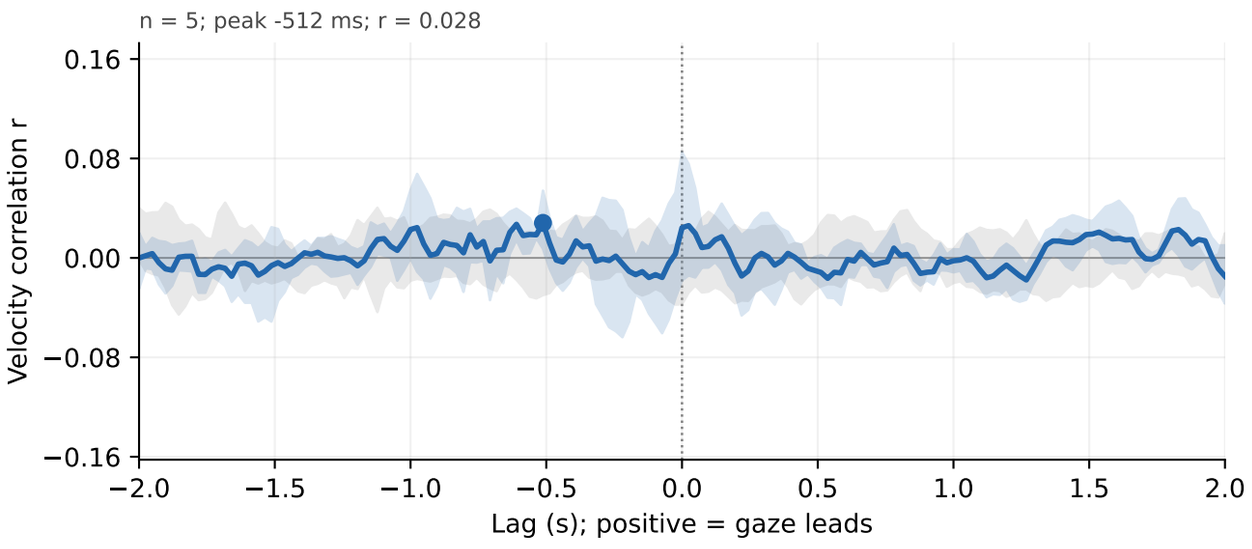
drop ring / psm_stage



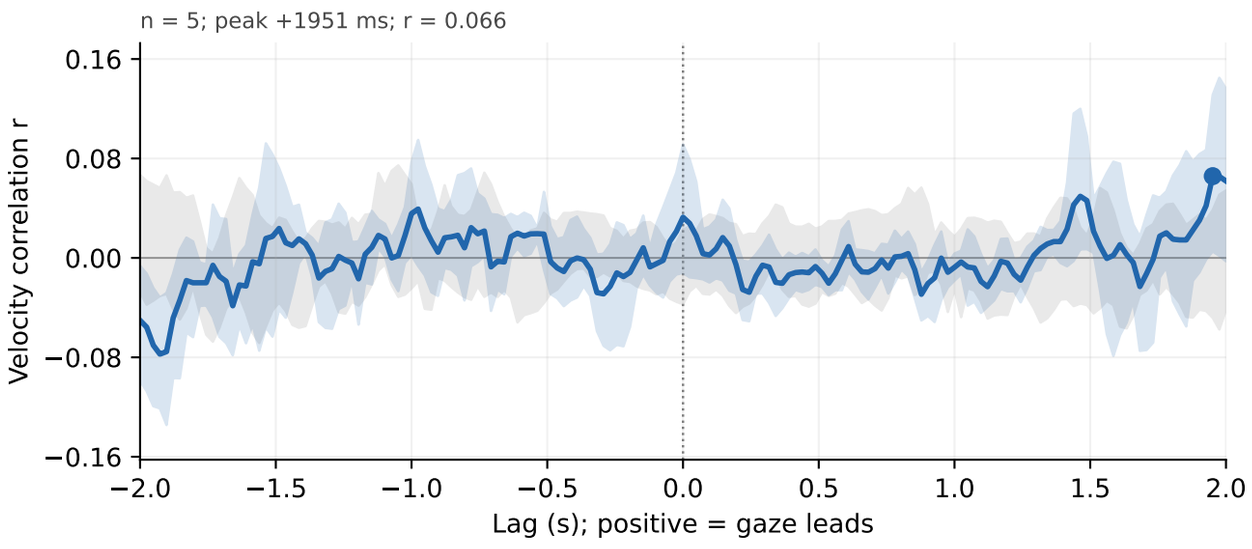
drop ring / within_stage



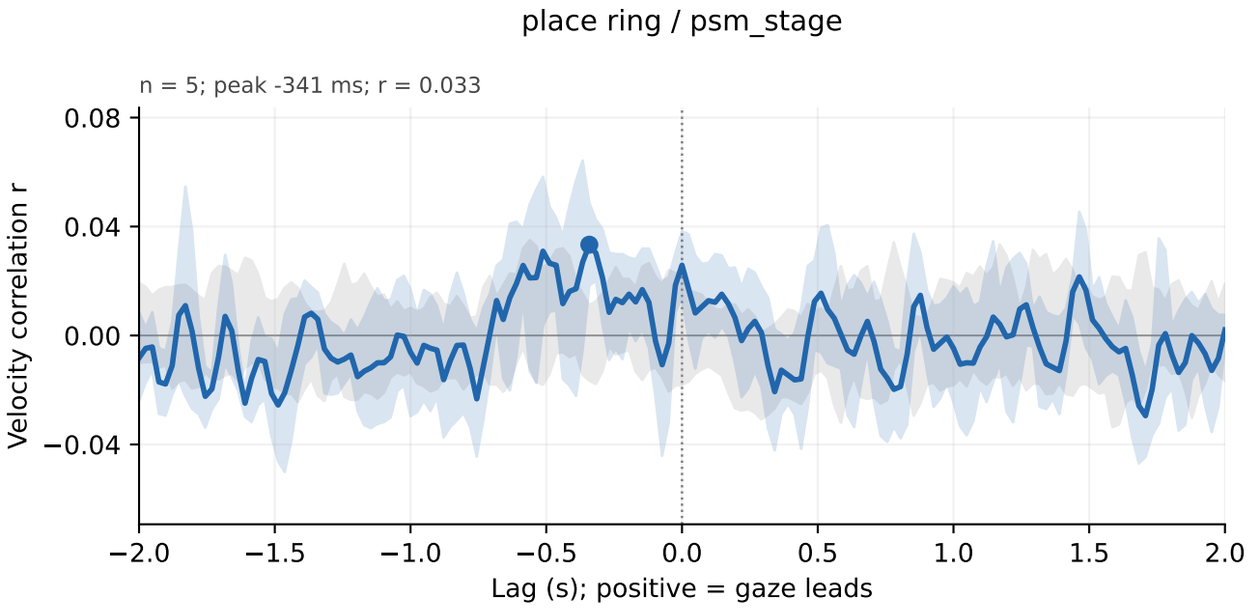
grab ring / psm_stage



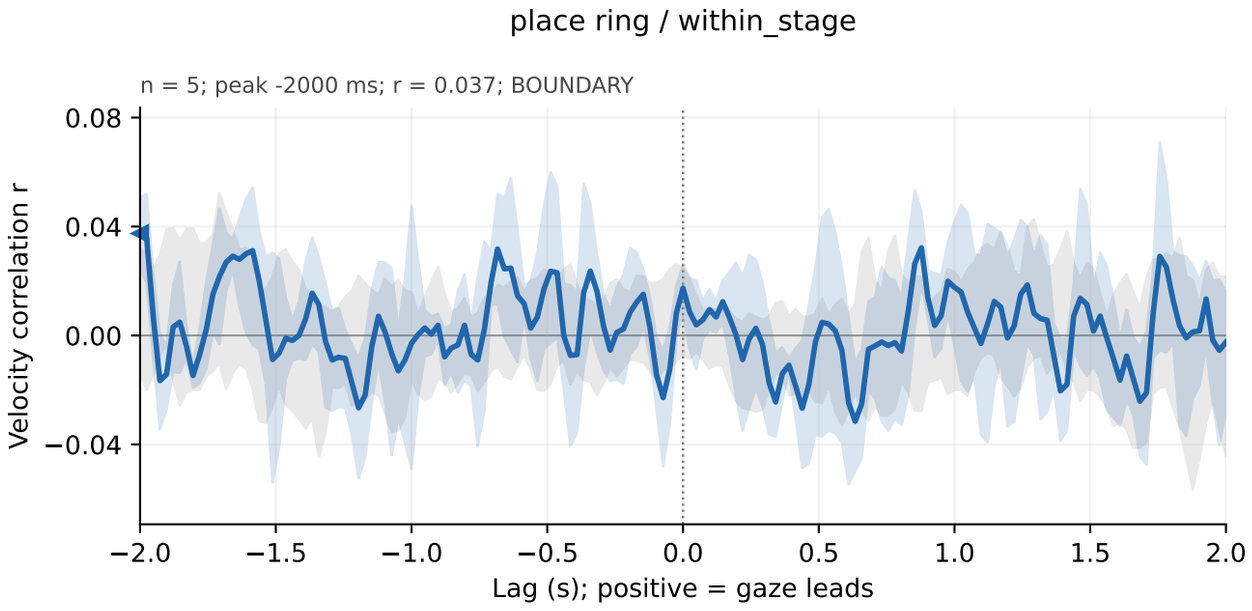
grab ring / within_stage



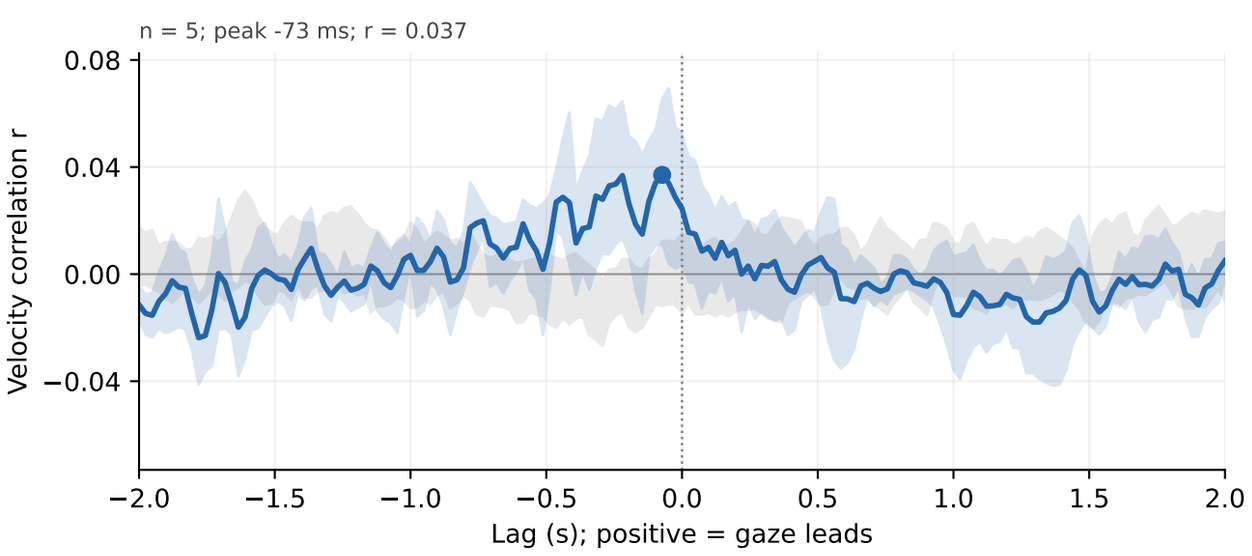
place ring / psm_stage



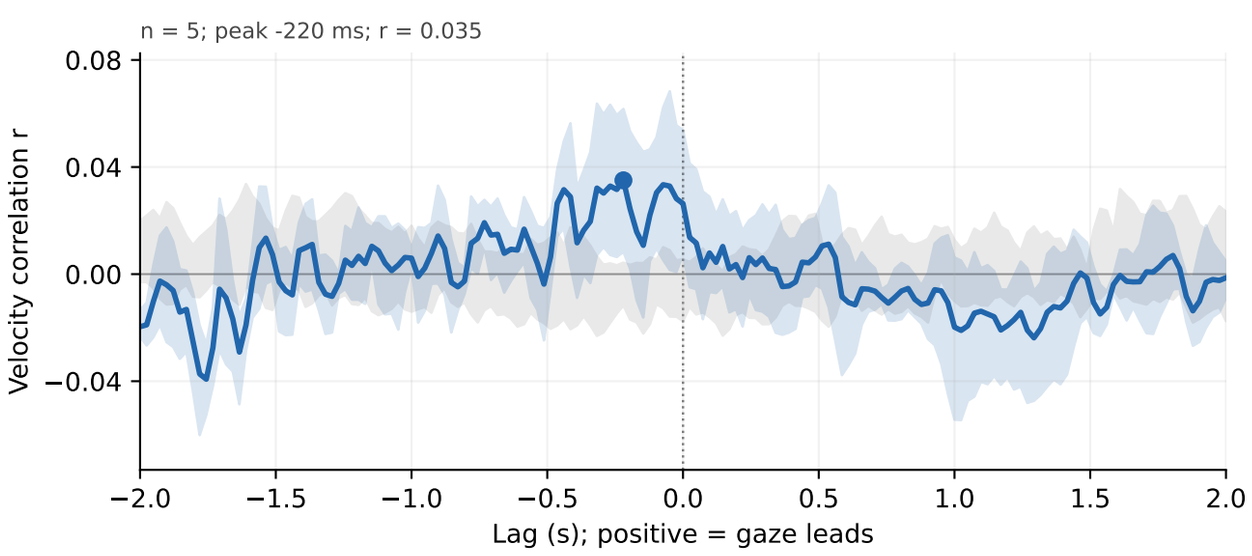
place ring / within_stage



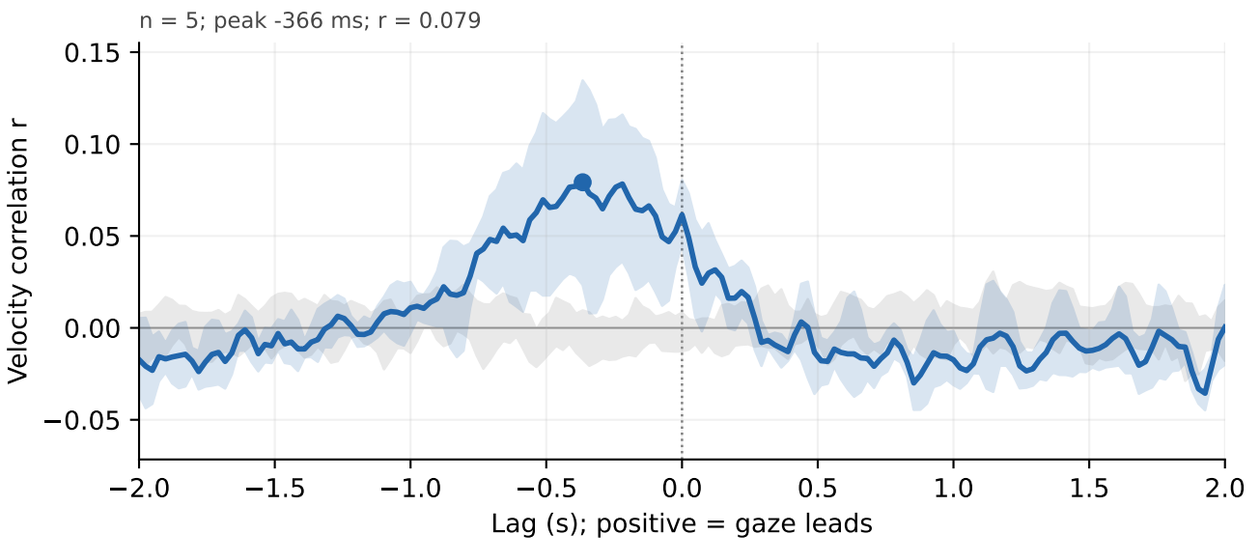
reach ring / psm_stage



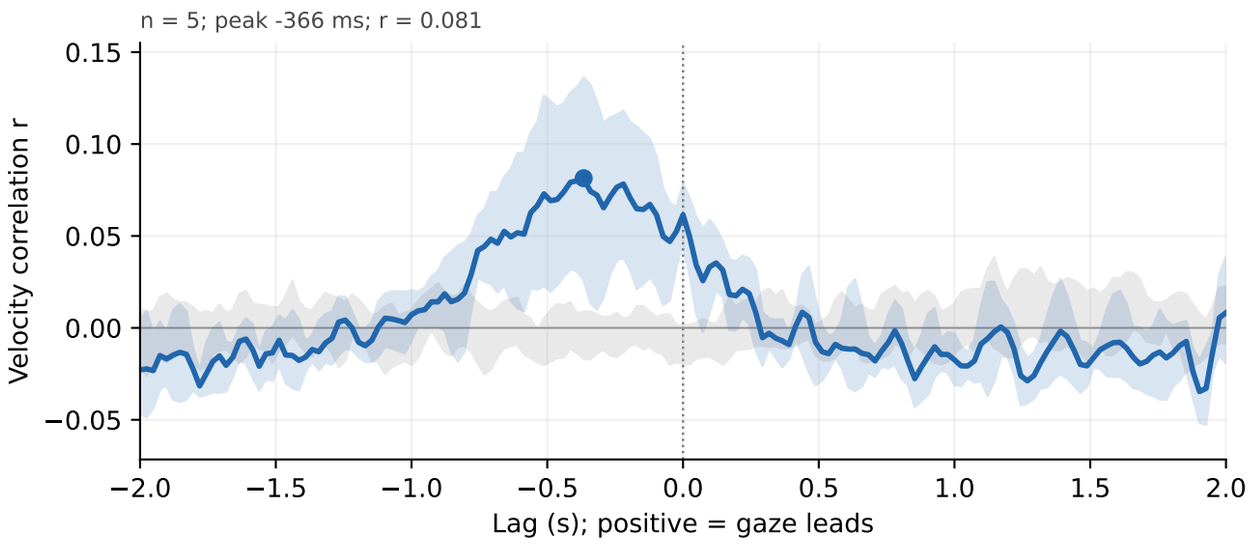
reach ring / within_stage



transfer ring / psm_stage



transfer ring / within_stage

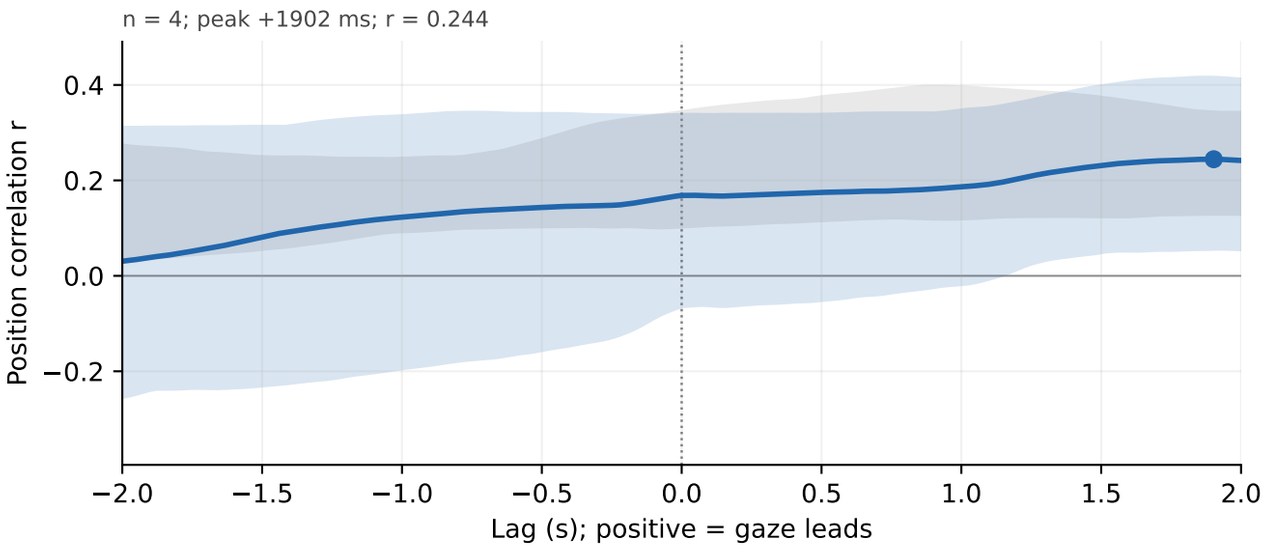


One PSM selected per recording/stage by largest signed xcorr peak; selection does not identify the operated instrument.

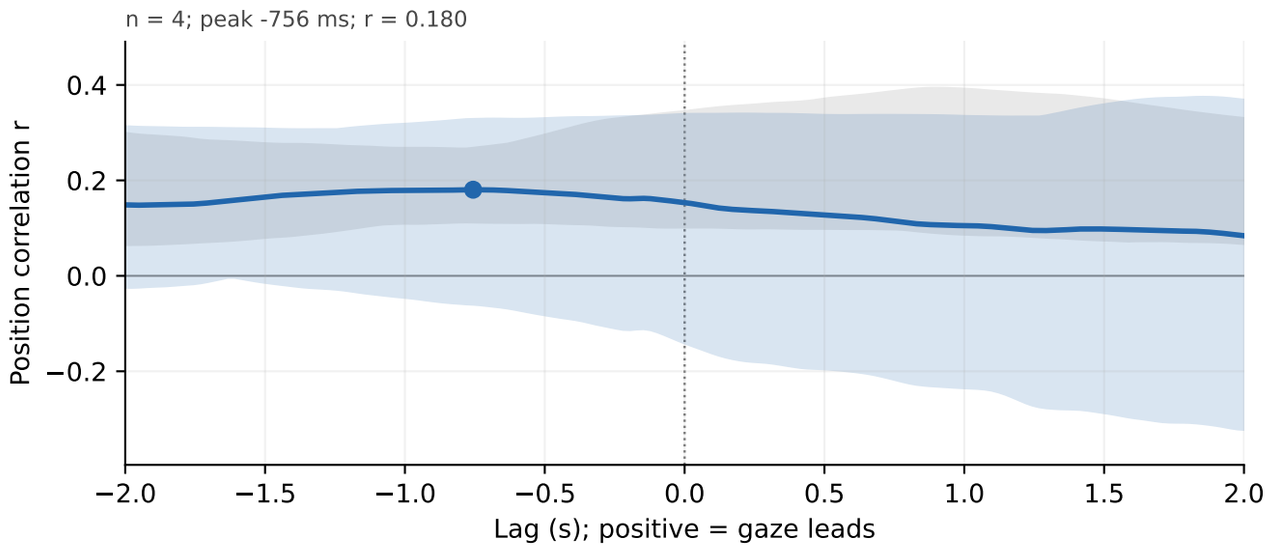
Task C: position, maximum-xcorr PSM per stage

Unpaired reference (95%) | Participant bootstrap (95%) | Group mean

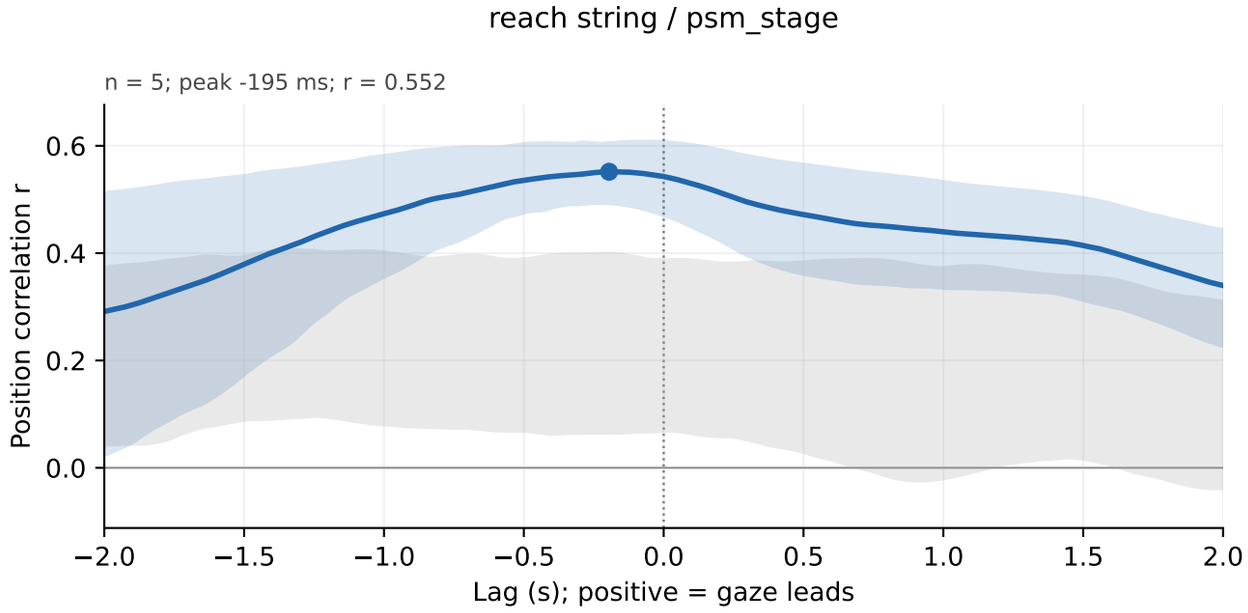
pull through and cinch / psm_stage



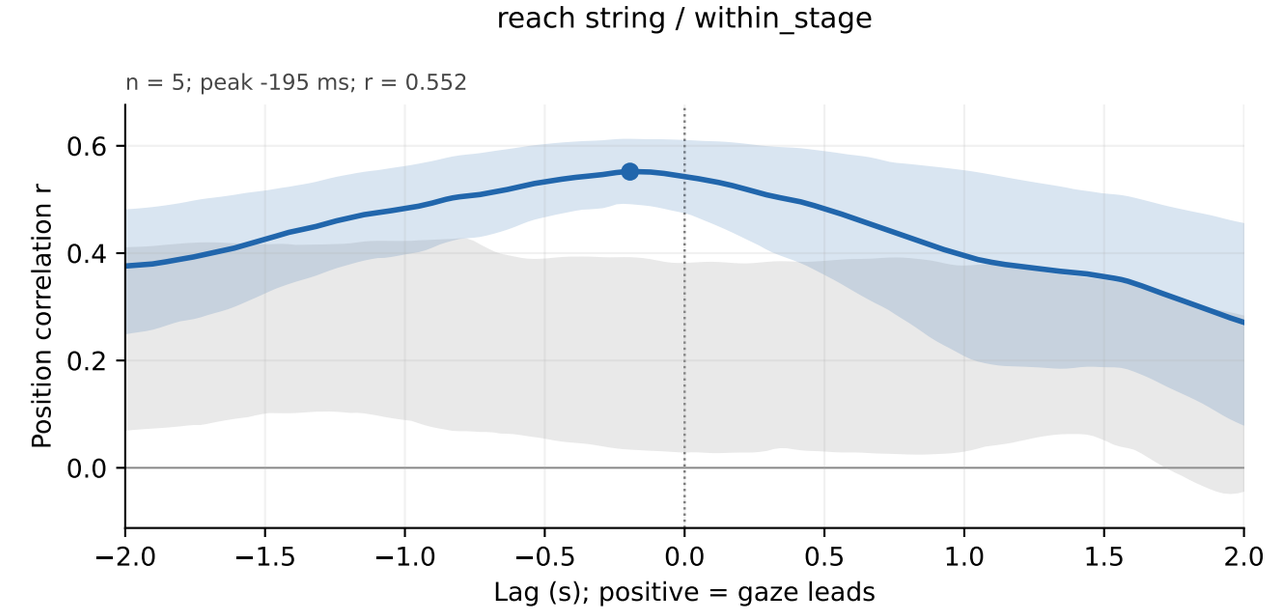
pull through and cinch / within_stage



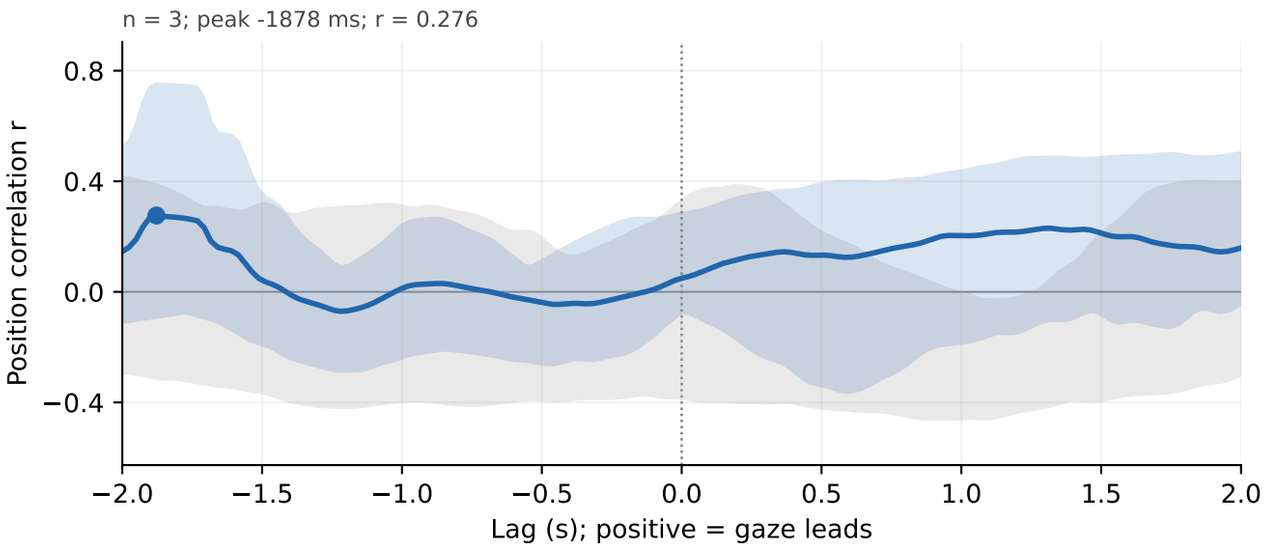
reach string / psm_stage



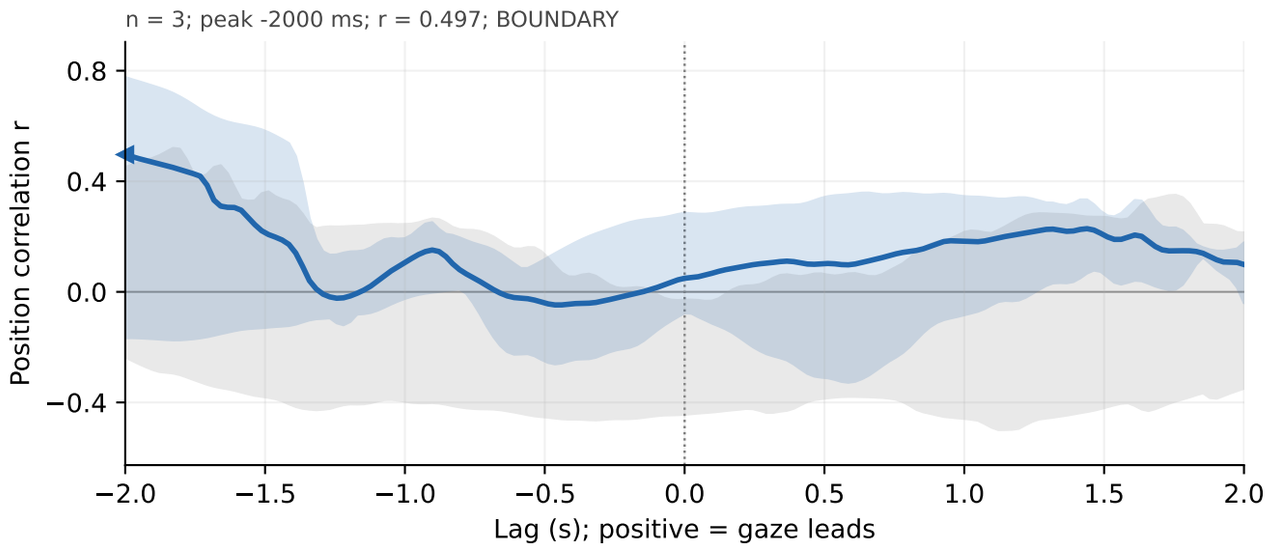
reach string / within_stage



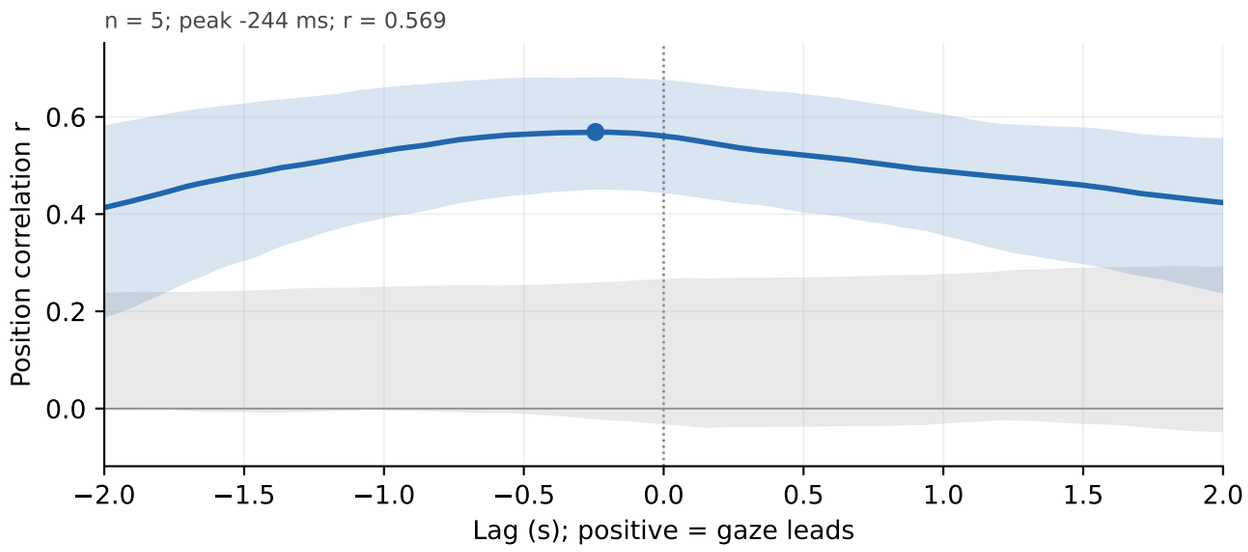
threading string through tower / psm_stage



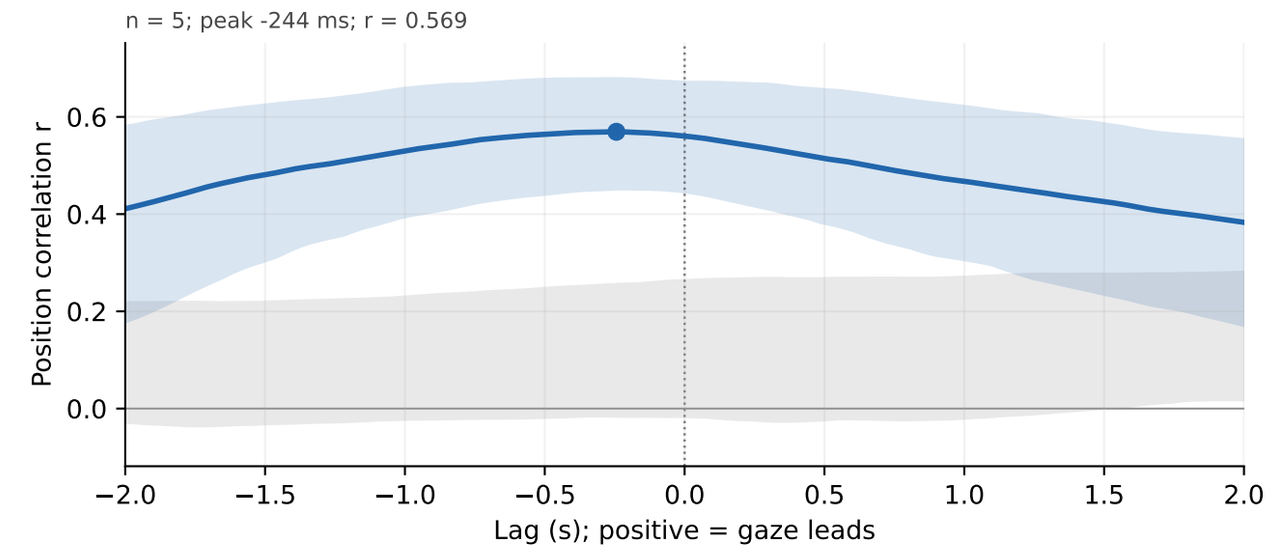
threading string through tower / within_stage



wrap around psm / psm_stage



wrap around psm / within_stage

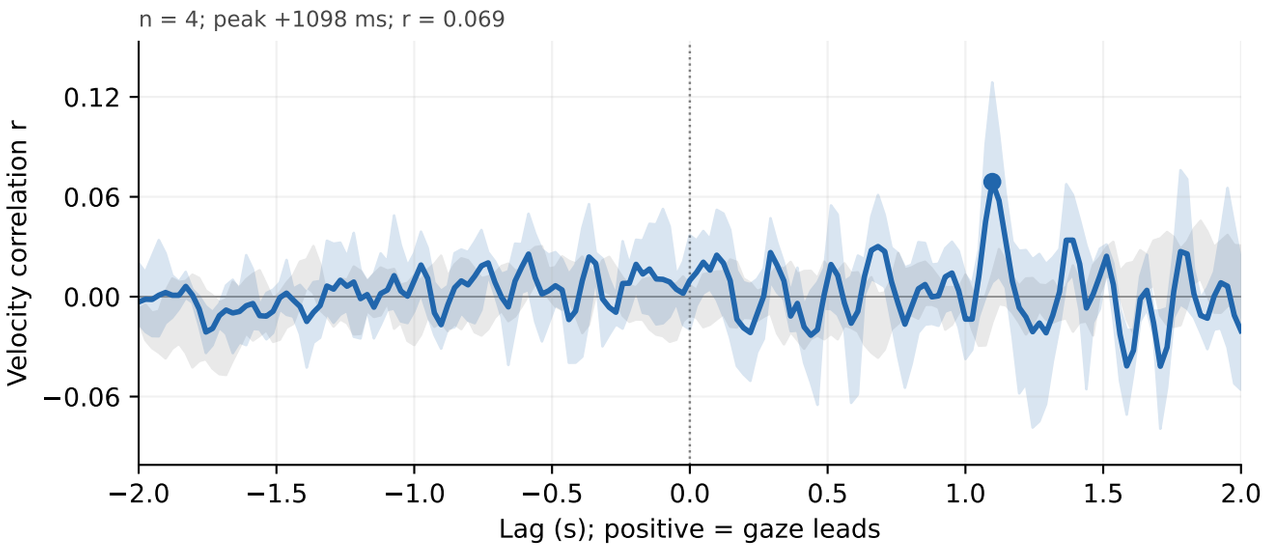


One PSM selected per recording/stage by largest signed xcorr peak; selection does not identify the operated instrument.

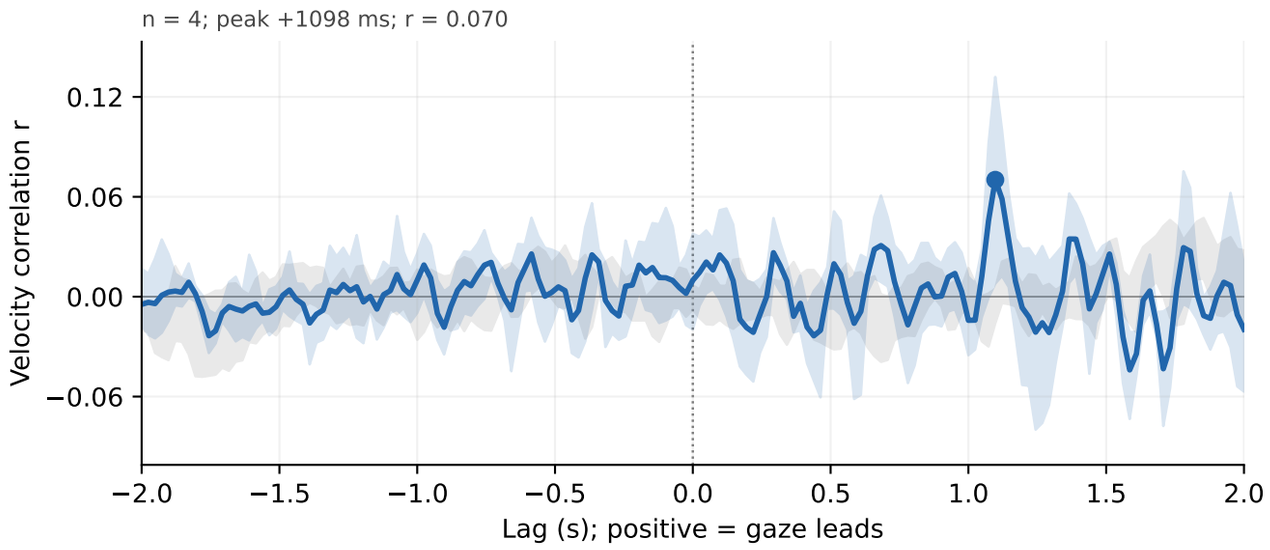
Task C: velocity, maximum-xcorr PSM per stage

Unpaired reference (95%) | Participant bootstrap (95%) | Group mean

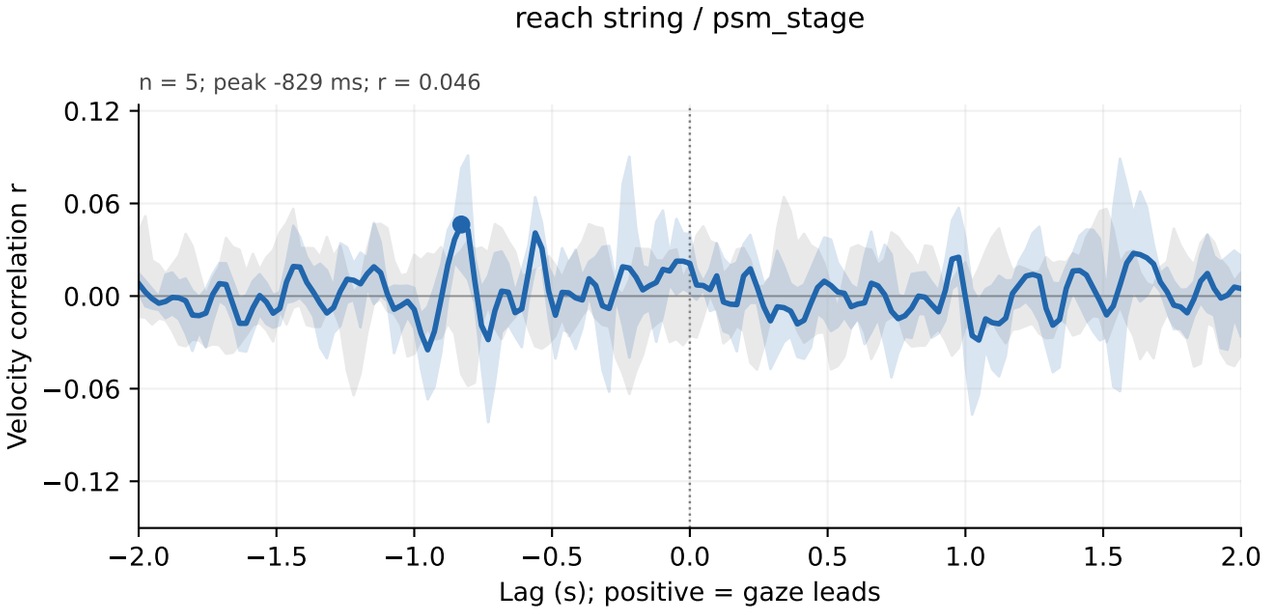
pull through and cinch / psm_stage



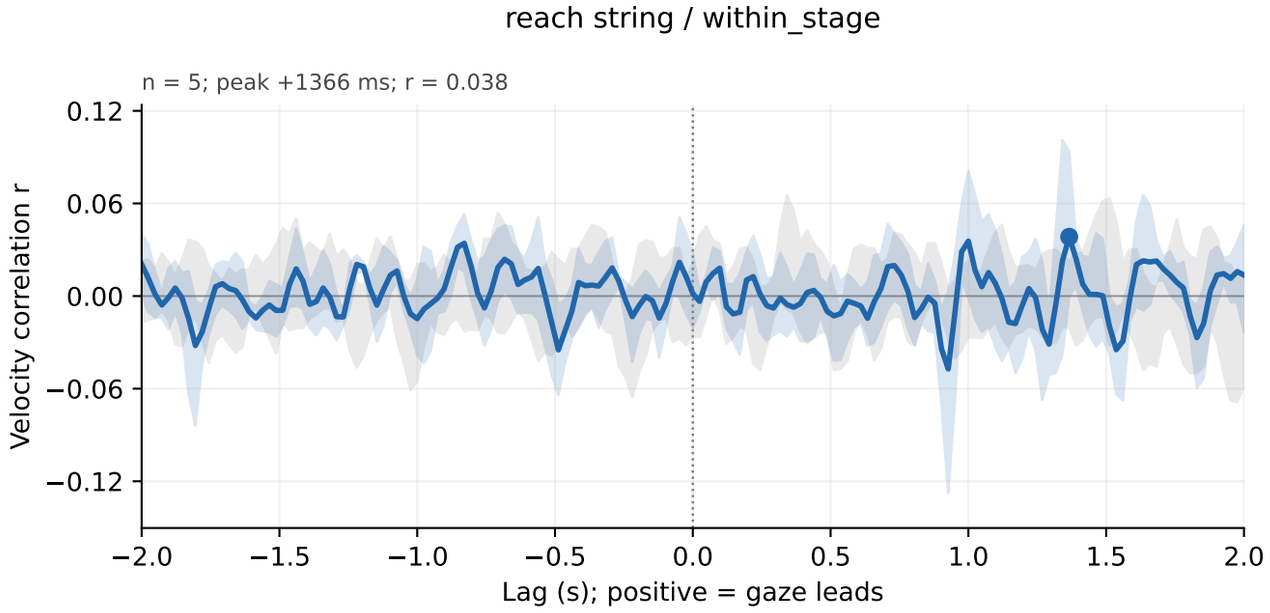
pull through and cinch / within_stage



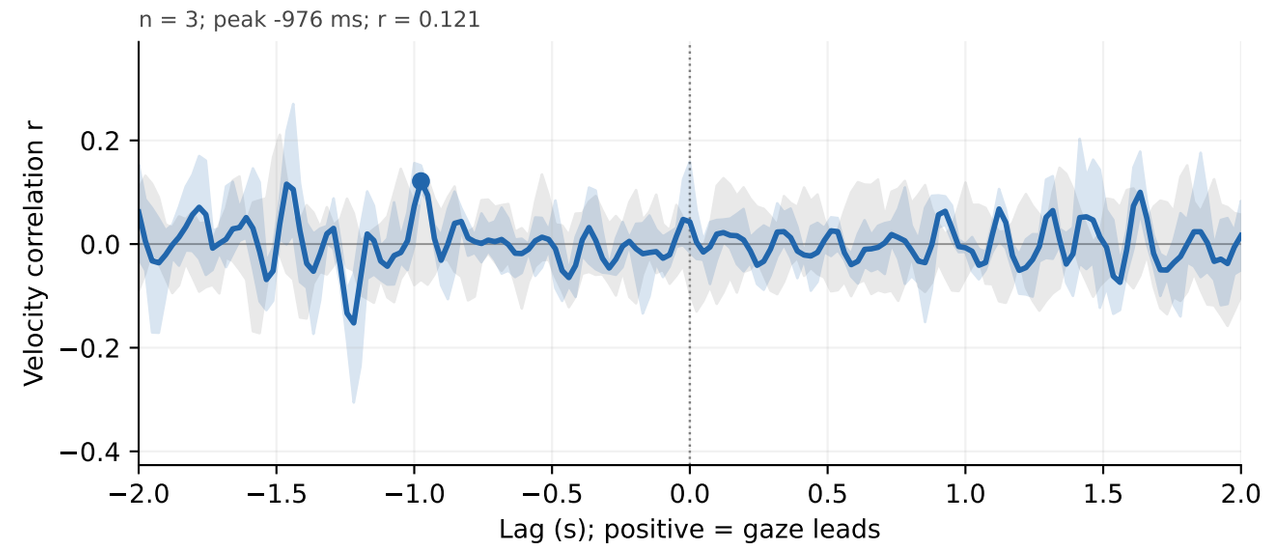
reach string / psm_stage



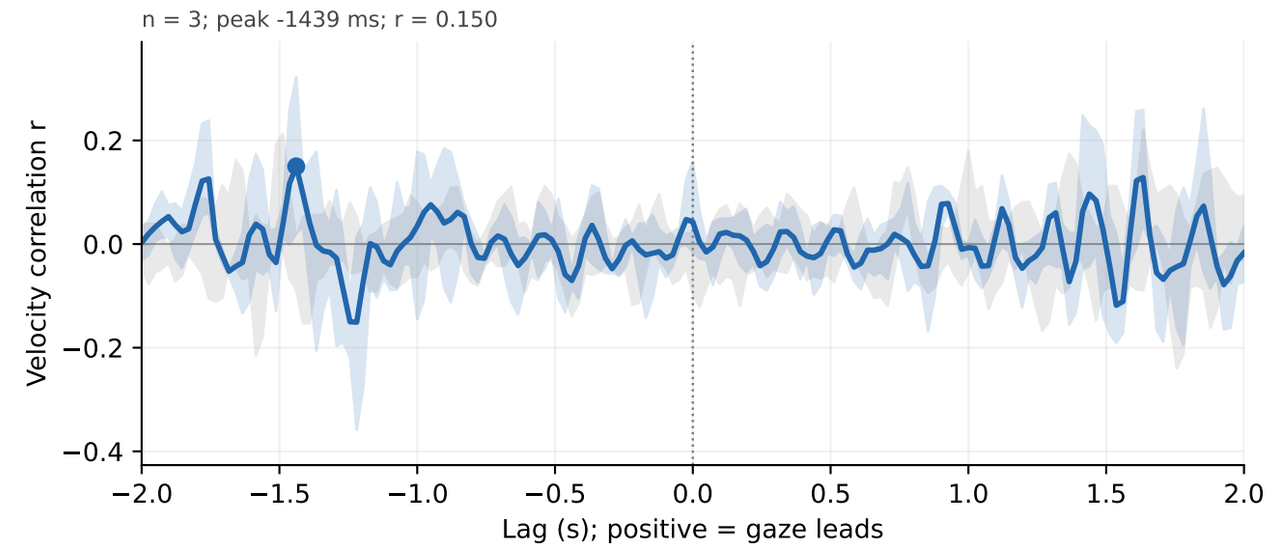
reach string / within_stage



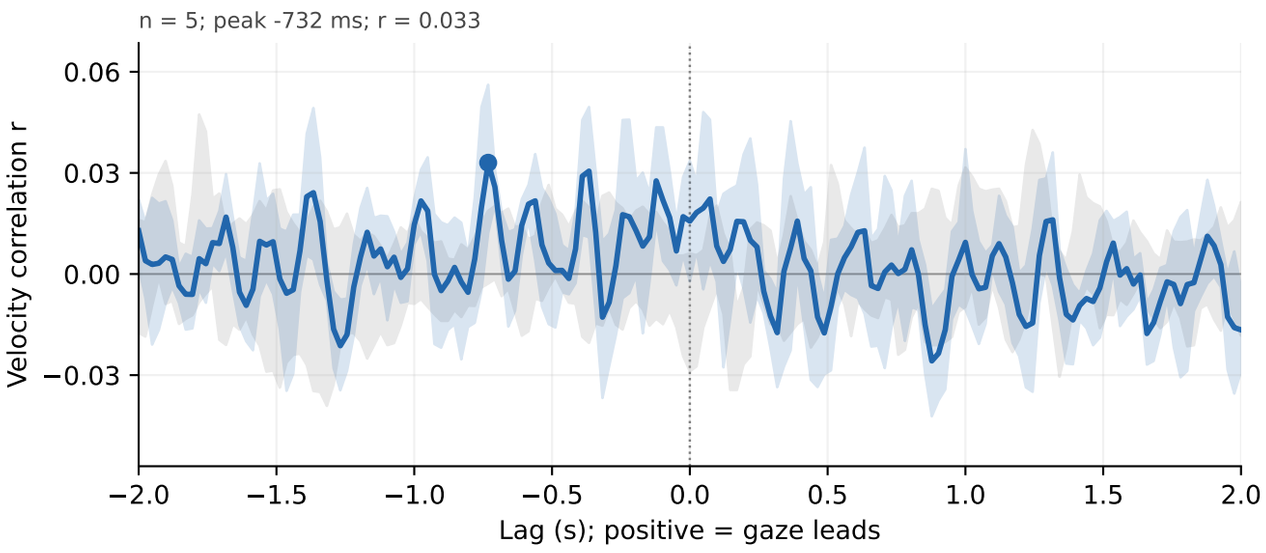
threading string through tower / psm_stage



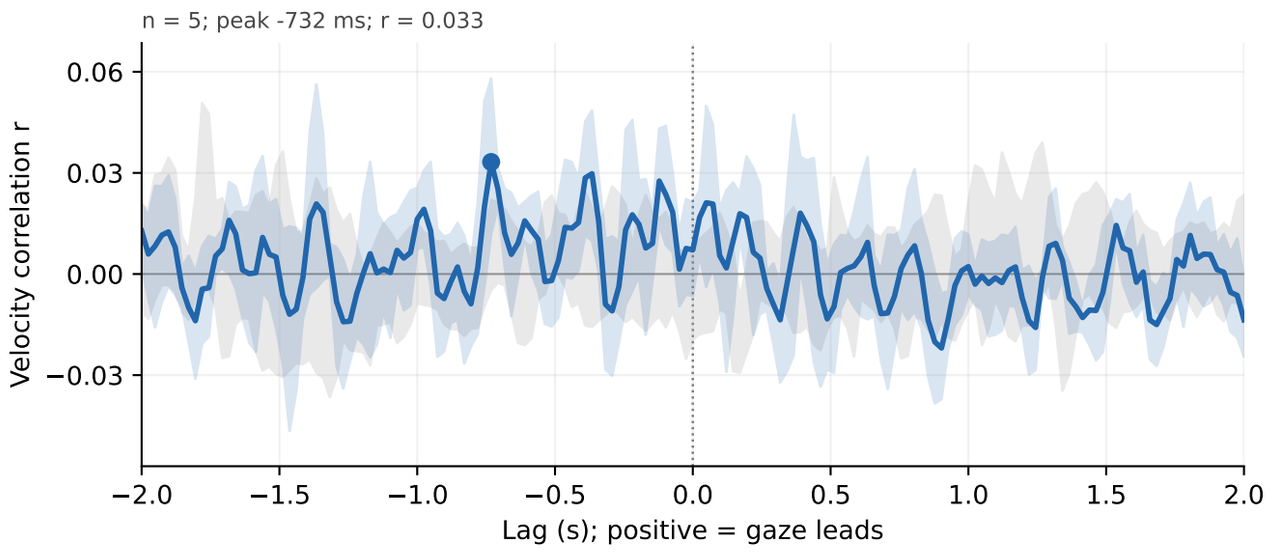
threading string through tower / within_stage



wrap around psm / psm_stage



wrap around psm / within_stage



One PSM selected per recording/stage by largest signed xcorr peak; selection does not identify the operated instrument.